We should start with administrative stuff.

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- Text: Concentration inequalities: a nonasymptotic theory of independence by Boucheron, Lugosi, Massart (BLM). With various other references, possibly including Ledoux's monograph.

1 Examples and basic results

We will be trudging through the BLM book, so these notes will be more or less an explanation of their text. We will begin with Chapter 2, with some additional remarks and examples.

Given a random variable X, we will be interested in how well X may be approximated by its expected value $\mathbb{E}X$. This is a question about how deterministic X is. In many applications, this greatly simplifies the problem, and hopefully we will see some of these applications. One way to phrase it is: what is an optimal bound for

$$\mathbb{P}\left(|X - \mathbb{E}X| \geq \epsilon\right) \text{ ?}$$

An equivalent question related to probability measure μ is: how well concentrated is μ around a particular value? In the setting above, μ is simply the distribution of X and this value is $\mathbb{E}X$. Generally, for a Borel probability measure μ on a metric space, we seek a bound for

$$\mu(B_\epsilon(a)^c) \ .$$

This will generally turn out to be possible when X is a well-behaved function of many independent random variables.

1. Take $X_1, X_2, \ldots$ to be i.i.d. bounded random variables and set, for $n \geq 1$, $A_n = \sum_{i=1}^n X_n/n$. Then $\mathbb{E}A_n = \mathbb{E}X_1$ and by the (weak) law of large numbers,

   $$\mathbb{P}\left(|A_n - \mathbb{E}A_n| \geq \epsilon\right) \to 0 \text{ for } \epsilon > 0 \ .$$

   But how quickly does this approach zero? In fact, one can show the upper bound

   $$c \exp\left(-cn\epsilon^2\right) \ .$$

   This is actually a concentration result for the sequence of measures given by the distributions of the A_n's, all about the same point, $\mathbb{E}X_1$. We can prove this relatively easily later.

2. As an application, take $X_1, X_2, \ldots$ to be uniformly distributed on $[-1,1]$, so that $X = (X_1, \ldots, X_n)$ is uniformly distributed on $[-1,1]^n$. Then the square of the ℓ^2 norm:
$$\|X\|_2^2 = X_1^2 + \cdots + X_n^2$$
is a sum of i.i.d. random variables with
$$\mathbb{E}X_1^2 = \frac{1}{2}\int_{-1}^{1} x^2 \, \mathrm{d}x = \frac{1}{6}\left[1^3 - (-1)^3\right] = \frac{1}{3} \, .$$
Therefore
$$\mathbb{P}\left(\left|\|X\|_2^2/n - \frac{1}{3}\right| > \epsilon\right) \leq c\exp\left(-cn\epsilon^2\right) \, .$$
Therefore the distribution of $X = (X_1, \ldots, X_n)$ is concentrated near the points where $\|X\|_2 = \sqrt{\frac{n}{3}}$.

3. Take $X_1, X_2, \ldots$ to be i.i.d. random variables supported in $[0,1]$ and, for each n, let f_n be a bounded 1-Lipschitz function on $\mathbb{R}^n$ (relative to ℓ^1 norm). Then
$$\mathbb{P}\left(|f_n - \mathbb{E}f_n| \geq \epsilon\right) \leq 2\exp\left(-\frac{2\epsilon^2}{n}\right) \, .$$
This is a generalization of the first example.

We will begin by studying concentration properties of the most basic function of independent variables: the sum.

1.1 Basic estimates

For a little while, we will give some of the basic concentration results from graduate probability. They are generally ways to obtain upper bounds for deviations of random variables given bounds on their moments (or expectation of certain functions of the variable).

1. (Markov's inequality) If $X \geq 0$ has finite mean then
$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}X}{a} \text{ for } a > 0 \, .$$
Proof. Write
$$\mathbb{E}X = \mathbb{E}\int_0^X 1 \, \mathrm{d}t = \mathbb{E}\int_0^\infty \mathbf{1}_{\{X \geq t\}} \, \mathrm{d}t = \int_0^\infty \mathbb{P}(X \geq t) \, \mathrm{d}t \, .$$
So
$$\mathbb{E}X \geq \int_0^a \mathbb{P}(X \geq t) \, \mathrm{d}t \geq a\mathbb{P}(X \geq a) \, .$$
□

2. (Chebyshev's inequality) If X has $\mathbb{E}X^2 < \infty$ then

$$\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \frac{\text{Var } X}{a^2} \text{ for } a > 0 \ .$$

Proof. Apply Markov to $(X - \mathbb{E}X)^2$:

$$\mathbb{P}(|X - \mathbb{E}X| \geq a) = \mathbb{P}((X - \mathbb{E}X)^2 \geq a^2) \leq \frac{\text{Var } X}{a^2} \ .$$

□

3. (Cramer-Chernoff bound) Applying Markov to the exponential and taking infimum over $\lambda > 0$,

$$\mathbb{P}(X \geq t) = \mathbb{P}(e^{\lambda X} \geq e^{\lambda t}) \leq \inf_{\lambda > 0} e^{-\lambda t} \mathbb{E} e^{\lambda X}$$

This also holds for $\lambda = 0$. If we write $\psi_X(\lambda) = \log \mathbb{E} e^{\lambda X}$, then this reads

$$\begin{aligned} \mathbb{P}(X \geq t) &\leq \exp\left(- \sup_{\lambda \geq 0} (\lambda t - \psi_X(\lambda)) \right) \\ &= \exp\left(-\psi_X^*(t)\right) \ , \end{aligned}$$

where for a function $f : \mathbb{R} \to \mathbb{R}$, f^* is the transform

$$f^*(y) = \sup_{x \geq 0} (yx - f(x)) \ .$$

4. Although the above bound will be perhaps the most useful for us, it is actually theoretically inferior to a power bound. Indeed, for a nonnegative X, and $t \geq 0$,

$$\inf_n t^{-n} \mathbb{E} X^n \leq \inf_{\lambda \geq 0} e^{-\lambda t} \mathbb{E} e^{\lambda X} \ .$$

To see why, call the left side a and compute for $\lambda \geq 0$,

$$\mathbb{E} e^{\lambda X} = \sum_{n=0}^{\infty} \frac{\mathbb{E} X^n}{n!} \lambda^n \geq a \sum_{n=0}^{\infty} \frac{t^n}{n!} \lambda^n = a e^{\lambda t} \ .$$

From item 3, we are quite interested in cumulant generating functions; that is, the functions $\psi_X(t) = \log \mathbb{E} e^{tX}$, or moment generating functions, given by $e^{\psi_X(t)}$. From them, we will be able to derive some concentration properties, so often we will analyze them as a proxies for the tails of distributions. Let's go through some cases.

1.2 Gaussian

Recall that a centered Gaussian variable with variance σ^2 has density function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right) .$$

We can compute its moment generating function by a change of variable. If X is Gaussian with mean zero and variance 1 (written $N(0,1)$),

$$\begin{aligned}
\mathbb{E}e^{tX} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{x^2}{2}} \, \mathrm{d}x &= e^{\frac{t^2}{2}} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{x^2-2tx+t^2}{2}} \, \mathrm{d}x \\
&= e^{\frac{t^2}{2}} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-t)^2}{2}} \, \mathrm{d}x \\
&= e^{\frac{t^2}{2}} .
\end{aligned}$$

So we can extend this to $N(0,\sigma^2)$ (variance σ^2) by noting the scaling identity

$$X_\sigma = \sigma X_1 \text{ in distribution} ,$$

where X_σ has distribution $N(0,\sigma^2)$ and X_1 has distribution $N(0,1)$. Therefore an $N(0,\sigma^2)$ variable has moment generating function

$$\mathbb{E}e^{tX_\sigma} = \mathbb{E}e^{t\sigma X_1} = e^{\frac{t^2\sigma^2}{2}} .$$

In conclusion, the cumulant generating function is

$$\psi_{X_\sigma}(t) = \log \mathbb{E}e^{tX_\sigma} = \frac{t^2\sigma^2}{2} .$$

Now let's compute the "Cramer transform" for a $N(0,\sigma^2)$ variable.

$$\psi_X^*(\lambda) = \sup_{t\in\mathbb{R}} (\lambda t - \psi_X(t)) = \sup_{t\in\mathbb{R}} \left(\lambda t - \frac{t^2\sigma^2}{2}\right) .$$

Just take the derivative and set it to 0:

$$0 = \lambda - \sigma^2 t \Rightarrow t = \frac{\lambda}{\sigma^2} ,$$

so since this is the location of the maximum (second derivative is negative),

$$\psi_X^*(\lambda) = \frac{\lambda^2}{\sigma^2} - \frac{\lambda^2}{2\sigma^2} = \frac{\lambda^2}{2\sigma^2} .$$

Putting this in Chernoff's bound, we obtain

$$\mathbb{P}(X \geq t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right) .$$

The same argument works for $\mathbb{P}(X < -t)$. This is not far off, as we know that

$$\mathbb{P}(X \geq t) = \frac{1}{\sqrt{2\pi}\sigma} \int_t^\infty \exp\left(-\frac{x^2}{2\sigma^2}\right)\, \mathrm{d}x \ .$$

Motivated by the form of the moment generating function, we can define a class of variables that are at least as concentrated as Gaussians. The definition is in terms of the moment generating function.

Definition 1.1. *A variable X is said to be* **sub-gaussian** *with variance factor ν if*

$$\psi_X(t) \leq \frac{t^2\nu}{2} \text{ for all } t \in \mathbb{R} \ .$$

We write $X \in \mathcal{G}(\nu)$.

Here are some useful properties of subgaussian variables.

- If X_1, X_2 are independent with $X_i \in \mathcal{G}(\nu_i)$ for $i = 1, 2$ then $X_1 + X_2 \in \mathcal{G}(\nu_1 + \nu_2)$.

  Proof. By independence,

$$\log \mathbb{E}e^{t(X_1+X_2)} = \log \mathbb{E}e^{tX_1} + \log \mathbb{E}e^{tX_2} \leq \frac{t^2(\nu_1+\nu_2)}{2} \ .$$

  □

- If $X \in \mathcal{G}(\nu)$ then

$$\max\left\{\mathbb{P}(X > x), \mathbb{P}(X < -x)\right\} \leq \exp\left(-\frac{x^2}{2\nu}\right) \text{ for all } x > 0 \ . \tag{1}$$

  Proof. We showed this above for a Gaussian using the Chernoff bound. The proof only relied on a Gaussian being subgaussian. □

- Conversely, if $\mathbb{E}X = 0$ and (1) holds for some $\nu > 0$ then the even moments of X are bounded as follows:

$$\mathbb{E}X^{2n} \leq 2n!(2\nu)^n \text{ for all } n \geq 1 \ ,$$

  and $X \in \mathcal{G}(16\nu)$. (Roughly speaking, the even moments of order n behave like $(n/2)!$.)

  Proof. First consider $\nu = 1$. Compute the $2n$-th moment by Fubini:

$$\begin{aligned}\mathbb{E}X^{2n} = 2n\mathbb{E}\int_0^{|X|} x^{2n-1}\, \mathrm{d}x &= 2n\mathbb{E}\int_0^\infty \mathbf{1}_{\{x>|X|\}}x^{2n-1}\, \mathrm{d}x \\ &= 2n\int_0^\infty x^{2n-1}\mathbb{P}(|X| > x)\, \mathrm{d}x \\ &\leq 4n\int_0^\infty x^{2n-1}e^{-x^2/2}\, \mathrm{d}x \ .\end{aligned}$$

Change variables, setting $u = x^2/2$, so that $\mathrm{d}u = x\ \mathrm{d}x$. This becomes

$$4n \int_0^\infty (2u)^{n-1} e^{-u}\ \mathrm{d}u = 4n \cdot 2^{n-1}\Gamma(n) = 2n!2^n \ ,$$

which is the result. In the case that $\nu \neq 1$ but is positive, set $Y = X/\nu$, so that (1) holds for Y in place of X and $\nu = 1$. By what we have proved, then

$$\mathbb{E}X^{2n} = (\nu)^{2n}\mathbb{E}Y^{2n} \leq 2n!(2\nu)^n \ .$$

Now to show that $X \in \mathcal{G}(4\nu)$, we would like to apply the moment bound to each term of the series

$$\mathbb{E}e^{tX} = \sum_{n=0}^\infty \frac{\mathbb{E}X^n}{n!} t^n \ .$$

However we only know bounds for even moments. It would be nice if the odd ones were zero, and there is a standard way to force this to happen: we symmetrize. So let X, X' be independent copies and consider $Y = X - X'$. Then

$$\mathbb{E}e^{tY} = \mathbb{E}e^{tX}\mathbb{E}e^{-tX}$$

and by Jensen,

$$\mathbb{E}e^{-tX} \geq \exp\left(\mathbb{E}tX\right) = 1 \ ,$$

so

$$\mathbb{E}e^{tX} \leq \mathbb{E}e^{tY} = \sum_{n \text{ even}} \frac{\mathbb{E}Y^n}{n!} t^n = \sum_{n=0}^\infty \frac{\mathbb{E}Y^{2n}}{(2n)!} t^{2n} \ .$$

By Jensen again,

$$\mathbb{E}Y^{2n} = 2^{2n}\mathbb{E}\left(\frac{1}{2}(|X| + |X'|)\right)^{2n} \leq 2^{2n-1} \cdot 2\mathbb{E}X^{2n} = 2^{2n}\mathbb{E}X^{2n} \ .$$

Therefore

$$\mathbb{E}e^{tX} \leq \sum_{n=0}^\infty \frac{2^{2n}\mathbb{E}X^{2n}}{(2n)!} t^{2n} \leq \sum_{n=0}^\infty \frac{2^{2n}2n!(2\nu)^n}{(2n)!} t^{2n} = \sum_{n=0}^\infty \frac{2^n n!}{(2n)!}\left(\frac{t^2(16\nu)}{2}\right)^n \ .$$

Last,

$$\frac{2^n n!}{(2n)!} \leq \frac{1}{n!} \ ,$$

since this is equivalent to

$$\frac{2n(2n-1)\cdots(n+1)}{2n(2n-2)\cdots 2} \geq 1 \ .$$

So

$$\mathbb{E}e^{tX} \leq \sum_{n=0}^\infty \left(\frac{t^2(16\nu)}{2}\right)^n \frac{1}{n!} = e^{t^2(16\nu)/2} \ ,$$

giving $X \in \mathcal{G}(16\nu)$. $\square$

In fact there is yet another standard equivalent formulation of subgaussian. For $\alpha > 0$,

$$\mathbb{E}e^{\alpha X^2} < 2 \text{ iff } X \in \mathcal{G}(\nu) \text{ for some } \nu \in [2/\alpha, 4/\alpha] \ .$$

1.3 Examples

1. Bounded variables. Suppose X has mean zero and is bounded. That is, X takes only values in $[a,b]$ with $\mathbb{E}X = 0$. To show that X is subgaussian, we will bound its cumulant generating function, but we accomplish this by bounding its derivatives and integrating.

   Then, $\psi_X(t)$ is infinitely differentiable (since X if bounded) and

$$\psi_X'(t) = \frac{\mathbb{E}Xe^{tX}}{\mathbb{E}e^{tX}}$$

$$\psi_X''(t) = \frac{\mathbb{E}X^2e^{tX}}{\mathbb{E}e^{tX}} - \left(\frac{\mathbb{E}Xe^{tX}}{\mathbb{E}e^{tX}}\right)^2 \ .$$

   You may recognize these expressions as the mean and variance of a different random variable. Indeed, if Y is a variable with distribution

$$\mathbb{P}(Y \in A) = \frac{\mathbb{E}\mathbf{1}_{\{X \in A\}}e^{tX}}{\mathbb{E}e^{tX}} \ ,$$

   then $\psi_X'(t) = \mathbb{E}Y$ and $\psi_X''(t) = \text{Var } Y$. Note that Y is also supported in the interval $[a,b]$, so we just need the following lemma.

   Lemma 1.2. *If Y is supported in $[a,b]$ then* $\text{Var } Y \leq \left(\frac{b-a}{2}\right)^2$.

   Proof. By an alternative characterization of variance (that you can check)

$$\text{Var } Y = \min_{y \in \mathbb{R}} \mathbb{E}(Y-y)^2 \leq \mathbb{E}\left(Y - \frac{b-a}{2}\right)^2 \ .$$

   Since Y is supported in $[a,b]$, $|Y - (b-a)/2| \leq (b-a)/2$, so our upper bound is $((b-a)/2)^2$. □

   By the lemma,

$$|\psi_X''(t)| \leq \frac{(b-a)^2}{4} \text{ for all } t \ .$$

   Furthermore,

$$\psi_X(0) = 0 \text{ and } \psi_X'(0) = \mathbb{E}X = 0 \ .$$

   So by Taylor's theorem, for some c between 0 and t,

$$|\psi_X(t)| = |\psi_X(t) - \psi_X(0) - \psi_X'(0)t| \leq \frac{t^2}{2}|\psi_X''(c)| \leq t^2\frac{(b-a)^2}{8} \ .$$

We conclude **Hoeffding's Lemma**, that

$$X \text{ supported in } [a,b] \Rightarrow X \in \mathcal{G}\left(\frac{(b-a)^2}{4}\right) .$$

2. Back to previous examples. In the beginning of the section we took $X_1, X_2, \ldots$ i.i.d. and bounded. Suppose now that they have mean zero (which we can accomplish just by subtracting the mean) and, allowing for different distributions, assume that the X_i's are only independent with support in $[a_i, b_i]$. Then by the previous example and independence,

$$S_n := X_1 + \cdots + X_n \in \mathcal{G}\left(\sum_{i=1}^n \frac{(b_i - a_i)^2}{4}\right) .$$

We obtain then **Hoeffding's inequality**:

$$\mathbb{P}\left(|S_n| > x\right) \leq 2\exp\left(-\frac{2x^2}{\sum_{i=1}^n (b_i - a_i)^2}\right) .$$

In the i.i.d. case, one would say that the sequence $(X_1, X_2, \ldots)$ obeys Gaussian concentration on scale $\sqrt{n}$. The reason can be seen by changing variables:

$$\mathbb{P}\left(|S_n| > x\sqrt{n}\right) \leq 2\exp\left(-\frac{2x^2}{(b-a)^2}\right) .$$

Returning also to the "uniform points in the n-dimensional cube" example, we took $X_1, \ldots$ i.i.d. uniform on the interval $[-1,1]$, so that the variables $X_1^2, \ldots, X_n^2$ are i.i.d. and supported in $[0,1]$ (with mean $1/3$). So using the last example, setting $X = (X_1, \ldots, X_n)$,

$$\mathbb{P}\left(\left|\|X\|_2^2 - \frac{n}{3}\right| > x\right) \leq 2\exp\left(-\frac{2x^2}{n}\right) .$$

1.4 Bennett's inequality

How close to optimal is Hoeffding's inequality? Suppose, for instance, that $X_1, X_2, \ldots$ are i.i.d. with support in $[a,b]$ and have mean zero. Then if $\sigma^2 = \text{Var } X_1$, by the central limit theorem,

$$\frac{X_1 + \cdots + X_n}{\sqrt{n}} \Rightarrow N(0, \sigma^2) ,$$

so

$$\lim_n \mathbb{P}\left(|X_1 + \cdots + X_n| \geq x\sqrt{n}\right) = \frac{2}{\sqrt{2\pi}\sigma}\int_x^\infty \exp\left(-\frac{t^2}{2\sigma^2}\right)\,\mathrm{d}t .$$

Now we will argue non-rigorously to relate this to Hoeffding. Suppose that the right side were not an integral, and that this expression held for all n with an inequality. Then we would obtain

$$\mathbb{P}\left(|X_1 + \cdots + X_n| \geq x\sqrt{n}\right) \leq 2\exp\left(-\frac{x^2}{2\sigma^2}\right) .$$

(For large x, this is roughly the order of magnitude of the integral anyway.) But Hoeffding gives

$$\exp\left(-\frac{x^2}{2\left(\frac{b-a}{2}\right)^2}\right) \text{ instead of } \exp\left(-\frac{x^2}{2\sigma^2}\right) .$$

So in a sense, we are using the upper bound

$$\sigma^2 \le \left(\frac{b-a}{2}\right)^2 ,$$

which we actually used in Hoeffding. Therefore Hoeffding is not a strong upper bound when $\sigma^2 \ll \left(\frac{b-a}{2}\right)^2$.

For an example with this property, take

$$X_i = \begin{cases} \pm 1 & \text{with probability } .49 \\ \pm 10 & \text{with probability } .01 \end{cases} .$$

If we wanted to use Hoeffding, we would need to use the interval $[a,b] = [-10,10]$, giving

$$\mathbb{P}\left(|X_1 + \cdots + X_n| \ge x\sqrt{n}\right) \le 2\exp\left(-\frac{x^2}{200}\right) ,$$

whereas our heuristic calculation with the CLT would give the upper bound (with $\sigma^2 = 2.98$)

$$2\exp\left(-\frac{x^2}{5.96}\right) .$$

We can try to get a better bound by modifying the proof of Hoeffding. Instead of using Hoeffding's lemma, we will try to introduce the variance of the summands into the bound. Furthermore, we will only require an upper bound on the variables and a second moment assumption. After the theorem we will explain how it gives a better bound than does Hoeffding.

Theorem 1.3. *Let $X_1, \ldots, X_n$ be independent with $X_i \le 1$ almost surely and $\mathbb{E}X_i^2 < \infty$ for all i. Then for $t > 0$,*

$$\psi_S(t) \le \nu\phi(t) ,$$

where $S = \sum_{i=1}^n (X_i - \mathbb{E}X_i)$, $\nu = \sum_{i=1}^n \mathbb{E}X_i^2$ and $\phi(t) = e^t - 1 - t$.

Proof. The function $\phi(t)$ will help to introduce the variance, since it is the Taylor expansion of e^t starting at degree two. We first derive the inequality

$$\phi(tX_i) \le \phi(t)X_i^2 \text{ for } t \ge 0 . \tag{2}$$

Write

$$\phi(tX_i) = \phi(t(X_i)_+ - t(X_i)_-) = \phi(t(X_i)_+) + \phi(-t(X_i)_-) .$$

Next use the inequality

$$\phi(x) = \sum_{n=2}^{\infty} \frac{x^n}{n!} \leq x^2/2 \text{ when } x \leq 0 \ .$$

This gives

$$\phi(tX_i) \leq \phi(t(X_i)_+) + \frac{1}{2}t^2(X_i)_-^2 \ .$$

Next use the inequality

$$\phi(x) = \sum_{n=2}^{\infty} \frac{x^n}{n} \geq x^2/2 \text{ when } x \geq 0 \ ,$$

so

$$\phi(tX_i) \leq \phi(t(X_i)_+) + \phi(t)(X_i)_-^2 \ .$$

Last, since $(X_i)_+ \in [0,1]$,

$$\phi(t(X_i)_+) = \sum_{n=2}^{\infty} \frac{(t(X_i)_+)^n}{n!} = (X_i)_+^2 \sum_{n=2}^{\infty} \frac{t^n(X_i)_+^{n-2}}{n!} \leq (X_i)_+^2\phi(t) \ .$$

Therefore

$$\phi(tX_i) \leq \phi(t)((X_i)_+^2 + (X_i)_-^2) = \phi(t)X_i^2 \ .$$

Returning to (2), since $\phi(tX_i)$ is integrable,

$$\mathbb{E}\phi(tX_i) \leq \mathbb{E}X_i^2\phi(t) \ .$$

or

$$\psi_{X_i}(t) \leq \log\left(1 + t\mathbb{E}X_i + \mathbb{E}X_i^2\phi(t)\right) \ .$$

Sum over i for

$$\psi_S(t) \leq \sum_{i=1}^{n} \log\left(1 + t\mathbb{E}X_i + \mathbb{E}X_i^2\phi(t)\right) - t\sum_{i=1}^{n} \mathbb{E}X_i \ .$$

Now use $\log(1+u) \leq u$ for

$$\psi_S(t) \leq \sum_{i=1}^{n} \mathbb{E}X_i^2\phi(t) = \nu\phi(t) \ .$$

□

Remarks.

1. If instead $X_i \leq b$ for all b, then $\psi_S(t) \leq \frac{\nu}{b^2}\phi(bt)$.

2. We can relate the bound in the above inequality to Poisson variables. Let X be a Poisson variable with parameter λ:

$$\mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!} \text{ for } k \geq 0 \ .$$

Then

$$\mathbb{E}X = e^{-\lambda}\sum_{k=0}^{\infty} k\frac{\lambda^k}{k!} = \lambda e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = \lambda$$

and

$$\mathbb{E}e^{tX} = e^{-\lambda}\sum_{k=0}^{\infty} e^{tk}\frac{\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(e^t\lambda)^k}{k!} = \exp\left(\lambda(e^t - 1)\right) \ .$$

Therefore writing $X' = X - \mathbb{E}X$,

$$\psi_{X'}(t) = \log \mathbb{E}e^{tX} - t\mathbb{E}X = \lambda(e^t - 1) - \lambda t = \lambda\phi(t) \ .$$

Therefore in the case $b = 1$, we have bounded the cumulant generating function of S by that of a centered Poisson variable with parameter ν.

3. As usual, we can compute the Cramer transform of this bound $f(t) = \frac{\nu}{b^2}\phi(bt)$. We set the derivative equal to t:

$$t = \frac{\nu}{b}\phi'(by) = \frac{\nu}{b}(e^{by} - 1)$$
$$y = \frac{1}{b}\log\left(1 + \frac{bt}{\nu}\right)$$

and plug this back in for

$$\begin{aligned} f^*(t) = ty - f(y) &= \frac{t}{b}\log\left(1 + \frac{bt}{\nu}\right) - \frac{\nu}{b^2}\phi\left(\log\left(1 + \frac{bt}{\nu}\right)\right) \\ &= \frac{t}{b}\log\left(1 + \frac{bt}{\nu}\right) - \frac{\nu}{b^2}\left(\frac{bt}{\nu} - \log\left(1 + \frac{bt}{\nu}\right)\right) \\ &= \frac{\nu}{b^2}\left(\left(1 + \frac{bt}{\nu}\right)\log\left(1 + \frac{bt}{\nu}\right) - \frac{bt}{\nu}\right) \\ &= \frac{\nu}{b^2}h\left(\frac{bt}{\nu}\right) \ , \end{aligned}$$

where $h(u) = (1+u)\log(1+u) - u$. So we obtain **Bennett's inequality**: if $X_1, \ldots, X_n$ are independent with $X_i \leq b$ and $\mathbb{E}X_i^2 < \infty$ then for $t > 0$,

$$\mathbb{P}(S \geq t) \leq \exp\left(-\frac{\nu}{b^2}h\left(\frac{bt}{\nu}\right)\right) \ .$$

4. For $u \geq 0$, one has the inequality

$$h(u) \geq \frac{u^2}{2(1+u/3)} \ .$$

(You can try to check this.) Therefore, one has **Bernstein's inequality**: if $X_1, \dots, X_n$ are independent with $X_i \leq b$ and $\mathbb{E}X_i^2 < \infty$ for all i, then for $t \geq 0$,

$$\mathbb{P}(S \geq t) \leq \exp\left(-\frac{\nu}{b^2} \cdot \frac{(bt)^2/\nu^2}{2(1+bt/3\nu)}\right) = \exp\left(-\frac{t^2}{2(\nu + bt/3)}\right) \ .$$

Note that when t is small (say $t \ll \nu/b$), this upper bound behaves as $\exp\left(-\frac{t^2}{2\nu}\right)$, which is what we expect from the CLT (if $\mathbb{E}X_i = 0$). When t is large, it behaves as $\exp\left(-3t/(2b)\right)$.

To show an example where this inequality is much better than Hoeffding, let $X_1, X_2, \dots$ be i.i.d. with mean zero, variance σ^2 and $|X_i| \leq b$ with $\sigma^2 < b^2$. Then Bernstein reads

$$\mathbb{P}\left(X_1 + \cdots + X_n \geq t\right) \leq \exp\left(-\frac{t^2}{2(n\sigma^2 + bt/3)}\right) \ ,$$

or

$$\begin{aligned}\mathbb{P}\left(X_1 + \cdots + X_n \geq t\sqrt{n}\sigma\right) &\leq \exp\left(-\frac{t^2 n\sigma^2}{2(n\sigma^2 + bt\sqrt{n}\sigma/3)}\right) \\ &= \exp\left(-\frac{t^2}{2\left(1+\frac{bt}{3\sqrt{n}\sigma}\right)}\right) \ .\end{aligned}$$

Thus when $t \ll \sqrt{n}\sigma/b$, this is nearly a Gaussian bound with variance factor σ^2. However, if t is larger order than this, say if $\sqrt{n} = o(t)$, then this bound is no longer Gaussian, and for very large t, it is only exponential. So for these cases, Hoeffding can be better.

1.5 Gamma (exponential concentration)

Next let's look at a distribution with a tail that does not decay quite as quickly as that of the Gaussian. The Gamma distribution has density function

$$f(x) = \frac{1}{\Gamma(a)b^a} x^{a-1} e^{-x/b} \text{ for } x \geq 0 \ .$$

We can see this as a generalization of the exponential distribution, which we get for the choice $a = 1$. Note that $f(x)$ decays exponentially. We can compute the moment generating function: if X is gamma distributed,

$$\mathbb{E}e^{tX} = \frac{1}{\Gamma(a)b^a} \int_0^\infty x^{a-1} e^{-x(1/b-t)} \, \mathrm{d}x \ .$$

Setting $u = x(1/b - t)$, so that $\mathrm{d}u = (1/b - t)\ \mathrm{d}x$, the integral becomes for $t < 1/b$,

$$\frac{1}{\Gamma(a)b^a}\int_0^\infty \left(\frac{u}{1/b-t}\right)^{a-1} e^{-u}\ \frac{\mathrm{d}u}{1/b-t} = \frac{1}{(1/b-t)^a}\cdot\frac{1}{\Gamma(a)b^a}\int_0^\infty u^{a-1}e^{-u}\ \mathrm{d}u\ .$$

The integral here is just the definition of $\Gamma(a)$, so we obtain

$$\mathbb{E}e^{tX} = \frac{1}{(1-bt)^a},\ \text{ or } \psi_X(t) = -a\log(1-bt) \text{ for } t < 1/b\ .$$

Of course when we look at the cumulant generating function, we want to center X by its mean, which is

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{(1-bt)^a}\bigg|_{t=0} = \frac{ab}{(1-bt)^{a+1}}\bigg|_{t=0} = ab\ .$$

So we obtain for $Y = X - \mathbb{E}X$,

$$\psi_Y(t) = \log\mathbb{E}e^{t(X-\mathbb{E}X)} = -abt - a\log(1-bt)\ .$$

We wish to define subgamma distributions, but using this cumulant generating function is a bit unwieldy. So we give a simple upper bound for it. Note that for $|x| < 1$,

$$\begin{aligned}-x - \log(1-x) = -x + \sum_{n=1}^\infty \frac{x^n}{n} &= \frac{x^2}{2} + \frac{x^3}{3} + \cdots \\ &= \frac{x^2}{2}\left(1 + (2/3)x + (2/4)x^2 + \cdots\right)\ .\end{aligned}$$

If $x \in (0,1)$ then we obtain the upper bound

$$-x - \log(1-x) \le \frac{x^2}{2}\left(1 + x + x^2 + \cdots\right) = \frac{x^2}{2(1-x)}\ . \tag{3}$$

Applying this to the cumulant generating function of Y,

$$\psi_Y(t) \le \frac{t^2(ab^2)}{2(1-bt)} \text{ for } t \in (0, 1/b)\ .$$

This motivates the definition

Definition 1.4. *A random variable X is said to be* **sub-gamma** *on the right tail with variance factor ν and scale parameter c if*

$$\psi_X(t) \le \frac{t^2\nu}{2(1-ct)} \text{ for } t \in (0, 1/c)\ .$$

We write $X \in \Gamma_+(\nu, c)$. A similar definition is made for the left tail if $-X$ satisfies this bound and we write $X \in \Gamma_-(\nu, c)$.

Note that this definition is slightly different than that of sub-gaussian, in that it has an extra factor of $(1-ct)^{-1}$:

$$\frac{t^2\nu}{2} \cdot \frac{1}{1-ct} \ .$$

The main difference is that this blows up as $t \uparrow 1/c$, and this allows the tail of X to decay less quickly.

As before, we can derive concentration bounds and moment characterizations.

- We can compute the Cramer transform

$$\psi_X^*(t) = \sup_{y \geq 0} \left(yt - \psi_X(y)\right)$$

by setting the derivative equal to t:

$$\psi_X'(y) = \nu y \frac{1}{1-cy} + \frac{y^2\nu}{2} \cdot \frac{c}{(1-cy)^2} = t$$

$$\nu y(1-cy) + y^2\nu c/2 = t(1-cy)^2$$

$$y^2(-c\nu/2 - c^2 t) + y(\nu + 2ct) - t = 0 \ .$$

$$\begin{aligned} y &= \frac{-\nu - 2ct \pm \sqrt{\nu^2 + 4\nu ct + 4c^2t^2 - 2ct\nu - 4c^2t^2}}{-c\nu - 2c^2 t} \\ &= \frac{-\nu - 2ct \pm \sqrt{\nu^2 + 2ct\nu}}{-c\nu - 2c^2t} \\ &= \frac{1}{c}\left(1 \pm \frac{\sqrt{\nu^2 + 2ct\nu}}{\nu + 2ct}\right) \ . \end{aligned}$$

Because $\psi_X(y)$ is only defined for $y < 1/c$, we are forced to take the root

$$y = \frac{1}{c}\left(1 - \frac{\sqrt{\nu^2 + 2ct\nu}}{\nu + 2ct}\right) = \frac{1}{c}\left(1 - \frac{1}{\sqrt{1 + 2ct/\nu}}\right) \ .$$

Therefore

$$\begin{aligned} \psi_X(y) &= \frac{\nu}{2c^2}\left(1 - \frac{1}{\sqrt{1+2ct/\nu}}\right)^2 \cdot \sqrt{1+2ct/\nu} \\ &= \frac{\nu}{2c^2}\left(\frac{2+2ct/\nu}{1+2ct/\nu} - \frac{2}{\sqrt{1+2ct\nu}}\right) \cdot \sqrt{1+2ct/\nu} \\ &= \frac{\nu}{2c^2}\left(\frac{2+2ct/\nu}{\sqrt{1+2ct/\nu}} - 2\right) \end{aligned}$$

and

$$
\begin{aligned}
\psi_X^*(t) &= \frac{t}{c} - \frac{t/c}{\sqrt{1+2ct/\nu}} - \frac{\nu}{c^2}\frac{1}{\sqrt{1+2ct/\nu}} - \frac{t/c}{\sqrt{1+2ct/\nu}} + \nu/c^2 \\
&= \frac{t}{c} + \frac{\nu}{c^2} - \frac{2t/c + \nu/c^2}{\sqrt{1+2ct/\nu}} \\
&= \frac{\nu}{c^2}\left(1 + \frac{ct}{\nu} - \sqrt{1+2ct/\nu}\right) \ .
\end{aligned}
$$

Written succinctly, with $h(u) = 1 + u - \sqrt{1+2u}$,

$$
\psi_X^*(t) = \frac{\nu}{c^2} h\left(\frac{ct}{\nu}\right) \ .
$$

Therefore if $X \in \Gamma_+(c,\nu)$ then

$$
\mathbb{P}\left(X > t\right) \leq \exp\left(-\frac{\nu}{c^2} h\left(\frac{ct}{\nu}\right)\right) \ .
$$

We can write this conclusion in a nicer way if we compute the inverse function of h. To see that there is one, compute

$$
h'(u) = 1 - \frac{1}{\sqrt{1+2u}} > 0 \text{ for } u > 0 \ ,
$$

so that h is strictly increasing on $[0,\infty)$ and is thus invertible there. If $v = h^{-1}(u)$ then

$$
\begin{gathered}
u = 1 + v - \sqrt{1+2v} \\
(1 + v - u)^2 = 1 + 2v \\
1 + v - u + v + v^2 - vu - u - vu + u^2 = 1 + 2v \\
v^2 - 2vu - 2u + u^2 = 0 \ .
\end{gathered}
$$

So

$$
v = \frac{2u \pm \sqrt{4u^2 - 4u^2 + 8u}}{2} = u \pm \sqrt{2u} \ .
$$

We take the positive root, since it is the one which gives an increasing function of u. Returning to the probability bound, setting $a = ct/\nu$,

$$
\mathbb{P}(X > a\nu/c) \leq \exp\left(-\frac{\nu}{c^2} h(a)\right) \ ,
$$

or

$$
\mathbb{P}\left(X > \frac{\nu}{c}\left(u + \sqrt{2u}\right)\right) \leq \exp\left(-\frac{\nu}{c^2} u\right) \ .
$$

Last, putting $t = \nu u/c^2$,

$$X \in \Gamma_+(\nu, c) \Rightarrow \mathbb{P}\left(X > ct + \sqrt{2\nu t}\right) \leq e^{-t} \text{ for all } t \geq 0 \ .$$

(For large t, the square root term is negligible, so this is a precise version of an exponential tail bound.) Furthermore, if X is also in $\Gamma_-(\nu, c)$, written $X \in \Gamma(\nu, c)$, then

$$\max\left\{\mathbb{P}(X > ct + \sqrt{2\nu t}), \mathbb{P}(X < -ct - \sqrt{2\nu t})\right\} \leq e^{-t} \text{ for } t \geq 0 \ . \tag{4}$$

- In fact, if X has mean zero and satisfies (4), then

$$\mathbb{E}X^{2n} \leq n!(8\nu)^n + (2n)!(4c)^{2n} \ . \tag{5}$$

(To compare to Gaussian, this say roughly that even moments of X of order n behave like $n!$ instead of $(n/2)!$. This is due to the less quickly decaying tail.)

Proof. Just as in the Gaussian case,

$$\mathbb{E}X^{2n} = 2n \int_0^\infty x^{2n-1}\mathbb{P}(|X| > x) \ \mathrm{d}x \ .$$

Now change variables, so that $x = ct + \sqrt{2\nu t}$. This gives $\mathrm{d}x = c + \frac{\nu}{\sqrt{2\nu t}} \ \mathrm{d}t$. This becomes

$$\begin{aligned}
&2n \int_0^\infty \left(ct + \sqrt{2\nu t}\right)^{2n-1} \mathbb{P}\left(|X| > ct + \sqrt{2\nu t}\right) \left(\frac{2ct + \sqrt{2\nu t}}{2t}\right) \ \mathrm{d}t \\
&\leq 2n \int_0^\infty \left(2ct + \sqrt{2\nu t}\right)^{2n} \frac{e^{-t}}{t} \ \mathrm{d}t \\
&\leq 2^{2n} n \int_0^\infty ((2ct)^{2n} + (2\nu t)^n) \frac{e^{-t}}{t} \ \mathrm{d}t \\
&= (4c)^{2n} n \int_0^\infty t^{2n-1} e^{-t} \ \mathrm{d}t + (8\nu)^n n \int_0^\infty t^{n-1} e^{-t} \ \mathrm{d}t \\
&= (4c)^{2n} n(2n-1)! + (8\nu)^n n(n-1)! \\
&\leq (4c)^{2n}(2n)! + (8\nu)^n n! \ .
\end{aligned}$$

□

- Conversely, if $\mathbb{E}X = 0$ and for some A, B,

$$\mathbb{E}X^{2n} \leq n!A^n + (2n)!B^{2n}$$

then $X \in \Gamma(4(A + B^2), 2B)$. This implies that the exponential tail bound (4) holds with $\nu = 4(A + B^2)$ and $c = 2B$.

Proof. We will again symmetrize, defining $Y = X - X'$, where X' is an independent copy of X. Then all even moments of Y are zero, and as in the subgaussian section, for $t \in \mathbb{R}$, since $\mathbb{E}X = 0$, $\mathbb{E}e^{-tX} \geq 1$ and

$$\mathbb{E}e^{tX} \leq \mathbb{E}e^{tY} = \sum_{n=0}^{\infty} \frac{\mathbb{E}(X-X')^{2n}}{(2n)!} t^{2n} \leq \sum_{n=0}^{\infty} 2^{2n} \frac{\mathbb{E}X^{2n}}{(2n)!} t^{2n}$$
$$\leq 1 + \sum_{n=1}^{\infty} \frac{n!A^n + (2n)!B^{2n}}{(2n)!} (2t)^{2n} \ .$$

Split the sum into two pieces. The second is

$$\sum_{n=1}^{\infty} (4B^2t^2)^n = \frac{4B^2t^2}{1-4B^2t^2} \text{ for } |t| < 1/(4B^2) \ .$$

Using $2^n n!/(2n)! \leq 1/n!$, the first is

$$1 + \sum_{n=1}^{\infty} \frac{n!A^n}{(2n)!} (2t)^{2n} = \sum_{n=0}^{\infty} \frac{n!2^n}{(2n)!} (2At^2)^n \leq \sum_{n=0}^{\infty} \frac{(2At^2)^n}{n!} = e^{2At^2} \ .$$

Therefore

$$\mathbb{E}e^{tX} \leq e^{2At^2} + \frac{4B^2t^2}{1-4B^2t^2} \ .$$

Next use

$$\frac{x^2}{1-x^2} \leq \frac{x/2}{1-x} \text{ for } x \in [0,1)$$

to obtain

$$\mathbb{E}e^{tX} \leq e^{2At^2} + \frac{2B^2t^2}{1-2B|t|} \text{ for } |x| < 1/(2B) \ .$$

Last, use the inequality

$$e^x + y \leq e^{x+y} \text{ for } x, y > 0 \ .$$

This can be checked by the mean value theorem: it is equivalent to

$$\frac{e^{x+y} - e^x}{y} \geq 1 \ .$$

So we obtain the upper bound

$$\exp\left(2At^2 + \frac{2B^2t^2}{1-2B|t|}\right) \leq \exp\left(\frac{4(A+B^2)t^2}{2(1-2B|t|)}\right) \text{ for } |t| < 1/(2B) \ .$$

This implies $X \in \Gamma(4(A+B^2), 2B)$. $\square$

1.6 Another Bernstein inequality

The conditions of Bernstein's inequality can be quite restrictive; after all we need $X_i \le b$ for all i. It is more convenient to assume instead a moment bound for our variables X_i. To see what we should assume, consider the setup again. Let $X_1, \dots, X_n$ be independent with $X_i \le b$ for all i and $\mathbb{E}X_i^2 < \infty$. We will consider only moments of $(X_i)_+$, the positive part of X_i (this is analogous to assuming an upper bound for the variables). For $n \ge 2$,

$$\mathbb{E}(X_i)_+^n \le b^{n-2}\mathbb{E}(X_i)_+^2 \le b^{n-2}\mathbb{E}X_i^2 \ .$$

This is still quite restrictive, as even the moments of an exponential distribution go to infinity like $n!$. So note that for $n \ge 3$,

$$b^{n-2} = 3^{n-2}(b/3)^{n-2} \le \frac{n!}{2}(b/3)^{n-2} \ .$$

So our moment bound becomes

$$\mathbb{E}(X_i)_+^n \le n!\frac{\mathbb{E}X_i^2}{2}(b/3)^{n-2} \text{ for } n \ge 3 \ .$$

Theorem 1.5 (Bernstein)**.** *Let $X_1, \dots, X_n$ be independent such that for some $c, \nu > 0$,*

1. $\sum_{i=1}^n \mathbb{E}X_i^2 \le \nu$ *and*
2. $\sum_{i=1}^n \mathbb{E}(X_i)_+^k \le k!\frac{\nu}{2}c^{k-2}$ *for* $k \ge 3$.

Then, setting $S = \sum_{i=1}^n X_i - \mathbb{E}X_i$, one has $S \in \Gamma_+(c, \nu)$.

Proof. We need to show that $\psi_S(t) \le \frac{\nu t^2}{2(1-ct)}$ for $t \in (0, 1/c)$. As in the last Bernstein inequality, we will involve the variance by considering the function $\phi(t) = e^t - 1 - t$. For $t \le 0$ one has $\phi(t) \le t^2/2$, so for $t \ge 0$,

$$\begin{aligned}
\phi(tX_i) = \phi(t((X_i)_+ - (X_i)_-)) &= \phi(t(X_i)_+) + \phi(-t(X_i)_-) \\
&\le \sum_{k=2}^\infty \frac{(X_i)_+^k}{k!}t^k + t^2(X_i)_-^2/2 \\
&= \sum_{k=3}^\infty \frac{(X_i)_+^k}{k!}t^k + t^2X_i^2/2 \ .
\end{aligned}$$

So using our assumption on the growth of moments and monotone convergence,

$$\begin{aligned}
\sum_{i=1}^n \mathbb{E}\phi(tX_i) &\le \sum_{i=1}^n\sum_{k=3}^\infty \frac{\mathbb{E}(X_i)_+^k}{k!}t^k + t^2\sum_{i=1}^n \mathbb{E}X_i^2/2 \\
&\le \frac{\nu}{2}\sum_{k=3}^\infty c^{k-2}t^k + t^2\nu/2 \\
&\le t^2\left(\frac{\nu}{2}\cdot\frac{ct}{1-ct} + \nu/2\right) = \frac{t^2\nu}{2(1-ct)} \ ,
\end{aligned}$$

as long as $t \in (0, 1/c)$. We finish as in the proof of Bennett with

$$\psi_S(t) = \sum_{i=1}^n \left[\log \mathbb{E}e^{tX_i} - t\mathbb{E}X_i\right] ,$$

so that, using the inequality $\log u \le u - 1$ for $u \ge 1$, our upper bound is

$$\psi_S(t) \le \sum_{i=1}^n \mathbb{E}[e^{tX_i} - 1 - tX_i] = \sum_{i=1}^n \mathbb{E}\phi(tX_i) \le \frac{t^2\nu}{2(1-ct)} .$$

□

Remarks

- As we saw, in the case of sub-gamma, one can give the following exponential tail bound (under the assumptions of Bernstein's inequality above):

$$\mathbb{P}\left(S > ct + \sqrt{2\nu t}\right) \le e^{-t} \text{ for } t \ge 0 .$$

- Furthermore, one can recover the old Bernstein inequality from this one. Recall the setup is that $X_1, \ldots, X_n$ are independent with $X_i \le b$ and $\mathbb{E}X_i^2 < \infty$. From before the theorem, we should take $\nu = \sum_{i=1}^n \mathbb{E}X_i^2$ and $c = b/3$. This gives $S \in \Gamma_+(b/3, \nu)$. By our computation of the Cramer transform of a sub-gamma variable, we obtain

$$\mathbb{P}(S > t) \le \exp\left(-\frac{\nu}{c^2} h\left(\frac{ct}{\nu}\right)\right) ,$$

where $h(u) = 1 + u - \sqrt{1+2u}$. Now, one can show the inequality

$$h(u) \ge \frac{u^2}{2(1+u)} \text{ for } u > 0 .$$

Indeed, this is equivalent to

$$\begin{aligned}
\sqrt{1+2u} &\le \frac{2(1+u)^2 - u^2}{2(1+u)} \\
(1+2u)4(1+u)^2 &\le (2+4u+2u^2)^2 \\
(4+8u)(1+2u+u^2) &\le 4 + 8u + 4u^2 + 8u + 16u^2 + 8u^3 + 4u^2 + 8u^3 + 4u^4 . \\
4 + 8u + 4u^2 + 8u + 16u^2 + 8u^3 &\le 4 + 16u + 24u^2 + 16u^3 + 4u^4 \\
0 &\le 4u^4 + 8u^3 + 8u^2 + 8u ,
\end{aligned}$$

which is true.

Therefore we obtain

$$\begin{aligned}
S \in \Gamma_+(c,\nu) \Rightarrow \mathbb{P}(S > t) \le \exp\left(-\frac{\nu}{c^2}\frac{(ct/\nu)^2}{2(1+ct/\nu)}\right) &= \exp\left(-\frac{t^2/\nu}{2(1+ct/\nu)}\right) \\
&= \exp\left(-\frac{t^2}{2(\nu+ct)}\right) . \qquad (6)
\end{aligned}$$

Putting $c = b/3$ gives the old Bernstein inequality.

1.7 Example: Gaussian chaos

Before moving to the example of Section 2.8 of the text, let us mention two other types of sub- variables.

1. A nonnegative variable has a *sub-exponential* distribution if for some $a > 0$, one has

$$\mathbb{E}e^{tX} \leq \frac{1}{1 - t/a} \text{ for } t \in (0, a) \ .$$

   This definition is due to the fact that the exponential distribution with parameter a (whose density is ae^{-ax} for $x \geq 0$) has moment generating function $1/(1 - t/a)$ for $t \in (0, a)$. If X is sub-exponential, then $\mathbb{E}X^n \leq 2^{n+1} \frac{n!}{a^n}$.

2. At times one can get a bound $\mathbb{P}(X > t) \leq e^{-t^p}$ for $p \in (1, 2)$. This is between exponential and gaussian. Typically such a bound is called *stretched exponential*.

Let $\vec{X} = (X_1, \ldots, X_n)$ be an n-dimensional standard (i.i.d.) Gaussian vector. For $\alpha > 0$, an **α-order Gaussian chaos** is $f(\vec{X})$, where $f : \mathbb{R} \to \mathbb{R}$ is α-homogeneous. (This means that for $t \in \mathbb{R}, x \in \mathbb{R}^n$, $f(tx) = |t|^\alpha f(x)$.) This example concerns order $\alpha = 2$ in the special case that f is a polynomial.

From linear algebra, any such f is a quadratic form on $\mathbb{R}^n$ and can be written as $f(x) = Ax \cdot x$, where A is an $n \times n$ symmetric matrix. Letting the (i, j)-th entry of A be called $a_{i,j}$, we will here assume that $a_{i,i} = 0$ for all i. The goal is to give a good upper bound for $\mathbb{P}(f(\vec{X}) > t)$. Note that since $a_{i,i} = 0$,

$$\mathbb{E}f(\vec{X}) = \mathbb{E} \sum_{i \neq j} a_{i,j} X_i X_j = 0 \ ,$$

so the bound we want can be alternatively phrased as $\mathbb{P}(f(\vec{X}) - \mathbb{E}f(\vec{X}) > t)$.

First we can use our linear algebra skills to diagonalize the quadratic form: there is an orthogonal matrix O and a diagonal matrix D such that $A = O^T DO$, and therefore

$$f(\vec{X}) = D(O\vec{X}) \cdot (O\vec{X}) \ .$$

However the standard Gaussian distribution is invariant under orthogonal transformations, so this has the same distribution as $h(\vec{X}) = D\vec{X} \cdot \vec{X}$. Writing the (diagonal) entries of D as $d_1, \ldots, d_n$, we then obtain

$$\mathbb{P}(f(\vec{X}) > t) = \mathbb{P}(d_1 X_1^2 + \cdots + d_n X_n^2 > t) \ .$$

Using the fact that

$$0 = \sum_i a_{i,i} = \text{Tr } A = \text{Tr } D = \sum_i d_i \ ,$$

we obtain

$$\sum_i d_i X_i^2 = \sum_i d_i (X_i^2 - 1) \ .$$

If $td_i < 1/2$,

$$\mathbb{E}e^{td_i(X_i^2-1)} = \frac{e^{-td_i}}{\sqrt{2\pi}} \int e^{td_i x^2} e^{-x^2/2} \, \mathrm{d}x = \frac{e^{-td_i}}{\sqrt{2\pi}} \int \exp\left(-\frac{x^2}{2(1-2td_i)^{-1}}\right) \, \mathrm{d}x = \frac{e^{-td_i}}{\sqrt{1-2td_i}} \ ,$$

giving

$$\psi_{d_i(X_i^2-1)}(t) \le -\frac{1}{2}\left[2td_i + \log(1-2td_i)\right] \ .$$

Recall the inequality from (3)

$$-u - \log(1-u) \le \frac{u^2}{2(1-u)} \text{ for } u \in (0,1) \ .$$

So putting $\|A\| = \max_i |d_i|$, for $t \in (0, 1/(2\|A\|))$ and $d_i > 0$,

$$\psi_{d_i(X_i^2-1)}(t) \le \frac{t^2 d_i^2}{1-2td_i} \le \frac{t^2 d_i^2}{1-2t\|A\|} \ .$$

On the other hand, if $d_i \le 0$ we use the inequality for $u \in (-1, 0)$

$$-u - \log(1-u) = -u + \sum_{k=1}^{\infty} \frac{u^k}{k} \le u^2/2$$

so for $t \in (0, 1/(2\|A\|))$, we obtain

$$\psi_{d_i(X_i^2-1)}(t) = -\frac{1}{2}[2td_i + \log(1-2td_i)] \le t^2 d_i^2 \le \frac{t^2 d_i^2}{1-2t\|A\|} \ .$$

Combining these cases, for t as above and setting $S = \sum_{i=1}^n d_i(X_i^2 - 1)$,

$$\psi_S(t) \le \sum_{i=1}^n \frac{t^2 d_i^2}{1-2t\|A\|} = \frac{t^2 \|A\|_{HS}^2}{1-2t\|A\|} \ ,$$

where $\|A\|_{HS}^2 = \sum_i d_i^2$ is the Hilbert-Schmidt norm. This means $S \in \Gamma_+(2\|A\|, 2\|A\|_{HS}^2)$. In particular,

$$\mathbb{P}\left(f(\vec{X}) > 2\|A\|_{HS}\sqrt{t} + 2\|A\|t\right) \le e^{-t} \ .$$

To elucidate this we can again use (6) for

$$\begin{aligned}\mathbb{P}\left(f(\vec{X}) > t\right) \le \exp\left(-\frac{t^2}{2(\nu + ct)}\right) &= \exp\left(-\frac{t^2}{2(2\|A\|_{HS}^2 + 2\|A\|t)}\right) \\ &= \exp\left(-\frac{t^2}{4(\|A\|_{HS}^2 + \|A\|t)}\right) \ .\end{aligned}$$

1 Efron-Stein and applications

We have seen that to get some more optimized concentration inequalities, it is useful to involve the variance of our variables. So this brings the question: how do we bound the variance?

We will consider a function $f : \mathbb{R}^n \to \mathbb{R}$ and a sequence $X_1, \ldots, X_n$ of independent random variables. Our goal is to

$$\text{estimate Var } Z \text{, where } Z = f(X_1, \ldots, X_n) \; .$$

In the special case that $f = x_1 + \cdots + x_n$, we can decompose Z using independence as $\text{Var } Z = \text{Var } X_1 + \cdots + \text{Var } X_n$. This is an exact decomposition of the variance into "local" variations of the X_i's. The Efron-Stein inequality is a generalization of this.

Theorem 1.1 (Efron-Stein inequality)**.** *Let $X_1, \ldots, X_n$ be independent and $f : \mathbb{R}^n \to \mathbb{R}$ be such that $\mathbb{E} f(X)^2 < \infty$, where $X = (X_1, \ldots, X_n)$. Let $X_1', \ldots, X_n'$ be independent copies of $X_1, \ldots, X_n$ and define*

$$Z_i' = f(X_1, \ldots, X_{i-1}, X_i', X_{i+1}, \ldots, X_n) \; .$$

Then, putting $Z = f(X)$,

$$\text{Var } Z \leq \nu := \frac{1}{2} \sum_{i=1}^n \mathbb{E}(Z - Z_i')^2 \; .$$

Proof. Write

$$\Sigma_k = \sigma(X_1, \ldots, X_k) \text{ for } k = 1, \ldots, n$$

and $\Sigma_0 = \{\emptyset, \Omega\}$ for the trivial sigma-algebra. Then

$$\text{Var } Z = \mathbb{E}\left(\mathbb{E}[Z \mid \Sigma_n] - \mathbb{E}[Z \mid \Sigma_0]\right) = \mathbb{E}\left(\sum_{k=1}^n \Delta_k\right)^2 \; ,$$

where $\Delta_k = \mathbb{E}[Z \mid \Sigma_k] - \mathbb{E}[Z \mid \Sigma_{k-1}]$. As usual, the Δ_k's are orthogonal in L^2: if $j < k$ then

$$\mathbb{E}\Delta_j \Delta_k = \mathbb{E}\left[\mathbb{E}[\Delta_j \Delta_k \mid \Sigma_j]\right] = \mathbb{E}\left[\Delta_j \mathbb{E}[\Delta_k \mid \Sigma_j]\right] = 0 \; .$$

Thus we obtain the standard martingale decomposition of variance

$$\text{Var } Z = \sum_{k=1}^n \mathbb{E}\Delta_k^2 \; .$$

Note this only requires $\mathbb{E}Z^2 < \infty$.

To get to Efron-Stein, we will simply apply Jensen's inequality. To do this, we need a little bit of notation. Let $\mathbb{E}_{\geq k}$ be expectation over $X_k, \ldots, X_n$ and $\mathbb{E}_k$ be expectation over

X_k. Similarly define $\mathbb{E}'_{\geq k}$ and $\mathbb{E}'_k$ for the variables $X'_k, \ldots, X'_n$ and X'_k. By independence and symmetry,

$$\begin{aligned}
\Delta_k(X) &= \mathbb{E}_{\geq k+1} Z(X) - \mathbb{E}_{\geq k} Z(X) \\
&= \mathbb{E}'_{\geq k+1} Z(X_1, \ldots, X_k, X'_{k+1}, \ldots, X'_n) - \mathbb{E}'_k \mathbb{E}'_{\geq k+1} Z(X_1, \ldots, X_{k-1}, X'_k, \ldots, X'_n) \\
&= \mathbb{E}'_{\geq k+1} \left[Z(X_1, \ldots, X_k, X'_{k+1}, \ldots, X'_n) - \mathbb{E}'_k Z(X_1, \ldots, X_{k-1}, X'_k, \ldots, X'_n) \right] \\
&= \mathbb{E}_{\geq k+1} \left[Z(X) - \mathbb{E}'_k Z(X_1, \ldots, X_{k-1}, X'_k, X_{k+1} \ldots, X_n) \right] \\
&= \mathbb{E}_{\geq k+1} \left[Z - \mathbb{E}'_k Z'_k \right] \ .
\end{aligned}$$

Therefore by Jensen,

$$\mathbb{E}\Delta_k^2 \leq \mathbb{E} \left[Z - \mathbb{E}'_k Z'_k \right]^2 \ .$$

This is a rephrasing of Efron-Stein. Write it as

$$\mathbb{E}\mathbb{E}_k [Z - \mathbb{E}_k Z_k]^2 = \mathbb{E}\mathrm{Var}_k Z \ ,$$

where Var_k is variance relative to X_k only. By the formula $\mathrm{Var}\ Y = \frac{1}{2}\mathbb{E}[Y - Y']^2$, where Y and Y' are i.i.d. (you can check this), we obtain

$$\mathbb{E}\Delta_k^2 \leq \frac{1}{2}\mathbb{E}[Z - Z'_k]^2 \ .$$

Sum over k to complete the proof. $\square$

There are a couple of equivalent formulations of Efron-Stein.

- $\mathrm{Var}\ Z \leq \sum_{k=1}^n \mathbb{E}[Z - \mathbb{E}'_k Z'_k]^2 = \nu$, which we saw in the proof.
- Writing

  $$\mathbb{E}[Z - Z'_k]^2 = \mathbb{E} \left[(Z - Z'_k)_+ - (Z - Z'_k)_- \right]^2 = \mathbb{E}(Z - Z'_k)_+^2 + \mathbb{E}(Z - Z'_k)_-^2 \ ,$$

  and using symmetry, we also obtain

  $$\mathrm{Var}\ Z \leq \mathbb{E} \sum_{k=1}^n (Z - Z'_k)_+^2 = \mathbb{E} \sum_{k=1}^n (Z - Z'_k)_-^2 = \nu \ .$$

- Efron-Stein even holds for L^2 functions of sequences of independent random variables. The reason is that if $\mathbb{E} f(X_1, X_2, \ldots)^2 < \infty$, then L^2-martingale convergence still gives

  $$\mathrm{Var}\ Z = \mathbb{E} \left(\sum_{i=1}^\infty \Delta_i \right)^2 = \sum_{i=1}^\infty \mathbb{E}\Delta_i^2 \ .$$

  Then we apply Jensen as above.

1.1 Examples

First assume that f has **bounded differences**. That is, there exist constants $c_1, \ldots, c_n$ such that for all i,

$$|f(x_1, \ldots, x_n) - f(x_1, \ldots, x_{i-1}, x_i', x_{i+1}, \ldots, x_n)| \leq c_i \text{ for all } x_1, x_1', \ldots, x_n, x_n' \in \mathbb{R} \ .$$

Then if $X_1, \ldots, X_n$ are independent one has

$$\text{Var } Z \leq \sum_{k=1}^{n} \mathbb{E}[Z - \mathbb{E}_k' Z_k']^2 = \sum_{k=1}^{n} \mathbb{E}\text{Var}_k Z \ ,$$

where Var_k is variance relative to X_k only. By the bounded differences condition, for given $X_1, \ldots, X_{k-1}, X_{k+1}, \ldots, X_n$, the function $x_k \mapsto f(X_1, \ldots, X_{k-1}, x_k, X_{k+1}, \ldots, X_n)$ takes values in an interval $[a, b]$ of length c_k. Therefore

$$\text{Var}_k Z = \min_{c \in \mathbb{R}} \mathbb{E}_k [Z - c]^2 \leq \mathbb{E}_k \left[Z - \frac{b-a}{2} \right]^2 \leq c_k^2/4 \ .$$

Therefore we obtain the **bounded differences estimate** for variance:

$$\text{Var } Z \leq \frac{1}{4} \sum_{k=1}^{n} c_k^2 \ .$$

Here are some examples from the text.

1. (Bin packing.) Given numbers $x_1, \ldots, x_n \in [0, 1]$, we imagine that we have n objects with sizes $x_1, \ldots, x_n$. Let $f(x_1, \ldots, x_n)$ be the minimum number of bins we need to pack these objects, given that each bin can accommodate a total size 1 of objects. For example $f(.5, .5) = 1$ whereas $f(.5, .51) = 2$.

   For $X_1, \ldots, X_n$ independent, taking values in $[0, 1]$, we ask the variance of $Z = f(X_1, \ldots, X_n)$. f satisfies the bounded differences inequality with $c_i = 1$ for all i. This is because if we change the size of one object, we need at most one more bin to accommodate the new objects. By the bounded differences estimate,

   $$\text{Var } Z \leq \frac{n}{4} \ .$$

2. (Longest common subsequence.) Let $\vec{x} = (x_1, \ldots, x_n)$ and $\vec{y} = (y_1, \ldots, y_n)$ be two sequences with elements from $\{0, 1\}$. Set $L = L(\vec{x}, \vec{y})$ to be the length of the longest subsequence common to both $\vec{x}$ and $\vec{y}$. For example,

   $$L((1, 1, 1), (1, 1, 0)) = 2$$

   since the sub sequence $(1, 1)$ is common to both.

Now let $\vec{X} = (X_1, \ldots, X_n)$ and $\vec{Y} = (Y_1, \ldots, Y_n)$ be two vectors of i.i.d. Bernoulli(1/2) variables and $L = L(\vec{X}, \vec{Y})$. Note that L satisfies the bounded differences inequality with $c_i = 1$ for all i. The reason is that changing one of the X_i's or Y_i's only results in L increasing by at most one. To see why, suppose for example that we change X_1 to $X_1' = 1 - X_1$. Then a maximal common subsequence C of $\vec{X}$ and $\vec{Y}$ either contains X_1 or it does not. If not, then C is also a common subsequence of $\vec{X}'$ and $\vec{Y}$, where $\vec{X}' = (X_1', X_2, \ldots, X_n)$, giving $L' := L(\vec{X}', \vec{Y}) \geq L$. Otherwise if C contains X_1 then the subsequence obtained from C by removing X_1 is a common subsequence of $\vec{X}'$ and $\vec{Y}$, giving $L' \geq L - 1$. In total, $L' \geq L - 1$. By symmetry, $L \geq L' - 1$, so $|L - L'| \leq 1$.

By the bounded differences estimate, we conclude that

$$\text{Var } L \leq \frac{1}{4} \sum_{k=1}^{2n} 1 = \frac{n}{2} \ .$$

3. (Number of distinct values in a discrete sample.) Let $X_1, X_2, \ldots$ be i.i.d. with

$$\mathbb{P}(X_1 = k) = p_k \text{ for } k \geq 1$$

and define for each $n \geq 1$, N_n to be the size of the set $\{X_1, \ldots, X_n\}$ (number of distinct elements). Then $N_n = N_n(X_1, \ldots, X_n)$ satisfies the bounded differences condition with $c_i = 1$ for all i and therefore Var $N_n \leq n/4$. However this is far from optimal.

We can compute $\mathbb{E}N_n$ and see that it is $o(n)$. Indeed, the size of $\{X_1, \ldots, X_n\}$ is equal to the number of $i = 1, \ldots, n$ such that X_i is not equal to any of the X_j's that came before it: $N_n = \sum_{k=1}^n \mathbf{1}_{\{X_k \neq X_1, \ldots, X_k \neq X_{k-1}\}}$. Therefore

$$\begin{aligned} \mathbb{E}N_n = \sum_{k=1}^n \mathbb{P}(X_k \neq X_1, \ldots, X_k \neq X_{k-1}) &= \sum_{k=1}^n \sum_{j=1}^\infty \mathbb{P}(X_1 \neq j, \ldots, X_{k-1} \neq j)\mathbb{P}(X_k = j) \\ &= \sum_{k=1}^n \sum_{j=1}^\infty p_j(1-p_j)^{k-1} \ . \end{aligned}$$

Note that for each $k \geq 1$,

$$\max_{x \in [0,1]} x(1-x)^k = \frac{k^k}{(k+1)^{k+1}}$$

as

$$\frac{\mathrm{d}}{\mathrm{d}x} x(1-x)^k = (1-x)^k - kx(1-x)^{k-1} = 0 \Leftrightarrow x = \frac{1}{k+1} \ .$$

Therefore for fixed j, $p_j(1-p_j)^{k-1} \to 0$ as $k \to \infty$. By dominated convergence (dominating by p_j), $\sum_{j=1}^\infty p_j(1-p_j)^{k-1} \to 0$ as $k \to \infty$. Therefore

$$\frac{1}{n}\mathbb{E}N_n = \frac{1}{n} \sum_{k=1}^n \sum_{j=1}^\infty p_j(1-p_j)^{k-1} \to 0 \ ,$$

or $\mathbb{E}N_n = o(n)$. Because of this, we might expect Var $N_n = o(n)$ as well. We will use the following facts in the proof:

(a) For each $k = 1, \ldots, n$, let $N_n^{(k)} = N_{n-1}(X_1, \ldots, X_{k-1}, X_{k+1}, \ldots, X_n)$ be the number of distinct values in $\{X_1, \ldots, X_n\} \setminus \{X_k\}$. Then $0 \leq N_n - N_n^{(k)} \leq 1$ almost surely.

(b) One has the inequality

$$\sum_{k=1}^{n} (N_n - N_n^{(k)}) \leq N_n \ ,$$

since the sum on the left is equal to the number of entries of $(X_1, \ldots, X_n)$ which appear only once.

In the case of the last example, we say that $f = N_n$ is self-bounding:

Definition 1.2. *A nonnegative $f : \mathbb{R}^n \to \mathbb{R}$ is* **self-bounding** *if for each $i = 1, \ldots, n$, there is a function $f_i : \mathbb{R}^{n-1} \to \mathbb{R}$ such that both of the following hold.*

1. $0 \leq f(x_1, \ldots, x_n) - f_i(x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n) \leq 1$ *and*
2. $\sum_{i=1}^n (f(x_1, \ldots, x_n) - f_i(x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n)) \leq f(x_1, \ldots, x_n)$.

Theorem 1.3. *Let $X_1, \ldots, X_n$ be independent and $Z = f(X_1, \ldots, X_n)$ be square integrable with f self-bounding. Then* $\mathrm{Var}\ Z \leq \mathbb{E}Z$.

Proof. We apply Efron-Stein in the form

$$\mathrm{Var}\ Z \leq \sum_{k=1}^{n} \mathbb{E}\left(Z - \mathbb{E}_k' Z_k'\right)^2 = \sum_{k=1}^{n} \mathbb{E}\mathrm{Var}_k Z \ ,$$

where Var_k is variance relative to only X_k (with all other X_i's fixed). Note that

$$\begin{aligned}\mathrm{Var}_k Z &= \min_{a \in \mathbb{R}} \mathbb{E}_k [Z - a]^2 \\ &\leq \mathbb{E}_k (f(X_1, \ldots, X_n) - f_k(X_1, \ldots, X_{k-1}, X_{k+1}, \ldots, X_n))^2 \\ &\leq \mathbb{E}_k (f(X_1, \ldots, X_n) - f_k(X_1, \ldots, X_{k-1}, X_{k+1}, \ldots, X_n))\end{aligned}$$

so that

$$\mathrm{Var}\ Z \leq \sum_{k=1}^{n} \mathbb{E}\mathbb{E}_k (f(X_1, \ldots, X_n) - f_k(X_1, \ldots, X_{k-1}, X_{k+1}, \ldots, X_n)) \leq \mathbb{E}Z \ .$$

□

Because N_n from the last example (number of distinct values ...) is self-bounding and bounded by n (therefore square integrable), we conclude

$$\mathrm{Var}\ N_n \leq \mathbb{E}N_n = o(n) \ .$$

Why would it be useful to know that Z is self-bounding? From the inequality $\text{Var } Z \leq \mathbb{E}Z$ we obtain by a straight-forward application of Chebyshev,

$$\mathbb{P}\left(|Z - \mathbb{E}Z| \geq \epsilon \mathbb{E}Z\right) \leq \frac{\text{Var } Z}{\epsilon^2 (\mathbb{E}Z)^2} = \leq \frac{1}{\epsilon^2 \mathbb{E}Z} \ .$$

In most applications, $Z = Z_n$ has $\mathbb{E}Z \to \infty$, so this shows that the fluctuations of Z about its mean are small compared to the size of the mean. In BLM, this is referred to as *relative stability*.

More examples.

1. (First-passage percolation.) Let (t_e) be a collection of i.i.d. nonnegative random variables with continuous distribution, each assigned to a nearest-neighbor edge of the integer lattice $\mathbb{Z}^d$. That is, each edge has integer endpoints and is of the form $\{x, y\}$ for $x, y \in \mathbb{Z}^d$. A path P from x to y is a sequence $x = x_0, e_0, x_1, \ldots, x_{n-1}, e_{n-1}, x_n = y$ of alternating vertices and edges such that for each $i = 1, \ldots, n$, e_{i-1} has endpoints x_{i-1} and x_i. The passage time of a path P is defined as $T(P) = \sum_{e \in P} t_e$ and the passage time between x and y is defined $T(x, y) = \inf_{P: x \to y} T(P)$, where the infimum is over all paths from x to y.

   We are interested in the random variable $T(0, ne_1)$, where $e_1 = (1, 0, \ldots, 0)$. Here we will record the order of its mean and variance. We will make two assumptions, the second of which is a strong form of a necessary assumption:

   $$\mathbb{E}t_e^2 < \infty \text{ and } \mathbb{P}(t_e \geq 1) = 1 \ . \tag{1}$$

   This ensures, for example, that $T(0, ne_1) \geq n$, since every path from 0 to ne_1 contains at least n edges and each of their weights is at least 1. Furthermore, taking γ to be the deterministic path from 0 to ne_1 along the e_1 axis (consisting of n edges), we obtain

   $$T(0, ne_1) \leq T(\gamma) = \sum_{k=1}^{n} t_{\{(k-1)e_1, ke_1\}} \ ,$$

   so $\mathbb{E}T(0, ne_1) \leq n\mathbb{E}t_e$. This means that

   $$\mathbb{E}T(0, ne_1) \text{ is of order } n \ .$$

   What about $\text{Var } T(0, ne_1)$? In dimensions $d \geq 2$ it is actually expected to be of order $n^{2\chi}$ where $\chi < 1/2$ (for $d = 2$, $\chi = 1/3$ is expected), but this is far from proved. Here we will show that $\text{Var } T(0, ne_1) \leq Cn$ for some constant $C > 0$.

   If we let $T = T(0, ne_1)$ and T_e be the passage time from 0 to ne_1 when the edge weight t_e is replaced by an independent copy t_e', then Efron-Stein gives the bound

   $$\text{Var } T \leq \sum_e \mathbb{E}(T - T_e)_-^2 = \sum_e \mathbb{E}(T_e - T)_+^2 \ .$$

Note that $(T_e - T)_+$ is nonzero only when $T_e > T$, and this is only possible if the resampled weight t_e' is larger than the original weight t_e. In this case, the passage time can only be raised from T to T_e if e is in an optimal path (that is, a geodesic) from 0 to ne_1 in the original edge weights. The fact that the edge weights are continuously distributed implies that there is almost surely only one geodesic P from 0 to ne_1 (think about why this is true!) Furthermore, if $e \in P$ then $T_e - T \leq t_e'$. Therefore

$$\mathrm{Var}\ T \leq \sum_e \mathbb{E}(t_e')^2 \mathbf{1}_{e \in P}\ .$$

By independence of t_e' and the original weights, this equals

$$\sum_e \mathbb{E} t_e^2 \mathbb{P}(e \in P) = \mathbb{E} t_e^2 \mathbb{E} \# P\ .$$

Since all edge weights are at least 1, we have $T(P) \leq T$, so

$$\mathrm{Var}\ T \leq \mathbb{E} t_e^2 \mathbb{E} T \leq \mathbb{E} t_e^2 \mathbb{E} t_e n = Cn\ .$$

(This is a result of Kesten (1993) and does not require continuous weights or $t_e \geq 1$ almost surely.)

2. (Random symmetric matrix.) Let $(X_{i,j} : 1 \leq i \leq j \leq n)$ be a collection of i.i.d. random variables with $|X_{i,j}| \leq 1$. We aim to understand the order of the largest eigenvalue λ of the symmetric matrix A whose entries are given by

$$A_{i,j} = \begin{cases} X_{i,j} & \text{if } i \leq j \\ X_{j,i} & \text{if } j \leq i \end{cases}\ .$$

From linear algebra, the largest eigenvalue is given by

$$\lambda = \sup_{v : \|v\| = 1} Av \cdot v\ ,$$

where $\|v\|$ is the ℓ^2-norm and this supremum is attained at an eigenvector v_λ for λ. $\lambda = \lambda(A)$ is a function of $n(n-1)$ variables $(X_{i,j})$ and we can bound its variance using Efron-Stein:

$$\mathrm{Var}\ \lambda \leq \sum_{1 \leq i \leq j \leq n} \mathbb{E}(\lambda(A) - \lambda(A_{i,j}'))_+^2\ ,$$

where $A_{i,j}'$ is the $n \times n$ matrix obtained by replacing the (i,j)-th entry of A by $X_{i,j}'$, an independent copy of $X_{i,j}$, and the (j,i)-th entry of A by $X_{i,j}'$.

Let v be an eigenvector for A of ℓ^2-norm 1 corresponding to eigenvalue λ and bound

$$\lambda(A) - \lambda(A_{i,j}') = Av \cdot v - \sup_{u : \|u\| = 1} A_{i,j}' u \cdot u \leq Av \cdot v - A_{i,j}' v \cdot v = (A - A_{i,j}') v \cdot v\ .$$

Therefore

$$\begin{aligned}(\lambda(A)-\lambda(A'_{i,j}))_+^2 &= (\lambda(A)-\lambda(A'_{i,j}))^2\mathbf{1}_{\lambda(A)\geq\lambda(A'_{i,j})}\\ &\leq ((A-A'_{i,j})v\cdot v)^2\mathbf{1}_{\lambda(A)\geq\lambda(A'_{i,j})}\\ &\leq ((A-A'_{i,j})v\cdot v)^2 .\end{aligned}$$

However $A-A'_{i,j}$ is a matrix with at most one nonzero entry when $i=j$ (it equals $X_{i,j}-X'_{i,j}$) and at most two when $i\neq j$. So we get the bound

$$\begin{aligned}(\lambda(A)-\lambda(A'_{i,j}))_+^2 &\leq \begin{cases}(X_{i,i}-X'_{i,i})^2v_i^4 & \text{if } i=j\\ 4(X_{i,j}-X'_{i,j})^2v_i^2v_j^2 & \text{if } i\neq j\end{cases}\\ &\leq \begin{cases}4v_i^4 & \text{if } i=j\\ 16v_i^2v_j^2 & \text{if } i\neq j\end{cases} .\end{aligned}$$

This means

$$\sum_{1\leq i\leq j\leq n}(\lambda(A)-\lambda(A'_{i,j}))_+^2 \leq (2(v_1^2+\cdots+v_n^2))^2 \leq 16 .$$

By Efron-Stein, $\operatorname{Var}\lambda(A)\leq 16$.

The interesting thing here is that the variance is bounded, whereas for many examples of particular distributions of the $X_{i,j}$'s satisfying the above assumptions, $\mathbb{E}\lambda(A)\to\infty$ as $n\to\infty$ (even at the rate of $n^{1/2}$ or n). See Section 3 of Alon-Krivelevich-Vu (2002).

1.2 Exponential concentration using Efron-Stein

Bounding the variance can be more useful than just for fun. These techniques can lead to concentration inequalities. We will show two different methods. Our setting will be just as before, that $Z=f(X_1,\ldots,X_n)$ is a function of independent variables (we do not assume square integrability for now). We will make a strong type of bounded differences assumption, mostly for simplicity: if $X'_1,\ldots,X'_n$ are independent copies of $X_1,\ldots,X_n$, then

$$\sum_{i=1}^n (Z-Z'_i)_+^2 \leq \nu \text{ almost surely for some } \nu , \tag{2}$$

where $Z'_i=f(X_1,\ldots,X_{i-1},X'_i,X_{i+1},\ldots,X_n)$. This assumption is valid, for instance, in the random matrix example above.

In fact, under the strong assumption (2), one can show Gaussian concentration on scale $\sqrt{\nu}$, and not just exponential concentration. But the methods of BLM require entropy techniques. We may return to this.

Before we move on, let us note that if (2) holds, then Z must be bounded a.s. To see why, first note that we can write

$$\begin{aligned}&f(X_1,\ldots,X_n)-f(X'_1,\ldots,X'_n)\\ &=\sum_{i=1}^n\left[f(X_1,\ldots,X_i,X'_{i+1},\ldots,X'_n)-f(X_1,\ldots,X_{i-1},X'_i,\ldots,X'_n)\right] .\end{aligned}$$

Then if Z is unbounded, at least one of the summands is unbounded. But the i-th summand has the same distribution (under symmetry) as $Z - Z_i'$. So for some i, $Z - Z_i'$ must be unbounded. Therefore at least one of the terms $(Z - Z_i')_+$ or $(Z - Z_i')_-$ is unbounded. These actually have the same distribution, so $(Z - Z_i')_+$ is unbounded and finally so is $(Z - Z_i')_+^2$, which is a contradiction, since

$$(Z - Z_i')_+^2 \le \sum_{i=1}^n (Z - Z_i')_+^2 \le \nu \ .$$

What we learn from this is that the condition (2) is quite strong and has a built-in assumption of, for instance, finite moment generating function for Z for all parameter values. However, such a strong assumption is not needed for the techniques presented here to be applicable; they just simplify the exposition and more clearly show the ideas.

1.2.1 Using quantiles

The connection between variance bounds and concentration perhaps most clearly seen by applying Efron-Stein (or other inequalities, like a Poincaré inequality, which we will see soon) to the following function. For $a < b$ define $g : \mathbb{R}^n \to \mathbb{R}$ by

$$g_{a,b}(x) = \begin{cases} a & \text{if } f(x) \le a \\ f(x) & \text{if } f(x) \in [a,b] \\ b & \text{if } f(x) \ge b \end{cases} \ .$$

This is a "cutoff" of the function f, truncated below a and above b. The effect of $g_{a,b}$ will be that it isolates the level set $\{x : f(x) \in [a,b]\}$ and we will be able to bound its probability.

Here we will give a bound assuming a is chosen such that

$$\mathbb{P}(Z \le a) \ge 1/2 \ . \tag{3}$$

In this case, we can complement the variance upper bound we will obtain by Efron-Stein with a lower bound (setting $X' = (X_1', \ldots, X_n')$)

$$\begin{aligned} \text{Var } g_{a,b}(X) = \frac{1}{2}\mathbb{E}\left(g_{a,b}(X) - g_{a,b}(X')\right)^2 &\ge \frac{(b-a)^2}{2}\mathbb{P}(g_{a,b}(X) = a,\ g_{a,b}(X') = b) \\ &\ge \frac{(b-a)^2}{4}\mathbb{P}(Z \ge b) \ . \end{aligned}$$

By giving a corresponding upper bound, we can solve for $\mathbb{P}(Z \ge b)$ and get a tail bound for Z.

Set $X'(i) = (X_1, \ldots, X_{i-1}, X_i', X_{i+1}, \ldots, X_n)$ and use Efron-Stein (note that $g_{a,b}(X)$ is bounded, so is square integrable):

$$\text{Var } g_{a,b}(X) \le \sum_{i=1}^n \mathbb{E}\left(g_{a,b}(X) - g_{a,b}(X'(i))\right)_+^2$$

Note that if $g_{a,b}(X) \leq a$ then $g_{a,b}(X) - g_{a,b}(X'(i)) \leq 0$ and so it does not contribute to the sum above. This means the right side equals

$$\mathbb{E}\sum_{i=1}^{n}(g_{a,b}(X) - g_{a,b}(X'(i)))_+^2 \mathbf{1}_{\{Z>a\}} \leq \mathbb{E}\sum_{i=1}^{n}(Z - Z_i')_+^2 \mathbf{1}_{\{Z>a\}} \leq \nu\mathbb{P}(Z > a) \ .$$

Combining with the lower bound, one has

$$\frac{(b-a)^2}{4}\mathbb{P}(Z \geq b) \leq \nu\mathbb{P}(Z > a) \ ,$$

or

$$b - a \leq \sqrt{4\nu\frac{\mathbb{P}(Z > a)}{\mathbb{P}(Z \geq b)}} \ . \tag{4}$$

To obtain concentration from (4), we choose specific values of b and a: the quantiles. For $\alpha \in [0, 1)$, let the α-th quantile be defined as

$$Q_\alpha = \inf\{y \in \mathbb{R} : \mathbb{P}(Z \leq y) \geq \alpha\} \ .$$

(Since $\alpha < 1$, the set on the right is nonempty.) Note the following property of the α-th quantile.

- $Q_\alpha \leq x$ if and only if $\mathbb{P}(Z \leq x) \geq \alpha$.

  Proof. First observe that by choosing $x_n \downarrow Q_\alpha$ to be a sequence of elements in the set $S_\alpha = \{y \in \mathbb{R} : \mathbb{P}(Z \leq y) \geq \alpha\}$ then by right continuity of the distribution function of Z, $\mathbb{P}(Z \leq Q_\alpha) \geq \alpha$.

  If $\mathbb{P}(Z \leq x) \geq \alpha$ then x is in the set whose infimum we take in the definition of Q_α, so $Q_\alpha \leq x$. Conversely, if $\mathbb{P}(Z \leq x) < \alpha$ then $x \notin S_\alpha$, so $x \neq Q_\alpha$. However x is a lower bound for the set and so $Q_\alpha \geq x$, giving $Q_\alpha > x$. □

We now choose $a = Q_{1-2^{-n}}$ and $b = Q_{1-2^{-(n+1)}}$ for $n \geq 0$. Then by the above property,

$$\mathbb{P}(Z > a) = 1 - \mathbb{P}(Z \leq a) \leq 1 - (1 - 2^{-n}) = 2^{-n}$$

and

$$\mathbb{P}(Z \geq b) = 1 - \mathbb{P}(Z < b) = 1 - \lim_{c \uparrow b}\mathbb{P}(Z \leq c) \geq 2^{-(n+1)} \ .$$

Returning to (4),

$$Q_{1-2^{-(n+1)}} - Q_{1-2^{-n}} \leq \sqrt{4\nu\frac{2^{-n}}{2^{-(n+1)}}} = \sqrt{8\nu} \ . \tag{5}$$

Intuitively speaking, this upper bound on the distance between quantiles should give us a concentration bound because it is giving us a bound on the x-distance that the distribution function $F(x)$ of Z is allowed to take to raise from near $1 - 2^{-n}$ to $1 - 2^{-(n+1)}$. The shorter distance this takes, the faster the decay of the tail of Z. To formalize this,

Theorem 1.4 (Exponential concentration)**.** *Suppose that $Z = f(X_1, \ldots, X_n)$ satisfies* (2)*. Then letting $MZ = Q_{1/2}$ be the median of Z,*

$$\mathbb{P}(Z > MZ + t) \leq 2^{-\frac{t}{\sqrt{8\nu}}} \text{ for } t > 0 \ .$$

Proof. Sum (5) from $n = 1$ (for which $Q_{1-2^{-n}} = MZ$) to n for

$$Q_{1-2^{-(n+1)}} - MZ \leq n\sqrt{8\nu} \ .$$

So for $t \geq 0$, putting $n = \lfloor t/\sqrt{8\nu} \rfloor$,

$$\begin{aligned}\mathbb{P}(Z - MZ > t) \leq \mathbb{P}(Z - MZ > n\sqrt{8\nu}) \leq \mathbb{P}(Z > Q_{1-2^{-(n+1)}}) &\leq 2^{-(n+1)} \\ &= 2^{-\lfloor t/\sqrt{8\nu} \rfloor - 1} \\ &\leq 2^{-t/\sqrt{8\nu}} \ .\end{aligned}$$

□

1.2.2 Using moment generating functions

We saw above that we can get exponential concentration by applying Efron-Stein to the function $g_{a,b}(Z)$. Another method is to apply Efron-Stein to the function $e^{\lambda Z}$.

What do we get when we try this and assume (2)? For $t > 0$, assuming $\mathbb{E}e^{2tZ} < \infty$,

$$\text{Var } e^{tZ} \leq \sum_{i=1}^{n} \mathbb{E}\left(e^{tZ} - e^{tZ_i'}\right)_+^2 \ .$$

To bound the difference of exponentials, we use the mean value theorem. The quantity $(e^{tZ} - e^{tZ_i'})_+$ is nonzero only if $Z > Z_i'$. In this case we apply the mean value theorem on the interval $[Z_i', Z]$ to find c in this interval such that

$$e^{tZ} - e^{tZ_i'} = (Z - Z_i')\frac{\mathrm{d}}{\mathrm{d}x}e^{tx}\bigg|_{x=c} = (Z - Z_i')te^{\lambda c} \leq t(Z - Z_i')e^{tZ} \ .$$

Therefore

$$\text{Var } e^{tZ} \leq t^2 \sum_{i=1}^{n} \mathbb{E}e^{2tZ}(Z - Z_i')_+^2 \ .$$

Using assumption (2), we obtain the upper bound

$$t^2\mathbb{E}e^{2tZ} \sum_{i=1}^{n} (Z - Z_i')_+^2 \leq t^2\nu\mathbb{E}e^{2tZ} \ .$$

The typical way to write this uses $t/2$ instead of t:

$$\text{Var } e^{tZ/2} \leq \frac{t^2\nu}{4}\mathbb{E}e^{tZ} \text{ whenever } \mathbb{E}e^{tZ} < \infty \ .$$

To go from this bound to exponential concentration, we can use the following result.

Proposition 1.5 (Iteration method)**.** *If $\mathbb{E}Z = 0$ and for some constants B, C satisfying $0 < C \le B$,*

$$\text{Var } e^{tZ/2} \le Ct^2 \mathbb{E}e^{tZ} < \infty \text{ for } t \in (0, B^{-1/2}) ,$$

then

$$\psi_Z(t) \le -2\log(1 - Ct^2) \text{ for } t \in (0, B^{-1/2}) .$$

Proof. Beginning with

$$\mathbb{E}e^{tZ} - \left(\mathbb{E}e^{tZ/2}\right)^2 \le Ct^2 \mathbb{E}e^{tZ} ,$$

we obtain

$$\mathbb{E}e^{tZ} \le \frac{1}{1 - Ct^2} \left(\mathbb{E}e^{tZ/2}\right)^2 ,$$

or

$$\psi_Z(t) \le -\log(1 - Ct^2) + 2\psi_Z(t/2) .$$

Iterating,

$$\begin{aligned}\psi_Z(t) &\le -\log(1 - Ct^2) + 2\left(-\log(1 - C(t/2)^2) + 2\psi_Z(t/4)\right) \\ &= -\sum_{k=1}^{2} 2^{k-1} \log(1 - C(t/2^{k-1})^2) + 4\psi_Z(t/4) .\end{aligned}$$

By induction, for $n \ge 1$,

$$\psi_Z(t) \le -\sum_{k=1}^{n} 2^{k-1} \log(1 - C(t/2^{k-1})^2) + 2^n \psi_Z(t/2^n) .$$

Because $\mathbb{E}Z = 0$,

$$2^n \psi_Z(t/2^n) = t \frac{\psi_Z(t/2^n) - \psi_Z(0)}{t/2^n} \to t\psi_Z'(0) = 0 ,$$

so

$$\psi_Z(t) \le -\sum_{k=1}^{\infty} 2^{k-1} \log(1 - C(t/2^{k-1})^2) . \tag{6}$$

To bound these terms, we use the fact that $-u^{-1}\log(1-u)$ is non-decreasing in $u \in (0,1)$. This is due to convexity of the function $u \mapsto -\log(1-u)$. In other words $-u^{-1}\log(1-u)$ is the slope of the secant to the graph of $-\log(1-u)$ from 0 to u and this is nondecreasing. Therefore

$$-C^{-1}(t/2^{k-1})^{-2} \log(1 - C(t/2^{k-1})^2) \le -C^{-1}t^{-2}\log(1 - Ct^2) \text{ for } k \ge 1 .$$

Rewritten,

$$-2^{k-1}\log(1 - C(t/2^{k-1})^2) \le -2^{-(k-1)}\log(1 - Ct^2) .$$

Placing this in (6),

$$\psi_Z(t) \le -\log(1 - Ct^2) \sum_{k=1}^{\infty} 2^{-(k-1)} = -2\log(1 - Ct^2) .$$

□

To make the last bound more clear, we can use the inequality

$$-\log(1-u) \leq \frac{u}{1-u} \text{ for } u \in [0,1) ,$$

which follows from the mean value theorem: for some $c \in [0,u]$,

$$\frac{-\log(1-u) - 0}{u} = \frac{\mathrm{d}}{\mathrm{d}u} - \log(1-u)\bigg|_{u=c} = \frac{1}{1-c} \leq \frac{1}{1-u} .$$

So we obtain

$$\psi_Z(t) \leq \frac{2Ct^2}{1-Ct^2} \leq \frac{4Ct^2}{2(1-\sqrt{C}t)} \text{ for } t \in (0, B^{-1/2}) .$$

If $B = C$, this means that $Z \in \Gamma_+(c, \hat{\nu})$ for $\hat{\nu} = 4C$ and $c = \sqrt{C}$:

Theorem 1.6 (Exponential concentration)**.** *Suppose that $Z = f(X_1, \ldots, X_n)$ satisfies* (2)*. Then $Z \in \Gamma_+(\sqrt{\nu/4}, \nu)$. In other words,*

$$\mathbb{P}\left(Z - \mathbb{E}Z \geq \sqrt{2\nu t} + \frac{\sqrt{\nu}}{2}t\right) \leq e^{-t} \text{ for } t > 0 .$$

2 Poincaré inequalities

Poincaré inequalities are a type of functional inequality similar to Sobolev inequalities. Given a function $f : \mathbb{R}^n \to \mathbb{R}$, they give an upper bound of the type

$$\mathbb{E}f(X)^2 \leq C\mathbb{E}\|\nabla f(X)\|^2 \text{ if } \mathbb{E}f(X) = 0 .$$

Here $X = (X_1, \ldots, X_n)$ is some independent vector and the right side is

$$\|\nabla f(X)\|^2 = \sum_{i=1}^n \left(\frac{\partial}{\partial x_i} f\right)^2 (X) .$$

The left side is Var f, so they give us a variance bound on f in terms of the norm of the gradient. In particular, if f is 1-Lipschitz, we get a variance bound of C.

We will show below two types of Poincaré inequalities, one which holds with essentially no restrictions on f for Gaussian vectors X, and the second holds for bounded variables, but for only convex f.

2.1 Gaussian Poincaré inequality

Here we will prove:

Theorem 2.1. *Let $X = (X_1, \ldots, X_n)$ be a standard Gaussian vector and $f : \mathbb{R}^n \to \mathbb{R}$ be C^1. Then*

$$\text{Var } f(X) \leq \mathbb{E}\|\nabla f(X)\|_2^2 .$$

Proof. First consider $n = 1$, so X is just a standard Gaussian variable. The idea is to approximate X by sums of Bernoulli variables and take a limit with the CLT. We will apply Efron-Stein to the variance of a function of the Bernoullis.

We may assume that $\mathbb{E}\|\nabla f(X)\|_2^2 < \infty$ or the statement is trivial. Therefore by a standard approximation argument, we may assume that f is C^2 with compact support. So let

$$S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \epsilon_i \ ,$$

where ϵ_i are Rademacher variables (that is, they are ± 1 with probability 1/2). By Efron-Stein,

$$\text{Var } f(S_n) \leq \sum_{i=1}^{n} \mathbb{E}\left[f(S_n) - \mathbb{E}_i f(S_n)\right]^2 = \sum_{i=1}^{n} \mathbb{E}\text{Var}_i f(S_n) \ , \tag{7}$$

where Var_i is variance relative to the variable ϵ_i only. We can compute this variance using the identity $\text{Var } X = \frac{1}{2}\mathbb{E}[X - X']$, where X, X' are i.i.d.: setting ϵ_i' to be an independent copy of ϵ_i and $\mathbb{E}_i$ expectation over both ϵ_i, ϵ_i',

$$\text{Var}_i f(S_n) = \frac{1}{2}\mathbb{E}_i \left[f\left(\frac{\epsilon_1 + \cdots + \epsilon_n}{\sqrt{n}}\right) - f\left(\frac{\epsilon_1 + \cdots + \epsilon_{i-1} + \epsilon_i' + \epsilon_{i+1} + \cdots + \epsilon_n}{\sqrt{n}}\right)\right]^2 \ .$$

There are four equally likely choices for ϵ_i, ϵ_i'. Only two of them contribute, and they contribute the same amount, so this equals

$$\frac{1}{4}\left(f\left(\frac{\epsilon_1 + \cdots + \epsilon_{i-1} + 1 + \epsilon_{i+1} + \cdots + \epsilon_n}{\sqrt{n}}\right) - f\left(\frac{\epsilon_1 + \cdots + \epsilon_{i-1} - 1 + \epsilon_{i+1} + \cdots + \epsilon_n}{\sqrt{n}}\right)\right)^2 \ .$$

This can be rewritten

$$\text{Var}_i f(S_n) = \frac{1}{4}\left(f\left(S_n + \frac{1-\epsilon_i}{\sqrt{n}}\right) - f\left(S_n - \frac{1+\epsilon_i}{\sqrt{n}}\right)\right)^2 \ ,$$

and we obtain from (7)

$$\text{Var } f(S_n) \leq \frac{1}{4}\sum_{i=1}^{n} \mathbb{E}\left(f\left(S_n + \frac{1-\epsilon_i}{\sqrt{n}}\right) - f\left(S_n - \frac{1+\epsilon_i}{\sqrt{n}}\right)\right)^2 \ . \tag{8}$$

To estimate these differences we use Taylor's theorem. For each i there exists $c_i^{(1)}$ between $S_n - \frac{1+\epsilon_i}{\sqrt{n}}$ and S_n such that

$$f(S_n) - f\left(S_n - \frac{1+\epsilon_i}{\sqrt{n}}\right) = -\frac{1+\epsilon_i}{\sqrt{n}} f'(S_n) - \frac{(1+\epsilon_i)^2}{2n} f''(c_i^{(1)}) \ .$$

Applying a similar estimate for S_n and $S_n + \frac{1-\epsilon_i}{\sqrt{n}}$, we obtain for some $c_i^{(2)}$,

$$f\left(S_n + \frac{1-\epsilon_i}{\sqrt{n}}\right) - f(S_n) = f'(S_n)\frac{1-\epsilon_i}{\sqrt{n}} + f''(c_i^{(2)})\frac{(1-\epsilon_i)^2}{2n} \ .$$

Therefore taking K to be the supremum of f'',

$$\left| f\left(S_n + \frac{1-\epsilon_i}{\sqrt{n}}\right) - f\left(S_n - \frac{1+\epsilon_i}{\sqrt{n}}\right)\right| \leq 2|f'(S_n)|\frac{1}{\sqrt{n}} + \frac{2K}{n} \ ,$$

giving

$$\text{Var } f(S_n) \leq \frac{1}{4}\sum_{i=1}^n \mathbb{E}\left[2\frac{|f'(S_n)|}{\sqrt{n}} + \frac{2K}{n}\right]^2 \leq \mathbb{E}(f'(S_n))^2 + \frac{K^2}{n} \ .$$

Now apply the CLT, noting that $S_n \to X$ in distribution implies that $\mathbb{E}(f'(S_n))^2 \to \mathbb{E}(f'(X))^2$ (since f' is continuous and bounded). Therefore

$$\limsup_n \text{Var } f(S_n) \leq \limsup_n \mathbb{E}(f'(S_n))^2 = \mathbb{E}(f'(X))^2 \ .$$

By the same reasoning, $\text{Var } f(S_n) \to \text{Var } f(X)$, so we finish the case $n = 1$ with

$$\text{Var } f(X) = \limsup_n \text{Var } f(S_n) \leq \mathbb{E}(f'(X))^2 \ .$$

In the case of general n, we apply Efron-Stein directly to the variables $X_1, \ldots, X_n$ to obtain

$$\text{Var } f(X) \leq \sum_{i=1}^n \mathbb{E}\text{Var}_i f(X) \ ,$$

where Var_i is variance relative to X_i only. We want to apply the $n = 1$ result to the inner variance, so we need only verify that f is C_2 with compact support in the variable x_i, and this is true. Therefore

$$\text{Var } f(X) \leq \sum_{i=1}^n \mathbb{E}\mathbb{E}_i \left(\frac{\partial}{\partial x_i} f\right)^2 (X) = \mathbb{E}\|\nabla f(X)\|_2^2 \ .$$

□

2.2 Convex Poincaré inequality

In this section we will only assume that the X_i's are bounded in $[0, 1]$, but that f is separately convex: when all arguments except one are fixed, then f is convex in the free argument. Because of this assumption, the proof is much easier than in the last section and follows in a straightforward way from Efron-Stein.

Theorem 2.2. *Let $X_1, \ldots, X_n$ be independent with $\mathbb{P}(X_i \in [0, 1]) = 1$ for all i. Let $f : \mathbb{R}^n \to \mathbb{R}$ be separately convex whose partial derivatives exist. Then*

$$\text{Var } f(X) \leq \mathbb{E}\|\nabla f(X)\|_2^2 \ .$$

Proof. Since f is continuous it is bounded on $[0,1]^n$ so the variable $f(X)$ has two moments. By Efron-Stein,

$$\text{Var } f(X) \le \sum_{i=1}^n \mathbb{E}\text{Var}_i f(X) = \sum_{i=1}^n \mathbb{E} \min_{a\in\mathbb{R}} \mathbb{E}_i (f(X)-a)^2 \ , \tag{9}$$

where $\mathbb{E}_i$ and Var_i are expectation and variance relative to X_i only. We choose particular value of a. Since f is continuous, the function $u \mapsto f(x_1,\ldots,x_{i-1},u,x_{i+1},\ldots,x_n)$ attains its minimum on $[0,1]^n$ at some point x_i'. So we take a to be this minimum, $a = f(X_1,\ldots,X_{i-1},x_i',X_{i+1}\ldots,X_n)$. We claim that for this choice of a, one has

$$(f(X)-a)^2 \le \left(\frac{\partial}{\partial x_i} f\right)^2 (X) \ . \tag{10}$$

Translating the problem back to one dimension, we must show that if $g : [0,1] \to \mathbb{R}$ is differentiable and convex then if we set $a = \min_{x\in[0,1]} g(x)$, one has $(g(x)-a)^2 \le (g'(x))^2$. By the mean value theorem, if $a = g(u)$,

$$|g(x)-a| \le |x-u| \max\{|g'(x)|,|g'(u)|\} \le \max\{|g'(x)|,|g'(u)|\} \ .$$

If u is in the interior $(0,1)$, then $g'(u)=0$ and the maximum is $|g'(x)|$. Otherwise if $u=0$, since $g(u)$ is minimal, $g'(u)\ge 0$ and by convexity, $g'(x)\ge g'(u)$, making the maximum $|g'(x)|$ again. Last, if $u=1$ then by minimality of $g(u)$, one has $g'(u)\le 0$, and again by convexity, $g'(x)\le g'(u)\le 0$. Therefore

$$|g(x)-a| \le |g'(x)|$$

and this shows (10).

To finish the proof we combine (9) and (10):

$$\text{Var } f(X) \le \sum_{i=1}^n \mathbb{E}\mathbb{E}_i \left(\frac{\partial}{\partial x_i} f\right)^2 (X) = \mathbb{E}\|\nabla f(X)\|_2^2 \ .$$

□

We end by showing a Poincaré inequality for the two-sided exponential distribution, due to Ledoux. It has density function $g(x) = \frac{1}{2}e^{-|x|}$ for $x \in \mathbb{R}$. We begin by integrating by parts. For f smooth with compact support and X with two-sided exponential distribution,

$$\begin{aligned}
\mathbb{E}f(X) = \frac{1}{2}\int f(x)e^{-|x|}\ \mathrm{d}x &= \frac{1}{2}\int_0^\infty f(x)e^{-x}\ \mathrm{d}x + \frac{1}{2}\int_{-\infty}^0 f(x)e^{x}\ \mathrm{d}x \\
&= \frac{1}{2} - f(x)e^{-x}\Big|_0^\infty + \frac{1}{2}\int_0^\infty f'(x)e^{-x}\ \mathrm{d}x \\
&+ \frac{1}{2}f(x)e^{x}\Big|_{-\infty}^0 - \frac{1}{2}\int_{-\infty}^0 f'(x)e^{x}\ \mathrm{d}x \\
&= f(0) + \frac{1}{2}\int \text{sgn}(x)f'(x)e^{-|x|}\ \mathrm{d}x \\
&= f(0) + \mathbb{E}\left(\text{sgn}(X)f'(X)\right) \ .
\end{aligned}$$

Therefore putting $g(x) = f(x) - f(0)$,

$$\mathbb{E}g^2(X) = g^2(0) + 2\mathbb{E}\left(\operatorname{sgn}(X)g'(X)g(X)\right) = 2\mathbb{E}\left(\operatorname{sgn}(X)g'(X)g(X)\right) \\ \leq 2\sqrt{\mathbb{E}(g'(X))^2\mathbb{E}g^2(X)}\ .$$

Dividing and squaring, we obtain

$$\mathbb{E}g^2(X) \leq 4\mathbb{E}(g'(X))^2 = 4\mathbb{E}(f'(X))^2\ .$$

Last,

$$\operatorname{Var}\ f(X) = \min_{a\in\mathbb{R}} \mathbb{E}(X-a)^2 \leq \mathbb{E}g^2(X) \leq 4\mathbb{E}(f'(X))^2\ .$$

This proves:

Proposition 2.3 (Poincaré inequality for exponential)**.** *Let X have two-sided exponential distribution. If f is differentiable and satisfies* $\operatorname{Var}\ f(X) < \infty$ *then*

$$\operatorname{Var}\ f(X) \leq 4\mathbb{E}(f'(X))^2\ .$$

2.3 Relation between Poincaré and concentration

Here we will see that if the distribution of a random variable satisfies a Poincaré inequality, its tail must decay no slower than exponential. So assume that X satisfies

$$\operatorname{Var}\ f(X) \leq C\mathbb{E}(f'(X))^2$$

for all differentiable f such that $\operatorname{Var}\ f(X) < \infty$. We again choose $f = g_{a,b}$, where $a = Q_{1-2^{-n}}$ and $b = Q_{1-2^{-(n+1)}}$ are the quantiles. Although $g_{a,b}$ is not differentiable everywhere, we can use the above inequality after an approximation by smooth functions (say, convolutions with a gaussian). As before,

$$\operatorname{Var}\ g_{a,b}(X) \geq \frac{(b-a)^2}{4}\mathbb{P}(X \geq b)\ .$$

On the other hand,

$$\operatorname{Var}\ g_{a,b}(X) \leq C\mathbb{E}(g'_{a,b}(X))^2 \leq C\mathbb{P}(X \geq a)\ .$$

We then reproduce the inequality

$$b - a \leq \sqrt{4C\frac{\mathbb{P}(X > a)}{\mathbb{P}(X \geq b)}}\ ,$$

which as before let to

$$\mathbb{P}(X > MX + t) \leq 2^{-\frac{t}{\sqrt{8C}}}\ .$$

We can generalize this to metric spaces. For this we should define the concentration function of a Borel probability measure. The following material is taken from Ledoux's monograph, Sections 1.1 and 3.1.

Definition 2.4. *Let (X, d) be a metric space with a Borel probability measure μ. The concentration function α_μ is defined*

$$\alpha_\mu(r) = \sup\{1 - \mu(A_r) : A \subset X,\ \mu(A) \geq 1/2\} \text{ for } r > 0 \ .$$

Here $A_r = \{x \in X : d(x, A) < r\}$.

Here are some basic properties of α_μ.

1. $\alpha_\mu(r)$ is monotone in r with $\alpha_\mu(r) \leq 1/2$ and $\lim_{r\uparrow\infty} \alpha_\mu(r) = 0$.

   Proof. Each number $\mu(A_r)$ is monotone in r, so $\alpha_\mu(r)$ is also. Also if $\mu(A) \geq 1/2$, by monotonicity, $1 - \mu(A_r) \leq 1 - 1/2 = 1/2$. To prove the limit, let $\epsilon \in (0, 1/2)$ and $x \in X$ and note by countable additivity,

   $$1 = \mu(X) = \lim_{r\uparrow\infty} \mu(B_r(x)) \ ,$$

   so we can choose r such that $\mu(B_r(x)) > 1 - \epsilon$. If A satisfies $\mu(A) \geq 1/2$ then

   $$\mu(A \cap B_r(x)) = 1 - \mu(A^c \cup B_r(x)^c) \geq 1 - 1/2 - \epsilon > 0 \ ,$$

   so A and $B_r(x)$ contain a common point z. Now A_{2r} must cover $B_r(x)$: if $y \in B_r(x)$ then

   $$d(y, A) \leq d(y, z) \leq d(y, x) + d(x, z) < 2r \ .$$

   This means that $1 - \mu(A_{2r}) \leq 1 - \mu(B) < \epsilon$ and $\mu_\alpha(2r) < \epsilon$. By monotonicity, $\mu_\alpha(s) \leq \mu_\alpha(2r) < \epsilon$ for all $s \geq 2r$. □

2. One need not have $\lim_{r\downarrow 0} \mu_\alpha(r) = 1/2$. Take $X = \mathbb{R}$ with the usual metric and $\mu = \frac{3}{4}\delta_0 + \frac{1}{4}\delta_1$. Then if A has $\mu(A) \geq 1/2$, it must contain 0 and therefore $1 - \mu(A_r) \leq 1 - \mu(A) \leq 1/4$ for all $r \geq 0$. So in this example, $\alpha_\mu(r) \leq 1/4$ for all $r > 0$.

3. We can relate the abstract concentration function to our study of concentration of random variables. Indeed, if Z is a random variable, let MZ be a median of Z; that is, any number such that

   $$\mathbb{P}(Z \geq MZ) \geq 1/2 \text{ and } \mathbb{P}(Z \leq MZ) \leq 1/2 \ .$$

   Let μ be the distribution of Z and take $A = (-\infty, MZ]$. For $r > 0$,

   $$\mathbb{P}(Z \geq MZ + r) = 1 - \mu(A_r) \leq \alpha_\mu(r) \ .$$

   Note this is true for every median MZ.

For our promised relation between concentration and Poincaré, define the generalized length of the gradient of a function $f : X \to \mathbb{R}$ at a point $x \in X$ by

$$|\nabla f(x)| := \limsup_{y \to x} \frac{|f(x) - f(y)|}{d(x,y)} \ .$$

Note this will only be finite everywhere if f is locally Lipschitz. Consider the following Poincaré inequality

$$\text{Var } f \leq C \int |\nabla f|^2 \, \mathrm{d}\mu \ , \tag{11}$$

where variance is evaluated relative to μ.

Theorem 2.5. *Assume that for all locally Lipschitz real-valued functions f on X,* (11) *holds. Then*

$$\alpha_\mu \leq \exp\left(-\frac{r}{20\sqrt{C}}\right) \ \textit{for } r > 0 \ .$$

Proof. Let A be an open subset of X satisfying $\mu(A) \geq 1/2$ and let $\epsilon > 0$. Define B to be the open set

$$B = \{x \in X : d(x, A) > \epsilon\} \ .$$

Note that $B \supset A^c_{2\epsilon}$. Furthermore, set $a = \mu(A)$ and $b = \mu(B)$. We will apply the Poincaré inequality to a function that is equal to $1/a$ on A, $-1/b$ on B and linear in between:

$$f(x) = \frac{1}{a} - \frac{1}{\epsilon}\left(\frac{1}{a} + \frac{1}{b}\right) \min\{d(x, A), \epsilon\} \ .$$

So we need to bound both sides of the inequality.

First $|\nabla f(x)| = 0$ when $x \in A \cup B$ and

$$|\nabla f(x)| \leq \frac{1}{\epsilon}\left(\frac{1}{a} + \frac{1}{b}\right) \ \text{for } x \in (A \cup B)^c \ .$$

To prove this, put $c = \frac{1}{\epsilon}\left(\frac{1}{a} + \frac{1}{b}\right)$ and compute

$$\limsup_{y \to x} \frac{|f(y) - f(x)|}{d(y,x)} = \limsup_{y \to x} \frac{c|\min\{d(x,A), \epsilon\} - \min\{d(y,A), \epsilon\}|}{d(x,y)} \leq c \ .$$

Therefore

$$\int |\nabla f(x)|^2 \, \mathrm{d}\mu \leq \frac{1}{\epsilon^2}\left(\frac{1}{a} + \frac{1}{b}\right)^2 (1 - a - b) \ .$$

On the other hand, we can bound the variance as follows:

$$\begin{aligned}
\text{Var } f = \mathbb{E}(f - \mathbb{E}f)^2 &\geq \int_A \left(f - \int f \, \mathrm{d}\mu\right)^2 \, \mathrm{d}\mu + \int_B \left(f - \int f \, \mathrm{d}\mu\right)^2 \, \mathrm{d}\mu \\
&= a\left(\frac{1}{a} - \mathbb{E}f\right)^2 + b\left(-\frac{1}{b} - \mathbb{E}f\right)^2 \\
&= \frac{1}{a} - 2\mathbb{E}f + a(\mathbb{E}f)^2 + \frac{1}{b} + 2\mathbb{E}f + b(\mathbb{E}f)^2 \\
&\geq \frac{1}{a} + \frac{1}{b} \ .
\end{aligned}$$

Thus Poincaré gives

$$\frac{1}{a} + \frac{1}{b} \leq \frac{C}{\epsilon^2} \left(\frac{1}{a} + \frac{1}{b} \right)^2 (1 - a - b) \ ,$$

or

$$\frac{\epsilon^2}{C} \leq \left(\frac{1}{a} + \frac{1}{b} \right) (1 - a - b) \leq \frac{1 - a - b}{ab} \ .$$

Select $\epsilon = \sqrt{2C}$ to get

$$2ab \leq 1 - a - b \text{ or } b \leq \frac{1 - a}{1 + 2a} = \frac{\mu(A^c)}{1 + 2a} \ .$$

Using $a = \mu(A) \geq 1/2$, we obtain $b \leq \mu(A^c)/2$, and since $B \supset A^c_{2\epsilon}$,

$$\mu((A_{2\epsilon})^c) \leq \mu(A^c)/2 \ .$$

Iterating this,

$$\mu((A_{2k\epsilon})^c) \leq \mu(A^c)/2^k \leq 2^{-(k+1)} \ .$$

So given A open with $\mu(A) \geq 1/2$ and $r > 0$, set $k = \lfloor r/(2\epsilon) \rfloor$, so that

$$\begin{aligned} 1 - \mu(A_r) \leq \mu((A_{2k\epsilon})^c) \leq 2^{-(k+1)} &= 2^{-\lfloor r/(2\epsilon) \rfloor - 1} \\ &\leq 2^{-r/(2\epsilon)} = \exp\left(-r \frac{\log 2}{2\sqrt{2C}} \right) \leq \exp\left(-r/(10\sqrt{C}) \right) \ . \end{aligned}$$

To deal with non-open A with $\mu(A) \geq 1/2$, apply the above result to the open set $A_{r/2}$ for $r > 0$ to obtain

$$1 - \mu(A_r) = 1 - \mu(A_{r/2 + r/2}) \leq \exp\left(-r/(20\sqrt{C}) \right) \ \text{ for } r > 0 \ .$$

This gives $\alpha_\mu(r) \leq \exp\left(-r/(20\sqrt{C}) \right)$. □

Remarks

- We can phrase a Poincaré inequality in terms of the spectral gap for the Laplace operator. For example, taking Ω to be a smooth bounded domain in $\mathbb{R}^d$ with volume 1. Then an eigenvector of $-\Delta$ is given by a function u satisfying

  $$-\Delta u = \lambda u \ ,$$

  where λ is the eigenvalue. In this case, all eigenvalues are nonnegative. The smallest is 0, and constants are eigenvectors. By the method of Rayleigh quotient, one can obtain the further eigenvalues by

  $$\lambda_j = \inf\{ \langle -\Delta u, u \rangle : \|u\|_2 = 1, \ u \perp U_0, \ldots, U_{j-1} \} \ ,$$

where we have written the eigenvalues as $0 = \lambda_0 < \lambda_1 < \cdots$ with eigenspaces $U_0, U_1, \ldots$. In the case of $j = 1$, we obtain

$$\lambda_1 = \inf\{\langle -\Delta u, u\rangle : \|u\|_2 = 1,\ u \perp 1\}\ ,$$

or using $\langle -\Delta u, v\rangle = \langle \nabla u, \nabla v\rangle$,

$$\lambda_1 = \inf_{u\neq 0,\ u\perp 1} \frac{\int |\nabla u|^2}{\int |u|^2}\ .$$

The condition $u \perp 1$ is the same as $\int u = 0$, so

$$\lambda_1 = \inf_{u\neq 0} \frac{\int |\nabla u|^2}{\text{Var } u}\ .$$

This means for all nonzero u,

$$\int |\nabla u|^2 \geq \lambda_1 \text{Var } u\ ,$$

which is our Poincaré inequality with $\lambda_1 = 1/C$.

- Poincaré inequalities (and so the spectral gap for the discrete Laplacian) are also related to convergence rates for Markov chains. See Ledoux, Section 3.1 for more details.

1 Entropy and friends

1.1 What is entropy?

Let $\Omega = \{1, \ldots, n\}$ be a finite probability space and $\mathbb{P}$ be a probability measure on Ω with probabilities $p_k = \mathbb{P}(\{k\})$. If $X_1, X_2, \ldots$ are i.i.d. variables with distribution $\mathbb{P}$, we can look at the probability that $(X_1, \ldots, X_m)$ is equal to a given sequence:

$$\mathbb{P}((X_1, \ldots, X_m) = (a_1, \ldots, a_m)) = p_{a_1} \cdots p_{a_m} = \exp\left(\sum_{i=1}^{m} \log p_{a_i}\right) .$$

If $\mathbb{P}$ is not a point mass, then this expression will typically be exponentially small, so it is reasonable to write it as

$$\exp\left(-m\left(-\frac{1}{m}\sum_{i=1}^{m} \log p_{a_i}\right)\right) .$$

The quantity $-\frac{1}{m}\sum_{i=1}^{m} \log p_{a_i}$ is a measure of the likelihood of a particular sequence $(a_1, \ldots, a_m)$. It will be larger if the sequence is less likely.

Now we can ask: if we sample $X_1, \ldots, X_m$ from $\mathbb{P}$, what is the typical probability under $\mathbb{Q}$ of the sequence $X_1, \ldots, X_m$? As in the above discussion, this can be measured by

$$-\frac{1}{m}\sum_{i=1}^{m} \log q_{X_i} ,$$

where $q_1, \ldots, q_n$ are the probabilities of $\mathbb{Q}$. By the law of large numbers, one has

$$-\frac{1}{m}\sum_{i=1}^{m} \log q_{X_i} \to -\mathbb{E} \log q_X ,$$

where X has distribution $\mathbb{Q}$. We can evaluate this limit as

$$-\sum_{k=1}^{n} p_i \log q_i .$$

When $\mathbb{P} = \mathbb{Q}$, this is a measure of the likelihood under $\mathbb{P}$ of a typical sequence sampled from $\mathbb{P}$. It will be larger if $\mathbb{P}$ is less deterministic. So, in a sense, it measures the amount of information given in an i.i.d. sequence sampled from $\mathbb{P}$, or the amount of uncertainty we have about $\mathbb{P}$.

Definition 1.1. *The entropy of $\mathbb{P}$ is defined as $H(X) = -\sum_{i=1}^{n} p_i \log p_i$, where X has distribution $\mathbb{P}$. Here $0 \log 0$ is defined as 0.*

When $\mathbb{P} \neq \mathbb{Q}$, we sample a sequence according to $\mathbb{P}$ and measure the difference in likelihood of sequences when we compute the probabilities using $\mathbb{P}$ and $\mathbb{Q}$:

Definition 1.2. *For $\mathbb{P}$ absolutely continuous relative to $\mathbb{Q}$, the relative entropy (or Kullback-Leibler divergence) of $\mathbb{P}$ and $\mathbb{Q}$ is defined*

$$D(\mathbb{P}||\mathbb{Q}) = \sum_{i=1}^{n} p_i \log \frac{p_i}{q_i} = -\sum_{i=1}^{n} p_i \log q_i - \left(-\sum_{i=1}^{n} p_i \log p_i \right) .$$

Otherwise $D(\mathbb{P}||\mathbb{Q})$ is defined as ∞.

Example. If $\Omega = \{1, 2\}$ and $\mathbb{P}(1) = p$, $\mathbb{P}(2) = q$, then we can compute

$$H(X) = -p \log p - q \log q = -p \log p - (1-p) \log(1-p) .$$

In the symmetric case that $p = q = 1/2$, we obtain $H(X) = \log 2$. When $p < q$, the most likely sequence of m terms is

$$\mathbb{P}((X_1, \ldots, X_m) = (2, \ldots, 2)) = q^m = \exp(-m \log q) ,$$

so we might expect the entropy to equal $\log q$. However, there is only one sequence with this probability, and most sequences have smaller probability. This makes the entropy lower than $\log q$ (it is a convex combination of $\log p$ and $\log q$). Therefore the entropy represents a tug-of-war between the number of sequences with certain probabilities and the probabilities themselves.

Continuing this reasoning, we can ask how many sequences of length m have this typical probability $e^{-mH(X)}$ (or close to it). We know that the set Ω_m of such sequences has large probability by the law of large numbers and they all have roughly the same probability. Therefore we should have $\#\Omega_m e^{-mH(X)} \sim 1$, or $\#\Omega_m \sim e^{mH(X)}$. This means that we can also view $H(X)$ as the exponential rate of growth of the number of sequences with the typical probability $e^{-mH(X)}$.

In a sense, at least on an exponential scale, the distribution of the i.i.d. sequence $(X_1, \ldots)$ becomes uniformly distributed on a set whose cardinality grows like $e^{mH(X)}$. This in itself is a form of concentration, so it is not a surprise that entropy is related to concentration.

Precisely,

Theorem 1.3 (MacMillan Theorem)**.** *Let $X, X_1, X_2, \ldots$ be i.i.d. variables taking values in $S = \{1, \ldots, n\}$ with $H(X) > 0$. Given $\epsilon > 0$, for all sufficiently large m, one may find a set of sequences Ω_m of length m with elements from S such that*

1. $e^{m(H(X)-\epsilon)} \leq \#\Omega_m \leq e^{m(H(X)+\epsilon)}$,
2. $\lim_m \mathbb{P}((X_1, \ldots, X_m) \in \Omega_m) = 1$ *and*
3. *for each $(a_1, \ldots, a_m) \in \Omega_m$, one has*

$$e^{-m(H(X)+\epsilon)} \leq \mathbb{P}((X_1, \ldots, X_m) = (a_1, \ldots, a_m)) \leq e^{-m(H(X)-\epsilon)} .$$

Proof. This proof is from Koralov and Sinai, p. 28. Write as before $p_j = \mathbb{P}(X = j)$. For a sequence $\mathbf{a} = (a_1, \ldots, a_m)$ and $j \in S$, let $f_j(\mathbf{a})$ be the fraction of elements of the sequence equal to j:

$$f_j(\mathbf{a}) = \frac{\#\{k = 1, \ldots, m : a_k = j\}}{m} \ .$$

Given $\delta > 0$, define

$$\Omega_m = \{\mathbf{a} = (a_1, \ldots, a_m) : |f_j(\mathbf{a}) - p_j| < \delta \text{ for all } j = 1, \ldots, m\} \ .$$

This is the set of sequences of length m such that the frequency of appearance of the number j is close to p_j. From the law of large numbers, $\mathbb{P}((X_1, \ldots, X_m) \in \Omega_m) \to 1$ as $m \to \infty$. This is the second statement of the theorem.

Assume that all p_j's are strictly positive (otherwise we remove those j's from the set S). Then

$$\mathbb{P}((X_1, \ldots, X_m) = (a_1, \ldots, a_m)) = p_1^{mf_1(\mathbf{a})} \cdots p_n^{mf_n(\mathbf{a})} = \exp\left(\sum_{j=1}^n mf_j(\mathbf{a}) \log p_j\right) \ .$$

We can further rewrite this probability as

$$\begin{aligned} &\exp\left(m \sum_{j=1}^n p_j \log p_j\right) \exp\left(m \sum_{j=1}^n (f_j(\mathbf{a}) - p_j) \log p_j\right) \\ = \ &\exp\left(m \left(-H(X) + \sum_{j=1}^n (f_j(\mathbf{a}) - p_j) \log p_j\right)\right) \ . \end{aligned}$$

If δ is chosen small enough, and $\mathbf{a} \in \Omega_m$ then

$$\left|\sum_{j=1}^n (f_j(\mathbf{a}) - p_j) \log p_j\right| \le \epsilon \ ,$$

and this shows the third statement.

For the first, write

$$1 \ge \mathbb{P}(\Omega_m) = \sum_{\mathbf{a} \in \Omega_m} \mathbb{P}((X_1, \ldots, X_m) = \mathbf{a}) \ge e^{-m(H(X)+\epsilon)} \#\Omega_m \ ,$$

giving $\#\Omega_m \le e^{m(H(X)+\epsilon)}$. On the other hand, for large m,

$$\frac{1}{2} \le \mathbb{P}(\Omega_m) \le e^{-m(H(X)-\epsilon/2)} \#\Omega_m \ ,$$

giving $\#\Omega_m \ge \frac{1}{2} e^{m(H(X)-\epsilon/2)} \ge e^{n(H(X)-\epsilon)}$ for large m. $\square$

1.2 Properties of entropy

We take $\Omega = \mathbb{N}$ and $\mathbb{P}, \mathbb{Q}$ two probability measures on Ω with $p_k = \mathbb{P}(\{k\})$ and $q_k = \mathbb{Q}(\{k\})$. The definition of entropy is exactly the same on this countably infinite sample space as it was on $\{1, \dots, n\}$.

- If $\Omega = \{1, \dots, n\}$ and X has distribution $\mathbb{P}$ then $H(X) \in [0, \log n]$ with $H(X) = \log n$ if and only if X is uniformly distributed and $H(X) = 0$ if and only if X is deterministic. Therefore a delta mass is the "most deterministic" and uniform distribution is the "least deterministic."

  Proof. The function $x \mapsto x \log x$ is convex, so by Jensen's inequality,

  $$\sum_{k=1}^{n} p_k \log p_k = n \frac{1}{n} \sum_{k=1}^{n} p_k \log p_k \geq n \left(\frac{1}{n} \sum_{k=1}^{n} p_k \right) \log \left(\frac{1}{n} \sum_{k=1}^{n} p_k \right) = -\log n \ ,$$

  giving $H(X) \leq \log n$ with equality if and only if the function $k \mapsto p_k \log p_k$ is constant; that is, that X is uniformly distributed. For the other statement, $H(X)$ is a sum of nonnegative terms $-p_k \log p_k$ and so is nonnegative, zero if and only if each term is zero. This means that $p_k \in \{0, 1\}$ for all k and this is true if and only if X is deterministic. □

- $D(\mathbb{P}||\mathbb{Q}) \geq 0$ with equality if and only if $\mathbb{P} = \mathbb{Q}$. This means that when we sample sequences from $\mathbb{P}$, their probabilities are maximized when we compute them under $\mathbb{P}$.

  Proof. Use the inequality $\log x \leq x - 1$ for $x \geq 0$ with equality if and only if $x = 1$:

  $$D(\mathbb{P}||\mathbb{Q}) = -\sum_{k} p_k \log \frac{q_k}{p_k} \geq \sum_{k} p_k \left(\frac{q_k}{p_k} - 1 \right) = 0 \ .$$

  Equality holds if and only if $p_k = q_k$ for all k. □

If X and Y are two random variables taking values in Ω then we can consider the entropy $H(X, Y)$ of the vector (X, Y). This is just the entropy computed as a measure on the product space $\Omega \times \Omega$ and represents the information contained in this joint distribution. If we subtract the information contained in the distribution of X, what remains is the conditional entropy of Y given X.

Definition 1.4. *The conditional entropy of Y given X is*

$$H(Y \mid X) = H(X, Y) - H(X) \ .$$

Just as in the definition of conditional probability, we imagine that we "know" the value of X. Here we simply subtract the uncertainty we have about X, since we know its value.

- Joint entropy is sub-additive:

$$H(X,Y) \leq H(X) + H(Y) \ .$$

Therefore $H(Y \mid X) \leq H(Y)$. This means we may lose some uncertainty about the value of Y when we know X.

Proof. We write

$$H(X) + H(Y) - H(X,Y) = -\sum_j p_j \log p_j - \sum_k q_k \log q_k + \sum_{k,j} p(j,k) \log p(j,k) \ ,$$

where $p(j,k) = \mathbb{P}(X = j, Y = k)$. We can rewrite this as

$$\sum_{k,j} p(j,k) \log \frac{p(j,k)}{p_j q_k} \ ,$$

which is the relative entropy of the distribution of (X,Y) and the product distribution of X with Y (note the absolute continuity here, so the relative entropy is defined – whenever $p_j = 0$ or $q_k = 0$ take 0 for the above summand). This is nonnegative by the above properties of relative entropy. $\square$

- Conditional entropy can be written relative to the conditional measure

$$H(X \mid Y) = -\sum_{k,j}^{n} p(j,k) \log \frac{p(j,k)}{q_k} = -\sum_{k,j}^{n} q_k p(j|k) \log p(j|k) \ ,$$

where $p(j|k)$ is the conditional probability $\mathbb{P}(X = j \mid Y = k)$. This is

$$-\sum_k q_k \left[\sum_j p(j|k) \log p(j|k) \right] = \mathbb{E}(H_{\mathbb{P}_k}(X)) \ ,$$

where $\mathbb{P}_k$ is the conditional measure $\mathbb{P}(X = \cdot \mid Y = k)$. Therefore conditional entropy is nonnegative. Combining with the previous remark,

$$0 \leq H(X \mid Y) = H(X,Y) - H(Y) \leq H(X) \ .$$

- (Conditional version of subadditivity) If Z is another variable, then $H(X, Y \mid Z) \leq H(X \mid Z) + H(Y \mid Z)$. Therefore $H(Y \mid X, Z) \leq H(Y \mid Z)$.

Proof. Apply subadditivity to entropy relative to the conditional measure $\mathbb{P}_l(\cdot) = \mathbb{P}(\cdot \mid Z = l)$:

$$H_{\mathbb{P}_l}(X,Y) \leq H_{\mathbb{P}_l}(X) + H_{\mathbb{P}_l}(Y)$$

and take expectation for the first statement. For the second, argue as in the unconditioned case:

$$H(Y \mid Z) \geq H(X, Y \mid Z) - H(X \mid Z) = H(X,Y,Z) - H(X,Z) = H(Y \mid X, Z) \ .$$

$\square$

- The mutual information between X and Y is defined as

$$H(X) + H(Y) - H(X,Y) \ .$$

This is nonnegative and zero if and only if X and Y are independent.

Proof. We saw this is the relative entropy between the joint distribution of (X, Y) and the product distribution of X and Y. This is nonnegative and zero if and only if these distributions are equal; that is, X and Y are independent. □

- (Chain rule) If $X_1, \ldots, X_n$ are random variables taking values in Ω then

$$H(X_1, \ldots, X_n) = H(X_1) + H(X_2 \mid X_1) + \cdots + H(X_n \mid X_1, \ldots, X_{n-1}) \ .$$

Proof. Write for $i = 2, \ldots, n$,

$$H(X_i \mid X_1, \ldots, X_{i-1}) = H(X_1, \ldots, X_i) - H(X_1, \ldots, X_{i-1}) \ ,$$

so that the right side of the claimed equation is a telescoping sum. □

- (Han's inequality) If $X_1, \ldots, X_n$ are random variables taking values in Ω then

$$H(X_1, \ldots, X_n) \le \frac{1}{n-1} \sum_{i=1}^{n} H(X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n) \ .$$

Proof. For $i = 1, \ldots, n$, use conditional subadditivity for

$$\begin{aligned} H(X_1, \ldots, X_n) &= H(X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n) + H(X_i \mid X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n) \\ &\le H(X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n) + H(X_i \mid X_1, \ldots, X_{i-1}) \ . \end{aligned}$$

Summing over i,

$$\begin{aligned} nH(X_1, \ldots, X_n) &\le \sum_{i=1}^{n} [H(X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n) + H(X_i \mid X_1 \ldots, X_{i-1})] \\ &= \sum_{i=1}^{n} H(X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n) + H(X_1, \ldots, X_n) \ . \end{aligned}$$

□

1.3 Application: influences on the hypercube

We can apply these ideas to get good bounds for total influences for subsets of the hypercube. Let $\mathbb{P}$ be the uniform measure on $\{0,1\}^n$ and $X = (X_1, \ldots, X_n)$ a vector distributed as $\mathbb{P}$ (the X_i's are then i.i.d. Bernoulli(1/2)). For $i = 1, \ldots, n$, write $X_+(i)$ for the vector X after replacing the i-th coordinate with 1 and $X_-(i)$ for the vector X after replacing the i-th coordinate with 0.

Definition 1.5. *Let A be a subset of $\{-1,+1\}^n$. The influence of the i-th coordinate on A is defined as*

$$I_i(A) = \mathbb{P}\left(X_i \text{ is pivotal for } A\right) \ ,$$

where X_i is pivotal means that

$$\mathbf{1}_A(X_+(i)) \neq \mathbf{1}_A(X_-(i)) \ .$$

The total influence is defined as

$$I(A) = \sum_{i=1}^{n} I_i(A) \ .$$

The event that X_i is pivotal for A is the event that changing the i-th element of X influences whether or not $X \in A$. For instance, if A is the event that the sum of the X_k's equals 0, then $I_i(A)$ is the probability that the sum of the X_k's for $k \neq i$ is equal to either $+1$ or -1.

As we will see later, influences are related to the sharp threshold phenomenon, but they are also related to isoperimetric inequalities. Such an inequality is of the type

$$\text{Area } \partial A \leq \phi\left(\text{Vol } A\right)$$

for some function ϕ. In our case, A is a discrete subset of the hypercube, so volume is analogous to cardinality. We can define a version of boundary as the *edge boundary* of A:

$$\partial_E(A) = \{(\{x,y\} : x \in A, \ y \in A^c, \ d_H(x,y) = 1\} \ ,$$

where d_H is the Hamming distance

$$d_H(x,y) = \sum_{i=1}^{n} |x_i - y_i|$$

and $x = (x_1, \ldots, x_n)$, $y = (y_1, \ldots, y_n)$.

The edge boundary is related to influences as follows:

$$I(A) = \frac{2\#\partial_E(A)}{2^n} \ . \tag{1}$$

To prove this, let $\partial_E^i(A)$ be the set of edges $\{x,y\}$ in $\partial_E(A)$ such that x and y differ only in their i-th coordinate. Then X is an endpoint of an edge in $\partial_E^i(A)$ if and only if X_i is pivotal for A. Since each such edge has two endpoints,

$$\frac{2\#\partial_E(A)}{2^n} = \sum_{i=1}^{n} \frac{2\#\partial_E^i(A)}{2^n} = \sum_{i=1}^{n} \mathbb{P}(X_i \text{ is pivotal for } A) = I(A) \ .$$

Given relation (1), an isoperimetric inequality would be of the form $\#A \leq \phi\left(\partial_E(A)\right)$ or, noting that $\mathbb{P}(A) = \#A/2^n$,

$$\mathbb{P}(A) \leq \psi(I(A))$$

for some other function ψ.

Bounding influences. (Isoperimetry). An easy way to give a lower bound for the total influence is to use Efron-Stein. We claim:

$$I(A) \geq 4\mathbb{P}(A)(1-\mathbb{P}(A)) \ . \tag{2}$$

To prove this, apply Efron-Stein to the random variable $Z = \mathbf{1}_A(X)$:

$$\text{Var } Z \leq \frac{1}{2}\sum_{i=1}^{n} \mathbb{E}(Z - Z_i')^2 \ .$$

Recall that $Z_i' = \mathbf{1}_A(X_1, \ldots, X_{i-1}, X_i', X_{i+1}, \ldots, X_n)$, where X_i' is an independent copy of X_i. Note that this difference is only nonzero when i is pivotal, and in this case an upper bound for the difference is 1. Therefore

$$\begin{aligned}\mathbb{P}(A)(1-\mathbb{P}(A)) = \text{Var } Z &\leq \frac{1}{2}\sum_{i=1}^{n} \mathbb{E}(Z - Z_i')^2 \mathbf{1}_{i \text{ is pivotal}}(X) \\ &\leq \frac{1}{2}\sum_{i=1}^{n} \mathbb{P}(X_i \neq X_i', i \text{ is pivotal}) \ .\end{aligned}$$

However the event that i is pivotal does not depend on X_i or X_i'. So we obtain the bound

$$\frac{1}{4}\sum_{i=1}^{n} \mathbb{P}(i \text{ is pivotal}) = \frac{I(A)}{4} \ .$$

This shows (2).

To get a better lower bound (at least when $\mathbb{P}(A)$ is small), we can use entropy and Han's inequality.

Theorem 1.6. *For any $A \subset \{-1,+1\}^n$,*

$$I(A) \geq 2\mathbb{P}(A)\log_2 \frac{1}{\mathbb{P}(A)} \ .$$

Proof. We first derive an auxiliary result. We claim that

$$\#E(A) \leq \frac{\#A}{2}\log_2 \#A \ . \tag{3}$$

Let $Y = (Y_1, \ldots, Y_n)$ be a vector uniformly distributed on A, so that for each $y \in A$, $\mathbb{P}(Y = y) = 1/\#A$. Write $Y^{(i)} = (Y_1, \ldots, Y_{i-1}, Y_{i+1}, \ldots, Y_n)$ for the vector Y with the entry Y_i removed. Then $H(Y) = H(Y_i, Y^{(i)})$ and

$$H(Y) - H(Y^{(i)}) = H(Y_i \mid Y^{(i)}) = -\sum_{y\in A} p(y)\log p(y_i \mid y^{(i)}) = -\frac{1}{\#A}\sum_{y\in A}\log p(y_i \mid y^{(i)}) \ ,$$

where $p(y) = \mathbb{P}(Y = y)$ and $p(y_i \mid y^{(i)})$ is the conditional probability $\mathbb{P}(Y_i = y_i \mid Y^{(i)} = y^{(i)})$. As in the properties above, we take this conditional probability to be zero when $\mathbb{P}(Y^{(i)} = y^{(i)}) = 0$. Note that for $y \in A$, this conditional probability is either 1 or 1/2 depending on the following:

$$p(y_i \mid y^{(i)}) = \begin{cases} 1 & \text{if } \hat{y} \notin A \\ 1/2 & \text{if } \hat{y} \in A \end{cases},$$

where $\hat{y} = (y_1, \ldots, y_{i-1}, 1 - y_i, y_{i+1}, \ldots, y_n)$. The first case is equivalent to $\{y, \hat{y}\} \in \partial_E(A)$, whereas the second is equivalent to $\{y, \hat{y}\} \notin \partial_E(A)$. So we obtain

$$-\sum_{y \in A} \log p(y_i \mid y^{(i)}) = \log 2 \ \#\{y \in A : \{y, \hat{y}\} \notin \partial_E(A)\} \ .$$

The number $\#\partial_E(A)^c$ is equal to $\#E(A)$, where $E(A)$ is the set of edges between points in A. Therefore

$$H(Y) - H(Y^{(i)}) = \frac{\log 2}{\#A} \ \#\{y \in A : \{y, \hat{y}\} \in E(A)\} = 2 \log 2 \ \frac{\#E_i(A)}{\#A} \ ,$$

where $E_i(A)$ is the set of edges between points in A of the form $\{y, \hat{y}\}$. Summing over all i, we obtain

$$\sum_{i=1}^n (H(Y) - H(Y^{(i)})) = 2 \log 2 \ \frac{\#E(A)}{\#A} \ .$$

On the other hand, Han's inequality gives

$$\begin{aligned} \sum_{i=1}^n (H(Y) - H(Y^{(i)})) &= nH(Y) - (n-1) \cdot \frac{1}{n-1} \sum_{i=1}^n H(Y^{(i)}) \\ &\leq nH(Y) - (n-1)H(Y) = H(Y) \ . \end{aligned}$$

Therefore

$$\log \#A = H(Y) \geq 2 \log 2 \ \frac{\#E(A)}{\#A} \ ,$$

which implies (3).

To relate this result to what we want to show for influences, note that each point of A is an endpoint of n edges. So

$$\begin{aligned} n\#A &= \sum_{y \in A} \#\{y' : d_H(y, y') = 1\} \\ &= \sum_{y \in A} (\#\{y' : d_H(y, y') = 1, \{y, y'\} \in E(A)\} + \#\{y' : d_H(y, y') = 1, \{y, y'\} \notin E(A)\}) \\ &= 2\#E(A) + \#\partial_E(A) \ . \end{aligned}$$

Using (3), one then has

$$n\#A \leq \#A \log_2 \#A + \#\partial_E(A) \ ,$$

or $\#\partial_E(A) \geq \#A(n - \log_2 \#A) = -\#A \log_2 \left(\frac{\#A}{2^n}\right)$. Dividing by 2^n,

$$2\frac{\#\partial_E(A)}{2^n} \geq \frac{\#A}{2^n} \log_2 \frac{2^n}{\#A} ,$$

which is the statement of the theorem. □

- The above bound is much better than Efron-Stein when $\#A$ is small. Take for instance A to have one element. Then $I(A) = 2\#\partial_E(A)/2^n = 2n/2^n$ whereas $2\mathbb{P}(A) \log_2 \frac{1}{\mathbb{P}(A)} = 2n/2^n$, so the bounds match. If we had used Efron-Stein, we would have $2n/2^n \geq \frac{1}{2^n}\frac{2^n-1}{2^n}$, and the left side can be much larger than the right.
- The intermediate result that for every $A \subset \{-1,+1\}^n$, one has

$$\#E(A) \leq \frac{\#A}{2} \log_2 \#A$$

  is clear in the case $A = \{-1,+1\}^n$. In this case, $\#E(A) = n\#A/2 = n2^{n-1}$ and $\#A = 2^n$, so the result reads $n2^{n-1} \leq n2^{n-1}$. Our inequality tells us that this holds for all A, not just the full space.

1.4 Subadditivity of entropy

In one common case, we can use subadditivity to bound the entropy of a random variable. Consider our setup from the Efron-Stein section, in which we have a random variable Z which is a function of several independent random variables $X_1, \ldots, X_n$. In the text, BLM introduce subadditivity of entropy as a type of strengthening of the Efron-Stein inequality. To see how, let's recall the inequality:

$$\text{Var } Z \leq \frac{1}{2} \sum_{i=1}^n \mathbb{E}\left(Z - Z_i'\right)^2 ,$$

where Z_i' is the variable $Z(X_1, \ldots, X_{i-1}, X_i', X_{i+1}, \ldots, X_n)$, and X_i' is an independent copy of X_i. We have seen before that this can be rewritten as

$$\text{Var } Z \leq \sum_{i=1}^n \mathbb{E}\text{Var}_i Z ,$$

where Var_i is the variance computed only relative to the variable X_i (with all other variables fixed). Rewriting this once again and setting $\Phi(x) = x^2$,

$$\begin{aligned} \mathbb{E}\Phi(Z) - \Phi\left(\mathbb{E}Z\right) = \text{Var } Z &\leq \sum_{i=1}^n \mathbb{E}\text{Var}_i Z \\ &\leq \sum_{i=1}^n \mathbb{E}\left(\mathbb{E}_i\Phi(Z) - \Phi\left(\mathbb{E}_i Z\right)\right) . \end{aligned}$$

Here we have written $\mathbb{E}_i$ as expectation relative only to X_i. This can be viewed as a type of subadditivity statement. The same statement holds for entropy.

Theorem 1.7 (Subadditivity of entropy)**.** *Let* $\Phi(x) = x \log x$ *for* $x > 0$ *and* $\Phi(0) = 0$. *If* $X_1, \ldots, X_n$ *are independent random variables taking values in a countable set* $\mathcal{X}$ *and* $f : \mathcal{X}^n \to [0, \infty)$, *set* $Z = f(X_1, \ldots, X_n)$. *Then*

$$Ent\ Z := \mathbb{E}\Phi(Z) - \Phi(\mathbb{E}Z) \leq \sum_{i=1}^n \mathbb{E}\left(\mathbb{E}_i\Phi(Z) - \Phi\left(\mathbb{E}_i Z\right)\right) = \sum_{i=1}^n \mathbb{E} Ent_i\ Z\ ,$$

where $Ent_i\ Z = \mathbb{E}_i\Phi(Z) - \Phi\left(\mathbb{E}_i Z\right)$.

Here we are using $Ent\ Z$ as notation for $\mathbb{E}\Phi(Z) - \Phi\left(\mathbb{E}Z\right)$, the entropy of a nonnegative random variable. This definition is different from that of Shannon entropy defined earlier, but in some ways it is similar. For example, if $Z \geq 0$ has $\mathbb{E}Z = 1$ (and therefore is a density of another measure), $Ent\ Z = \mathbb{E}Z \log Z - \mathbb{E}Z \log \mathbb{E}Z = \mathbb{E}Z \log Z$. In fact, we can view this entropy of a type of relative entropy, and this will appear in the proof.

Proof. First assume that we have proved the result for all Z such that $\mathbb{E}Z = 1$. If $Z \geq 0$ then if $\mathbb{E}Z = 0$, then $Z = 0$ almost surely and then both sides are zero. Otherwise $\mathbb{E}Z > 0$ and we can set $Y = Z/\mathbb{E}Z$. Applying the result for Y,

$$Ent\ Y \leq \sum_{i=1}^n \mathbb{E} Ent_i\ Y\ .$$

Note, however, that for $c > 0$,

$$Ent\ cZ = \mathbb{E}cZ \log(cZ) - c\mathbb{E}Z \log(c\mathbb{E}Z) = cEnt\ Z\ ,$$

and similarly for Ent_i, so the above inequality implies that $Ent\ Z \leq \sum_{i=1}^n \mathbb{E} Ent_i\ Z$.

So assume that $\mathbb{E}Z = 1$ and define a probability measure Q on $\mathcal{X}^n$ using f for the density:

$$q(x) := Q(\{x\}) = f(x)p(x)\ ,$$

where $p(x) = \mathbb{P}(X_1 = x_1, \ldots, X_n = x_n)$ and set P to be the distribution of $(X_1, \ldots, X_n)$. Then $Ent\ Z$ is just the relative entropy of Q and P:

$$\begin{aligned} Ent\ Z = \mathbb{E}Z \log Z &= \sum_{x \in \mathcal{X}^n} p(x) f(x) \log f(x) \\ &= \sum_{x \in \mathcal{X}^n} q(x) \log q(x) - \sum_{x \in \mathcal{X}^n} q(x) \log p(x) = D(Q||P)\ . \end{aligned}$$

On the other hand, we can $\sum_{i=1}^n \mathbb{E} Ent_i\ Z$ in terms of a different relative entropy. For $i = 1, \ldots, n$, if $Y = (Y_1, \ldots, Y_n)$ is distributed as Q, then write $Q^{(i)}$ for the distribution of $Y^{(i)} = (Y_1, \ldots, Y_{i-1}, Y_{i+1}, \ldots, Y_n)$ and $P^{(i)}$ for the distribution of $X^{(i)} = (X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n)$. Then putting $p_i(x_i) = \mathbb{P}(X_i = x_i)$,

$$p^{(i)}(x^{(i)})\mathbb{E}_i Z(x) = \sum_{x_i \in \mathcal{X}} p^{(i)}(x^{(i)}) p_i(x_i) f(x_1, \ldots, x_n) = \sum_{x_i \in \mathcal{X}} q(x_1, \ldots, x_n) = q^{(i)}(x^{(i)})\ .$$

From this we see

$$\begin{aligned}
\mathbb{E}\Phi\left(\mathbb{E}_i Z\right) &= \sum_{x\in\mathcal{X}^n} p^{(i)}(x^{(i)})\mathbb{E}_i Z \log \mathbb{E}_i Z \\
&= \sum_{x\in\mathcal{X}^n} q^{(i)}(x^{(i)}) \log \frac{q^{(i)}(x^{(i)})}{p^{(i)}(x^{(i)})} \\
&= \sum_{x\in\mathcal{X}^n} q^{(i)}(x^{(i)}) \log q^{(i)}(x^{(i)}) - \sum_{x\in\mathcal{X}^n} q^{(i)}(x^{(i)}) \log p^{(i)}(x^{(i)}) \\
&= D(Q^{(i)}||P^{(i)}) \ .
\end{aligned}$$

So subadditivity will be established once we show:

Lemma 1.8 (Han's inequality for relative entropy)**.** *Let P, Q be probability measures on $\mathcal{X}^n$ such that P is a product measure. Then*

$$D(Q||P) \le \sum_{i=1}^{n} (D(Q||P) - D(Q^{(i)}||P^{(i)})) \ .$$

Equivalently,

$$D(Q||P) \ge \frac{1}{n-1} \sum_{i=1}^{n} D(Q^{(i)}||P^{(i)}) \ .$$

Proof. The left side is

$$D(Q||P) = \sum_{x\in\mathcal{X}^n} q(x)\log q(x) - \sum_{x\in\mathcal{X}^n} q(x)\log p(x) \ .$$

Apply Han's inequality to the first term

$$\sum_{x\in\mathcal{X}^n} q(x)\log q(x) \ge \frac{1}{n-1} \sum_{i=1}^{n} \sum_{x^{(i)}\in\mathcal{X}^{n-1}} q^{(i)}(x^{(i)}) \log q^{(i)}(x^{(i)}) \ .$$

For the second term,

$$\begin{aligned}
\sum_{x\in\mathcal{X}^n} q(x)\log p(x) &= \frac{1}{n} \sum_{i=1}^{n} \sum_{x\in\mathcal{X}^n} q(x)\log p(x) \\
&= \frac{1}{n} \sum_{i=1}^{n} \left[\sum_{x\in\mathcal{X}^n} q(x)(\log p^{(i)}(x^{(i)}) + \log p_i(x_i)) \right] \\
&= \frac{1}{n} \sum_{i=1}^{n} \sum_{x\in\mathcal{X}^n} q(x) \log p^{(i)}(x^{(i)}) + \frac{1}{n} \sum_{x\in\mathcal{X}^n} q(x)\log p(x) \ ,
\end{aligned}$$

giving

$$\sum_{x \in \mathcal{X}^n} q(x) \log p(x) \leq \frac{1}{n-1} \sum_{i=1}^{n} \sum_{x \in \mathcal{X}^n} q(x) \log p^{(i)}(x^{(i)})$$
$$= \frac{1}{n-1} \sum_{i=1}^{n} \sum_{x^{(i)} \in \mathcal{X}^{n-1}} q^{(i)}(x^{(i)}) \log p^{(i)}(x^{(i)}) \ .$$

Combining these, we obtain

$$D(Q||P) \geq \frac{1}{n-1} \sum_{i=1}^{n} \left[\sum_{x^{(i)} \in \mathcal{X}^{n-1}} q^{(i)}(x^{(i)}) \log q^{(i)}(x^{(i)}) - q^{(i)}(x^{(i)}) \log p^{(i)}(x^{(i)}) \right] \ ,$$

which equals $\frac{1}{n-1} \sum_{i=1}^{n} D(Q^{(i)}||P^{(i)})$. □

□

Application. BLM give an generalization of the edge isoperimetric inequality (influence inequality) from the last section. Instead of considering uniform measure on $\{0,1\}^n$, we will take product measure with some parameter $p \in (0,1)$. For reasons having to do with notation, we will change the space to $\{-1,+1\}^n$ and take $\mathbb{P}$ the product measure with $\mathbb{P}(x_i = 1) = p$ and $\mathbb{P}(x_i = -1) = 1 - p$ for all i. Not only is p allowed to be biased, but we will consider both positive and negative influences.

First some notation. If X is distributed as $\mathbb{P}$, we set $X_i^- = (X_1, \ldots, X_{i-1}, -1, X_{i+1}, \ldots, X_n)$ and $X_i^+ = (X_1, \ldots, X_{i-1}, 1, X_{i+1}, \ldots, X_n)$. The positive and negative influences of coordinate i on the set $A \subset \{-1,+1\}^n$ is

$$I_i^-(A) = \mathbb{P}\left(X_i^- \in A \text{ but } X_i^+ \notin A\right), \ I_i^+(A) = \mathbb{P}\left(X_i^- \notin A \text{ but } I_i^+ \in A\right) \ .$$

Last, the positive and negative total influences are

$$I^-(A) = \sum_{i=1}^{n} I_i^-(A) \text{ and } I^+(A) = \sum_{i=1}^{n} I_i^+(A) \ .$$

We will prove that for any set $A \subset \{-1,+1\}^n$, one has

$$\mathbb{P}(A) \log \frac{1}{\mathbb{P}(A)} \leq I^+(A) p \log \frac{1}{p} + I^-(A)(1-p) \log \frac{1}{1-p} \ . \tag{4}$$

If we take $p = 1/2$ then this inequality reduces to $I_+(A) + I^-(A) \geq 2\mathbb{P}(A) \log \frac{1}{\mathbb{P}(A)}$, which was our previous result.

To prove (4), apply subadditivity of entropy to the function $Z = f(X) = \mathbf{1}_A(X)$. Then

$$Ent\ Z = 0 - \mathbb{P}(A) \log \mathbb{P}(A) = \mathbb{P}(A) \log \frac{1}{\mathbb{P}(A)} \ .$$

On the other hand,

$$\mathbb{E}_i Z(X) = \begin{cases} p & \text{if } X_i^+ \in A \text{ but } X_i^- \notin A \\ 1-p & \text{if } X_i^+ \notin A \text{ but } X_i^- \in A \\ 0 & \text{otherwise} \end{cases} ,$$

giving

$$Ent_i\ Z = 0 - \Phi\left(\mathbb{E}_i Z\right) = p \log p \mathbf{1}_{X_i^+ \in A,\ X_i^- \notin A} + (1-p)\log(1-p)\mathbf{1}_{X_i^+ \notin A,\ X_i^- \in A}$$

and

$$\sum_{i=1}^n \mathbb{E} Ent_i\ Z = p \log p I^+(A) + (1-p)\log(1-p) I^-(A)\ .$$

By subadditivity of entropy, we obtain (4).

The expression simplifies if, for example, $I^-(A) = 0$. This is true if A is monotone; that is, whenever $X_i^- \in A$, it follows that $X_i^+ \in A$. In these cases, one has

$$I(A) \geq \frac{\mathbb{P}(A) \log \frac{1}{\mathbb{P}(A)}}{p \log \frac{1}{p}}\ .$$

2 Entropy of general random variables

The general definition of entropy is similar to that given in the last section.

Definition 2.1. *If X is a nonnegative random variable with finite mean, the entropy of X is defined as*

$$Ent\ X = \mathbb{E}\Phi(X) - \Phi\left(\mathbb{E}X\right)\ ,$$

where $\Phi(x) = x \log x$.

In other words, $Ent\ X = \mathbb{E}X \log X - \mathbb{E}X \log \mathbb{E}X$, or

$$Ent\ X = \mathbb{E}X \log \frac{X}{\mathbb{E}X} \text{ if } \mathbb{E}X > 0\ .$$

Note that $\Phi(x) \geq -e^{-1}$ for all x, so if $\mathbb{E}\Phi(X) = \infty$, then $Ent\ X = \infty$.

- Since Φ is convex, Jensen's inequality states $\mathbb{E}\Phi(X) \geq \Phi(\mathbb{E}X)$, and so

$$Ent\ X \geq 0 \text{ for all } X\ .$$

- If $X \geq 0$ and $c > 0$ then

$$\begin{aligned} Ent\ cX &= \mathbb{E}cX \log(cX) - \mathbb{E}cX \log \mathbb{E}cX \\ &= c\mathbb{E}X \log X + c\mathbb{E}X \log c - c\mathbb{E}X \log \mathbb{E}X - c\mathbb{E}X \log c \\ &= c\ Ent\ X\ . \end{aligned}$$

- If $X \geq 0$ has $\mathbb{E}X = 1$ we can define a probability measure using X as a density:

$$\mathbb{Q}(A) = \mathbb{E}X\mathbf{1}_A \ .$$

We then define the relative entropy, or Kullback-Leibler divergence of $\mathbb{Q}$ with respect to $\mathbb{P}$ as

$$D(\mathbb{Q}||\mathbb{P}) = Ent\ X \ .$$

We claim that if $\mathbb{P}$ is a discrete distribution (that is, a countable sum of delta masses), then $D(\mathbb{Q}||\mathbb{P})$ is simply the relative entropy that we defined earlier. Take $\Omega = \mathbb{N}$ and $\mathbb{P}$ with probabilities $\mathbb{P}(\{k\}) = p_k$. If $f : \mathbb{N}$ is nonnegative with mean 1, we can define a probability measure $\mathbb{Q}$ on Ω by $q_k = \mathbb{Q}(\{k\}) = f(k)p_k = \int_{\{k\}} f(n)\mathbb{P}(\mathrm{d}n)$. Then

$$\begin{aligned} Ent\ f &= \sum_k p_k f(k) \log f(k) - \sum_k p_k f(k) \log \sum_k p_k f_k \\ &= \sum_k q_k \log q_k - \sum_k q_k \log p_k \\ &= D(\mathbb{Q}||\mathbb{P}) \ . \end{aligned}$$

There are some variational characterizations of entropy. These will be useful not only in deriving subadditivity for general entropy, but also for estimating entropy later when we use it to prove concentration.

Variational characterization. (Duality formula). This is similar to characterizations of norms in Banach spaces (for instance in L^p). Let $X \geq 0$ satisfy $\mathbb{E}\Phi(X) < \infty$. (This also implies $\mathbb{E}X < \infty$.) Then

$$Ent\ X = \sup\{\mathbb{E}XY : \mathbb{E}e^Y = 1\} \ .$$

Here the supremum is over $Y : \Omega \to [-\infty, \infty)$ and we define $0 \cdot -\infty$ and $e^{-\infty}$ as 0.

Proof. This formula is a consequence of the fact that the Legendre transform of the function $x \log x - x$ is e^x. To show this, recall that is $f(x) = x \log x - x$ then the Legendre transform is

$$f^*(y) = \sup_{x \in \mathbb{R}} (xy - f(x)) = \sup_x (xy - x \log x + x) \ .$$

Take derivative in x to obtain

$$0 = \frac{\mathrm{d}}{\mathrm{d}x}(xy - x \log x + x) = y - \log x \Rightarrow x = e^y \ .$$

Since the second derivative at this point is $-\frac{1}{e^y} < 0$, it is the location of the unique maximum and we obtain

$$f^*(y) = ye^y - e^y \log e^y + e^y = e^y \ .$$

We therefore obtain

$$xy \leq e^y + x \log x - x \text{ for all } x \geq 0,\ y \in [-\infty, \infty) \ . \tag{5}$$

Now take $X \geq 0$ and $Y : \Omega \to [-\infty, \infty)$ with $\mathbb{E}e^Y = 1$. If $\mathbb{E}X = 1$ then taking expectation in (5),

$$\mathbb{E}XY \leq \mathbb{E}X \log X = Ent\ X\ .$$

If $\mathbb{E}X = 0$ then both sides here are zero. If $\mathbb{E}X > 0$ then apply this inequality to $X/\mathbb{E}X$:

$$\mathbb{E}XY/\mathbb{E}X \leq Ent\ \frac{X}{\mathbb{E}X} = \frac{1}{\mathbb{E}X} Ent\ X\ ,$$

so we have shown one bound of the variational formula.

For the other bound, if $\mathbb{E}X > 0$, take $Y = \log \frac{X}{\mathbb{E}X}$. (If $X = 0$ then this is $-\infty$.) Then

$$\mathbb{E}e^Y = \mathbb{E}\frac{X}{\mathbb{E}X} = 1$$

and $\mathbb{E}XY = \mathbb{E}X \log \frac{X}{\mathbb{E}X} = Ent\ X$. If $\mathbb{E}X = 0$ then $X = 0$ almost surely and $XY = 0$ almost surely for any $Y : \Omega \to [-\infty, \infty)$ with $\mathbb{E}e^Y = 1$. $\square$

As a consequence, we obtain the **exponential Hölder inequality**: for X, Y with $X \geq 0$ and $\mathbb{E}X < \infty$,

$$\mathbb{E}XY \leq c\ Ent\ X + c\ \mathbb{E}X \log \mathbb{E}e^{Y/c} \text{ for } c > 0\ .$$

Proof. Take $c = 1$ first. We may assume that X and e^Y are integrable, or else the inequality is vacuous. Since $Z = Y - \log \mathbb{E}e^Y$ has $\mathbb{E}e^Z = 1$, we apply the variational characterization of entropy:

$$Ent\ X \geq \mathbb{E}XZ = \mathbb{E}XY - \mathbb{E}X \log \mathbb{E}e^Y\ .$$

If $c \neq 1$ then we just apply the inequality to cX and Y/c. $\square$

2.1 Subadditivity

Here we can give the form of subadditivity of entropy that is valid for general random variables. Sometimes this is called the "tensorization inequality" or the tensorization of entropy.

Theorem 2.2. *Let $X_1, \ldots, X_n$ be independent and $Z = f(X_1, \ldots, X_n)$ be nonnegative such that $\mathbb{E}\Phi(Z) < \infty$. Then*

$$Ent\ Z \leq \sum_{i=1}^n \mathbb{E} Ent_i\ Z\ ,$$

where $Ent_i\ Z$ is the entropy of Z relative to the variable X_i only (with all other variables fixed).

Proof. Analogously to the proof of Efron-Stein, we begin by expressing $Ent\ Z$ as a telescoping series:

$$Ent\ Z = \mathbb{E}Z \log Z - \mathbb{E}Z \log \mathbb{E}Z = \sum_{i=1}^n \left[\mathbb{E}Z \log \mathbb{E}_{>i} Z - \mathbb{E}Z \log \mathbb{E}_{>i-1} Z \right]\ ,$$

where $\mathbb{E}_{>i}$ is expectation over all variables $X_{i+1}, \ldots, X_n$ (with $\mathbb{E}_{>n}$ meaning that we take no expectation at all). This gives

$$Ent\ Z = \sum_{i=1}^{n} \mathbb{E}\mathbb{E}_i \Big[Z \log \mathbb{E}_{>i} Z - Z \log \mathbb{E}_{>i-1} Z \Big] \ ,$$

where $\mathbb{E}_i$ is expectation only over X_i.

If $\mathbb{E}_{>i-1}Z > 0$, we can write the summand as

$$Z \Big[\log \mathbb{E}_{>i} Z - \log \mathbb{E}_{>i-1} Z \Big] \ .$$

Note then that

$$\mathbb{E}_i \exp\left(\log \mathbb{E}_{>i} Z - \log \mathbb{E}_{>i-1} Z \right) = \frac{\mathbb{E}_i \mathbb{E}_{>i} Z}{\mathbb{E}_{>i-1} Z} = 1 \ ,$$

so by the variational characterization of entropy,

$$\mathbb{E}_i Z \left(\log \mathbb{E}_{>i} Z - \log \mathbb{E}_{>i-1} Z \right) \leq Ent_i\ Z \ .$$

Otherwise $\mathbb{E}_{>i-1}Z = 0$ then since Z is nonnegative, we must have $\mathbb{E}_{>i}Z = 0$ and $Z = 0$. In this case, the above summand is zero and also $Ent_i\ Z = 0$; therefore, the above inequality still holds.

Using this inequality in the telescoping sum,

$$Ent\ Z \leq \sum_{i=1}^{n} \mathbb{E} Ent_i\ Z \ .$$

$\square$

2.2 The transportation method

BLM has an entire chapter on the transportation method, but we will only give one application. The idea is to use a different variational characterization of entropy which relates the cumulant generating function to relative entropy between our given distribution $\mathbb{P}$ and another one $\mathbb{Q}$. To bound the cumulant generating function, we will have to estimate quantities related to the "cost" of computing the expectation of our variable Z relative to this other distribution $\mathbb{Q}$ rather than relative to $\mathbb{P}$. (See the term $t(\mathbb{E}_{\mathbb{Q}} Z - \mathbb{E}Z)$ in the corollary below.)

The variational characterization of entropy we will need is:

Proposition 2.3 (Gibbs variational principle)**.** *Let Z be integrable and put $Z' = Z - \mathbb{E}Z$. For every $t \in \mathbb{R}$,*

$$\psi_{Z'}(t) = \sup_{\mathbb{Q} \ll \mathbb{P}} \left[t(\mathbb{E}_{\mathbb{Q}} Z - \mathbb{E}Z) - D(\mathbb{Q}||\mathbb{P}) \right] \ .$$

The supremum is over all $\mathbb{Q}$ that are absolutely continuous relative to $\mathbb{P}$.

Proof. The inequality $\geq$ follows directly from the last characterization. To see why, we may assume that $\psi_{Z'}(t) < \infty$. (It is nonnegative since Z' has mean zero.) Let $\mathbb{Q}$ be absolutely continuous relative to $\mathbb{P}$ and set Y to be the Radon-Nikodym derivative $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}$. Then setting $U = tZ' - \psi_{Z'}(t)$,

$$\mathbb{E} \exp^U = \mathbb{E}e^{tZ'} / \exp \psi_{Z'}(t) = 1 \ .$$

Therefore by the duality formula,

$$\begin{aligned} D(\mathbb{Q}||\mathbb{P}) = \mathit{Ent}\ Y \geq \mathbb{E}UY &= t\mathbb{E}YZ' - \psi_{Z'}(t)\mathbb{E}Y \\ &= t(\mathbb{E}_{\mathbb{Q}}Z - \mathbb{E}Z) - \psi_{Z'}(t) \ , \end{aligned}$$

and this establishes the bound $\geq$ in the corollary.

Now we move to the other bound. The idea appears to be the following. Suppose for example that Z has exponential moments of all orders. Then we can set $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = \frac{e^{tZ'}}{\mathbb{E}e^{tZ'}}$. Using this measure, the right side is

$$\begin{aligned} t\mathbb{E}_{\mathbb{Q}}Z' - D(\mathbb{Q}||\mathbb{P}) &= \frac{t\mathbb{E}Z'e^{tZ'}}{\mathbb{E}e^{tZ'}} - \mathit{Ent}\ \frac{e^{tZ'}}{\mathbb{E}e^{tZ'}} \\ &= \frac{\mathbb{E}tZ'e^{tZ'}}{\mathbb{E}e^{tZ'}} - \frac{1}{\mathbb{E}e^{tZ'}}\left[\mathbb{E}tZ'e^{tZ'} - \mathbb{E}e^{tZ'}\log \mathbb{E}e^{tZ'}\right] \\ &= \psi_{Z'}(t) \ . \end{aligned}$$

This would show that the supremum is attained at $\psi_{Z'}(t)$.

Unfortunately, we are not guaranteed a priori that Z has exponential moments, so we will approximate in a seemingly roundabout way. Set a to be the right side of equation of the proposition. Note that if $a = \infty$ then by the other inequality, $\psi_{Z'}(t) = \infty$ and we are done. So we may assume that $a < \infty$. Note also that $a \geq 0$ since if we choose $\mathbb{P} = \mathbb{Q}$, then the right side is 0. So define

$$U = tZ' - a \ .$$

This allows us to rephrase the other inequality of the proposition as $\mathbb{E}e^U \leq 1$. We first claim that this follows if we can show that for all nonnegative Y with finite mean,

$$\mathbb{E}UY \leq \mathit{Ent}\ Y \ . \tag{6}$$

(This is somewhat of a converse to the duality formula from last section.) So assume (6). We will now truncate U, setting $U_n = \min\{U, n\}$ and $Y_n = e^{U_n}/\mathbb{E}e^{U_n}$. Then $\mathbb{E}e^Y = 1$ and by (6),

$$\begin{aligned} \frac{\mathbb{E}Ue^{U_n}}{\mathbb{E}e^{U_n}} = \mathbb{E}UY_n \leq \mathit{Ent}\ Y_n &= \mathbb{E}Y_n \log Y_n \\ &= \frac{1}{\mathbb{E}e^{U_n}}\left[\mathbb{E}U_n e^{U_n} - \log \mathbb{E}e^{U_n}\right] \ . \end{aligned}$$

Therefore

$$\log \mathbb{E}e^{U_n} \leq \mathbb{E}(U_n - U)e^{U_n} \leq 0 \ ,$$

or $\mathbb{E}e^{U_n} \le 1$. By monotone convergence, $\mathbb{E}e^U \le 1$ follows.

We are left to show (6). This follows from definition of a. Let Y be nonnegative with mean 1 and define $\mathbb{Q}$ by $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = Y$. Then

$$\begin{aligned} \mathbb{E}UY &= t(\mathbb{E}_{\mathbb{Q}}Z - \mathbb{E}Z) - a \\ &= t(\mathbb{E}_{\mathbb{Q}}Z - \mathbb{E}Z) - D(\mathbb{Q}||\mathbb{P}) - a + Ent\ Y \\ &\le Ent\ Y\ , \end{aligned}$$

completing the proof in this case. If $\mathbb{E}Y = 0$ then (6) is obvious, and if $\mathbb{E}Y > 0$ we just apply the above argument to $Y/\mathbb{E}Y$ to obtain

$$\mathbb{E}U\frac{Y}{\mathbb{E}Y} \le Ent\ \frac{Y}{\mathbb{E}Y}$$

and multiply by $\mathbb{E}Y$. □

From this characterization, we can derive some concentration inequalities. The simplest observation we can make is:

- **A transportation lemma.** If Z is a random variable with mean zero, then

$$\psi_Z(t) \le \frac{t^2\nu}{2} \text{ for } t > 0$$

if and only if

$$\mathbb{E}_{\mathbb{Q}}Z - \mathbb{E}Z \le \sqrt{2\nu D(\mathbb{Q}||\mathbb{P})}$$

and for all $\mathbb{Q}$ which are absolutely continuous relative to $\mathbb{P}$.

Proof. The variational characterization says

$$\psi_Z(t) = \sup_{\mathbb{Q}\ll\mathbb{P}} [t(\mathbb{E}_{\mathbb{Q}}Z - \mathbb{E}Z) - D(\mathbb{Q}||\mathbb{P})]\ .$$

So

$$\psi_Z(t) \le \frac{t^2\nu}{2} \Leftrightarrow t(\mathbb{E}_{\mathbb{Q}}Z - \mathbb{E}Z) - D(\mathbb{Q}||\mathbb{P}) - \frac{t^2\nu}{2} \le 0$$

for all $\mathbb{Q}$ that are absolutely continuous relative to $\mathbb{P}$. Note that we may assume that $\mathbb{E}_{\mathbb{Q}}Z - \mathbb{E}Z \ge 0$, otherwise the right inequality is always satisfied. In other words, the right inequality is equivalent to the same inequality but with $\mathbb{Q}$ restricted so that $\mathbb{E}_{\mathbb{Q}}Z - \mathbb{E}Z \ge 0$.

The left side is a polynomial in t of the form $a_1t - a_2 - a_3t^2$ for $a_1, a_2, a_3 \ge 0$ and so has its maximum at $t = a_1/(2a_3)$. The value of this maximum is then $\frac{a_1^2}{2a_3} - a_2 - \frac{a_1^2}{4a_3} = \frac{a_1^2}{4a_3} - a_2$. So the right is equivalent to $a_1^2 \le 4a_2a_3$, or $a_1 \le \sqrt{4a_2a_3}$. This proves $\psi_Z(t) \le \frac{t^2\nu}{2}$ for all $t > 0$ if and only if

$$\mathbb{E}_{\mathbb{Q}}Z - \mathbb{E}Z \le \sqrt{2\nu D(\mathbb{Q}||\mathbb{P})}$$

for all $\mathbb{Q}$ absolutely continuous relative to $\mathbb{P}$ for which the left side is positive. But this is equivalent to the same statement with $\mathbb{Q}$ required only to be absolutely continuous relative to $\mathbb{P}$. □

The last lemma gives us a way to prove Gaussian concentration. In fact a more general statement holds (and the proof is the same); see Lemma 4.18 in BLM. Let Z be integrable and ϕ be a convex and C^1 function on an interval $[0, b)$ with $0 < b \leq \infty$. Assume that $\phi(0) = \phi'(0) = 0$. Define for every $x \geq 0$

$$\phi^*(x) = \sup_{t \in (0,b)} (tx - \phi(t))$$

and let $(\phi^*)^{-1}(t) = \inf\{t \geq 0 : \phi^*(x) > t\}$ for $t \geq 0$. Then

$$\psi_Z(t) \leq \phi(t) \text{ for every } t \in (0, b)$$

if and only if

$$\mathbb{E}_{\mathbb{Q}} Z - \mathbb{E} Z \leq (\phi^*)^{-1}[D(\mathbb{Q}||\mathbb{P}))$$

for every probability measure $\mathbb{Q}$ absolutely continuous with respect to $\mathbb{P}$ such that $D(\mathbb{Q}||\mathbb{P}) < \infty$.

Our gaussian lemma was the case $\phi(t) = t^2\nu/2$ and $b = \infty$. Of course we can take other bounds, like a sub-gamma bound: $\phi(t) = t^2\nu/(2(1 - ct))$.

2.3 General setup

The general setup of the transportation method is that we have $X_1, \ldots, X_n$ independent and $f : \mathbb{R}^n \to \mathbb{R}$, as before. Now we take a measurable function $d : \mathbb{R} \times \mathbb{R} \to [0, \infty)$, which is normally a pseudo-metric and assume that f satisfies the regularity property

$$f(y) - f(x) \leq \sum_{i=1}^{n} c_i d(y_i, x_i) \text{ for } x, y \in \mathbb{R}^n \text{ and } c_i \geq 0 \ . \tag{7}$$

The example that we will consider is where

$$d(a, b) = \begin{cases} 1 & \text{if } a \neq b \\ 0 & \text{if } a = b \end{cases} ,$$

so that the regularity property is simply the bounded differences condition with constants $c_1, \ldots, c_n$.

By the transportation lemma, if we would like to show that $Z = f(X_1, \ldots, X_n)$ satisfies $\psi_Z(t) \leq \frac{t^2\nu}{2}$ for $t > 0$, we can show (10). In particular, we must bound $\mathbb{E}_{\mathbb{Q}} Z - \mathbb{E} Z$. To do this, we will use coupling. A coupling of $\mathbb{P}$ and $\mathbb{Q}$ is P, the distribution of a random vector (X, Y) such that X has distribution $\mathbb{P}$ and Y has distribution $\mathbb{Q}$. Call $\mathcal{P}(\mathbb{P}, \mathbb{Q})$ the set of all such joint distributions P.

In our case, X is the vector $(X_1, \ldots, X_n)$ and has distribution $\mathbb{P}$ and, if $\mathbb{Q} \ll \mathbb{P}$, then a vector with distribution $\mathbb{Q}$ is written $Y = (Y_1, \ldots, Y_n)$. Then for $P \in \mathcal{P}(\mathbb{P}, \mathbb{Q})$, we can use the regularity property

$$\begin{aligned}\mathbb{E}_{\mathbb{Q}} Z - \mathbb{E}Z = \mathbb{E}_P(Z(Y)) - \mathbb{E}_P(Z(X)) &= \mathbb{E}_P(Z(Y) - Z(X)) \\ &\leq \sum_{i=1}^n c_i \mathbb{E}_P d(X_i, Y_i) \\ &\leq \left(\sum_{i=1}^n c_i^2\right)^{1/2} \left(\sum_{i=1}^n (\mathbb{E}_P d(X_i, Y_i))^2\right)^{1/2} .\end{aligned}$$

Therefore to prove Gaussian concentration, we are reduced to proving

$$\inf_{P \in \mathcal{P}(\mathbb{P}, \mathbb{Q})} \sum_{i=1}^n (\mathbb{E}_P d(X_i, Y_i))^2 \leq \frac{2\nu}{C} D(\mathbb{Q}||\mathbb{P}) \text{ for } \mathbb{Q} \ll \mathbb{P} , \tag{8}$$

where $C = \sum_{i=1}^n c_i^2$. The quantity on the left is interpreted as the minimal "cost" to transport the vector X with distribution $\mathbb{P}$ to the vector Y with distribution $\mathbb{Q}$ on the same space. This transportation is accomplished using a coupling. Such an inequality is called a **transportation cost inequality**.

Whether or not (8) holds depends only on the choice of function d and the distribution $\mathbb{P}$ of the random variables $(X_1, \ldots, X_n)$. In other words, we have removed the function f and extracted from it the only information we need (the function d, or the regularity property of f) to prove a concentration inequality.

2.4 Application to bounded differences

We will now focus on proving (8) in the bounded differences case. It becomes

$$\inf_{P \in \mathcal{P}(\mathbb{P}, \mathbb{Q})} \sum_{i=1}^n (P(X_i \neq Y_i))^2 \leq \frac{2\nu}{C} D(\mathbb{Q}||\mathbb{P}) . \tag{9}$$

The left side is related to the total variation distance between measures.

Definition 2.4. *If $\mathbb{P}$ and $\mathbb{Q}$ are probability measures on the same space (Ω, Σ), the total variation distance between them is defined*

$$V(\mathbb{P}, \mathbb{Q}) = \sup_{A \in \Sigma} |\mathbb{P}(A) - \mathbb{Q}(A)| .$$

There are many equivalent formulations of the total variation distance, and we will give some later. Let us note however that

- There exists $A^* \in \Sigma$ such that $V(\mathbb{P}, \mathbb{Q}) = \mathbb{P}(A^*) - \mathbb{Q}(A^*)$.

Proof. Define $\lambda = \mathbb{P} + \mathbb{Q}$ and note that $\mathbb{P}$ and $\mathbb{Q}$ are absolutely continuous relative to λ, so set $f = \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\lambda}$ and $g = \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\lambda}$. We claim that $E = \{\omega : f(\omega) \geq g(\omega)\}$ satisfies the conditions. First, if $A \in \Sigma$ then

$$\begin{aligned}
|\mathbb{P}(A) - \mathbb{Q}(A)| &= \left| \int_A (f-g)\ \mathrm{d}\lambda \right| \\
&= \left| \int_{A\cap E} (f-g)\ \mathrm{d}\lambda - \int_{A\cap E^c} (g-f)\ \mathrm{d}\lambda \right|
\end{aligned}$$

is the absolute value of the difference of two nonnegative numbers. Therefore

$$\begin{aligned}
|\mathbb{P}(A) - \mathbb{Q}(A)| &\leq \max\left\{ \int_{A\cap E} (f-g)\ \mathrm{d}\lambda, \int_{A\cap E^c} (g-f)\ \mathrm{d}\lambda \right\} \\
&\leq \max\left\{ \int_{E} (f-g)\ \mathrm{d}\lambda, \int_{E^c} (g-f)\ \mathrm{d}\lambda \right\} \\
&= \max\left\{ \mathbb{P}(E) - \mathbb{Q}(E), \mathbb{Q}(E^c) - \mathbb{P}(E^c) \right\}\ .
\end{aligned}$$

But these two numbers are equal and so our upper bound is $\mathbb{P}(E) - \mathbb{Q}(E)$. Taking $A = E$ shows that the bound is attained. $\square$

Note that this argument actually gives

$$V(\mathbb{P}||\mathbb{Q}) = \int (f-g)_+\ \mathrm{d}\lambda = \int (g-f)_+\ \mathrm{d}\lambda\ ,$$

where λ is a measure relative to which $\mathbb{P}$ and $\mathbb{Q}$ are absolutely continuous, and f and g are their densities.

It turns out that total variation distance can be related to relative entropy.

Theorem 2.5 (Pinsker's inequality)**.** *Let $\mathbb{P}$ and $\mathbb{Q}$ be probability measures on (Ω, Σ) such that $\mathbb{Q} \ll \mathbb{P}$. Then*

$$V(\mathbb{P}, \mathbb{Q})^2 \leq \frac{1}{2} D(\mathbb{Q}||\mathbb{P})\ .$$

Proof. Let $A^* \in \Sigma$ be such that $V(\mathbb{P}, \mathbb{Q}) = \mathbb{Q}(A^*) - \mathbb{P}(A^*)$. Then since $Z = \mathbf{1}_{A^*}$ is bounded, Hoeffding's inequality gives

$$\psi_{Z-\mathbb{E}Z}(t) \leq \frac{t^2}{8} \text{ for all } t\ .$$

On the other hand, if we let $Y = \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}$ then the variational characterization of entropy gives

$$\begin{aligned}
\psi_{Z-\mathbb{E}Z}(t) &\geq t(\mathbb{E}_{\mathbb{Q}} Z - \mathbb{E}Z) - D(\mathbb{Q}||\mathbb{P}) \\
&= t(\mathbb{Q}(A^*) - \mathbb{P}(A^*)) - D(\mathbb{Q}||\mathbb{P}) \\
&= tV(\mathbb{Q}, \mathbb{P}) - D(\mathbb{Q}||\mathbb{P})\ .
\end{aligned}$$

Choose $t = 4V(\mathbb{Q}, \mathbb{P})$ and combine with the other bound for

$$\frac{16V(\mathbb{Q}, \mathbb{P})^2}{8} \geq 4V(\mathbb{Q}, \mathbb{P})^2 - D(\mathbb{Q}||\mathbb{P})\ ,$$

which reduces to the statement of the theorem. $\square$

Given Pinsker's inequality, we see that to show that $\psi_Z(t) \le t^2\nu/2$ for $t > 0$, it suffices to prove that

$$\inf_{P\in\mathcal{P}(\mathbb{P},\mathbb{Q})} \sum_{i=1}^{n} (P(X_i \neq Y_i))^2 \le \frac{4\nu}{C} V(\mathbb{Q},\mathbb{P})^2 \tag{10}$$

The case $n = 1$ is implied by:

Theorem 2.6. *If $\mathbb{P}$ and $\mathbb{Q}$ are probability distributions on the same space $\Omega = \mathbb{R}^k$,*

$$\min_{P\in\mathcal{P}(\mathbb{P},\mathbb{Q})} P(X \neq Y) = V(\mathbb{P},\mathbb{Q}) \ ,$$

where (X, Y) has distribution P.

Proof. First if $P \in \mathcal{P}(\mathbb{P},\mathbb{Q})$, then for $A \in \Sigma$,

$$|\mathbb{P}(A) - \mathbb{Q}(A)| = |P(X \in A) - P(Y \in A)| \le P(\{X \in A\}\Delta\{Y \in A\}) \le P(X \neq Y) \ .$$

This implies the inequality $\ge$ in the theorem.

For the other direction, we need to construct a coupling that attains the minimum. Set $a = V(\mathbb{P},\mathbb{Q})$ and note that we may assume that $a > 0$; otherwise, $\mathbb{P} = \mathbb{Q}$ and we can define $X = Y$. Furthermore if $a = 1$ then $\mathbb{P}$ and $\mathbb{Q}$ are mutually singular and so for any coupling P, $P(X \neq Y) = 1$.

If $a \in (0, 1)$, we want to define P on the product space $\Omega \times \Omega$ to be a combination $aP_1 + (1 - a)P_2$, where P_1 is concentrated on the set where $\{(x, y) : x \neq y\}$ and P_2 is concentrated on the set where $\{(x, y) : x = y\}$. If we do this in such a way that the marginal of x is $\mathbb{P}$ and the marginal of y is $\mathbb{Q}$, then

$$P(X \neq Y) = aP_1(X \neq Y) + (1 - a)P_2(X \neq Y) = aP_1(X \neq Y) = a \ .$$

So to define P, we again use Radon-Nikodym derivatives. Set $\lambda = \mathbb{P} + \mathbb{Q}$ and $f = \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\lambda}$ and $g = \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\lambda}$. The idea (for the intuition and pictures, consult Ramon van Handel's notes) will be to split the support of f and g into a "common part," which is the intersection of the supports of f and g, and on this portion we will choose the density $\phi_1 = \min\{f, g\}$. On the region where f is larger, we remove the common part and keep the density $\phi_2 = f - \min\{f, g\}$ and on the other region we take $\phi_3 = g - \min\{f, g\}$. Note that the supports of ϕ_2 and ϕ_3 are disjoint. We will define our coupling to be: with probability $1 - a$, we set $X = Y$, drawn from the density ϕ_1; with probability a, we choose X and Y independently from their disjoint supports ϕ_2 and ϕ_3. We will need to normalize these densities of course, since they do not necessarily integrate to 1.

Formally, define for $A, B \subset \Omega$,

$$P_1(A \times B) = \frac{1}{a^2} \int_{A\times B} (f(x) - g(x))_+ (g(y) - f(y))_+ \ \mathrm{d}\lambda(x) \ \mathrm{d}\lambda(y),$$

$$P_2(\{(x, x) : x \in A\}) = \frac{1}{1-a} \int_A \min\{f(x), g(x)\} \ \mathrm{d}\lambda(x) \ .$$

In particular, P_1 is supported on $\{x \neq y\}$ since the integrand vanishes on $\{x = y\}$ and P_2 is supported on $\{x = y\}$. Furthermore,

$$P_1(\Omega \times \Omega) = \frac{1}{a^2} \int (f(x) - g(x))_+ \, \mathrm{d}\lambda(x) \int (g(y) - f(y))_+ \, \mathrm{d}\lambda(y) = 1$$

and using $\min\{a, b\} = \frac{a+b-(a-b)_+-(a-b)_-}{2}$,

$$\begin{aligned} P_2(\Omega \times \Omega) &= \frac{1}{1-a} \int_\Omega \min\{f(x), g(x)\} \, \mathrm{d}\lambda(x) \\ &= \frac{1}{2(1-a)} \int f(x) + g(x) - (f(x) - g(x))_+ - (f(x) - g(x))_- \, \mathrm{d}\lambda(x) \\ &= \frac{1}{2(1-a)} (2 - 2V(\mathbb{P}, \mathbb{Q})) = 1 \ . \end{aligned}$$

Last the marginal of X is given by

$$P(A \times \Omega) = aP_1(A \times \Omega) + (1-a)P_2(A \times \Omega)$$

with

$$P_1(A \times \Omega) = \frac{1}{a} \int_A (f(x) - g(x))_+ \, \mathrm{d}\lambda(x)$$

and

$$P_2(A \times \Omega) = \frac{1}{1-a} \int_A \min\{f(x), g(x)\} \, \mathrm{d}\lambda(x) \ .$$

This gives

$$\begin{aligned} P(A \times \Omega) &= \int_A (f(x) - g(x))_+ + \min\{f(x), g(x)\} \, \mathrm{d}\lambda(x) \\ &= \int_A f(x) \, \mathrm{d}\lambda(x) = \mathbb{P}(A) \ . \end{aligned}$$

A similar argument gives $P(\Omega \times A) = \mathbb{Q}(A)$. □

The general case $n > 1$ requires an induction step and will prove:

Theorem 2.7 (Marton's transportation inequality)**.** *Let $X = (X_1, \dots, X_n)$ be an independent vector with distribution $\mathbb{P}$ and let $\mathbb{Q} \ll \mathbb{P}$ be the distribution of $Y = (Y_1, \dots, Y_n)$. Then*

$$\min_{P \in \mathcal{P}(\mathbb{P}, \mathbb{Q})} \sum_{i=1}^n (P(X_i \neq Y_i))^2 \leq \frac{1}{2} D(\mathbb{Q}||\mathbb{P}) \ .$$

Proof. The case $n = 1$ follows from Pinsker's inequality and Theorem 2.6:

$$\min_{P \in \mathcal{P}(\mathbb{P}, \mathbb{Q})} P(X_1 \neq Y_1)^2 = V(\mathbb{P}, \mathbb{Q})^2 \leq \frac{1}{2} D(\mathbb{Q}||\mathbb{P}) \ .$$

Now assume the statement holds for some $n - 1 \geq 1$; we will show it for n. Write $\mathbb{P} = \mathbb{P}_1 \times \cdots \times \mathbb{P}_n$ and suppose that $\mathbb{Q} \ll \mathbb{P}$. Let $\mathbb{Q}_n$ be the marginal distribution of Y_n, where $Y = (Y_1, \dots, Y_n)$ is distributed as $\mathbb{Q}$. With this notation, we claim:

Lemma 2.8 (Chain rule for relative entropy)**.** *With the above notation,*

$$D(\mathbb{Q}||\mathbb{P}) = \int D(\mathbb{Q}(\cdot \mid t)||\mathbb{P}_{<n}) \, \mathrm{d}\mathbb{Q}_n(t) + D(\mathbb{Q}_n||\mathbb{P}_n) \ . \tag{11}$$

Proof. To prove this, we make some definitions. Define $g = \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}$ and write $g(x,t)$, where $x \in \mathbb{R}^{n-1}$ and $t \in \mathbb{R}$. Then we can find the marginal density $g_n(t)$ of Y_n relative to $\mathbb{P}_n$ by

$$\begin{aligned}\mathbb{Q}(Y_n \in A) = \int_{\mathbb{R}^{n-1}\times A} g(x,t) \, \mathrm{d}\mathbb{P}(x,t) &= \int_A \left[\int_{\mathbb{R}^{n-1}} g(x,t) \, \mathrm{d}\mathbb{P}_{<n}(x) \right] \, \mathrm{d}\mathbb{P}_n(t) \\ &=: \int_A g_n(t) \, \mathrm{d}\mathbb{P}_n(t) \ .\end{aligned}$$

Furthermore the conditional density of $(Y_1, \ldots, Y_{n-1})$ given Y_n relative to $\mathbb{P}_{<n}$ is

$$g(x|t) = \frac{g(x,t)}{g_n(t)} \text{ when } g_n(t) \neq 0 \text{ and } 1 \text{ otherwise} \ .$$

So, recalling $\Phi(u) = u \log u$, the right side of (11) equals

$$\begin{aligned}&\int \left[\int \Phi(g(x|t)) \, \mathrm{d}\mathbb{P}_{<n}(x) \right] g_n(t) \, \mathrm{d}\mathbb{P}_n(t) + \int \Phi(g_n(t)) \, \mathrm{d}\mathbb{P}_n(t) \\ &= \int \int g(x|t) g_n(t) \log g(x|t) \, \mathrm{d}\mathbb{P}(x,t) + \int g_n(t) \log g_n(t) \, \mathrm{d}\mathbb{P}_n(t) \\ &= \int \int g(x,t) \log g(x|t) \, \mathrm{d}\mathbb{P}(x,t) + \int \int g(x,t) \log g_n(t) \, \mathrm{d}\mathbb{P}(x,t) \\ &= \int g(x,t) \log g(x,t) \, \mathrm{d}\mathbb{P}(x,t) \ ,\end{aligned}$$

which is $D(\mathbb{Q}||\mathbb{P})$. This verifies (11). □

Now we continue with the proof of Theorem 2.7. By induction, we can find, for each $t \in \mathbb{R}$, a coupling $P_t \in \mathcal{P}(\mathbb{Q}(\cdot \mid t), \mathbb{P}_{<n})$ such that

$$\sum_{i=1}^{n-1} (P_t(X_i \neq Y_i))^2 \leq \frac{1}{2} D(\mathbb{Q}(\cdot \mid t), \mathbb{P}_{<n}) \text{ for all } t \ . \tag{12}$$

Furthermore, from the case $n = 1$, we find a coupling $P_n \in \mathcal{P}(\mathbb{Q}_n, \mathbb{P}_n)$ such that

$$P_n(X_n \neq Y_n)^2 \leq \frac{1}{2} D(\mathbb{Q}_n, \mathbb{P}_n) \ . \tag{13}$$

To build our coupling of X and Y, we simply combine these two. Define $P \in \mathcal{P}(\mathbb{P}, \mathbb{Q})$ so that the distribution of (X_n, Y_n) is P_n, but conditional on (X_n, Y_n), the distribution of $(X_{<n}, Y_{<n})$ is P_{Y_n}. We can do this by a standard construction: for measurable $A_1, A_2 \subset \mathbb{R}^{n-1}$ and $B_1, B_2 \subset \mathbb{R}$, set

$$P(A_1 \times B_1 \times A_2 \times B_2) = \int_{B_1 \times B_2} P_{y_n}(A_1 \times A_2) \, \mathrm{d}P_n(x_n, y_n) \ .$$

(Actually to do this, we need to know that $y \mapsto P_y(A_1 \times A_2)$ can be chosen to be measurable. This is not addressed in BLM, but can be shown by induction in our case where $d(x,y) = \mathbf{1}_{x \neq y}$. Indeed, we know the form of the coupling for $n = 1$ from the proof of Theorem 2.6 and we can go from there. Apparently in many other cases it is possible by a similar argument.) Then for $i = 1, \ldots, n-1$,

$$P(X_i \neq Y_i) = \int P_y(X_i \neq Y_i) \, \mathrm{d}\mathbb{Q}_n(y) \tag{14}$$

and

$$P(X_n \neq Y_n) = P_n(X_n \neq Y_n) \ . \tag{15}$$

Now applying the chain rule (11), (12) and (13),

$$D(\mathbb{Q}, \mathbb{P}) \geq 2 \sum_{i=1}^{n-1} \int P_y(X_i \neq Y_i)^2 \, \mathrm{d}\mathbb{Q}_n(y) + 2P_n(X_n \neq Y_n)^2 \ .$$

By Jensen's inequality for the square function, (14) and (15), we get a lower bound of

$$\begin{aligned} & 2 \sum_{i=1}^{n-1} \left(\int P_y(X_i \neq Y_i) \, \mathrm{d}\mathbb{Q}_n(y) \right)^2 + 2P_n(X_n \neq Y_n)^2 \\ &= 2 \sum_{i=1}^{n-1} (P(X_i \neq Y_i))^2 + 2(P(X_n \neq Y_n))^2 \\ &= 2 \sum_{i=1}^{n} (P(X_i \neq Y_i))^2 \ , \end{aligned}$$

and this completes the induction step. □

We can finish by summing up the concentration inequality given by Marton's transportation bound.

Theorem 2.9 (Bounded differences by transportation)**.** *Let $X = (X_1, \ldots, X_n)$ be independent and $f : \mathbb{R}^n \to \mathbb{R}$ satisfy the bounded differences condition with constants $c_1, \ldots, c_n \geq 0$. Then for $Z = f(X)$, $\psi_{Z - \mathbb{E}Z}(t) \leq \frac{t^2 \nu}{2}$ for $t > 0$ with $\nu = C/4$. In other words,*

$$\mathbb{P}(f(X) - \mathbb{E}f(X) > t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2} \right) \ \textit{for } t > 0 \ .$$

Proof. By the transportation lemma, one has $\psi_Z(t) \leq \frac{t^2 \nu}{2}$ for $t > 0$ whenever

$$\mathbb{E}_{\mathbb{Q}} Z - \mathbb{E}Z \leq \sqrt{2\nu D(\mathbb{Q}||\mathbb{P})} \text{ for } \mathbb{Q} \ll \mathbb{P} \ .$$

By the remarks at the beginning of this section, under the bounded differences condition, if we set $C = \sum_{i=1}^n c_i^2$, one has

$$\mathbb{E}_{\mathbb{Q}} Z - \mathbb{E}Z \leq \sqrt{C} \inf_{P \in \mathcal{P}(\mathbb{P}, \mathbb{Q})} \left(\sum_{i=1}^n P(X_i \neq Y_i)^2 \right)^{1/2} \ .$$

By Marton's inequality, the right side is bounded by $\sqrt{\frac{C}{2} D(\mathbb{Q}||\mathbb{P})}$, so we obtain $\psi_Z(t) \leq \frac{t^2 \nu}{2}$ for $t > 0$ with $\nu = C/4$. $\square$

3 Remarks and extensions

- The above induction proof (Marton's transportation inequality) is written in BLM as a "general induction principle" in Sec. 8.6. Generally it is referred to as "tensorization" to draw analogy to Efron-Stein and subadditivity of entropy. It allows us to give the bound
$$\min_{P \in \mathcal{P}(\mathbb{P},\mathbb{Q})} \sum_{i=1}^n \phi(\mathbb{E}_P d(X_i, Y_i)) \leq D(\mathbb{Q}||\mathbb{P})$$
for a convex function $\phi : [0,\infty) \to [0,\infty)$ whenever we know the corresponding bound for the individual relative entropies. Therefore, to prove a Gaussian concentration inequality using the transportation method, it generally suffices to prove the transportation cost inequality above for $n = 1$. In the bounded differences case, this came from Pinsker's inequality and the coupling for total variation distance.

- Other transportation cost inequalities are proved in BLM. For example, Section 8.2 has Marton's conditional transportation inequality, which leads to a concentration inequality in which we assume a weakened version of bounded differences: what is called bounded differences in quadratic mean. In this case, the constants c_i are allowed to be functions of x such that $\mathbb{E} \sum_{i=1}^n c_i^2(X)$ is bounded.

- Talagrand's Gaussian transportation bound is a similar bound for the gaussian measure. Let $\mathbb{P}$ be a standard Gaussian measure on $\mathbb{R}^n$ and $\mathbb{Q} \ll \mathbb{P}$. Then
$$\inf_{P \in \mathcal{P}(\mathbb{P},\mathbb{Q})} \sum_{i=1}^n \mathbb{E}_P (X_i - Y_i)^2 \leq 2D(\mathbb{Q}||\mathbb{P}) \ .$$
(This is called a **quadratic transportation cost inequality**.) This leads to a Gaussian concentration inequality for Lipschitz functions of a gaussian vector. We will prove this soon using logarithmic Sobolev inequalities and the Herbst argument. But in the meantime, let us prove:

  Proposition 3.1 (QTCI implies gaussian concentration for Lipschitz functions)**.** *Take $n = 1$ and assume that*
$$\inf_{P \in \mathcal{P}(\mathbb{P},\mathbb{Q})} \mathbb{E}_P (X - Y)^2 \leq 2D(\mathbb{Q}||\mathbb{P}) \text{ for all } \mathbb{Q} \ll \mathbb{P} \ ,$$
where X has distribution $\mathbb{P}$ and Y has distribution $\mathbb{Q}$. (This is a condition on $\mathbb{P}$.) Then for all $f : \mathbb{R} \to \mathbb{R}$ that are L-Lipschitz, one has
$$\mathbb{P}(f(X) - \mathbb{E}f(X) > t) \leq \exp\left(-\frac{t^2}{2L^2}\right) \text{ for } t > 0 \ .$$

Proof. Let f be L-Lipschitz. Then for any $\mathbb{Q} \ll \mathbb{P}$ and $\epsilon > 0$ we can find a coupling $P \in \mathcal{P}(\mathbb{P}, \mathbb{Q})$ such that

$$\mathbb{E}_P(X - Y)^2 \leq 2D(\mathbb{Q}||\mathbb{P}) + \epsilon \ ,$$

where Y has distribution $\mathbb{Q}$ and X has distribution $\mathbb{P}$. Then

$$|\mathbb{E}_{\mathbb{Q}} f(X) - \mathbb{E} f(X)| = \mathbb{E}_P(f(Y) - f(X)) \leq L\mathbb{E}_P|Y - X|$$

and this is bounded by

$$L\sqrt{\mathbb{E}_P(Y - X)^2} \leq L\sqrt{2D(\mathbb{Q}||\mathbb{P}) + \epsilon} \ .$$

Take $\epsilon \to 0$ and we obtain

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}} f(X) - \mathbb{E} f(X) \leq |\mathbb{E}_{\mathbb{Q}} f(X) - \mathbb{E} f(X)| &\leq L\sqrt{2D(\mathbb{Q}||\mathbb{P})} \\ &= \sqrt{2L^2 D(\mathbb{Q}||\mathbb{P})} \ . \end{aligned}$$

By the transportation lemma, we obtain $\psi_{f(X)}(t) \leq \frac{t^2\nu}{2}$ for $t > 0$, where $\nu = L^2$. By Chernoff, this translates to the claimed concentration inequality. $\square$

As usual, we can use a tensorization argument to go from one to higher dimensions. That is, if the QTCI holds, one has Gaussian concentration for L-Lipschitz functions in n-dimensions relative to the ℓ^2-norm. That is, the above concentration inequality holds for $f : \mathbb{R}^n \to \mathbb{R}$ so long as

$$|f(x) - f(y)| \leq L\sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2} \ .$$

1 Log-Sobolev and the entropy method

1.1 Introduction

The type of Log-Soblev inequality (LSI) we will study is one which resembles the Poincaré inequality we introduced. It states that for a smooth function f of a random variable X,

$$Ent\ f^2(X) \leq C\mathbb{E}\|\nabla f(X)\|_2^2 \ . \tag{1}$$

We will show in this section that one can obtain Gaussian concentration (stronger than exponential) for certain functions of variables whose distributions obey LSI's.

We can show two ways in which this is a stronger condition than a Poincaré inequality:

1. If the LSI (1) holds then for every smooth f, one has

   $$\text{Var}\ f(X) \leq \frac{C}{2}\mathbb{E}\|\nabla f(X)\|_2^2 \ .$$

   To see why take a smooth f of compact support and apply the LSI to $1+\epsilon f$ for $\epsilon > 0$:

   $$Ent\ (1+\epsilon f(X))^2 \leq C\mathbb{E}\|\nabla(1+\epsilon f(X))\|_2^2 \ . \tag{2}$$

   The left side is

   $$\mathbb{E}(1+\epsilon f)^2 \log(1+\epsilon f)^2 - \mathbb{E}(1+\epsilon f)^2 \log \mathbb{E}(1+\epsilon f)^2 \ . \tag{3}$$

   Since f is bounded, for ϵ near zero one has

   $$\begin{aligned}(1+\epsilon f)^2 \log(1+\epsilon f)^2 &= 2(1+2\epsilon f)\log(1+\epsilon f) + 2(\epsilon f)^2 \log(1+\epsilon f) \\ &= 2(1+2\epsilon f)\left(\epsilon f - \frac{(\epsilon f)^2}{2}\right) + O(\epsilon^3) \\ &= 2\epsilon f + 3(\epsilon f)^2 + O(\epsilon^3) \ .\end{aligned}$$

   Therefore the first term of (3) is

   $$2\epsilon\mathbb{E}f + 3\epsilon^2\mathbb{E}f^2 + O(\epsilon^3) \ .$$

   The second term is written

   $$\begin{aligned}\mathbb{E}(1+2\epsilon f)\log\mathbb{E}(1+\epsilon f)^2 + O(\epsilon^3) &= \mathbb{E}(1+2\epsilon f)\log\mathbb{E}(1+2\epsilon f + (\epsilon f)^2) + O(\epsilon^3) \\ &= \mathbb{E}(1+2\epsilon f)\left(2\epsilon\mathbb{E}f + \epsilon^2\mathbb{E}f^2 - \frac{(2\epsilon\mathbb{E}f + \epsilon^2\mathbb{E}f^2)^2}{2}\right) \\ &+ O(\epsilon^3) \\ &= 2\epsilon\mathbb{E}f + \epsilon^2\mathbb{E}f^2 - 2\epsilon^2(\mathbb{E}f)^2 + 4\epsilon^2(\mathbb{E}f)^2 + O(\epsilon^3) \\ &= 2\epsilon\mathbb{E}f + \epsilon^2\mathbb{E}f^2 + 2\epsilon^2(\mathbb{E}f)^2 + O(\epsilon^3) \ .\end{aligned}$$

Therefore (3) is

$$2\epsilon^2\mathbb{E}f^2 - 2\epsilon^2(\mathbb{E}f)^2 + O(\epsilon^3) = 2\epsilon^2\text{Var } f + O(\epsilon^3) \ .$$

On the other hand, the right side of (2) is $C\epsilon^2\mathbb{E}\|\nabla f\|_2^2$, so we obtain

$$2\epsilon^2\text{Var } f + O(\epsilon^3) \leq C\epsilon^2\mathbb{E}\|\nabla f\|_2^2 \ .$$

Dividing by by $2\epsilon^2$ and taking $\epsilon \to 0$ gives the result for f with compact support. In the general case, assume that f is smooth with $\mathbb{E}\|\nabla f(X)\|_2^2 < \infty$ and approximate by smooth functions with compact support.

2. If Z is a nonnegative random variable, then Var $Z \leq Ent\ Z^2$. This is another way to see that the LSI implies Poincaré, but only in the case of $f \geq 0$.

   To show a counterexample when $f \geq 0$ does not hold, take Z to be a symmetric Bernoulli variable ($Z = \pm 1$). Then Var $Z = 1$ but $Ent Z^2 = Ent 1 = 0$.

   Proof. Assume $\mathbb{E}Z^2 < \infty$ and define for $p \in [1, 2)$,

$$g(p) = \frac{\|Z\|_2^2 - \|Z\|_p^2}{\frac{1}{p} - \frac{1}{2}} = 2p\frac{\|Z\|_2^2 - \|Z\|_p^2}{2-p} \ .$$

   Note that

$$g(1) = 2\text{Var } Z \ . \tag{4}$$

   We claim that

$$\lim_{p\uparrow 2} g(p) = 2Ent\ Z^2 \ . \tag{5}$$

   To prove this, we first compute

$$\frac{\mathrm{d}}{\mathrm{d}p}\mathbb{E}Z^p = \lim_{h\to 0}\mathbb{E}\frac{Z^{p+h} - Z^p}{h}$$

   and by the mean value theorem, we can bound for $h > 0$,

$$\left|\frac{Z^{p+h} - Z^p}{h}\right| \leq \begin{cases} Z^{p+h}\log Z & \text{if } Z \geq 1 \\ Z^{p-h}|\log Z| & \text{if } Z < 1 \end{cases} .$$

   Furthermore for $h < 0$,

$$\left|\frac{Z^{p+h} - Z_p}{h}\right| \leq \begin{cases} Z^{p-h}\log Z & \text{if } Z \geq 1 \\ Z^{p+h}|\log Z| & \text{if } Z < 1 \end{cases} .$$

   So if h is small, we can use the dominated convergence theorem to show

$$\frac{\mathrm{d}}{\mathrm{d}p}\mathbb{E}Z^p = \mathbb{E}Z^p\log Z \ .$$

Next,

$$\frac{\mathrm{d}}{\mathrm{d}p}\|Z\|_p = \frac{\mathrm{d}}{\mathrm{d}p}\exp\left(\frac{1}{p}\log\mathbb{E}Z^p\right) = \|Z\|_p\frac{\mathrm{d}}{\mathrm{d}p}\frac{\log\mathbb{E}Z^p}{p} = \|Z\|_p\frac{\frac{p}{\mathbb{E}Z^p}\mathbb{E}Z^p\log Z - \log\mathbb{E}Z^p}{p^2}$$
$$= \frac{\|Z\|_p^{1-p}}{p^2}Ent\ Z^p$$

and

$$\lim_{p\uparrow 2}\frac{\mathrm{d}}{\mathrm{d}p}\|Z\|_p^2 = \lim_{p\uparrow 2}\frac{2}{p^2}\|Z\|_p^{2-p}Ent\ Z^p = \frac{1}{2}Ent\ Z^2\ .$$

Therefore by L'Hopital, (5) holds.

Last we show that g is nondecreasing; this will complete the proof. To do this, we start with a claim:

$$\alpha \text{ given by } \alpha(t) = t\log\mathbb{E}Z^{1/t} \text{ is convex on } (1/2,1]\ . \tag{6}$$

This follows from Hölder. For $r,s\in(1/2,1]$ and $\lambda\in(0,1)$,

$$\mathbb{E}Z^{\frac{1}{\lambda r+(1-\lambda)s}} = \mathbb{E}Z^{\frac{\lambda}{\lambda r+(1-\lambda)s}}Z^{\frac{1-\lambda}{\lambda r+(1-\lambda)s}}$$
$$\leq \left\|Z^{\frac{\lambda}{\lambda r+(1-\lambda)s}}\right\|_{\frac{\lambda r+(1-\lambda)s}{\lambda r}}\cdot\left\|Z^{\frac{1-\lambda}{\lambda r+(1-\lambda)s}}\right\|_{\frac{\lambda r+(1-\lambda)s}{(1-\lambda)s}}\ .$$

Taking logarithms, this reduces to

$$\alpha(\lambda r+(1-\lambda)s) \leq \lambda\alpha(r)+(1-\lambda)\alpha(s)\ .$$

Since α is convex, so is $\beta(t)=e^{2\alpha(t)}$, which is

$$e^{2\alpha(t)} = \left(\mathbb{E}Z^{1/t}\right)^{2t}\ .$$

So the difference quotient

$$\frac{e^{2\alpha(t)}-e^{2\alpha(1/2)}}{t-1/2} = \frac{\|Z\|_{1/t}^2-\|Z\|_2^2}{t-1/2}$$

is non-decreasing on $(1/2,1]$. Therefore $\frac{\|Z\|_t^2-\|Z\|_2^2}{1/t-1/2}$ is non-increasing on $[1,2)$ and this is equivalent to g being non-decreasing. □

1.2 Distributions with log-Sobolev inequalities

Bernoulli. Consider the symmetric Bernoulli distribution and let X have this distribution: X is ±1 with probability $1/2$. To consider a log-Sobolev inequality, we need a version of the gradient, so define for any $f:\{-1,1\}\to\mathbb{R}$,

$$(\nabla f)(x) = \frac{1}{2}(f(1)-f(-1))\ .$$

Then the version we claim is:

$$Ent\ f^2(X) \leq 2\mathbb{E}\left[(\nabla f)(X)\right]^2\ . \tag{7}$$

Proof. In other words, writing $f(-1) = a$ and $f(1) = b$,

$$\frac{1}{2}\left(a^2 \log a^2 + b^2 \log b^2\right) - \frac{a^2+b^2}{2} \log \frac{a^2+b^2}{2} \leq \frac{1}{2}(b-a)^2 \ .$$

This is true for $a = b$. Viewed as a function of b, call the left side $g(b)$ and the right side $h(b)$. Then

$$\begin{aligned} g'(b) = 2b \log b + b - b \log \frac{a^2+b^2}{2} - \frac{b\frac{a^2+b^2}{2}}{\frac{a^2+b^2}{2}} &= 2b \log b - b \log \frac{a^2+b^2}{2} \\ &= b \log \frac{2b^2}{a^2+b^2} \end{aligned}$$

and $h'(b) = b - a$. Therefore

$$(h-g)'(a) = 0 - a \log \frac{2a^2}{a^2+a^2} = 0 \ .$$

Last,

$$\begin{aligned} g''(b) &= \log \frac{2b^2}{a^2+b^2} + \frac{b}{\frac{2b^2}{a^2+b^2}} \left[\frac{4b(a^2+b^2) - 4b^3}{(a^2+b^2)^2} \right] \\ &= \log \frac{2b^2}{a^2+b^2} + \frac{4a^2 b}{2b(a^2+b^2)} \\ &= \log \frac{2b^2}{a^2+b^2} - \frac{2b^2}{a^2+b^2} + 2 \ , \end{aligned}$$

giving

$$(h-g)''(b) = -1 - \log \frac{2b^2}{a^2+b^2} + \frac{2b^2}{a^2+b^2} \ .$$

Now use the inequality $\log x \leq x - 1$ for $x > 0$; this follows from concavity of $x \mapsto \log x$. We conclude that $(h-g)''(b) \geq 0$ as long as $b \neq 0$. To summarize, the function $G = h - g$ satisfies

$$G(a) = G'(a) = 0 \text{ and } G''(b) \geq 0 \text{ for } b \neq 0 \ .$$

So if $0 \leq a \leq b$, we conclude that $G(b) \geq 0$, and this is the claimed LSI.

For general a, b, just apply the above inequality to $|a|$ and $|b|$:

$$\begin{aligned} \frac{1}{2}(a^2 \log a^2 + b^2 \log b^2) - \frac{a^2+b^2}{2} \log \frac{a^2+b^2}{2} &\leq \frac{1}{2}(|b| - |a|)^2 \\ &\leq \frac{1}{2}(b-a)^2 \ . \end{aligned}$$

□

As a corollary, we can state the log-Sobolev inequality for product Bernoulli.

Theorem 1.1 (LSI for Bernoulli)**.** *Let X be uniformly distributed on $\Omega = \{-1, 1\}^n$ and $f : \Omega \to \mathbb{R}$. Then*

$$Ent\ f^2(X) \le 2\mathbb{E}\|\nabla f(X)\|_2^2\ ,$$

where

$$\|\nabla f(X)\|_2^2 = (\nabla_1 f(X))^2 + \cdots + (\nabla_n f(X))^2$$

and $\nabla_i f(X)$ is the discrete derivative relative to X_i (with all other variables fixed).

Before we begin the proof, let us note that the Efron-Stein inequality can be written for $Z = f(X)$ as

$$\begin{aligned} \mathrm{Var}\ f \le \sum_{i=1}^n \mathbb{E}\mathrm{Var}_i\ f &= \frac{1}{2}\sum_{i=1}^n \mathbb{E}(Z - Z_i')^2 \\ &= 2\sum_{i=1}^n \mathbb{E}\left(\frac{Z - Z_i'}{2}\right)^2 \mathbf{1}_{\{Z \ne Z_i'\}} \\ &= \mathbb{E}\|\nabla f(X)\|_2^2\ . \end{aligned}$$

Proof. Tensorize entropy for

$$Ent\ f^2(X) \le \sum_{i=1}^n \mathbb{E}Ent_i\ f^2(X)\ .$$

When computing Ent_i, we are considering $f(X)^2$ as a function of only the coordinate X_i, with all others fixed. Since that coordinate is a Bernoulli(1/2) variable, we can use the previous result for

$$Ent\ f^2(X) \le \sum_{i=1}^n \mathbb{E}2\mathbb{E}_i(\nabla_i f(X))^2 = 2\mathbb{E}\|\nabla f(X)\|_2^2\ .$$

□

There is an extension to the asymmetric case. If $\mathbb{P}$ is a product measure on $\{-1, 1\}^n$ with $\mathbb{P}(X_i = 1) = p$ and $\mathbb{P}(X_i = -1) = 1 - p$ (where $X = (X_1, \ldots, X_n)$ has distribution $\mathbb{P}$), one has

$$Ent\ f^2(X) \le c(p)\mathbb{E}\|f(X)\|_2^2\ ,$$

where $c(p) = \frac{1}{1-2p}\log\frac{1-p}{p}$.

Gaussian. Just as in the case of the Poincaré inequality, we can derive the log-Sobolev inequality for Gaussian using the one for Bernoulli and the CLT.

Theorem 1.2 (LSI for Gaussian)**.** *Let $X = (X_1, \ldots, X_n)$ be a standard normal vector and $f : \mathbb{R}^n \to \mathbb{R}$ be C^1. Then*

$$Ent\ f^2(X) \le 2\mathbb{E}\|\nabla f(X)\|_2^2\ .$$

Proof. Start with $n = 1$ and assume that $\mathbb{E}(f'(X))^2 < \infty$; otherwise, the statement is trivial. As in the case of Poincaré, we may approximate f by more regular functions, so assume that f is C^2 with compact support. Let $Y = (Y_1, \ldots, Y_n)$ be uniformly distributed on $\{-1,1\}^n$ for $n \geq 1$ and set $g_n : \{-1,1\}^n \to \mathbb{R}$ to be

$$g_n(Y) = f\left(\frac{Y_1 + \cdots + Y_n}{\sqrt{n}}\right) .$$

Just as in the proof of the Poincaré inequality,

$$\mathbb{E}\|\nabla_B g_n(Y)\|_2^2 \leq \frac{1}{2}\mathbb{E}\left[\sum_{j=1}^n \left(f\left(\frac{\sum_{i=1}^n Y_i}{\sqrt{n}}\right) - f\left(\frac{\sum_{i=1}^n Y_i}{\sqrt{n}} - \frac{2Y_j}{\sqrt{n}}\right)\right)^2\right] ,$$

where ∇_B denotes the discrete gradient on $\{-1,1\}^n$. Just as in that proof, this converges as $n \to \infty$ to $\mathbb{E}f'(X)^2$. Therefore by the log-Sobolev inequality for Bernoulli,

$$\limsup_n Ent\ g_n^2(Y) \leq 2\mathbb{E}\|\nabla_B g(Y)\|_2^2 \to 2\mathbb{E}f'(X)^2 .$$

On the other hand, since $(Y_1 + \cdots + Y_n)/\sqrt{n} \Rightarrow N(0,1)$, the fact that f is bounded and continuous implies that $\lim_n Ent\ g_n^2(Y) = \limsup_n Ent\ g_n^2(Y) = Ent\ f^2(X)$. □

Exponential. From the Gaussian LSI we can derive a type of LSI for the exponential. Let X be exponential with parameter one: $\mathbb{P}(X \geq x) = e^{-x}$. Then if $f : [0,\infty)$ is differentiable, we will show

$$Ent\ f(X)^2 \leq 4\mathbb{E}X\left(f'(X)\right)^2 .$$

Note here that the differential operator on the right is not simply the derivative. A general LSI has a certain type of differential operator on the right, and the theory of such inequalities is related to Markov semi-groups. We may get to this later, but you can see, for example, Ramon van Handel's notes for more details.

We can derive the above inequality directly from the Gaussian LSI. If X_1, X_2 are i.i.d. standard Gaussian, then $Y = \frac{X_1^2+X_2^2}{2}$ is exponential. (You can check this.) Therefore if we define $F : \mathbb{R}^2 \to \mathbb{R}$ by $F(x,y) = f\left(\frac{x^2+y^2}{2}\right)$, then by the Gaussian LSI,

$$\begin{aligned} Ent\ f^2(X) = Ent\ F^2(X_1,X_2) &\leq \mathbb{E}\|\nabla F(X_1,X_2)\|_2^2 \\ &= \mathbb{E}\left[\left(X_1 f'\left(\frac{X_1^2+X_2^2}{2}\right)\right)^2 + \left(X_2 f'\left(\frac{X_1^2+X_2^2}{2}\right)\right)^2\right] \\ &= 4\mathbb{E}X(f'(X))^2 . \end{aligned}$$

2 Concentration via log-Sobolev – the Herbst argument

Just as Efron-Stein (tensorization) was able to give us exponential concentration by considering the moment generating function of a variable and proving the inequality

$$\mathrm{Var}\ e^{t/2Z} \leq Ct^2 \mathbb{E}e^{tZ}$$

when Z is a mean zero function of independent variables $X_1, \ldots, X_n$, we can derive a similar inequality using tensorization of entropy (see (8) below). Recall that in the Efron-Stein argument we used an iteration method to bound the cumulant generating function of Z. Now we will apply a log-Sobolev inequality to the variable $e^{tZ/2}$ to obtain a differential inequality for the cumulant generating function. The method is known as the Herbst argument.

2.1 Herbst with Gaussian and Bernoulli

The first example of this method uses the Gaussian LSI. The argument is very simple.

Theorem 2.1 (Tsirelson-Ibragimov-Sudakov). *Let $X = (X_1, \ldots, X_n)$ be a standard normal vector and $f : \mathbb{R}^n \to \mathbb{R}$ an L-Lipschitz function (relative to the ℓ^2-norm). Then setting $Z = f(X)$,*

$$\psi_{Z-\mathbb{E}Z}(t) \leq \frac{t^2 L^2}{2} \text{ for } t \in \mathbb{R}\ .$$

Proof. By approximating, we may assume that f is C^1 with derivative bounded in absolute value by L. Then by the LSI for Gaussian,

$$Ent\ e^{tZ} \leq 2\mathbb{E}\|\nabla(e^{tf(X)/2})\|_2^2\ .$$

The norm of the gradient of the function h given by $h(x) = e^{tf(x)/2}$ satisfies

$$\|\nabla h(x)\|_2^2 = \sum_{i=1}^n \left(\frac{\partial}{\partial x_i} e^{tf(x)/2}\right)^2 = \frac{t^2}{4} e^{tf(x)} \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}(x)\right)^2 \leq \frac{t^2 L^2}{4} e^{tf(X)}\ .$$

Therefore

$$Ent\ e^{tZ} \leq \frac{t^2 L^2}{2} \mathbb{E}e^{tZ}\ . \tag{8}$$

Now we use the Herbst argument to transform (8) to the claimed bound. If we compute the entropy of e^{tZ} we obtain

$$Ent\ e^{tZ} = \mathbb{E}tZe^{tZ} - \mathbb{E}e^{tZ} \log \mathbb{E}e^{tZ}\ .$$

On the other hand, because a Gaussian has a finite moment generating function on all of $\mathbb{R}$, we can compute the derivative. Set $Z' = Z - \mathbb{E}Z$ and compute

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi_{Z'}(t) = \frac{\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}e^{tZ}}{\mathbb{E}e^{tZ}} - \mathbb{E}Z = \frac{\mathbb{E}tZe^{tZ}}{t\mathbb{E}e^{tZ}} - \mathbb{E}Z = \frac{Ent\ e^{tZ} + \mathbb{E}e^{tZ}\log \mathbb{E}e^{tZ}}{t\mathbb{E}e^{tZ}} - \mathbb{E}Z \text{ for } t \neq 0\ .$$

By (8), we obtain

$$\psi'_{Z'}(t) \le \frac{tL^2}{2} + \frac{\log \mathbb{E}e^{tZ}}{t} - \mathbb{E}Z = \frac{tL^2}{2} + \frac{\log \mathbb{E}e^{tZ} - \mathbb{E}tZ}{t} = \frac{tL^2}{2} + \frac{\psi_{Z'}(t)}{t} ,$$

or

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\psi_{Z'}(t)}{t}\right) = \frac{\psi'_{Z'}(t)}{t} - \frac{\psi_{Z'}(t)}{t^2} \le \frac{L^2}{2} .$$

Because $\mathbb{E}Z' = 0$, $\lim_{t\to 0} \frac{\psi_{Z'}(t)}{t} = \psi'_{Z'}(0) = 0$, so this differential inequality implies that

$$\psi_{Z'}(t)/t \le \frac{L^2 t}{2} \text{ or } \psi_{Z'}(t) \le \frac{t^2L^2}{2} .$$

□

Assume that (for simplicity) $X = X_1, \ldots, X_n$ is uniformly distributed on the hypercube $\{-1,1\}^n$ and that $f : \{-1,1\}^n \to \mathbb{R}$ is any function. For the next application of Herbst, we will, for convenience, use a slightly different form of the upper bound in the log-Sobolev inequality. Let $\overline{X}_i$ be the vector X but with the i-th coordinate flipped:

$$\overline{X}_i = (X_1, \ldots, X_{i-1}, -X_i, X_{i+1}, \ldots, X_n) .$$

Then we can rewrite the LSI as

$$Ent\ f^2(X) \le \frac{1}{2}\sum_{i=1}^{n} \mathbb{E}(f(X) - f(\overline{X}_i))^2 = \sum_{i=1}^{n} \mathbb{E}(f(X) - f(\overline{X}_i))_+^2 .$$

Now assume that

$$\sum_{i=1}^{n} (f(X) - f(\overline{X}_i))_+^2 \le \nu \text{ almost surely for some } \nu > 0 . \qquad (9)$$

(This is similar to the strong assumption we made in the iteration method: $\sum_{i=1}^n (Z - Z_i')_+^2 \le \nu$. It can be relaxed, and we will see this soon.)

Theorem 2.2. *Let X be uniformly distributed on $\{-1,1\}^n$ and take $f : \{-1,1\}^n \to \mathbb{R}$ such that (9) holds. Then $Z = f(X)$ satisfies*

$$\psi_{Z-\mathbb{E}Z}(t) \le \frac{t^2\nu}{4} \textit{ for all } t .$$

Proof. We again apply the log-Sobolev inequality but for Bernoulli:

$$Ent\ e^{tZ} \le \frac{1}{2}\sum_{i=1}^{n} \mathbb{E}\left(e^{tf(X)/2} - e^{tf(\overline{X}_i)/2}\right)_+^2 .$$

The summand is nonzero if and only if $tf(X) > tf(\overline{X}_i)$, so the term within the parentheses is of the form $e^{y/2} - e^{z/2}$ for $y > z$. In this case we can use the mean value theorem for

$$e^{y/2} - e^{z/2} \le e^{y/2}\frac{y-z}{2} \ .$$

We therefore obtain for $t \ge 0$

$$Ent\ e^{tZ} \le \frac{t^2}{4}\sum_{i=1}^n \mathbb{E}e^{tf(X)}(f(X) - f(\overline{X}_i))_+^2 = \frac{t^2}{4}\mathbb{E}e^{tf(X)}\sum_{i=1}^n (f(X) - f(\overline{X}_i))_+^2 \ ,$$

and by (9),

$$Ent\ e^{tZ} \le \frac{t^2\nu}{4}\mathbb{E}e^{tZ} \ . \tag{10}$$

Similarly, for $t < 0$ one has

$$Ent\ e^{tZ} \le \frac{t^2}{4}\sum_{i=1}^n \mathbb{E}e^{tf(X)}(f(\overline{X}_i) - f(X))_+^2 \le \frac{t^2\nu}{4}\mathbb{E}e^{tZ} \ ,$$

since the condition (9) is invariant under mapping x_i to $-x_i$. We are back in the scenario of the Herbst argument: (10) holds for all t, so as in the last proof we obtain

$$\psi_{Z'}(t) \le \frac{t^2\nu}{4} \text{ for all } t \ .$$

$\square$

2.2 Herbst on metric spaces

Just as in the setting of the Poincaré inequality, we will give a general version of the Herbst argument which shows that a log-Sobolev inequality implies Gaussian concentration. Let (X, d) be a metric space and μ be a probability measure on the Borel sets of X. Recall the definition of the concentration function α:

$$\alpha_\mu(r) = \sup\{1 - \mu(A_r) : \mu(A) \ge 1/2\} \ ,$$

where $A_r = \{x \in X : d(x, A) < r\}$. Also recall the definition of the length of the gradient, defined for locally Lipschitz functions:

$$|\nabla f(x)| = \limsup_{y \to x} \frac{|f(x) - f(y)|}{d(x,y)} \ .$$

Note the following property of the gradient. If $\phi : \mathbb{R} \to \mathbb{R}$ is smooth, then

$$|\nabla \phi \circ f(x)| \le |\phi'(f(x))||\nabla f(x)| \ . \tag{11}$$

Proof. Given $\epsilon > 0$, one can find $\delta > 0$ such that if y satisfies $|f(x) - f(y)| < \delta$ then

$$|\phi(f(x)) - \phi(f(y)) - \phi'(f(x))(f(x) - f(y))| \leq \epsilon |f(x) - f(y)| \ .$$

Therefore

$$|\phi(f(x)) - \phi(f(y))| \leq |\phi'(f(x))||f(x) - f(y)| + \epsilon |f(x) - f(y)| \ .$$

By definition of $|\nabla f(x)|$ and continuity of f, one can find $\eta > 0$ such that if $d(x,y) < \eta$ then (a) $|f(x) - f(y)| < \delta$ and $|f(x) - f(y)| \leq |\nabla f(x)| d(x,y) + \epsilon d(x,y)$. Combining these, for $d(x,y) < \eta$,

$$|\phi(f(x)) - \phi(f(y))| \leq |\phi'(f(x))||\nabla f(x)| d(x,y) + \epsilon |\phi'(f(x))| d(x,y) + \epsilon |f(x) - f(y)| \ .$$

Divide by $d(x,y)$, take $y \to x$ and use the fact that $|f(x) - f(y)|$ is bounded by a constant L times $d(x,y)$ (by the local Lipschitz property) to obtain

$$\limsup_{y \to x} \frac{|\phi(f(x)) - \phi(f(y))|}{d(x,y)} \leq |\phi' f(x)||\nabla f(x)| + \epsilon |\phi'(f(x))| + \epsilon L \ .$$

Take $\epsilon \to 0$ to finish the proof. □

Theorem 2.3 (LSI implies gaussian concentration)**.** *Assume that for some constant $C > 0$ and all locally Lipschitz functions $f : X \to \mathbb{R}$,*

$$Ent_\mu \ f^2 \leq C \int |\nabla f(x)|^2 \ \mathrm{d}\mu(x) \ .$$

Then

$$\alpha_\mu(r) \leq \exp\left(-\frac{r^2}{4C}\right) \ \textit{for } r > 0 \ .$$

Proof. Let $A \subset X$ have $\mu(A) \geq 1/2$ and define $f : X \to [0, \infty)$ by

$$f(x) = \min\{r, d(x,A)\} \ .$$

Then f is 1-Lipschitz and we can use (11) for

$$|\nabla e^{tf(x)/2}| \leq \frac{t}{2} e^{tf(x)/2} |\nabla f(x)| \leq \frac{t}{2} e^{tf(x)/2} \ .$$

So by the log-Sobolev inequality,

$$Ent \ e^{tf(x)} \leq \frac{Ct^2}{4} \int e^{tf(x)} \ \mathrm{d}\mu(x) \text{ for all } t \ .$$

By the Herbst argument (which we can apply since f is bounded),

$$\log \int e^{t(f - \int f)} \ \mathrm{d}\mu \leq \frac{Ct^2}{4} \text{ for all } t \ .$$

Therefore by Chernoff,

$$\mu\left(x : f(x) - \int f \, \mathrm{d}\mu \geq t\right) \leq \exp\left(-\frac{t^2}{C}\right) \text{ for all } t \ .$$

Note that $\int f \, \mathrm{d}\mu \leq r\mu(A^c)$, so

$$\mu(A_r^c) \leq \mu(x : f(x) \geq r) \leq \mu\left(x : f(x) \geq r\mu(A) + \int f \, \mathrm{d}\mu\right) \leq \exp\left(-\frac{r^2\mu(A)^2}{C}\right) \ .$$

Because $\mu(A) \geq 1/2$, we obtain

$$1 - \mu(A_r) \leq \exp\left(-\frac{r^2}{4C}\right) \ .$$

□

2.3 Applications

1. (Norm of a Gaussian vector). Let $X = (X_1, \ldots, X_n)$ be a Gaussian vector with mean zero and covariance matrix Σ. If $p \geq 1$, we are interested in concentration properties of

$$Z = \|X\|_p = \left(\sum_{i=1}^n |X_i|^p\right)^{1/p} \ .$$

By the elementary theory of Gaussian vectors, we may write $X = AY$ for an i.i.d. standard Gaussian vector Y and $n \times n$ matrix A such that $A^T A = \Sigma$. We can then consider A as a linear mapping from $\mathbb{R}^n \to \mathbb{R}^n$, with the first equipped with the ℓ^2-norm and the second equipped with the ℓ^p-norm. In this case, the operator norm is defined as

$$\|A\|_{2\to p} = \sup_{v \neq 0} \frac{\|Av\|_p}{\|v\|_2} \ .$$

Then, defining the function $f : \mathbb{R}^n \to \mathbb{R}$ by $f(v) = \|Av\|_p$, one has

$$|f(v) - f(w)| = |\|Av\|_p - \|Aw\|_p| \leq \|A(v-w)\|_p \leq \|A\|_{2\to p}\|v - w\|_2 \ .$$

This means f is L-Lipschitz with $L = \|A\|_{2\to p}$. So by Poincaré, applied to $f(Y)$, $\mathrm{Var}\ Z \leq \|A\|_{2\to p}^2$ and by the Gaussian concentration we have proved,

$$\mathbb{P}(Z \geq \mathbb{E}Z + t) \leq \exp\left(-\frac{t^2}{2\|A\|_{2\to p}^2}\right) \ .$$

2. (Maximum of a Gaussian vector.) Again take X a Gaussian vector with mean zero and covariance matrix Σ, so that $X = AY$ for a standard normal vector Y and A with $A^T A = \Sigma$. Define $f : \mathbb{R}^n \to \mathbb{R}$ by

$$f(x) = \max\{(Ax)_i : i = 1, \ldots, n\} \ .$$

To bound the Lipschitz constant of f, compute

$$|(Au)_i - (Av)_i| = \left|\sum_{j=1}^n A_{i,j}(u-v)_j\right| \leq \sqrt{\sum_{j=1}^n A_{i,j}^2}\ \|u-v\|_2 \ .$$

However,

$$\text{Var } X_i = \text{Var } (AY)_i = \text{Var } \sum_{j=1}^n A_{i,j} X_j = \sum_{j=1}^n A_{i,j}^2 \ ,$$

so, setting $\sigma^2 = \max\{\text{Var } X_i : i = 1, \ldots, n\}$, we obtain

$$|(Au)_i - (Av)_i| \leq \sigma^2 \ \|u-v\|_2 \text{ for all } i \ .$$

Therefore picking i_1 to maximize $(Au)_i$ and j_1 to maximize $(Av)_j$,

$$f(u) - f(v) = \max_{i=1,\ldots,n} (Au)_i - \max_{j=1,\ldots,n} (Av)_j \leq (Au)_{i_1} - (Av)_{i_1} \leq \sigma^2 \|u-v\|_2$$

and similarly for $f(v) - f(u)$. This implies that

$$|f(u) - f(v)| \leq \sigma^2 \|u-v\|_2 \ ,$$

and so the Lipschitz constant of f is bounded by σ^2. By Poincaré,

$$\text{Var } Z = \text{Var } \max_{i=1,\ldots,n} X_i \leq \max_{i=1,\ldots,n} \text{Var } X_i = \sigma^2$$

and

$$\mathbb{P}(Z - \mathbb{E}Z > t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right) \ .$$

3. By a limiting argument, one can prove (an even more general version of): let (X_t) be an almost surely continuous Gaussian process on $[0,1]$. Setting $\sigma^2 = \sup_{t\in[0,1]} \text{Var } X_t$, and $Z = \max_{t\in[0,1]} X_t$,

$$\text{Var } Z \leq \sigma^2 \text{ and } \mathbb{P}(Z - \mathbb{E}Z > t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right) \ .$$

(See Section 5.5 in BLM.)

2.4 LSI implies QTCI

Here we will show how a log-Sobolev inequality implies the quadratic transportation cost inequality:

$$\inf_{P\in\mathcal{P}(\mathbb{P},\mathbb{Q})} \mathbb{E}_P \|X - Y\|_2^2 \leq 2\nu D(\mathbb{Q}||\mathbb{P}) \text{ for } \mathbb{Q} \ll \mathbb{P} \ , \tag{12}$$

where the infimum is over all couplings P of $\mathbb{P}$ and $\mathbb{Q}$, probability measures on $\mathbb{R}^n$, X has distribution $\mathbb{P}$ and Y has distribution $\mathbb{Q}$.

Theorem 2.4. *Let $\mathbb{P}$ be an absolutely continuous probability measure on $\mathbb{R}^n$ such that for some $C > 0$ and all locally Lipschitz $f : \mathbb{R}^n \to \mathbb{R}$,*

$$Ent\ f^2 \leq C\ \mathbb{E}\|\nabla f\|_2^2\ .$$

Then (12) *holds for $\nu = C/2$.*

Proof. For the proof we need a detour into PDE's. We will break into two steps. The first involves obtaining an "infimum convolution inequality" and the second relates this inequality to a "dual" form of the left side of the QTCI.

Step 1. Given a bounded smooth Lipschitz function $f : \mathbb{R}^n \to \mathbb{R}$, consider the **infimum convolution** with quadratic cost

$$Q_c f(x) = \inf_{y\in\mathbb{R}^n}\left[f(y) + \frac{c}{2}\|x-y\|_2^2\right]\ .$$

For $\lambda > 0$,

$$\begin{aligned} Q_c(\lambda f)(x) &= \lambda c \inf_{y\in\mathbb{R}^n}\left[(1/c)f(y) + \frac{1}{2\lambda}\|x-y\|_2^2\right] \\ &=: \lambda c v(x,\lambda) \end{aligned}$$

satisfies

$$\frac{\partial}{\partial\lambda}(\lambda c v(x,\lambda)) = cv(x,\lambda) + c\lambda\frac{\partial}{\partial\lambda}v(x,\lambda)\ .$$

However, the function $v(x,\lambda)$ is Lipschitz and an a.s. solution of the Hamilton-Jacobi initial value PDE (see, for instance, Evans Ch. 3): one has

$$\begin{cases} \frac{\partial v}{\partial\lambda} + \frac{1}{2}\|\nabla v\|_2^2 = 0 & \text{in } \mathbb{R}^n\times(0,\infty) \\ v(x,0) = (1/c)f(x) & \text{in } \mathbb{R}^n\times\{0\} \end{cases}\ .$$

and v is continuous up to the boundary. So we obtain

$$cv = \frac{\partial}{\partial\lambda}(\lambda c v) + \frac{c\lambda}{2}\|\nabla v\|_2^2 \text{ a.s. in } \mathbb{R}^n\times(0,\infty)\ .$$

Or for $u(x,\lambda) = Q_c(\lambda f)(x)$,

$$u(x,\lambda) = \lambda\frac{\partial}{\partial\lambda}u(x,\lambda) + \frac{c}{2}\|\nabla u(x,\lambda)\|_2^2\ . \tag{13}$$

Define the function

$$M(\lambda) = \mathbb{E}\exp\left(u(x,\lambda)\right)\ .$$

Note that when λ is bounded and $x \in \mathbb{R}^d$, $u(x,\lambda)$ is bounded, so one can interchange the derivative and integral and use (13) for

$$\begin{aligned} \lambda M'(\lambda) &= \mathbb{E}\left[\lambda\frac{\partial}{\partial\lambda}u(x,\lambda)\exp\left(u(x,\lambda)\right)\right] \\ &= \mathbb{E}ue^u - \frac{c}{2}\mathbb{E}\left[\|\nabla u\|_2^2 e^u\right]\ . \end{aligned}$$

On the other hand, by the LSI,

$$\begin{aligned}\mathbb{E}ue^u - \mathbb{E}e^u \log \mathbb{E}e^u = Ent\ e^u &\le C\mathbb{E}\|\nabla e^{u/2}\|_2^2 \\ &= \frac{C}{4}\mathbb{E}\|\nabla u\|_2^2 e^u\ ,\end{aligned}$$

so if we pick $c = C/2$,

$$\lambda M'(\lambda) \le \mathbb{E}e^u \log \mathbb{E}e^u = M(\lambda)\log M(\lambda)\ .$$

This means that $N(\lambda) = \frac{1}{\lambda}\log M(\lambda)$ satisfies

$$N'(\lambda) = \frac{M'(\lambda)}{\lambda M(\lambda)} - \frac{\log M(\lambda)}{\lambda^2} \le 0\ .$$

Since

$$\lim_{\lambda\downarrow 0} N(\lambda) = \lim_{\lambda\downarrow 0}\frac{1}{\lambda}\log \mathbb{E}e^u = \lim_{\lambda\downarrow 0}\frac{1}{\lambda}\log \mathbb{E}e^{\lambda c v} = \mathbb{E}f\ ,$$

we obtain

$$N(\lambda) \le \mathbb{E}f,\ \text{ or } \mathbb{E}e^{Q_c(\lambda f)(x)} \le e^{\lambda \mathbb{E}f}\ .$$

The latter is known as an **infimum convolution inequality**.

Step 2. To show that this inequality implies the QTCI, we need to use the Monge-Kantorovich-Rubinstein dual representation. (See, for example, Dudley.) One has

$$\inf_{P\in\mathcal{P}(\mathbb{P},\mathbb{Q})} \mathbb{E}_P \frac{1}{2}\|X-Y\|_2^2 = \sup_{f,g}\left[\int g\ \mathrm{d}\mathbb{Q} - \int f\ \mathrm{d}\mathbb{P}\right]\ ,$$

where X has distribution $\mathbb{P}$, Y has distribution $\mathbb{Q}$, and the supremum on the right is over all bounded smooth Lipschitz functions $f, g : \mathbb{R}^n \to \mathbb{R}$ such that

$$g(x) \le f(y) + \frac{1}{2}\|x-y\|_2^2 \text{ for all } x, y \in \mathbb{R}^n\ .$$

Note that any such g must satisfy $g(x) \le \frac{1}{c}Q_c(cf)(x)$. Therefore (12) will be verified once we show that

$$\int g\ \mathrm{d}\mathbb{Q} - \int f\ \mathrm{d}\mathbb{P} \le \nu Ent_{\mathbb{P}}\left(\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right)\ .$$

Putting $\phi = \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}$, this is equivalent to

$$\int \nu^{-1}(g - \mathbb{E}f)\phi\ \mathrm{d}\mathbb{P} \le Ent\ \phi\ .$$

By the variational characterization of entropy, this is implied by the statement

$$\mathbb{E}\exp\left(\nu^{-1}(g - \mathbb{E}f)\right) \le 1\ ,$$

and, using our bound on g with $\nu = C/2$, this is implied by

$$\mathbb{E}\exp\left(\frac{2}{C}(\frac{1}{c}Q_c(cf) - \mathbb{E}f)\right) \le 1 \ . \tag{14}$$

Now applying the infimum convolution inequality to cf in place of λf, we obtain

$$\mathbb{E}\exp\left(Q_c(cf) - c\mathbb{E}f\right) \le 1 \ ,$$

so inequality (14) holds with the choice $c = 2/C$. □

3 The entropy method

We have seen how, via the Herbst argument, one can turn log-Sobolev inequalities into Gaussian concentration inequalities provided the function f is, say, Lipschitz. Of course we would like to lessen these constraints. For example, we would like to deal with more distributions than just those which satisfy LSI's.

The idea will be to modify the Herbst argument to derive more general differential inequalities. At times we will be able to do this by proving weaker versions of LSI's.

3.1 Example 1: bounded differences

Recall that $f : \mathbb{R}^n \to \mathbb{R}$ satisfies the bounded differences condition if

$$|f(x_1, \ldots, x_n) - f(x_1, \ldots, x_{i-1}, x_i', x_{i+1}, \ldots, x_n)| \le c_i \text{ for all } x_1, x_1', \ldots, x_n, x_n' \ .$$

We have shown before that if $X_1, \ldots, X_n$ are independent and $Z = f(X_1, \ldots, X_n)$ then

$$\text{Var } Z \le \nu \text{ and } \mathbb{P}\left(Z - \mathbb{E}Z \ge \sqrt{2\nu t} + \frac{\nu}{2}t\right) \le e^{-t} \ ,$$

where $\nu = \frac{1}{4}\sum_{i=1}^n c_i^2$. The first inequality followed from Efron-Stein and the second from the iteration method. We also showed via the transportation method (Marton's inequality) that Gaussian concentration holds. We will now reprove this result using the entropy method.

We will frequently make use of the identity

$$t\psi'(t) - \psi(t) = \frac{Ent\ e^{tZ}}{\mathbb{E}e^{tZ}} \ , \tag{15}$$

where Z is mean zero and $\psi(t)$ is the cumulant generating function of Z. In fact, we used this identity in the Herbst argument. We will also use the conclusion of Herbst: that if Z is a random variable such that

$$Ent\ e^{tZ} \le Ct^2\mathbb{E}e^{tZ} \text{ for } t \in (0, a) \Rightarrow \psi_{Z-\mathbb{E}Z}(t) \le Ct^2 \text{ for } t \in (0, a) \ .$$

Theorem 3.1 (Bounded differences inequality). *Let $X_1, \dots, X_n$ be independent with $Z = f(X_1, \dots, X_n)$ and f satisfying the bounded differences condition. With $\nu = \frac{1}{4}\sum_{i=1}^n c_i^2$,*

$$\mathbb{P}(Z - \mathbb{E}Z \geq t) \leq \exp\left(-\frac{t^2}{2\nu}\right) \ .$$

Proof. As mentioned before, bounded differences implies that the variable Z is bounded. Therefore we can tensorize entropy for $t > 0$:

$$Ent\ e^{tZ} \leq \sum_{i=1}^n \mathbb{E} Ent_i\ e^{tZ} \ . \tag{16}$$

With all arguments but X_i fixed, Z takes values in an interval $[a_i, b_i]$ of length c_i. In the proof of Hoeffding, we showed that if Y is any variable with this property and $\psi(t) := \psi_{Y-\mathbb{E}Y}(t)$ is the cumulant generating function of the centered Y, then

$$\psi''(t) \leq \frac{(b_i - a_i)^2}{4} = \frac{c_i^2}{4} \text{ for } t \geq 0 \ .$$

Therefore

$$t\psi'(t) - \psi(t) = \int_0^t s\psi''(s)\ \mathrm{d}s \leq \frac{c_i^2 t^2}{8} \ .$$

Combining this with (15) and (16), we obtain

$$Ent\ e^{tZ} \leq \sum_{i=1}^n \mathbb{E}\left(\frac{c_i^2 t^2}{8}\mathbb{E}_i e^{tZ}\right) = \frac{t^2\nu}{2}\mathbb{E}e^{tZ} \ .$$

Now the Herbst argument gives

$$\psi_{Z-\mathbb{E}Z}(t) \leq \frac{t^2\nu}{2} \text{ for all } t > 0$$

and we are done. □

3.2 Example 2: a boundedness condition

In this section we will consider a familiar setup: take $X_1, \dots, X_n$ independent with $f : \mathbb{R}^n \to \mathbb{R}$ and define $Z = f(X_1, \dots, X_n)$. Suppose there exists $\nu > 0$ such that

$$\sum_{i=1}^n (Z - Z_i)^2 \leq \nu \text{ almost surely} \ , \tag{17}$$

where $Z_i = \inf_{x_i'} f(X_1, \dots, X_{i-1}, x_i', X_{i+1}, \dots, X_n)$. Note this is a slightly different condition than what we considered before, in the Efron-Stein section. However it still implies that Z is almost surely bounded.

Theorem 3.2. *Assume* (17). *Then*

$$\mathbb{P}(Z - \mathbb{E}Z > t) \leq \exp\left(-\frac{t^2}{2\nu}\right) \text{ for } t > 0 \ .$$

Proof. Because we are not assuming bounded differences, but the weaker condition (17), we would like to compensate for this by using an LSI. However, not all distributions have LSI's – this is a rather strict condition, and we have assumed essentially nothing about the distribution of the X_i's. Therefore we will need to derive an approximate version of an LSI.

Lemma 3.3 (Modified LSI)**.** *Let* $\phi(x) = e^x - 1 - x$. *Then*

$$\text{Ent } e^{tZ} \leq \sum_{i=1}^{n} \mathbb{E}\left[e^{tZ}\phi\left(-t(Z - \hat{Z}_i)\right)\right] \text{ for } t \in \mathbb{R} \ ,$$

where $\hat{Z}_i$ *is any variable depending only on* $X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n$.

Before proving the lemma, let us give an indication how it is an approximate LSI. If we applied a genuine LSI to the function $g = e^{tZ/2}$, we would obtain

$$\begin{aligned} \text{Ent } e^{tZ} = \text{Ent } g^2 \leq \mathbb{E}\|\nabla g\|_2^2 &= \sum_{i=1}^{n} \mathbb{E}\left(e^{tZ/2}\frac{\partial f}{\partial x_i}\right)^2 \\ &= \sum_{i=1}^{n} \mathbb{E}\left[e^{tZ}\left(t\frac{\partial f}{\partial x_i}\right)^2\right] \ . \end{aligned}$$

Heuristically, we replace the partial derivative by $(Z - Z_i)$. This would be valid, for example, if Z were bounded with bounded partial derivatives. We then obtain

$$\text{Ent } e^{tZ} \lesssim \sum_{i=1}^{n} \mathbb{E}\left[e^{tZ}(-t(Z - Z_i))^2\right] \ .$$

Noting that $\phi(x) \sim x^2$ when x is small, we obtain

$$\text{Ent } e^{tZ} \lesssim \sum_{i=1}^{n} \mathbb{E}\left[e^{tZ}\phi(t(Z - Z_i))\right] \ .$$

Here we have used at least two approximations, but it shows the similarity to an LSI.

Proof of Lemma 3.3. Let $\Phi(x) = x \log x$. Then if $a, b > 0$, by convexity,

$$\Phi(b) - \Phi(a) - \Phi'(a)(b - a) \geq 0 \ .$$

Now take Y to be any positive random variable. Put $b = \mathbb{E}Y$ for

$$\Phi(\mathbb{E}Y) - \Phi(a) - \Phi'(a)(\mathbb{E}Y - a) \geq 0 \ .$$

This is equivalent to

$$\mathbb{E}\left[\Phi(Y) - \Phi(a) - \Phi'(a)(Y-a)\right] \geq \mathbb{E}\left[\Phi(Y) - \Phi(\mathbb{E}Y)\right] \ .$$

The right side is the entropy of Y, so using $\Phi'(x) = \log x + 1$,

$$\begin{aligned} Ent\ Y &\leq \mathbb{E}\left[Y \log Y - a \log a - (\log a + 1)(Y - a)\right] \\ &= \mathbb{E}\left[Y(\log Y - \log a) - (Y - a)\right] \ . \end{aligned}$$

Noting that if we place $a = \mathbb{E}\mathbb{Y}$, then both sides are equal, we obtain the variational formula

$$Ent\ Y = \inf_{a>0} \mathbb{E}\left[Y(\log Y - \log a) - (Y - a)\right] \ . \tag{18}$$

This variational formula is used conditionally. Setting $Y = e^{tZ}$, $a = e^{t\hat{Z}_i}$, and using tensorization of entropy,

$$\begin{aligned} Ent\ Y \leq \sum_{i=1}^{n} \mathbb{E} Ent_i\ Y &\leq \sum_{i=1}^{n} \mathbb{E}\mathbb{E}_i \left[Y(\log Y - \log e^{t\hat{Z}_i}) - (Y - e^{t\hat{Z}_i})\right] \\ &\leq \sum_{i=1}^{n} \mathbb{E}\left[e^{tZ}(tZ - tZ_i) - (e^{tZ} - e^{t\hat{Z}_i})\right] \ . \end{aligned}$$

Note this is valid because $e^{t\hat{Z}_i}$ is a constant function relative to X_i. To complete the proof, the last expression is rewritten as

$$\sum_{i=1}^{n} \mathbb{E}\left[e^{tZ}(t(Z - \hat{Z}_i) - 1 + e^{-t(Z-\hat{Z}_i)})\right] = \sum_{i=1}^{n} \mathbb{E}\left[e^{tZ}\phi\left(-t(Z - \hat{Z}_i)\right)\right] \ .$$

□

Returning to the proof of the concentration inequality, we use the lemma:

$$Ent\ e^{tZ} \leq \mathbb{E}\left[e^{tZ} \sum_{i=1}^{n} \phi\left(-t(Z - Z_i)\right)\right] \ ,$$

where we recall $Z_i = \inf_{x_i'} f(X_1, \ldots, X_{i-1}, x_i', X_{i+1}, \ldots, X_n)$. Note that in this case $-t(Z - Z_i) \leq 0$ whenever $t \geq 0$ and so we can use the inequality

$$\phi(x) = e^x - 1 - x \leq \frac{x^2}{2} \text{ for } x \leq 0 \ .$$

By this inequality and assumption (17),

$$Ent\ e^{tZ} \leq \mathbb{E}\left[e^{tZ} \sum_{i=1}^{n} \frac{t^2(Z - Z_i)^2}{2}\right] \leq \frac{t^2\nu}{2} \mathbb{E} e^{tZ} \text{ for } t \geq 0 \ .$$

The Herbst argument implies $\psi_{Z-\mathbb{E}Z}(t) \leq \frac{t^2\nu}{2}$ for $t \geq 0$ and finishes the proof. □

Application: random symmetric matrix. Recall the example from the Efron-Stein section. Let A be a random $n \times n$ symmetric matrix with (i,j)-th entry $X_{i,j}$ when $i \le j$ and $X_{j,i}$ when $i \ge j$. Here $(X_{i,j} : i \le j)$ is a collection of independent random variables with values in $[-1,1]$. In that section, we derived the following bound

$$\sum_{1 \le i \le j \le n} (\lambda(A) - \lambda(A'_{i,j}))_+^2 \le 16 \text{ almost surely },$$

where $A'_{i,j}$ is the matrix obtained by replacing $X_{i,j}$ by an independent copy $X'_{i,j}$, and $\lambda(A)$ is the largest eigenvalue. Exactly the same proof gives $\sum_{1 \le i \le j \le n} (Z - Z_{i,j})^2 \le 16$, where $Z_{i,j}$ is the infimum over $X'_{i,j}$ of $\lambda(A'_{i,j})$. So by the last concentration inequality,

$$\mathbb{P}\left(\lambda(A) - \mathbb{E}\lambda(A) > t\right) \le \exp\left(-\frac{t^2}{32}\right) .$$

Application: Gaussian concentration for convex Lipschitz functions. We will take here the same setup that we had for the convex Poincaré inequality, but now we will show Gaussian concentration. This will be a quick deduction from the last theorem.

Corollary 3.4. *Let $X_1, \ldots, X_n$ be independent variables taking values in $[0,1]$ and $f : [0,1]^n \to \mathbb{R}$ a separately convex function such that $|f(x) - f(y)| \le L\|x - y\|_2$ for all $x, y \in [0,1]^n$. Then $Z = f(X_1, \ldots, X_n)$ satisfies*

$$\mathbb{P}(Z - \mathbb{E}Z > t) \le e^{-\frac{t^2}{2L^2}} .$$

Note that we have proved such a result for the Gaussian distribution and all Lipschitz functions. Since we do not assume any particular distribution here (only bounded), we will have to use a modified LSI and restrict our class of functions. This shows us a general principle that we cannot have the best of both worlds: either the distribution is severely restricted (like requiring an LSI) or we must restrict the class of functions (for instance, convexity, or requiring certain boundedness properties).

Proof. Again we may approximate and assume that f has a partial derivatives. In the proof of the convex Poincaré inequality, we have already shown that if

$$Z_i = \inf_{x_i} f(X_1, \ldots, X_{i-1}, x_i, X_{i+1}, \ldots, X_n) ,$$

then

$$\sum_{i=1}^n (Z - Z_i)^2 \le \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}(X)\right)^2 = \|\nabla f\|_2^2 .$$

Using the Lipschitz condition,

$$\sum_{i=1}^n (Z - Z_i)^2 \le L^2 .$$

Therefore we can use Theorem 3.2 with $\nu = L^2$. □

Remark. The assumption of convexity cannot be dropped. For an example, define

$$A = \{(x_1, \ldots, x_n) \in \{0,1\}^n : x_1 + \cdots + x_n \le n/2\}$$

and $f(x) = \inf\{\|x - y\|_2 : y \in A\}$. Then f is 1-Lipschitz. Furthermore f is not convex. We can show that the preceding convex concentration inequality cannot hold for f.

Note that

$$f(x) = \sqrt{(x_1 + \cdots + x_n - n/2)_+} \ .$$

Therefore, putting $S_n = X_1 + \cdots + X_n$,

$$\frac{f(X)}{n^{1/4}} = \sqrt{\left(\frac{S_n - n/2}{\sqrt{n}}\right)_+} \ .$$

The function $x \mapsto \sqrt{x_+}$ is continuous, so by the central limit theorem and the continuous mapping theorem,

$$\frac{f(X)}{n^{1/4}} \Rightarrow \sqrt{Z_+} \ , \tag{19}$$

where Z is a standard normal variable.

Furthermore by Jensen,

$$\begin{aligned} \mathbb{E}f(X) = 2\mathbb{E}\sqrt{|S_n - n/2|} &= 2\mathbb{E}((S_n - n/2)^2)^{1/4} \\ &\le 2\left(\mathbb{E}(S_n - n/2)^2\right)^{1/4} = 2n^{1/4}\text{Var } X_1 \le n^{1/4} \ . \end{aligned}$$

So if the concentration inequality held, for $t > 0$,

$$\mathbb{P}(f(X) - n^{1/4} > t) \le \mathbb{P}(f(X) - \mathbb{E}f(X) > t) \le e^{-t^2/2} \ .$$

In other words,

$$\mathbb{P}\left(\frac{f(X)}{n^{1/4}} > 1 + t\right) \le e^{-t^2\sqrt{n}/2}$$

converges to zero for any $t > 0$. This contradicts (19).

3.3 Example 3: beyond boundedness

Both examples above required types of strong upper bounds for the quantity $\sum_{i=1}^n (Z - Z_i)^2$, which can be interpreted as related to Efron-Stein. One does not always have such almost sure upper bounds, and there are various weakenings possible. We saw one earlier, with the self-bounding condition. BLM explore this in more detail in Section 6.7, proving exponential bounds for self-bounding functions. A more significant weakening, however, is given by the "exponential Efron-Stein inequalities" given in Section 6.9. We will state the theorem without proof, and in return for your kindness, we will analyze in detail one case, first-passage percolation, where similar techniques are possible. The proof we will give comes from a paper of Damron-Hanson-Sosoe.

First exponential Efron-Stein. We will not require that $\sum_{i=1}^n (Z - Z_i)^2$ is bounded, but we will state concentration bounds for Z in terms of concentration of such random variables. Just as in Efron-Stein, we can define the following quantities, where $X_1, \ldots, X_n$ are independent, $X_1', \ldots, X_n'$ are independent copies, $f : \mathbb{R}^n \to \mathbb{R}$ and $Z = f(X_1, \ldots, X_n)$. Write $\mathbb{E}_i'$ as expectation over only X_i' and define

$$V_+ = \sum_{i=1}^n \mathbb{E}_i'(Z - Z_i')_+^2 \text{ and } V_- = \sum_{i=1}^n \mathbb{E}_i'(Z - Z_i')_+^2 \ ,$$

where Z_I' is, as usual, $f(X_1, \ldots, X_{i-1}, X_i', X_{i+1}, \ldots, X_n)$. Note that Efron-Stein can be written as

$$\text{Var } Z \leq \mathbb{E}V_+ = \mathbb{E}V_- \ .$$

If the variables V_+ and V_- were almost surely bounded, we might expect to be able to apply the techniques we have already developed. So here we only assume that they have exponential moments, but can be unbounded.

The first inequality below applies when V_+ can be bounded; the second does when V_- can be bounded.

Theorem 3.5 (Exponential Efron-Stein)**.** *Let Z and V_+, V_- be as above.*

1. *Let $\theta, t > 0$ be such that*

$$\theta t < 1 \text{ and } \mathbb{E} \exp\left(\frac{tV_+}{\theta}\right) < \infty \ .$$

   Then

$$\psi_{Z - \mathbb{E}Z}(t) \leq \frac{t\theta}{1 - t\theta} \psi_{V_+}(t/\theta) \ .$$

2. *Next assume that Z is such that*

$$Z_i' - Z \leq 1 \text{ for all } i \ .$$

   Then for all $t \in (0, 1/2)$,

$$\psi_{Z - \mathbb{E}Z}(t) \leq \frac{2t}{1 - 2t} \psi_{V_-}(t) \ .$$

The proofs of these results follow the same general strategy from before. We tensorize entropy of e^{tZ} and then try to obtain a differential inequality we can integrate in the same manner in which the Herbst argument proceeds. However, in this case, we cannot directly bound above terms like $\sum_{i=1}^n (Z - Z_i)^2$ like we could in the proof of, say, Theorem 3.2. Instead we must decouple them from $e^{\lambda Z}$ using the exponential Hölder inequality. One might think to use this because the terms V_+ and V_- are "almost" bounded almost surely – they have exponential moments. Indeed, exponential Hölder is typically useful when we would like to apply Hölder with p close to ∞ but q close to 1. However if the first variable is unbounded, our estimate will blow up as $p \to \infty$. So we think of exponential Hölder as a sort of "nice" limiting case of Hölder.

4 Application to first-passage percolation (FPP)

Here we will give the concentration argument from the appendix of "Subdiffusive concentration in first-passage percolation" (2014) by Damron-Hanson-Sosoe (DHS). The goal will be to give an exponential concentration inequality for the passage time above its mean assuming exponential moments for the edge-weights. This is not the main result of that paper, but an auxiliary one used to obtain the main one. A similar bound was derived in Damron-Kubota (2014) for the lower tail under the assumption of low moments for the edge-weights.

Recall the setup. Letting $\mathbb{Z}^d$ be the d-dimensional integer lattice, let $\mathcal{E}^d$ be the set of nearest neighbor edges:

$$\mathcal{E}^d = \{\{x,y\} : x,y \in \mathbb{Z}^d \text{ and } \|x-y\|_1 = 1\} .$$

Let (t_e) be a collection of i.i.d. non-negative random variables, one assigned to each edge $e \in \mathcal{E}^d$ and assume the conditions

$$\mathbb{P}(t_e = 0) < p_c(d) \text{ and } \mathbb{E}e^{\alpha t_e} < \infty \text{ for all } \alpha > 0 . \tag{20}$$

Here $p_c(d)$ is the bond percolation threshold in d-dimensions. It is not so important right now what this means, but we just need to know that it is a fixed number (depending on d) in the interval $(0,1)$. The moment condition can be reduced to $\mathbb{E}e^{\alpha t_e} < \infty$ for some $\alpha > 0$.

A lattice path γ from x to y is a sequence of alternating vertices and edges of the form $\{x_0, e_0, x_1, \ldots, e_{n-1}, x_n\}$ such that for all $i = 0, \ldots, n-1$, $e_i = \{x_i, x_{i+1}\}$ and $x_0 = x$, $x_n = y$. Any such γ is associated a passage time

$$T(\gamma) = \sum_{e \in \gamma} t_e ,$$

where this sum is over the edges $e_0, \ldots, e_{n-1}$ of γ. Between any two lattice points $x, y \in \mathbb{Z}^d$, we define the passage time

$$T(x,y) = \inf_{\gamma : x \to y} T(\gamma) .$$

Any path γ from x to y that achieves this infimum – that is, $T(x,y) = T(\gamma)$ – is called a geodesic from x to y. Under assumption (20) (actually only under $\mathbb{P}(t_e = 0) < p_c$), Kesten has shown

$$\mathbb{P}(\exists \text{ a geodesic from } x \to y \text{ for all } x, y \in \mathbb{Z}^d) = 1 .$$

Although geodesics exist under our assumptions, they do not have to be unique. For instance, take a distribution with atoms, say with $\mathbb{P}(t_e = 1) > 0$ but $\mathbb{P}(t_e < 1) = 0$. Then in any configuration of edge-weights in which

$$t_{\{0,e_1\}} = t_{\{0,e_2\}} = t_{\{e_1, e_1+e_2\}} = t_{\{e_2, e_2+e_1\}} = 1 ,$$

then there are at least two geodesics from 0 to $e_1 + e_2$: one that takes one step right and one step up, and one that takes one step up and one step right.

Our aim will be to prove the following result. It was initially established by Talagrand, using different methods.

Theorem 4.1. *Assuming* (20)*, there exist* $C_1, C_2 > 0$ *such that*

$$\mathbb{P}\left(T(0,x) - \mathbb{E}T(0,x) \geq t\sqrt{\|x\|_1}\right) \leq e^{-C_1 t^2} \text{ for } t \in \left(0, C_2\sqrt{\|x\|_1}\right) \ .$$

Proof. Write $T = T(0,x)$. We first tensorize entropy. Note that since $\mathbb{E}e^{\alpha t_e} < \infty$ for all $\alpha > 0$, then $Ent\ e^{\lambda T} = \mathbb{E}\lambda T e^{\lambda T} - \mathbb{E}e^{\lambda T} \log \mathbb{E}e^{\lambda T}$ exists. Therefore if we enumerate the edge variables (in any deterministic order) as $(t_{e_1}, t_{e_2}, \ldots\}$, then

$$Ent\ e^{\lambda T} \leq \sum_{i=1}^{\infty} \mathbb{E} Ent_i\ e^{\lambda T}\ , \tag{21}$$

where Ent_i is entropy relative to only the edge-weight t_{e_i}. (There is a small point we have glossed over here. We proved tensorization for functions of finitely many variables and here we have infinitely many. However, this can be fixed in our case in any number of ways, in particular a simple limiting argument.) We now apply a different modified LSI of BLM.

Lemma 4.2 (Symmetrized LSI)**.** *Let* $q(x) = x(e^x - 1)$*. If* X *is a random variable and* X' *is an independent copy, then for all* $t \in \mathbb{R}$*,*

$$Ent\ e^{tX} \leq \mathbb{E}\left[e^{tX} q(\lambda(X' - X)_+)\right]\ .$$

Note that for x small, $q(x) \sim x^2$ so, by the reasoning given near the other modified LSI, it is reasonable to call this a type of LSI.

Proof. We again use the variational formula (18): if $Y \geq 0$ then

$$Ent\ Y = \inf_{a>0} \mathbb{E}\left[Y(\log Y - \log a) - (Y - a)\right]\ .$$

Here $\mathbb{E}$ is expectation over Y. Let $Y = e^{tX}$ and introduce an independent copy of X, written X', we set $a = e^{tX'}$. Then

$$Ent\ Y \leq \mathbb{E}_X\left[e^{tX}(tX - tX') - (e^{tX} - e^{tX'})\right]\ .$$

Integrate both sides over X' for

$$\begin{aligned} Ent\ Y &\leq \mathbb{E}\left[e^{tX}(tX - tX') - (e^{tX} - e^{tX'})\right] \\ &= \mathbb{E}\left[e^{tX}(-t(X' - X) - 1 + e^{t(X'-X)})\right] \\ &= \mathbb{E}e^{tX}\phi(t(X' - X))\ , \end{aligned}$$

where $\phi(x) = e^x - x - 1$. At this point, all we have done is derive the old modified LSI but using an independent copy in place of Z_i. Now we want to take advantage of the fact that X' and X have the same distribution. To do so, write

$$\phi(t(X' - X)) = \phi(t(X' - X)_+) + \phi(-t(X' - X)_-)\ ,$$

giving

$$Ent\ Y \leq \mathbb{E}\left[e^{tX}\phi(t(X'-X)_+)\right] + \mathbb{E}\left[e^{tX}\phi(-t(X'-X)_-)\right] \ . \tag{22}$$

By symmetry of $(X,X') \mapsto (X',X)$, the second term can be rewritten as

$$\begin{aligned}\mathbb{E}e^{tX'}\phi(-t(X-X')_-) &= \mathbb{E}\left[e^{tX}e^{t(X'-X)}\phi(-t(X'-X)_+)\right] \\ &= \mathbb{E}\left[e^{tX}e^{t(X'-X)_+}\phi(-t(X'-X)_+)\right] \ .\end{aligned}$$

Placing this back in (22),

$$Ent\ Y \leq \mathbb{E}\left[e^{tX}\left[\phi(t(X'-X)_+) + e^{t(X'-X)_+}\phi(-t(X'-X)_+)\right]\right] \ .$$

The function in the interior is

$$\phi(z) + e^z\phi(-z) = (e^z - 1 - z) + e^z(e^{-z} - 1 + z) = ze^z - z = q(z) \ ,$$

so we obtain

$$Ent\ Y \leq \mathbb{E}e^{tX}q(t(X'-X)_+) \ .$$

□

Next use the symmetrized LSI in (21). By monotone convergence,

$$Ent\ e^{\lambda T} \leq \sum_{i=1}^{\infty} \mathbb{E}e^{\lambda T}q(\lambda(T_i' - T)_+) \ . \tag{23}$$

Here T_i' is the passage time from 0 to x in the edge-weight configuration in which the weight t_{e_i} is replaced by an independent copy t'_{e_i}, and all other edges remain the same (that is, they are equal to t_e). As in the FPP variance proof, we know that $T_i' - T$ is only positive if e_i is in a geodesic from 0 to x in the original edge-weights (t_e). We claim more here: that if we define $\mathcal{G}(0,x)$ as the collection of edges in the intersection of all geodesics from 0 to x (since there need not be a unique one), then

$$T_i' - T > 0 \Rightarrow e_i \in \mathcal{G}(0,x) \text{ in the original weights } (t_e) \ .$$

To argue this, assume that $T_i' - T > 0$. Then let γ be a geodesic from 0 to x in the edge-weights (t_e). If $e_i \notin \gamma$ then γ must have the same passage time in the new weights, since we only replace t_{e_i} by an independent copy. In other words, all edge-weights for edges on γ have the same value in both configurations. Therefore if τ denotes the passage time in the new weights (with only t_{e_i} replaced),

$$T_i' = \tau(0,x) \leq \tau(\gamma) = T(\gamma) = T(0,x) = T \ ,$$

contradicting that $T_i' - T > 0$.

Returning to (23), we can give the upper bound

$$Ent\ e^{\lambda T} \leq \sum_{i=1}^{\infty} \mathbb{E}e^{\lambda T} q(\lambda(T_i' - T)_+)\mathbf{1}_{\{e_i \in \mathcal{G}(0,x)\}}\ .$$

The function $x \mapsto q(x)$ is monotone increasing for $x \geq 0$ so using the bound $T_i' - T \leq t_{e_i}'$ (which we established during the FPP variance bound) and independence,

$$Ent\ e^{\lambda T} \leq \sum_{i=1}^{\infty} \mathbb{E}e^{\lambda T} q(\lambda t_{e_i}')\mathbf{1}_{\{e_i \in \mathcal{G}(0,x)\}} = \mathbb{E}q(\lambda t_e)\ \mathbb{E}\left[e^{\lambda T} \# \mathcal{G}(0,x)\right]\ . \tag{24}$$

To apply the Herbst argument we would love to decouple $e^{\lambda T}$ from $\#\mathcal{G}(0,x)$. Unfortunately, the variable $\#\mathcal{G}(0,x)$ is not bounded, so we cannot just pull it out. So we are in a prime position to use exponential Hölder. We write for arbitrary $a > 0$,

$$\mathbb{E}e^{\lambda T}\#\mathcal{G}(0,x) \leq a\ Ent\ e^{\lambda T} + a\ \mathbb{E}e^{\lambda T} \log \mathbb{E} \exp\left(\frac{\#\mathcal{G}(0,x)}{a}\right)\ .$$

Combining with (24), if $a\mathbb{E}q(\lambda t_e) < 1$,

$$Ent\ e^{\lambda T}(1 - a\mathbb{E}q(\lambda t_e)) \leq a\mathbb{E}q(\lambda t_e)\mathbb{E}e^{\lambda T} \log \mathbb{E}\exp\left(\frac{\#\mathcal{G}(0,x)}{a}\right)\ ,$$

or

$$Ent\ e^{\lambda T} \leq \frac{c\mathbb{E}q(\lambda t_e)}{1 - a\mathbb{E}q(\lambda t_e)} \log \mathbb{E}\exp\left(\frac{\#\mathcal{G}(0,x)}{a}\right) \times \mathbb{E}e^{\lambda T}\ .$$

To control these terms we will need a lemma from DHS. We will omit the proof, as it brings us too far into percolation, and we are here interested in the application of concentration ideas.

Lemma 4.3. *Assuming* (20), *there exist $a, c_1 > 0$ such that*

$$\log \mathbb{E}\exp\left(\frac{\#\mathcal{G}(0,x)}{a}\right) \leq c_1 \|x\|_1 \text{ for all } x \in \mathbb{Z}^d\ .$$

Applying the lemma,

$$Ent\ e^{\lambda T} \leq c_1\|x\|_1 \frac{a\mathbb{E}q(\lambda t_e)}{1 - a\mathbb{E}q(\lambda t_e)}\ \mathbb{E}e^{\lambda T} \text{ if } a\mathbb{E}q(\lambda t_e) < 1\ ,$$

or

$$Ent\ e^{\lambda T} \leq 2c_1\|x\|_1 q\mathbb{E}q(\lambda t_e)\mathbb{E}e^{\lambda T} \text{ if } a\mathbb{E}q(\lambda t_e) < 1/2\ .$$

Note that by dominated convergence,

$$\lim_{\lambda \downarrow 0} \frac{\mathbb{E}q(\lambda t_e)}{\lambda^2} = \lim_{\lambda \downarrow 0} \mathbb{E}\left(\frac{t_e(e^{\lambda t_e} - 1)}{\lambda}\right) = \mathbb{E}t_e^2\ .$$

So we can find $c_2 > 0$ such that if $\lambda \in (0, c_2)$ then $\mathbb{E}q(\lambda t_e) \leq 2\lambda^2 \mathbb{E}t_e^2$. For such λ, note that $a\mathbb{E}q(\lambda t_e) < 1/2$ if $2a\lambda^2 \mathbb{E}t_e^2 < 1/2$, which occurs if λ is smaller than some positive $c_3 < c_2$. So we obtain for some $c_4 > 0$,

$$Ent\ e^{\lambda T} \leq 4c_1 \|x\|_1 \lambda^2 \mathbb{E}t_e^2 \mathbb{E}e^{\lambda T} \leq \lambda^2 c_4 \|x\|_1 \mathbb{E}e^{\lambda T} \text{ for } \lambda \in (0, c_3) \ .$$

Now we apply Herbst to obtain

$$\psi_{T-\mathbb{E}T}(t) \leq c_4 \|x\|_1 t^2 \text{ for } t \in (0, c_3) \ . \tag{25}$$

To go to the probability bound we just apply Chernoff: setting $S = T - \mathbb{E}T$, and $\lambda > 0$,

$$\mathbb{P}(S \geq t) = \mathbb{P}(e^{\lambda S} \geq e^{\lambda t}) \leq e^{-\lambda t + \psi_S(\lambda)} \ .$$

Now use (25) for

$$\mathbb{P}(S \geq t) \leq \exp\left(-\lambda t + c_4 \|x\|_1 \lambda^2\right) \text{ for } \lambda \in (0, c_3) \ .$$

For the subgaussian bound, we would like to set $\lambda = t/(2c_4\|x\|_1)$, but we can only do this if this number is in the interval $(0, c_3)$. If we do this, we obtain

$$\mathbb{P}(T - \mathbb{E}T \geq t) \leq \exp\left(-\frac{t^2}{4c_4\|x\|_1}\right) \text{ if } t \in (0, 2c_3 c_4 \|x\|_1) \ .$$

Substituting $t\sqrt{\|x\|_1}$ for t, we obtain the bound of the theorem. $\square$

1 Influences and concentration on the hypercube

For a while now, we will focus on the hypercube $\{-1,+1\}^n$. Recall that if $A \subset \{-1,+1\}^n$ and $X = (X_1, \ldots, X_n) \in \{-1,+1\}^n$, then we say that X_i is pivotal for A (in X) if

$$\mathbf{1}_A(X^+(i)) \neq \mathbf{1}_A(X^-(i)) \ ,$$

where $X^+(i) = (X_1, \ldots, X_{i-1}, 1, X_{i+1}, \ldots, X_n)$ and $X^-(i) = (X_1, \ldots, X_{i-1}, -1, X_{i+1}, \ldots, X_n)$. If X has distribution $\mathbb{P}$, then the influence of X_i on A is

$$I_i(A) = \mathbb{P}(X_i \text{ is pivotal for } A)$$

and the total influence on A is

$$I(A) = \sum_{i=1}^n I_i(A) \ .$$

For X uniformly distributed, we wish to understand the possible values of $I(A)$ (as the set A varies), and the relation between this question and concentration.

We have already shown

$$I(A) \geq 4\mathbb{P}(A)(1 - \mathbb{P}(A)) \tag{1}$$

by Efron-Stein and the improvement

$$I(A) \geq 2\mathbb{P}(A) \log_2 \frac{1}{\mathbb{P}(A)} \tag{2}$$

using entropy. We identify any A with a **Boolean** function f on $\{-1,+1\}^n$ – that is, its range is contained in $\{0,1\}$ – by setting $f = \mathbf{1}_A$. For a general $f : \{-1,+1\}^n \to \mathbb{R}$, we define influence similarly:

$$I_i(f) = \mathbb{P}(X_i \text{ is pivotal for } f) \text{ and } I(f) = \sum_{i=1}^n I_i(f) \ ,$$

where "X_i is pivotal for f" means that $f(X^+(i)) \neq f(X^-(i))$.

Examples.

1. As we have described before, if A is a singleton set $\{x\}$, then $\mathbb{P}(A) = 2^{-n}$ and $I_i(A) = 2/2^n$. This equality for influence holds because X_i is pivotal if and only if $X = x$ or X agrees with x in all but the i-th coordinate. Therefore $I(A) = 2n/2^n$.

2. If we require that $\mathbb{P}(A) = 1/2$ then (2) implies that $I(A) \geq 1$. On the other hand, we always have $I(A) \leq n$, since $I_i(A) \leq 1$ for all i and A (even if $\mathbb{P}(A) \neq 1/2$). Therefore

$$\mathbb{P}(A) = 1/2 \Rightarrow I(A) \in [1, n] \ .$$

3. (Dictatorship). The lower bound is achieved. For any $i = 1, \ldots, n$, let f be the boolean function
$$f(x) = \frac{1 + x_i}{2} \ .$$
That is, $f = \mathbf{1}_{\{x_i = 1\}}$. Then the influence of X_j for $j \neq i$ is zero and the influence of X_i is 1. Therefore $I(A) = 1$ for $A = \{x_i = 1\}$ and A has the minimal total influence.

4. (Junta). If f is a boolean function determined by exactly k of the variables $X_1, \ldots, X_n$ then f is called a k-junta. The dictatorship is a 1-junta.

5. (Parity function). Let f be the boolean function
$$f(x) = \frac{1 + \prod_{i=1}^n (-1)^{x_i}}{2} = \begin{cases} 1 & \text{if } \#\{i : x_i = -1\} \text{ is even} \\ 0 & \text{if } \#\{i : x_i = -1\} \text{ is odd} \end{cases} \ .$$
Then if $A = \{\#\{i : x_i = -1\} \text{ is even}\}$, one has $I_i(A) = 1$ for all i. Therefore $I(A) = n$. In this case, every variable X_i has the maximum amount of influence on A, and the total influence is maximized.

6. (Tribes). We will see soon that the maximal influence of any variable satisfies
$$\mathbb{P}(A) = 1/2 \Rightarrow \max_{i=1,\ldots,n} I_i(A) \geq \frac{\log n}{4n} \ .$$
Note that although the dictatorship minimizes total influence, its maximal influence (which equals 1) does not achieve this bound. However, we can construct an example which satisfies $\max_{i=1,\ldots,n} I_i(A) = O(\log n / n)$.

Imagine that n people are split into n/l distinct tribes of size l. Each person votes yes or no, and the total vote of the population is computed as follows. If there is a tribe that votes yes unanimously, then the total vote is yes. Otherwise it is no. In other words, $f : \{-1, +1\}^n \to \{0, 1\}$ equals 1 if $X_{(k-1)l+1} = \cdots = X_{kl} = +1$ for some $k = 0, \ldots, n/l$ and f is zero otherwise.

First the probability of the event $A = \{f = 1\}$ under the uniform measure can be computed as
$$1 - \left(1 - 2^{-l}\right)^{n/l} \ .$$
If l is chosen so that $n/l \sim 2^l$, this converges to
$$\mathbb{P}(A) \to 1 - e^{-1} \ .$$
(By modifying l to be cl we can ensure convergence to 1/2.)

We can compute the influence of X_i. X_i is only pivotal if $X_j = 1$ for all $j \neq i$ in the same tribe as X_i and no other tribe votes unanimously yes. The probability of this under uniform measure is
$$I_i(f) = (1/2)^{l-1} \left(1 - (1/2)^l\right)^{(n/l)-1} = (1/2) \frac{(1/2)^l}{1 - (1/2)^l} \left(1 - (1/2)^l\right)^{n/l} \ .$$

We can write this as

$$\frac{1}{2}\cdot\frac{1}{2^l-1}\cdot\left(1-\frac{1}{2^l}\right)^{n/l} .$$

For the last term to converge to e^{-1}, we should choose $n/l \sim 2^l$. Then $\log_2 n - \log_2 l \sim l$, meaning $\log_2 n \sim l + \log_2 l$. Using $l \le \log_2 n$, we obtain $\log_2 n \le l + \log_2\log_2 n$, or $l \ge \log_2 n - \log_2\log_2 n$. In other words,

$$\frac{1}{2^l-1} \sim 2^{-l} \le 2^{-\log_2 n + \log_2\log_2 n} = \frac{\log_2 n}{n} .$$

We conclude with

$$I_i(f) \sim C\frac{\log n}{n}$$

and by symmetry $I(f) \sim C\log n$.

7. (Majority function). The last example was symmetric in the coordinate variables X_i. Another symmetric function is given by

$$f = \mathbf{1}_A, \text{ where } A = \left\{x : \sum_{i=1}^n x_i > 0\right\} .$$

If n is odd, then X_i is pivotal for A if and only if $\sum_{j\neq i} X_j = 0$. In this case $f(X^+(i)) = 1$ and $f(X^-(i)) = -1$. By standard estimates on random walks, for $i = 1,\dots,n$,

$$I_i(A) = \mathbb{P}(X_i \text{ is pivotal}) = \binom{n-1}{\frac{n-1}{2}}2^{-(n-1)} \sim \sqrt{\frac{2}{n\pi}} .$$

Furthermore, $I(A) \sim \sqrt{2/\pi}\cdot\sqrt{n}$.

1.1 Falik-Samorodnitsky: an improved Efron-Stein

We have seen already that our previous techniques (Efron-Stein, Poincaré, LSI) can give information about influences. For example, the above inequality (2) can be seen as a consequence of the LSI for Bernoulli variables. Letting $f = \mathbf{1}_A$,

$$\begin{aligned}\mathbb{P}(A)\log\frac{1}{\mathbb{P}(A)} = Ent\ f^2 \le 2\mathbb{E}\|\nabla f\|_2^2 &= \frac{1}{2}\sum_{i=1}^n \mathbb{E}(\mathbf{1}_A(X^+(i)) - \mathbf{1}_A(X^-(i)))^2 \\ &= \frac{1}{2}\sum_{i=1}^n \mathbb{P}(X_i \text{ is pivotal for } A) \\ &= \frac{1}{2}I(A) .\end{aligned}$$

(Actually this is slightly weaker than (2), since we have natural logarithm.)

Our goal now is to use information about influences in combination with our previous techniques to obtain more refined concentration inequalities. The first is a refinement of Efron-Stein. The main theorem is due to Falik and Samorodnitsky (a similar inequality was derived by Rossignol).

Theorem 1.1 (Falik-Samorodnitsky)**.** *Let $f : \{-1,+1\}^n \to \mathbb{R}$ and $\mathbb{P}$ be uniform measure. For $i = 1, \ldots, n$, let*

$$\Delta_i = \mathbb{E}[f \mid X_1, \ldots, X_i] - \mathbb{E}[f \mid X_1, \ldots, X_{i-1}] \ .$$

Then

$$\text{Var } f \ \log \frac{\text{Var } f}{\sum_{i=1}^n (\mathbb{E}|\Delta_i|)^2} \leq 2 \sum_{i=1}^n \mathbb{E}\text{Var}_i \ f \ .$$

Let us give some remarks and explanation before we prove the theorem.

- As we saw in the last section, in our case of the hypercube, $\sum_{i=1}^n \mathbb{E}\text{Var}_i \ f = \mathbb{E}\|\nabla f\|_2^2$. Therefore we can rewrite this inequality as

$$\text{Var } f \ \log \frac{\text{Var } f}{\sum_{i=1}^n (\mathbb{E}|\Delta_i|)^2} \leq 2\mathbb{E}\|\nabla f\|_2^2 \ .$$

- By the martingale decomposition of variance,

$$\text{Var } f = \sum_{i=1}^n \mathbb{E}\Delta_i^2 \ .$$

  Applying Jensen's inequality,

$$\text{Var } f \geq \sum_{i=1}^n (\mathbb{E}|\Delta_i|)^2 \ .$$

  Therefore the logarithm term is nonnegative. Furthermore, this is an improvement on Efron-Stein when the ratio of these two terms is at least e^2.

- The FS inequality is generally useful when $\sum_{i=1}^n (\mathbb{E}|\Delta_i|)^2 \ll \text{Var } f$. In many applications (like in first-passage percolation, as we will show), the ratio can be shown to be at least order n^α for some $\alpha > 0$. In such a situation, we obtain a logarithmic improvement over Efron-Stein:

$$\text{Var } f \leq C \ \frac{\sum_{i=1}^n \mathbb{E}\text{Var}_i \ f}{\log n} \ .$$

- We can bound the denominator in the logarithm in terms of influence when f is boolean. Indeed, if $f : \{-1,+1\}^n \to \{0,1\}$, then if $X_1', \ldots, X_n'$ are i.i.d. copies of the X_i's,

$$\begin{aligned}
\mathbb{E}|\Delta_i| = \mathbb{E}\left|\mathbb{E}_{>i} f - \mathbb{E}_{\geq i} f\right| &= \mathbb{E}\left|\mathbb{E}_{>i} f(X_{<i}, X_i', X_{>i}) - \mathbb{E}_{\geq i} f(X_{<i}, X_i, X_{>i})\right| \\
&= \mathbb{E}\left|\mathbb{E}_i' \mathbb{E}_{\geq i} f(X_{<i}, X_i', X_{>i}) - f(X_{<i}, X_i, X_{>i})\right| \\
&\leq \mathbb{E}|Z - Z_i'| \\
&= \mathbb{P}(X_i \text{ is pivotal}, \ X_i \neq X_i') \\
&= \frac{I_i(f)}{2} \ .
\end{aligned}$$

  Therefore in the boolean case, the FS inequality gives a good improvement over Efron-Stein when $\sum_i I_i^2(f) \ll \text{Var } f$.

We now move to the proof. We begin with a lemma.

Lemma 1.2. *Let f be a nonnegative function on a probability space such that Ent $f^2 < \infty$. Then*

$$Ent\ f^2 \geq \mathbb{E}f^2 \log \frac{\mathbb{E}f^2}{(\mathbb{E}f)^2}\ .$$

Proof. If $\mathbb{E}f^2 = 0$ then both sides are 0. If $\mathbb{E}f^2 = 1$ then the inequality reads

$$\mathbb{E}f^2 \log f^2 \geq -\mathbb{E}f^2 \log(\mathbb{E}f)^2\ ,$$

or

$$\mathbb{E}f^2 \log \frac{1}{f\mathbb{E}f} \leq 0\ .$$

Use the inequality $\log x \leq x - 1$ for $x > 0$ (that is, when $f > 0$) to obtain

$$\mathbb{E}f^2 \log \frac{1}{f\mathbb{E}f} \leq \mathbb{E}f^2 \left[\frac{1}{f\mathbb{E}f} - 1\right] \leq 0\ .$$

□

Proof of Falik-Samorodnitsky. Use the lemma on the Δ_i's:

$$\sum_{i=1}^n Ent\ \Delta_i^2 \geq \sum_{i=1}^n \mathbb{E}\Delta_i^2 \log \frac{\mathbb{E}\Delta_i^2}{(\mathbb{E}|\Delta_i|)^2} = -\text{Var}\ f \sum_{i=1}^n \frac{\mathbb{E}\Delta_i^2}{\text{Var}\ f} \log \frac{(\mathbb{E}|\Delta_i|)^2}{\mathbb{E}\Delta_i^2}\ .$$

Note that $\sum_{i=1}^n \mathbb{E}\Delta_i^2 = \text{Var}\ f$, so we can apply Jensen's inequality with the function $x \mapsto -\log x$ to get the lower bound

$$-\text{Var}\ f \log \left[\sum_{i=1}^n \frac{\mathbb{E}\Delta_i^2}{\text{Var}\ f} \cdot \frac{(\mathbb{E}|\Delta_i|)^2}{\mathbb{E}\Delta_i^2}\right]\ .$$

We therefore obtain the inequality

$$\text{Var}\ f \log \frac{\text{Var}\ f}{\sum_{i=1}^n (\mathbb{E}|\Delta_i|)^2} \leq \sum_{i=1}^n Ent\ \Delta_i^2\ . \tag{3}$$

Next we apply the Bernoulli LSI to $Ent\ \Delta_i^2$:

$$\sum_{i=1}^n Ent\ \Delta_i^2 \leq 2 \sum_{i=1}^n \mathbb{E}\|\nabla\Delta_i\|_2^2 \tag{4}$$

and

$$\nabla_j \Delta_i = \nabla_j \left(\mathbb{E}[f \mid \mathcal{F}_i] - \mathbb{E}[f \mid \mathcal{F}_{i-1}]\right)\ ,$$

where $\mathcal{F}_i = \sigma(X_1, \ldots, X_i)$. We can compute

$$\nabla_j \Delta_i = \begin{cases} 0 & \text{if } j > i \\ \mathbb{E}[\nabla_i f \mid \mathcal{F}_i] & \text{if } j = i \\ \mathbb{E}[\nabla_j f \mid \mathcal{F}_i] - \mathbb{E}[\nabla_j f \mid \mathcal{F}_{i-1}] & \text{if } j < i \end{cases} .$$

The first follows because when $j > i$ then Δ_i does not depend on X_j since this variable is integrated out. A similar idea works for the second, noting that $\nabla_j \mathbb{E}[f \mid \mathcal{F}_{i-1}] = 0$. The third is straightforward. We then deduce

$$\begin{aligned} \sum_{i=1}^n \|\nabla \Delta_i\|_2^2 &= \sum_{j=1}^n \sum_{i=1}^n (\nabla_j \Delta_i)^2 \\ &= \sum_{j=1}^n \left[(\mathbb{E}[\nabla_j f \mid \mathcal{F}_i])^2 + \sum_{i=j+1}^n (\mathbb{E}[\nabla_j f \mid \mathcal{F}_i] - \mathbb{E}[\nabla_j f \mid \mathcal{F}_{i-1}])^2 \right] . \end{aligned}$$

When we take expectation we can use L^2 orthogonality of martingale differences to obtain

$$\mathbb{E} \sum_{i=1}^n \|\nabla \Delta_i\|_2^2 = \mathbb{E} \sum_{j=1}^n (\mathbb{E}[\nabla_j f \mid \mathcal{F}_n])^2 = \sum_{j=1}^n \mathbb{E}(\nabla_j f)^2 = \mathbb{E}\|\nabla f\|_2^2 .$$

Returning to (4),

$$\sum_{i=1}^n Ent\ \Delta_i^2 \le 2\mathbb{E}\|\nabla f\|_2^2$$

and with (3), this completes the proof. □

In the proof we have used the following general identity. Set $\mathcal{E}(f) = \mathbb{E}\|\nabla f\|_2^2$. Then

$$\mathcal{E}(f) = \sum_{i=1}^n \mathcal{E}(\Delta_i) . \tag{5}$$

Applications.

1. Take f to be Boolean. Then using $\mathbb{E}|\Delta_i| \le \frac{I_i(f)}{2}$ and FS, we obtain

$$\text{Var } f \log \frac{4\text{Var } f}{\sum_{i=1}^n I_i^2(A)} \le 2\mathbb{E}\|\nabla f\|_2^2 = \frac{I(f)}{2} .$$

   This implies the following lower bound

$$\sum_{i=1}^n I_i^2(f) \ge 4\text{Var } f \exp\left(-\frac{I(f)}{2\text{Var } f} \right) . \tag{6}$$

On the other hand we have the simple bound by Jensen:

$$\sum_{i=1}^{n} I_i^2(f) = n \cdot \frac{1}{n} \sum_{i=1}^{n} I_i^2(f) \geq n \cdot \left(\frac{1}{n} \sum_{i=1}^{n} I_i(f) \right)^2 = \frac{1}{n} I^2(f) \; . \tag{7}$$

Both of these bounds are somewhat similar when $I(f) \sim 2 \log n \text{Var } f$, so we will consider two cases to prove:

Theorem 1.3. *The following bounds hold for* $f : \{-1, +1\}^n \to \{0, 1\}$.

(a)

$$\sum_{i=1}^{n} I_i^2(f) \geq \frac{(\log n \text{Var } f)^2}{n} \; .$$

(b)

$$\max\{I_i(f) : i = 1, \ldots, n\} \geq \frac{\log n \text{Var } f}{n} \; .$$

Proof. Let $\epsilon \in (0, 1)$ and consider two cases. If $I(f) \geq (2 - \epsilon) \log n \text{Var } f$, we use (7) for

$$\sum_{i=1}^{n} I_i^2(f) \geq \frac{((2 - \epsilon) \log n \text{Var } f)^2}{n} \geq \frac{(\log n \text{Var } f)^2}{n} \; .$$

If instead $I(f) < (2 - \epsilon) \log n \text{Var } f$ then we use (6) for

$$\sum_{i=1}^{n} I_i^2(f) \geq 4 \text{Var } f \exp\left(- \frac{(2 - \epsilon) \log n \text{Var } f}{2 \text{Var } f} \right) = 4 \frac{\text{Var } f}{n^{1 - \frac{\epsilon}{2}}} \; .$$

We simply choose ϵ to match these bounds. That is, put

$$\epsilon = \frac{2 \log \frac{\text{Var } f}{4} + 4 \log \log n}{\log n} \; ,$$

and this gives the first result. The second follows directly from the first, using

$$\sum_{i=1}^{n} I_i^2(f) \leq n \left(\max\{I_i(f) : i = 1, \ldots, n\} \right)^2 \; .$$

□

2. From the above we see that there must be a variable whose influence is at least $\frac{\log n \text{Var } f}{n}$. In particular, if f is a symmetric function of the coordinates (that is, each variable has the same influence), then $I(f) \geq \log n \text{Var } f$. In other words,

$$f \text{ symmetric and boolean } \Rightarrow I(f) \geq \log n \text{Var } f \; .$$

This is a huge improvement over $I(f) \geq 2\mathbb{P}(A) \log_2 \frac{1}{\mathbb{P}(A)}$ and shows a clear difference between the dictatorship (or junta) and any symmetric function. Because the tribes function is symmetric and has influence of order $\log n$, the above bound cannot be significantly improved.

3. (Sublinear variance in FPP on a torus) Let $\mathbb{T}(n)$ be the d-dimensional discrete torus of side-length n. It is the set $V = \{1, \ldots, n\}^d$ with periodic boundary conditions. In other words, we place an edge between pairs of vertices v and $v + e_i$ for $i = 1, \ldots, d$ (the e_i's are the standard coordinate vectors) when they are both in V. If $v \in V$ but $v + e_i \notin V$ then we place an edge between v and $v - (n-1)e_i$.

   Let (t_e) be a set of i.i.d. nonnegative weights, one associated to each edge $e \in \mathbb{T}(n)$ and define the passage time of a path γ as before: $T(\gamma) = \sum_{e \in \gamma} t_e$. Set

$$T_n = \min\{T(\gamma) : \gamma \text{ winds once}\} \ .$$

   Here γ winds once around the torus if it is the projection of a path on $\mathbb{Z}^d$ from a vertex v to $v + ne_i$ for some i.

   We will prove the bound

$$\text{Var } T_n \leq C \frac{n}{\log n} \ .$$

   This was first proved by Benjamini-Kalai-Schramm (2003). In the case of $\mathbb{Z}^d$, it was proved by BKS and then by Benaim-Rossignol in 2008, and last by Damron-Hanson-Sosoe in 2013. Each of these results used progressively larger classes of edge distributions. BKS used Bernoulli edges, then BR used "nearly gamma" edges and DHS used only the condition $\mathbb{E}t_e^2(\log t_e)_+ < \infty$.

   The argument we give here is essentially that of BR. The original BKS argument used the Bonami-Beckner inequality, which is a type of "hypercontractivity" result that only holds for a small class of distributions.

   As in the BKS argument, we will take our edge weights to be Bernoulli distributed:

$$\mathbb{P}(t_e = 1) = 1/2 = \mathbb{P}(t_e = 2) \ .$$

   Then our space is $\{1, 2\}^{N_n}$, where N_n is the number of edges in $\mathbb{T}(n)$. We can map the space $\{-1, +1\}^{N_n}$ with uniform measure bijectively to this one in a measure-preserving way (by just sending $-1 \mapsto 1$ and $+1 \mapsto 2$), so all the results we have developed hold on this space.

   By the FS inequality,

$$\text{Var } T_n \log \frac{\text{Var } T_n}{\sum_{i=1}^{N_n} (\mathbb{E}|\Delta_i|)^2} \leq \sum_{i=1}^{N_n} \mathbb{E}\text{Var}_i \ T_n \ .$$

   The right side is exactly the bound of Efron-Stein. Exactly as in the proof of the variance bound in FPP on $\mathbb{Z}^d$, we can show that

$$\sum_{i=1}^{N_n} \mathbb{E}\text{Var}_i \ T_n \leq 2\mathbb{E}\#Geo_n \leq 2\mathbb{E}T_n \ ,$$

where Geo_n is the set of edges in the union of all minimal paths in the definition of T_n. However T_n is almost surely bounded by $2n$, since we can construct a deterministic path γ going from 0 to ne_1. Then $T_n \leq T(\gamma) \leq 2n$. Therefore

$$\text{Var } T_n \log \frac{\text{Var } T_n}{\sum_{i=1}^{N_n} (\mathbb{E}|\Delta_i|)^2} \leq 4n \ . \tag{8}$$

We now give an upper bound for

$$\sum_{i=1}^{N_n} (\mathbb{E}|\Delta_i|)^2 \leq \left(\max_j \mathbb{E}|\Delta_j| \right) \sum_{i=1}^{N_n} \mathbb{E}|\Delta_i| \ .$$

Note arguing just as before the proof of FS,

$$\mathbb{E}|\Delta_i| \leq \mathbb{E}|T_n - T_n'(i)| \ ,$$

where $T_n'(i)$ is the variable T_n in the edge-weight configuration in which t_{e_i} is replaced by an independent copy t_{e_i}'. This is bounded above by $\mathbb{E}t_{e_i}' \mathbf{1}_{\{e_i \in Geo_n\}}$. Therefore

$$\sum_{i=1}^{N_n} \mathbb{E}|\Delta_i| \leq 2n \ .$$

Since each Δ_i has the same distribution,

$$\max_j \mathbb{E}|\Delta_j| \leq 2n/N_n \leq C/n^{d-1} \ .$$

Therefore we obtain

$$\sum_{i=1}^{N_n} (\mathbb{E}|\Delta_i|)^2 \leq Cn^{2-d} \leq C \text{ for } d \geq 2 \ . \tag{9}$$

Now we return to (8). If Var $T_n \leq n^{1/2}$ then the sub-linear variance inequality holds. Otherwise Var $T_n \geq n^{1/2}$ and we use this, along with (9), in (8):

$$\text{Var } T_n \log \frac{\sqrt{n}}{C} \leq 4n \ ,$$

or

$$\text{Var } T_n \leq C \frac{n}{\log n} \ .$$

4. (Approximation by junta). The FS inequality implies an approximation theorem for boolean functions by juntas. Take $f : \{-1, +1\}^n \to \{0, 1\}$.

Proposition 1.4. *Given $\epsilon > 0$, there exists a k-junta g such that $\|f - g\|_2^2 < \epsilon$ so long as*

$$k > I(f) \max \left\{ \exp\left(\frac{I(f)}{2\epsilon} \right), e/2 \right\} \ .$$

Proof. Without loss in generality, assume that the coordinates are ordered so that

$$I_1(f) \geq I_2(f) \geq I_3(f) \geq \cdots \ .$$

For a given k, we approximate f by the conditional expectation $g = \mathbb{E}[f \mid \mathcal{F}_k]$, where $\mathcal{F}_i = \sigma(X_1, \ldots, X_i)$. Note that f and g have the same mean, so $\|f - g\|_2^2 = \text{Var}\ (f - g)$. This allows us to use the FS inequality to do the approximation. Using the notation from (5),

$$\text{Var}\ (f-g) \log \frac{\text{Var}\ (f-g)}{\sum_{i=1}^n (\mathbb{E}|\Delta_i|)^2} \leq 2\mathcal{E}(f-g)\ ,$$

where Δ_i is defined relative to the function $f - g$. Since $f - g = \sum_{i=k+1}^n \Delta_i$, one has

$$\mathcal{E}(f-g) \leq \sum_{i=k+1}^n \mathcal{E}(\Delta_i) \leq \sum_{i=1}^n \mathcal{E}(\Delta_i) = \mathcal{E}(f)\ .$$

Therefore

$$\begin{aligned}
2\text{Var}\ (f-g) \log \frac{4\text{Var}\ (f-g)}{\sum_{i=k+1}^n I_i(f)^2} &\leq 2\text{Var}\ (f-g) \log \frac{\text{Var}\ (f-g)}{\sum_{i=k+1}^n (\mathbb{E}|\Delta_i|)^2} \\
&= 4\mathcal{E}(f-g) \\
&\leq 4\mathcal{E}(f) \\
&\leq I(f)\ .
\end{aligned}$$

Now set $m_k = \max\{I_{k+1}(f), \ldots, I_n(f)\}$. Then $\sum_{i=k+1}^n I_i(f)^2 \leq m_k I(f)$ and the above inequality gives

$$\frac{4\text{Var}\ (f-g)}{I(f)m_k} \log \frac{4\text{Var}\ (f-g)}{I(f)m_k} \leq \frac{2}{m_k}\ .$$

Because $x \log x \leq y$ implies $x \leq 2y/\log y$ when $y \geq e$ and $x > 0$, we have

$$\frac{4\text{Var}\ (f-g)}{I(f)m_k} \leq \frac{4/m_k}{\log(2/m_k)} \text{ when } m_k < 2/e\ .$$

Therefore

$$\|f-g\|_2^2 = \text{Var}\ (f-g) \leq \frac{I(f)}{\log(2/m_k)}\ .$$

This is $< \epsilon$ so long as $m_k \leq \exp\left(-\frac{I(f)}{2\epsilon}\right)$ and $m_k < 2/e$. For this to be guaranteed, we need

$$k \exp\left(-\frac{I(f)}{2\epsilon}\right) > I(f) \text{ and } 2k/e > I(f)\ ,$$

or

$$k > I(f) \max\left\{\exp\left(\frac{I(f)}{2\epsilon}\right), e/2\right\}\ .$$

$\square$

1.2 Concentration via Falik-Samorodnitsky

We will now apply the FS inequality to obtain a stronger exponential concentration inequality, by again considering quantiles. (When we talked about Efron-Stein, we gave two approaches to exponential concentration: quantiles and moment generating functions. The FS inequality can be used in the moment generating function approach as well. See Benaim-Rossignol or Damron-Hanson-Sosoe for an example.) To recall, if $f : \{-1,+1\}^n \to \mathbb{R}$ is given, and X is uniformly distributed on $\{-1,+1\}^n$, then the α-th quantile of the distribution of $Z = f(X)$ is defined as

$$Q_\alpha = \inf\{y \in \mathbb{R} : \mathbb{P}(Z \leq y) \geq \alpha\} \text{ for } \alpha \in [0,1) \ .$$

Our main assumption will be that for some constant ν,

$$\sum_{i=1}^n (Z - Z_i')_+^2 \leq \nu < \infty \text{ almost surely },$$

where $Z_i' = f(X'(i))$ and $X'(i) = (X_1, \ldots, X_{i-1}, X_i', X_{i+1}, \ldots, X_n)$, with X_i' an independent copy of X_i.

Theorem 1.5 (Local concentration). *For all $n \geq 1$,*

$$Q_{1-2^{-(n+1)}} - Q_{1-2^{-n}} \leq 8 \cdot \sqrt{\frac{2\nu}{n}} \ .$$

Again some remarks before the proof.

- I do not see a reason to believe that these constants are optimal.
- This result can be seen a sort of local version of gaussian concentration, and it implies a gaussian tail. Indeed, assume that for all $n \geq 1$,

  $$Q_{1-2^{-(n+1)}} - Q_{1-2^{-n}} \leq \sqrt{\frac{c}{n}} \ .$$

  Then

  $$Q_{1-2^{-n}} - Q_{1/2} \leq \sqrt{c} \sum_{k=1}^{n-1} k^{-1/2} \leq \sqrt{c} \int_0^{n-1} x^{-1/2} \, \mathrm{d}x \leq 2\sqrt{cn} \ .$$

  So if $t > 0$, choose $n_t = \lfloor t^2/(4c) \rfloor$ so that

  $$Q_{1-2^{-n_t}} - Q_{1/2} \leq 2\sqrt{cn_t} \leq t \ .$$

  Then writing $MZ = Q_{1/2}$,

  $$\mathbb{P}(Z - MZ > t) \leq \mathbb{P}(Z > Q_{1-2^{-n_t}}) \leq 2^{-n_t} \leq 2 \cdot 2^{-t^2/(4c)} = 2 \exp\left(-\frac{\log 2}{4c} t^2\right) \ .$$

Proof. Letting $a < b$ be such that $\mathbb{P}(Z \leq a) \geq 1/2$, we again set

$$g_{a,b}(x) = \begin{cases} a & \text{if } f(x) \leq a \\ f(x) & \text{if } f(x) \in [a,b] \\ b & \text{if } f(x) \geq b \end{cases} .$$

We claim that

$$A \log \frac{A}{4\mathbb{P}(Z > a)} \leq 1 \ , \tag{10}$$

where

$$A = \frac{(b-a)^2 \mathbb{P}(Z \geq b)}{8\nu \mathbb{P}(Z > a)} \ .$$

This is a relation between b and a that we will use afterward to give a form of concentration. In the Efron-Stein section, we have already seen that

$$\text{Var } g_{a,b}(X) \geq \frac{(b-a)^2}{4} \mathbb{P}(Z \geq b)$$

and

$$\sum_{i=1}^n \mathbb{E}\text{Var}_i \ g_{a,b}(X) \leq \nu \mathbb{P}(Z > a) \ .$$

So to use the FS inequality, we must give a useful bound for $\sum_{i=1}^n (\mathbb{E}|\Delta_i|)^2$. Just as when we bounded this by influences in the boolean case, we can still give the bound

$$\mathbb{E}|\Delta_i| \leq \mathbb{E}|g_{a,b}(X) - g_{a,b}(X'(i))| \ ,$$

In turn we can use symmetry for

$$2\mathbb{E}(g_{a,b}(X) - g_{a,b}(X'(i)))_+ = 2\mathbb{E}(g_{a,b}(X) - g_{a,b}(X'(i)))_+ \mathbf{1}_{\{f(X) > a\}} \ .$$

The equality holds because if $f(X) \leq a$ then $g_{a,b}(X) = a$ and therefore $g_{a,b}(X'(i)) \geq g_{a,b}(X)$. By Cauchy-Schwarz, we obtain the bound

$$\begin{aligned} & 2\left(\mathbb{E}(g_{a,b}(X) - g_{a,b}(X'(i)))_+^2\right)^{1/2} \cdot (\mathbb{P}(f(X) > a))^{1/2} \\ = \ & \left(2\mathbb{E}(g_{a,b}(X) - g_{a,b}(X'(i)))^2\right)^{1/2} \cdot (\mathbb{P}(f(X) > a))^{1/2} \ . \end{aligned}$$

Therefore

$$\begin{aligned} \sum_{i=1}^n (\mathbb{E}|\Delta_i|)^2 &\leq 2\mathbb{P}(f(X) > a) \sum_{i=1}^n \mathbb{E}(g_{a,b}(X) - g_{a,b}(X'(i)))^2 \\ &\leq 2\mathbb{P}(Z > a) \sum_{i=1}^n \mathbb{E}(f(X) - f(\overline{X}_i))^2 \\ &\leq 4\nu \mathbb{P}(Z > a)^2 \ . \end{aligned}$$

By the FS inequality,

$$\frac{(b-a)^2}{4}\mathbb{P}(Z \geq b)\log\frac{\frac{(b-a)^2}{4}\mathbb{P}(Z \geq b)}{8\nu\mathbb{P}(Z > a)^2} \leq 2\nu\mathbb{P}(Z > a) ,$$

and this implies (10).

Just as in the Efron-Stein section, we choose $a = Q_{1-2^{-n}}$ and $b = Q_{1-2^{-(n+1)}}$, so that

$$\mathbb{P}(Z > a) \leq 2^{-n} \text{ and } \mathbb{P}(Z \geq b) \geq 2^{-(n+1)} .$$

Therefore

$$A \log 2^{n-2} A \leq 1 ,$$

where

$$A \geq \frac{(b-a)^2}{16\nu} .$$

We then obtain $B \log B \leq 2^{n-2}$, where $B = 2^{n-2}A$. Again use the fact that $x \log x \leq y$ implies $x \leq 2y/\log y$ when $y \geq e$ and $x > 0$. Taking $y = 2^{n-2}$ and $x = B$, we obtain $B \leq 2^{n-1}/\log 2^{n-2}$ for $n \geq 4$. This bound also holds for $n = 3$. So

$$A \leq 2/\log 2^{n-2} = \frac{2}{(n-2)\log 2} \text{ for } n \geq 3 .$$

So

$$b - a \leq \sqrt{\frac{32\nu}{(n-2)\log 2}} \leq 8 \cdot \sqrt{\frac{2\nu}{n}} \text{ for } n \geq 3 .$$

For $n = 1, 2$, one has $B \leq 2$, so $b - a \leq 4 \cdot \sqrt{2\nu} \leq 8 \cdot \sqrt{\frac{2\nu}{n}}$. $\square$

2 Fourier-Walsh decomposition

The Falik-Samorodnitsky inequality gives us a better way to bound the variance of a function on the hypercube. A different way to represent the variance is through the Fourier-Walsh decomposition. The idea is that if $f : \{-1, +1\}^n \to \mathbb{R}$ and $\mathbb{P}$ is the uniform measure, then the variance of f equals $\|f - \mathbb{E}f\|_2^2$. This suggests we should look at the space $L^2(\mathbb{P})$, a Hilbert space of dimension 2^n. If we find an orthonormal basis $\{f_i\}$ then we can represent

$$f - \mathbb{E}f = \sum_i \langle f - \mathbb{E}f, f_i\rangle f_i$$

and use orthogonality (Parseval) to have

$$\text{Var } f = \|f - \mathbb{E}f\|_2^2 = \sum_i |\langle f - \mathbb{E}f, f_i\rangle|^2 .$$

This decomposition of variance is analogous to the martingale decomposition we used for Efron-Stein and FS.

A natural way to construct an orthonormal basis is to use orthogonal polynomials. We can start with the constant 1, all monomials (functions of the type $x \mapsto x_i$ for $i = 1, \ldots, n$) and so on, and perform Gram-Schmidt to them. In our case, with the uniform measure, we obtain

Definition 2.1. *The Fourier-Walsh basis for L^2 is the set $\{\chi_S : S \subset [n]\}$, where $[n] = \{1, \ldots, n\}$ and χ_S is defined as*

$$\chi_S(x) = \prod_{i \in S} x_i \ .$$

Proposition 2.2. *The set $\{\chi_S : S \subset [n]\}$ is an orthonormal basis for $L^2(\Omega, \mathbb{P})$, where $\Omega = \{-1, +1\}^n$ and $\mathbb{P}$ is the uniform measure.*

Proof. We only need to show the set is orthonormal, since it has 2^n elements. If $S, T \subset [n]$, then use independence for

$$\begin{aligned} \langle \chi_S, \chi_T \rangle = \mathbb{E}\left(\prod_{i \in S} x_i \prod_{k \in T} x_k\right) &= \mathbb{E} \prod_{i \in S \Delta T} x_i \ \mathbb{E} \prod_{j \in S \cap T} x_j^2 \\ &= \mathbb{E} \prod_{i \in S \Delta T} x_i \\ &= \begin{cases} 0 & \text{if } S \neq T \\ 1 & \text{if } S = T \end{cases} \ . \end{aligned}$$

□

Definition 2.3. *The S-th coefficient (for $S \subset [n]$) of a function $f : \{-1, +1\}^n \to \mathbb{R}$ is defined as*

$$\hat{f}(S) = \langle f, \chi_S \rangle \ .$$

Here are some properties of the Fourier-Walsh coefficients.

1. $\hat{f}(\emptyset) = \mathbb{E}f$. To see this, note that $\chi_\emptyset = 1$, so

$$\hat{f}(\emptyset) = \langle f, \chi_\emptyset \rangle = \mathbb{E} f \chi_\emptyset = \mathbb{E} f \ .$$

2. As usual, $f = \sum_S \hat{f}(S) \chi_S$.

   Proof. This is the standard representation of an element of a Hilbert space in terms of an orthonormal basis. Since $\{\chi_S\}$ is a basis, we can write $f = \sum_S a_S \chi_S$. Now use orthogonality to find

$$\hat{f}(T) = \langle f, \chi_T \rangle = \sum_S b_S \langle \chi_S, \chi_T \rangle = b_T \ .$$

□

3. $\|f\|_2^2 = \sum_S \hat{f}(S)^2$.

Proof. This is Parseval. Using orthogonality,

$$\|f\|_2^2 = \mathbb{E}\left(\sum_S \hat{f}(S)\chi_S \sum_T \hat{f}(T)\chi_T\right) = \sum_{S,T} \hat{f}(S)\hat{f}(T)\langle \chi_S, \chi_T\rangle = \sum_S \hat{f}(S)^2 \ .$$

□

4. $\mathrm{Var}\ f = \sum_{S\neq\emptyset} \hat{f}(S)^2$.

Proof.
$$\mathrm{Var}\ f = \|f\|_2^2 - (\mathbb{E}f)^2 = \sum_S \hat{f}(S)^2 - \hat{f}(\emptyset)^2 \ .$$

□

We can now get some information about functions by looking at the Fourier-Walsh decomposition of their derivatives. We define a discrete derivative here a bit differently than we have before.

Definition 2.4. *If $f : \{-1,+1\}^n \to \mathbb{R}$ then we set $D_i f$ to be the function given by $D_i f(x) = \frac{f(x)-f(\overline{x}_i)}{2}$, where $\overline{x}_i$ is the vector x but with the i-th coordinate flipped.*

- For $S \subset [n]$,
$$\langle D_i f, \chi_S\rangle = \mathbb{E}\left(D_i f \prod_{j\in S} x_j\right) \ .$$

  If $i \in S$ then write this as
$$\begin{aligned} &\frac{1}{2}\mathbb{E}\left(f(x)\prod_{j\in S} x_j\right) - \frac{1}{2}\mathbb{E}\left(f(\overline{x}_i)\prod_{j\in S} x_j\right) \\ &= \frac{1}{2}\mathbb{E}\left(f(x)\prod_{j\in S} x_j\right) + \frac{1}{2}\mathbb{E}\left(f(x)\prod_{j\in S} x_j\right) \\ &= \hat{f}(S) \ . \end{aligned}$$

  On the other hand if $i \notin S$, when we change variables above from $\overline{x}_i$ to x, we do not get a -1 contribution from $\prod_{j\in S} x_j$, so the terms cancel and we get 0. Thus
$$\langle D_i f, \chi_S\rangle = \begin{cases} \hat{f}(S) & \text{if } i \in S \\ 0 & \text{if } i \notin S \end{cases} \ .$$

- From the last item we can show:

$$\sum_S \#S \hat{f}(S)^2 = \sum_{i=1}^n \|D_i f\|_2^2 \ .$$

Therefore

$$\text{Var } f \le \sum_{i=1}^n \|D_i f\|_2^2 \ .$$

This is a version of Efron-Stein.

Proof. Write the right side as

$$\sum_S \left[\sum_{i=1}^n \mathbf{1}_{i \in S} \right] \hat{f}(S)^2 = \sum_{i=1}^n \sum_{S : i \in S} \hat{f}(S)^2 = \sum_{i=1}^n \sum_S |\langle D_i f, \chi_S \rangle|^2 = \sum_{i=1}^n \|D_i f\|_2^2 \ .$$

The inequality is justified as

$$\text{Var } f = \sum_{S \neq \emptyset} \hat{f}(S)^2 \le \sum_S \#S \hat{f}(S)^2 \ .$$

$\square$

- If $f : \{-1,+1\}^n \to \mathbb{R}$ is boolean (takes values in $\{0,1\}$) then

$$|D_i f(x)| = \frac{1}{2} \cdot \mathbf{1}_{\{X_i \text{ is pivotal for } f\}} \ .$$

So $\|D_i f\|_2^2 = \frac{I_i(f)}{4}$ and we obtain

$$I(f) = 4 \sum_{i=1}^n \|D_i f\|_2^2 = 4 \sum_S \#S \hat{f}(S)^2$$

and the upper bound for the variance (which we proved via Efron-Stein)

$$\text{Var } f \le \frac{I(f)}{4} \ .$$

3 Semigroups

Concentration is naturally related to Markov processes. We will try to explain the relation through an example, the noise process on the hypercube. In that context, we can make some nice simplifications but hopefully the main ideas will still come through. We will be following parts of Ramon van Handel's notes (starting in his Section 2.2).

Recall the Markov property: if $(X_t)_{t\geq 0}$ is a stochastic process taking values in a measurable space S, then it satisfies the Markov property if for each bounded measurable function $f : S \to \mathbb{R}$ and $t, s \geq 0$,

$$\mathbb{E}[f(X_{t+s}) \mid X_r : r \in [0,s]] = \mathbb{E}[f(X_{t+s}) \mid X_s] \ .$$

That is, the law of X_{t+s} (the future) depends on $X_r : r \in [0,s]$ (the past) only through X_s (the present). This can be rewritten (in the time-homogeneous case) as

$$\mathbb{E}[f(X_{t+s}) \mid X_r : r \in [0,s]] = (P_t f)(X_s) \tag{11}$$

for some bounded measurable function $P_t f : S \to \mathbb{R}$.

In our case, we construct a Markov process on $\{-1,+1\}^n$ as follows. (This construction will be informal, but you can easily write a rigorous definition.) Starting with a vector $X_0 = (X_1(0), \ldots, X_n(0))$ (or a random vector sampled from some measure μ), we construct n independent Poisson processes $(N_t^1, \ldots, N_t^n)$. We imagine that i-th process represents a Poisson clock associated with X_i. Each time the i-th clock rings (that is, the process N_t^i increments) then X_i resamples itself from uniform measure on $\{-1,+1\}$. In between rings, X_i remains constant. In this way we obtain a stochastic process $(X_t) = (X_1(t), \ldots, X_n(t))$.

Because the Poisson process in Markov, you can check that the process $(X_t)_{t\geq 0}$ is also Markov. Associated with any Markov process is the collection (P_t) of operators from (11) on bounded functions. When we have transition probabilities $\mathbb{P}_x(X_t \in A)$ (the probability $X_t \in A$ given that we start at $t = 0$ in state x), then we can write

$$(P_t f)(x) = \mathbb{E}_x[f(X_t)] \ .$$

With the noise process X_t above, the state space $\{-1,+1\}^n$ is finite, so we can easily write transition probabilities. Actually this equation is often taken as the definition of the operators (P_t), as many Markov processes are defined via transition probabilities or transition kernels.

(P_t) acts on distributions as well. Given an initial measure μ on S, we can push it through P_t to get a new measure $P_t \circ \mu$, defined either through functions:

$$\int f \ \mathrm{d}(P_t \circ \mu) = \int P_t f \ \mathrm{d}\mu$$

or by taking our initial vector X_0 to be distributed as μ and setting

$$P_t \circ \mu(A) = \mathbb{P}(X_t \in A \mid X_0) \ .$$

Definition 3.1. *A probability measure μ is stationary for (X_t) if it is invariant under P_t; that is, $P_t \circ \mu = \mu$.*

Proposition 3.2 (Lemma 2.7 of RvH)**.** *Let μ be a stationary measure. Then the following hold for all $p \geq 1$, $t, s \geq 0$, $\alpha, \beta \in \mathbb{R}$ and bounded measurable f, g:*

1. *(Contraction).* $\|P_t f\|_{L^p(\mu)} \leq \|f\|_{L^p(\mu)}$.

2. *(Linearity).* $P_t(\alpha f + \beta g) = \alpha P_t f + \beta P_t f$, *$\mu$ almost surely.*

3. *(Semigroup property).* $P_{t+s} f = P_t P_s f$, *μ almost surely.*

4. *(Conservativeness).* $P_t 1 = 1$, *μ almost surely.*

Proof. For the first, apply Jensen. Take X_0 distributed as μ and compute

$$\|P_t f\|_p^p = \mathbb{E}\left[\mathbb{E}[f(X_t) \mid X_0]^p\right] \leq \mathbb{E}\left[\mathbb{E}[|f(X_t)|^p \mid X_0]\right] = \|f\|_p^p .$$

Linearity is similar. The semigroup property follows from the Markov property:

$$\mathbb{E}[f(X_{t+s}) \mid X_0] = \mathbb{E}\left[\mathbb{E}[f(X_{t+s}) \mid X_r : r \in [0,t]] \mid X_0\right] = \mathbb{E}[P_s f(X_t) \mid X_0] .$$

In other words,

$$\mathbb{E}_{X_0} f(X_{t+s}) = \mathbb{E}_{X_0} P_s(X_t) \text{ for } \mu \text{ almost every } X_0 .$$

This means $P_{t+s} f = P_t P_s f$, μ almost surely.

The last property is trivial. □

In the example of the noise process, we note that the uniform measure is invariant, and we can compute a nice representation for the semigroup using the Fourier-Walsh basis.

Proposition 3.3. *For the noise process (X_t) with initial measure μ uniform on $\{-1,+1\}^n$ and f bounded and measurable,*

$$P_t f = \sum_S e^{-t\#S} \hat{f}(S) \chi_S .$$

Furthermore μ is invariant.

Proof. First use linearity

$$P_t f = \sum_S \hat{f}(S) P_t \chi_S .$$

By independence,

$$P_t \chi_S(x) = \mathbb{E}_x\left[\chi_S(X_t)\right] = \mathbb{E}_x \prod_{i \in S} X_i(t) = \prod_{i \in S} \mathbb{E}_x X_i(t) .$$

Let $A_i(k)$ be the event that the clock associated to X_i has incremented by k units at time t. Then

$$\mathbb{E}_x X_i(t) = \sum_{k \geq 0} \mathbb{E}_x\left[X_i(t) \mid A_i(k)\right] \mathbb{P}_x(A_i(k)) .$$

On $A_i(k)$ for $k \geq 1$, $X_i(t)$ is equal to a variable which has mean zero and is independent of $A_i(k)$ (since it has been resampled k times). Therefore,

$$\mathbb{E}_x X_i(t) = \mathbb{E}_x\left[X_i(t) \mid A_i(0)\right] \mathbb{P}_x(A_i(0)) = x_i e^{-t} ,$$

and we obtain

$$P_t\chi_S(x) = e^{-t\#S}\prod_{i\in S} x_i \ .$$

This gives

$$P_t f = \sum_S e^{-t\#S}\hat{f}(S)\chi_S \ .$$

To see that μ is invariant, compute

$$\int P_t f \ \mathrm{d}\mu = \sum_S e^{-t\#S}\hat{f}(S)\int \chi_S \ \mathrm{d}\mu = \hat{f}(\emptyset) = \mathbb{E}_\mu f \ .$$

$\square$

Here are some consequences of the last result.

- From the representation of the noise semigroup,

$$\text{Var } P_t f = \sum_{S\neq\emptyset} |\langle P_t f, \chi_S\rangle|^2 = \sum_{S\neq\emptyset} e^{-2t\#S}\hat{f}(S)^2 \ .$$

  Therefore Var $P_t f \to 0$. In other words,

$$P_t f \to \mathbb{E}f \text{ in } L^2(\mu) \text{ for all bounded measurable } f \ .$$

  In fact, this convergence occurs exponentially quickly. (We will see in the next section this is related to a Poincaré inequality.)

- Generally speaking, the variance is a decreasing function of time. If μ is stationary and $0 \leq s \leq t$,

$$\text{Var}_\mu P_t f = \|P_t f - \mathbb{E}_\mu f\|_2^2 = \|P_t(f - \mathbb{E}_\mu f)\|_2^2 = \|P_{t-s}P_s(f - \mathbb{E}_\mu f)\|_2^2 \ ,$$

  and by the contraction property, this is bounded by $\|P_s(f - \mathbb{E}_\mu f)\|_2^2 = \text{Var}_\mu P_s f$.

- From the same representation, if $f = \chi_T$ for some T,

$$P_t f = \sum_S e^{-t\#S}\hat{f}(S)\chi_S = e^{-t\#T}\chi_T \ .$$

  Therefore $\{\chi_S : S \subset [n]\}$ is an orthonormal set of eigenvectors for each P_t. The eigenvalue for χ_S is $e^{-t\#S}$.

3.1 Poincaré inequalities revisited

In this general setting, we would like to rephrase a Poincaré inequality as $\mathrm{Var}_\mu f \le \mathbb{E}_\mu \|\nabla f\|_2^2$ for some notion of gradient. The definition is somewhat obscure, but will be somewhat more motivated in our noise case.

Definition 3.4. *The generator $\mathcal{L}$ of a Markov process (X_t) with stationary distribution μ is defined on $f \in L^2(\mu)$ by*

$$\mathcal{L}f = \lim_{t\to 0} \frac{P_t f - f}{t} \ .$$

The limit is taken in $L^2(\mu)$ and the set of f for which the limit exists is called the domain of $\mathcal{L}$.

This generator is not usually defined on all of L^2, but in the noise case, the fact that our process has a finite state space gives us much more freedom. Hence it will be defined on all of L^2 in our case.

Proposition 3.5. *The generator of the noise process (X_t) is $\mathcal{L} = -\sum_{i=1}^n D_i$, where D_i is the discrete derivative defined as $D_i f(x) = \frac{f(x)-f(\overline{x}_i)}{2}$.*

Proof. The function $\frac{P_t f - f}{t}$ can be written in the Fourier-Walsh basis as

$$\frac{P_t f - f}{t} = \sum_S \frac{e^{-t\#S}-1}{t}\hat{f}(S)\chi_S \to -\sum_S \#S\hat{f}(S)\chi_S \ .$$

Recalling from the Fourier-Walsh section that $D_i f = \sum_{S:i\in S}\hat{f}(S)\chi_S$, one has

$$\sum_{i=1}^n D_i f = \sum_{i=1}^n \sum_S \mathbf{1}_{i\in S}\hat{f}(S)\chi_S = \sum_S \#S\hat{f}(S)\chi_S \ .$$

Therefore

$$\mathcal{L}f = -\sum_{i=1}^n D_i f \ .$$

□

Corollary 3.6. *The collection $\{\chi_S : S \subset [n]\}$ is an orthonormal set of eigenvectors for $-\mathcal{L}$. The eigenvalue associated to χ_S is $\lambda = \#S$. (Note the eigenvalue for P_t is $e^{-t\lambda}$.)*

Proof. Use the representation

$$\sum_{i=1}^n D_i \chi_T = \sum_S \#S\hat{\chi}_T(S)\chi_S = \#T\chi_T \ .$$

□

The usual way to relate a generator to a generalized gradient is to define

Definition 3.7. *The Dirichlet form for the Markov process (X_t) with invariant measure μ is defined on functions f, g as the bilinear form*

$$\mathcal{E}(f,g) = -\langle \mathcal{L}f, g\rangle \ .$$

We write $\mathcal{E}(f) = \mathcal{E}(f,f)$.

The reason for our interest in this definition can be seen through the noise example.

Proposition 3.8. *The Dirichlet form for the noise process satisfies*

$$\mathcal{E}(f) = \sum_{i=1}^n \|D_i f\|_2^2 \ .$$

Proof. For $f : \{-1,+1\}^n \to \mathbb{R}$,

$$\mathcal{E}(f) = \sum_{i=1}^n \mathbb{E} f D_i f$$

and

$$\begin{aligned} \mathbb{E} f D_i f &= \frac{1}{2}\left(\mathbb{E} f^2 - \mathbb{E} f(x) f(\overline{x}_i)\right) \\ &= \frac{1}{2}\left(\mathbb{E} f(\overline{x}_i)^2 - \mathbb{E} f(\overline{x}_i) f(x)\right) \ , \end{aligned}$$

Adding these,

$$\mathbb{E} f D_i f = \frac{1}{4}\mathbb{E}(f(x) - f(\overline{x}_i))^2 \ ,$$

giving

$$\mathcal{E}(f) = \sum_{i=1}^n \|D_i f\|_2^2 \ .$$

□

The previous proposition shows that we may want to use the Dirichlet form as a replacement for the norm of the gradient squared. This turns out to be appropriate in a wide variety of contexts.

Definition 3.9. *The probability measure μ satisfies a Poincaré inequality for the Dirichlet form $\mathcal{E}$ and constant $c \geq 0$ if*

$$\mathrm{Var}_\mu f \leq c\ \mathcal{E}(f) \textit{ for all } f \ .$$

We have already shown that Var $f \leq \sum_{i=1}^n \|D_i f\|_2^2$, so the uniform measure satisfies a Poincaré inequality with the noise Dirichlet form $\mathcal{E}$ for constant 1.

A Poincaré inequality is related to convergence to equilibrium for our Markov process. We have seen that in the noise example, Var $P_t f \to 0$ as $t \to \infty$. In other words, $P_t f \to \mathbb{E}_\mu f$ in L^2.

Definition 3.10. *The semigroup* (P_t) *is called ergodic if* $P_t f \to \mathbb{E}_\mu f$ *in* $L^2(\mu)$ *for all* $f \in L^2(\mu)$. *Here* μ *is an invariant measure.*

The relation between Poincaré and convergence will come from the following derivative calculation. It is Corollary 2.30 in RvH.

Proposition 3.11. *For each bounded measurable* f,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Var}_\mu P_t f = -2\mathcal{E}(P_t f)\ .$$

Proof. Compute

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left[\mathbb{E}_\mu (P_t f)^2 - (\mathbb{E}_\mu P_t f)\right]^2 &= \frac{\mathrm{d}}{\mathrm{d}t}\left[\mathbb{E}_\mu (P_t f)^2 - (\mathbb{E}_\mu f)^2\right] \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}_\mu (P_t f)^2\ .\end{aligned}$$

This derivative can be computed using the semigroup property as

$$\lim_{h\downarrow 0}\frac{\mathbb{E}_\mu (P_{t+h} f)^2 - \mathbb{E}_\mu (P_t f)^2}{h} = \lim_{h\downarrow 0}\mathbb{E}_\mu\left(\frac{P_h P_t f - P_t f}{h}(P_h P_t f + P_t f)\right)$$

By definition of the generator, $\frac{P_h P_t f - P_t f}{h} \to \mathcal{L} P_t f$ in L^2. This implies that $P_h P_t f \to P_t f$ in L^2 as well. Because f is bounded (and therefore so are $P_h P_t f$ and $P_t f$), we can take the limit to get

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Var}_\mu P_t f = \mathbb{E}_\mu\left(P_t f \mathcal{L} P_t f\right) = -2\mathcal{E}(P_t f)\ .$$

□

Theorem 3.12 (Poincaré and rate of convergence)**.** *Let* (P_t) *be a Markov semigroup with stationary measure* μ. *The following are equivalent for* $c \geq 0$.

1. $\mathrm{Var}_\mu\ f \leq c\ \mathcal{E}(f)$ *for all bounded measurable* f.
2. $\|P_t f - \mathbb{E}_\mu f\|_{L^2(\mu)} \leq e^{-t/c}\|f - \mathbb{E}_\mu f\|_{L^2(\mu)}$ *for all bounded measurable* f *and* $t \geq 0$.

Proof. Assume 1. Then using Proposition 3.11,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Var}_\mu P_t f = -2\mathcal{E}(P_t f, P_t f) \leq -\frac{2}{c}\mathrm{Var}_\mu P_t f\ .$$

Setting $g(t) = \mathrm{Var}_\mu P_t f$, this is a non-increasing function of $t \geq 0$ and is nonnegative. Therefore if it is zero for some t, it is also zero for $s \geq t$. Assuming for the moment that $g(t)$ is never zero, the above is rewritten as

$$\frac{\mathrm{d}}{\mathrm{d}t}\log g(t) \leq -2/c$$

and by integration,

$$g(t) \le g(0)e^{-2t/c}, \text{ or } \mathrm{Var}_\mu P_t f \le e^{-2t/c}\mathrm{Var}_\mu f \ .$$

In fact this conclusion remains true if $g(t)$ is zero. Therefore 2 holds.

Assume 2. We may take f so that $\mathrm{Var}_\mu f > 0$. By Proposition 3.11,

$$\begin{aligned} 2\mathcal{E}(f) = -\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Var}_\mu P_t f\Big|_{t=0} = \lim_{h\downarrow 0}\frac{\mathrm{Var}_\mu f - \mathrm{Var}_\mu P_h f}{h} &= \mathrm{Var}_\mu f \ \lim_{h\downarrow 0}\frac{1-\frac{\mathrm{Var}_\mu P_t f}{\mathrm{Var}_\mu f}}{h} \\ &\ge \mathrm{Var}_\mu f \lim_{h\downarrow 0}\frac{1-e^{-2h/c}}{h} \\ &= \frac{2}{c}\mathrm{Var}_\mu f \ . \end{aligned}$$

□

Here are some remarks.

- In the case of the noise semigroup, we already know a Poincaré inequality with constant $c = 1$. Therefore we have the convergence rate $\mathrm{Var}_\mu P_t f \le e^{-2t}\mathrm{Var}_\mu f$ for bounded measurable f.
- As we mentioned in the Efron-Stein section, the Poincaré inequality is related to spectral gap. Rewrite it as
$$\frac{\mathcal{E}(f)}{\mathrm{Var}_\mu f} \ge \frac{1}{c}$$
or by definition of the Dirichlet form, when $\mathbb{E}_\mu f = 0$,
$$\frac{\langle -\mathcal{L}f, f\rangle}{\langle f, f\rangle} \ge \frac{1}{c} \ .$$
This is stated as
$$\frac{1}{c} \le \inf\left\{\frac{\langle -\mathcal{L}f, f\rangle}{\langle f, f\rangle} : \langle f, 1\rangle = 0, \ f \ne 0\right\} \ . \tag{12}$$
For simplicity, let us restrict ourself to the finite-dimensional case; that is, $L^2(\mu)$ is assumed to have finite dimension. (Although with some functional analysis, this can be extended.) This is true, for instance, in the noise semigroup. To state the relation to the spectral gap, we need the notion of reversibility. The semigroup (P_t) with stationary measure μ is called *reversible* if the operators are self-adjoint on $L^2(\mu)$. In other words,
$$\langle f, P_t g\rangle = \langle P_t f, g\rangle \text{ for all } f, g \in L^2(\mu) \ .$$

Proposition 3.13. *If (P_t) is reversible with stationary measure μ then $\mathcal{L}$ is self-adjoint as well. The eigenvalues of $-\mathcal{L}$ are nonnegative. Listing them as $0 = \lambda_0 \le \lambda_1 \le \cdots$, one has*
$$\mathit{Var}_\mu f \le c\ \mathcal{E}(f)$$
with $c = 1/(\lambda_1 - \lambda_0) = 1/\lambda_1$.

Proof. To show that $\mathcal{L}$ is self-adjoint, take $f, g \in L^2(\mu)$ and use the fact that the convergence defining the generator is in L^2 to obtain

$$\langle \mathcal{L}f, g \rangle = \lim_{t \downarrow 0} \left\langle \frac{P_t f - f}{t}, g \right\rangle \ .$$

(Here we have used that if $f_n \to f$ in L^2 then for each $g \in L^2$, one has $\langle f_n, g \rangle \to \langle f, g \rangle$.) Since P_t is self-adjoint, the term in the limit is

$$\begin{aligned} \frac{1}{t}\left[\langle P_t f, g \rangle - \langle f, g \rangle\right] &= \frac{1}{t}\left[\langle f, P_t g \rangle - \langle f, g \rangle\right] \\ &= \left\langle f, \frac{P_t g - g}{t} \right\rangle \to \langle f, \mathcal{L}g \rangle \ . \end{aligned}$$

Since $\mathcal{L}$ is self-adjoint, all eigenvalues are real. Taking $f \in L^2(\mu)$,

$$\mathcal{E}(f) = -\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Var}_\mu P_t f \bigg|_{t=0} \ .$$

The function $t \mapsto \mathrm{Var}_\mu P_t f$ is non-increasing, so

$$\mathcal{E}(f) \geq 0 \text{ for all } f \in L^2(\mu) \ .$$

So if f is an eigenvector with unit norm and eigenvalue λ,

$$\lambda = \langle -\mathcal{L}f, f \rangle = \mathcal{E}(f) \geq 0 \ .$$

By the definition of $\mathcal{L}$, one sees that $\mathcal{L}1 = 0$, so 1 is an eigenvector for eigenvalue 0. To find the second lowest eigenvalue, we use the Raleigh quotient:

$$\lambda_1 = \inf\left\{ \frac{\langle -\mathcal{L}f, f \rangle}{\langle f, f \rangle} : \langle f, 1 \rangle = 0, \ f \neq 0 \right\} \ .$$

By (12), one has the Poincaré inequality with $1/c = \lambda_1$. □

- The above notion of reversibility mirrors that of Markov chains. Taking $f, g \in L^2(\mu)$ and starting the chain in the invariant measure μ,

$$\begin{aligned} \mathbb{E} g P_t f = \langle P_t f, g \rangle_\mu = \langle f, P_t g \rangle &= \mathbb{E}\left[f(X_0)\mathbb{E}[g(X_t) \mid X_0]\right] \\ &= \mathbb{E} f(X_0) g(X_t) \\ &= \mathbb{E}\left[g(X_t)\mathbb{E}[f(X_0) \mid X_t]\right] \ . \end{aligned}$$

Since X_t also has distribution μ, we obtain

$$\mathbb{E}_\mu g(x) P_t f(x) = \mathbb{E}_\mu g(x) \mathbb{E}[f(X_0) \mid X_t = x] \ .$$

This is true for all f, g, so

$$\mathbb{E}[f(X_t) \mid X_0 = x] = P_t f(x) = \mathbb{E}[f(X_0) \mid X_t = x] \ .$$

This is the usual notion of reversibility: the distribution of $(X_0 : 0 \leq s \leq t)$ equals that of $(X_{t-s} : 0 \leq s \leq t)$.

Example. Gaussian distribution (example 2.22 in RvH). If we want to prove concentration inequalities or functional inequalities in this framework for a given measure μ, we need to come up with a Markov process which has μ as its stationary measure. Here we will do this for the standard Gaussian, for which the Ornstein-Uhlenbeck (OU) process

$$X_t = e^{-t}X_0 + e^{-t}B_{e^{2t}-1} : t \geq 0$$

is invariant. Here $(B_t : t \geq 0)$ is a standard Brownian motion and X_0 is a standard normal variable independent of (B_t).

The following is Lemma 2.23 of RvH. We will take f below to be smooth with compact support, but this can of course be extended. In fact, a Dirichlet form $\mathcal{E}$ is typically defined for functions f, g contained in some dense subset $\mathcal{D}$ of $L^2(\mu)$. Then one can build the inner product $(f, g) = \langle f, g\rangle_{L^2(\mu)} + \mathcal{E}(f, g)$ and complete $\mathcal{D}$ to a Hilbert space relative to this inner product. In the case of the OU process below, one gets from the representation $\mathcal{E}(f, g) = \langle f', g'\rangle$ that this completed space is the Sobolev space

$$H^1(\mu) = \{f : \mathbb{E}f^2 + \mathbb{E}(f')^2 < \infty\}$$

relative to Gaussian measure μ.

Lemma 3.14. *The OU process is a Markov process with semigroup*

$$P_t f(x) = \mathbb{E}[f(e^{-t}x + \sqrt{1 - e^{-2t}}\xi)]$$

for ξ a standard normal variable. $\mu = N(0, 1)$ is the stationary measure and the process is ergodic. The generator is given by

$$\mathcal{L}f(x) = -xf'(x) + f''(x) \ ,$$

so that $\mathcal{E}(f, g) = \langle f', g'\rangle_\mu$. The OU process is reversible.

Proof. We start with the Markov property. If $s \leq t$ then

$$\begin{aligned} X_t &= e^{-(t-s)}(e^{-s}X_0 + e^{-s}B_{e^{2s-1}}) + e^{-t}(B_{e^{2t-1}} - B_{e^{2s}-1}) \\ &= e^{-(t-s)}X_s + e^{-t}(B_{e^{2t-1}} - B_{e^{2s}-1}) \\ &= e^{-(t-s)}X_s + \sqrt{1 - e^{-2(t-s)}} \cdot \frac{B_{e^{2t}-1} - B_{e^{2s}-1}}{\sqrt{e^{2t} - e^{2s}}} \ . \end{aligned}$$

Calling the ratio on the right ξ, note that it is a standard normal variable independent of X_s (due to (B_t) having independent increments). So we obtain

$$\mathbb{E}[f(X_t) \mid X_r : r \in [0, s]] = \mathbb{E}f(e^{-(t-s)}X_s + \sqrt{1 - e^{-2(t-s)}} \cdot \xi) = P_{t-s}f(X_s) \ ,$$

and this depends only on X_s, showing the Markov property.

Next $N(0, 1)$ is stationary, since, in distribution,

$$X_t = e^{-t}N(0, 1) + e^{-t}N(0, e^{2t-1}) = e^{-t}N(0, e^{2t}) = N(0, 1) \ .$$

The second equality holds due to the summands being independent.

To find the generator for the OU process, we need a lemma: Gaussian integration by parts.

Lemma 3.15. *Let X be a standard gaussian and $f : \mathbb{R} \to \mathbb{R}$ such that $Xf(X)$ and $f'(X)$ are L^1. Then*

$$\mathbb{E}Xf(X) = \mathbb{E}f'(X) \ .$$

Proof. First take f smooth with compact support. Then

$$\mathbb{E}Xf(X) = \frac{1}{\sqrt{2\pi}} \int xf(x)e^{-x^2/2} \ \mathrm{d}x = -\frac{1}{\sqrt{2\pi}} \int f(x) \left[\frac{\mathrm{d}}{\mathrm{d}x}e^{-x^2/2}\right] \ \mathrm{d}x \ .$$

Now integrate by parts (noting zero boundary terms) to obtain

$$\frac{1}{\sqrt{2\pi}} \int f'(x)e^{-x^2/2} \ \mathrm{d}x = \mathbb{E}f'(X) \ .$$

Now if $Xf(X)$ and $f'(X)$ are in L^1, we can approximate both sides by smooth functions with compact support. □

Returning to the derivation of the generator, for f smooth with compact support,

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}P_t f &= \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}f(e^{-t}x + \sqrt{1-e^{-2t}}\xi) \\
&= \mathbb{E}\left[-e^{-t}x \ f'(e^{-t}x + \sqrt{1-e^{-2t}}\xi)\right] \\
&+ \frac{e^{-2t}}{\sqrt{1-e^{-2t}}}\mathbb{E}\left[\xi \ f'(e^{-t}x + \sqrt{1-e^{-2t}}\xi)\right] \ .
\end{aligned}$$

For the second term, first integrate only over the normal variable ξ and use Gaussian integration by parts on the function $u \mapsto f'(e^{-t}x + \sqrt{1-e^{-2t}}u)$ to obtain

$$\mathbb{E}_\xi \left[\xi \ f'(e^{-t}x + \sqrt{1-e^{-2t}}\xi)\right] = \sqrt{1-e^{-2t}}\mathbb{E}_\xi \left[f''(e^{-t}x + \sqrt{1-e^{-2t}}\xi)\right] \ .$$

Putting this back into the above equation,

$$\frac{\mathrm{d}}{\mathrm{d}t}P_t f = \mathbb{E}\left[-xe^{-t}f'(e^{-t}x + \sqrt{1-e^{-2t}}\xi) + e^{-2t}f''(e^{-t}x + \sqrt{1-e^{-2t}}\xi)\right] \ .$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}P_t f(x) = -x\frac{\mathrm{d}}{\mathrm{d}x}P_t f(x) + \frac{\mathrm{d}^2}{\mathrm{d}x^2}P_t f(x) \ .$$

Take $t \downarrow 0$ and use the mean value theorem to find

$$\lim_{h\downarrow 0} \frac{P_h f(x) - f(x)}{h} = \lim_{t\downarrow 0} \frac{\mathrm{d}}{\mathrm{d}t}P_t f(x) = \left[-x\frac{\mathrm{d}}{\mathrm{d}x} + \frac{\mathrm{d}^2}{\mathrm{d}x^2}\right] f(x) \ .$$

This convergence occurs for fixed x, but since f is smooth with compact support, this convergence also occurs in L^2 (use the mean value theorem and dominated convergence). Therefore the claimed equation for the generator holds.

To find the Dirichlet form, write for f, g smooth of compact support,

$$\mathcal{E}(f,g) = -\langle \mathcal{L}f, g\rangle = \mathbb{E}\left[(\xi f'(\xi) - f''(\xi))g(\xi)\right] \ .$$

However by integration by parts,

$$\mathbb{E}\xi f'(\xi)g(\xi) = \mathbb{E}\left[f''(\xi)g(\xi) + f'(\xi)g(\xi)\right] \ ,$$

so that

$$\mathcal{E}(f,g) = \langle f', g'\rangle \ .$$

Since this is symmetric, we find that $\langle \mathcal{L}f, g\rangle = \langle f, \mathcal{L}g\rangle$, and therefore the operators (P_t) are self-adjoint. This means that OU is reversible.

Last, for ergodicity, if f is smooth with compact support then

$$\begin{aligned}\|P_t f - \mathbb{E}_\mu f\|_2^2 &= \|\mathbb{E}_\xi f(e^{-t}X_0 + \sqrt{1-e^{-2t}}\xi) - \mathbb{E}_\xi f(\xi)\|_{L^2(\mu)}^2 \\ &\leq \mathbb{E}\left(f(e^{-t}X_0 + \sqrt{1-e^{-2t}}\xi) - f(\xi)\right)^2 \to 0 \text{ as } t \to \infty \ .\end{aligned}$$

For general $f \in L^2(\mu)$ we approximate by g that is smooth with compact support:

$$\|P_t f - \mathbb{E}_\mu f\|_2 \leq \|P_t f - P_t g\|_2 + \|P_t g - \mathbb{E}_\mu g\|_2 + \mathbb{E}_\mu |f - g|$$

and use the contractive property of P_t on L^2. □

Remarks.

1. By our form of the Dirichlet form, $\mathcal{E}(f) = \|f'\|_2^2$, where the norm is taken relative to the Gaussian distribution. Therefore the Poincaré inequality reads exactly as that from the Efron-Stein section:

$$\text{Var } f(X) \leq \mathcal{E}(f) = \|f'\|_2^2 \ .$$

2. The standard OU process can also be defined by

$$\mathrm{d}X_t = -X_t \ \mathrm{d}t + \mathrm{d}B_t \ .$$

   In other words, it is a Brownian motion with a drift back toward 0. This drift is what keeps the Brownian motion from running away to infinity, and allows the normal distribution to be invariant for it.

3. The semigroup for the OU process can also be interpreted as a form of "noising," just as in the case of our noise process on the hypercube. The formula $e^{-t}x + \sqrt{1-e^{-2t}}\xi$ represents gradually replacing the original value x by an independent gaussian ξ (as $t \to \infty$). This is a form of resampling our Gaussian.

4. (From notes written to me by P. Sosoe) There is a corresponding decomposition of functions in terms of eigenvectors for the generator for OU (similarly to Fourier-Walsh). To find it, one standard way is to find orthogonal polynomials for the Gaussian measure μ. These are given by the Hermite polynomials:

$$H_n(x) = \frac{(-1)^n}{\sqrt{n!}} e^{x^2/2} \frac{\mathrm{d}^n}{\mathrm{d}x^n} e^{-x^2/2} \ .$$

One can show that H_n is a polynomial of degree n and the set $\{H_n\}$ is orthonormal relative to μ. Furthermore, any $f \in L^2(\mu)$ can be expanded as

$$P_t f = \sum_n e^{-tn} \langle H_n, f \rangle_\mu H_n \ .$$

Thus each H_n is an eigenvector for P_t with eigenvalue e^{-tn} and so H_n is an eigenvector for $\mathcal{L}$ with eigenvalue n.

We finish this section with a simple proof of the Poincaré inequality for the gaussian measure. It is Theorem 2.26 in RvH. Since we have developed semigroups, the proof is quite easy. In fact, it is generalizable to many other distributions and corresponding processes.

Theorem 3.16 (Poincaré inequality for Gaussian)**.** *For μ the standard Gaussian and f smooth with compact support,*

$$\mathrm{Var}_\mu f \leq \|f'\|^2_{L^2(\mu)} \ .$$

Proof. We first show that

$$\mathcal{E}(P_t f, P_t f) \leq e^{-2t} \mathcal{E}(f,f) \ . \tag{13}$$

To do this, first compute

$$\frac{\mathrm{d}}{\mathrm{d}x} P_t f(x) = \frac{\mathrm{d}}{\mathrm{d}x} \mathbb{E}[f(e^{-t}x + \sqrt{1-e^{-2t}}\xi] = e^{-t} P_t f'(x) \ .$$

Therefore by the form of the Dirichlet form for OU and the contraction property,

$$\mathcal{E}(P_t f, P_t f) = e^{-2t} \langle P_t f', P_t f' \rangle_\mu \leq e^{-2t} \langle f', f' \rangle = e^{-2t} \mathcal{E}(f,f) \ .$$

Now we can show that, quite generally speaking, (13) implies Poincaré since the OU process is ergodic. For this, we use the derivative formula in (3.11). By the fundamental theorem of calculus,

$$\mathrm{Var}_\mu f - \mathrm{Var}_\mu P_t f = 2 \int_0^t \mathcal{E}(P_s f, P_s f) \ \mathrm{d}s \leq 2\mathcal{E}(f,f) \int_0^t e^{-2s} \ \mathrm{d}s \ .$$

Take $t \to \infty$. One the left, ergodicity gives convergence to $\mathrm{Var}_\mu f$. On the right, we converge to $\mathcal{E}(f,f)$. $\square$

3.2 LSI and hypercontractivity

In the setting of Markov processes, the LSI reads:

Definition 3.17. *The measure μ satisfies a logarithmic Sobolev inequality for the Dirichlet form $\mathcal{E}$ and constant $c \geq 0$ if*

$$Ent_\mu \, f^2 \leq c \, \mathcal{E}(f) \text{ for all } f \ .$$

There is a corresponding convergence to equilibrium statement that is equivalent to a LSI. You can see this in RvH, Theorem 3.20, but it is actually for a slightly different LSI: $Ent_\mu f \leq c \, \mathcal{E}(\log f, f)$. This is shown to be equivalent to $Ent_\mu P_t f \leq e^{-t/c} Ent_\mu f$. This is the same convergence statement that we had with the Poincaré inequality, but with entropy in place of variance. Coupled with this result is Example 3.23 in RvH, where the Gaussian LSI is derived in a simple way using semigroup tools.

The above LSI also implies the Poincaré inequality from the last section. We will however move in a different direction, toward proving that the LSI is equivalent to so-called hypercontractivity of (P_t). We will follow parts of notes that P. Sosoe wrote for me, along with Gross's 1975 paper *Logarithmic Sobolev Inequalities* and the notes by Guionnet and Zegarlinski.

Definition 3.18. *The semigroup (P_t) with invariant measure μ is hypercontractive if there exists T such that*

$$\|P_T\|_{2 \to 4} \leq 1 \ .$$

In other words, for each $f \in L^2(\mu)$ with $\|f\|_2 \leq 1$, one has $\|P_T f\|_4 \leq 1$.

Note that (P_t) is hypercontractive if and only if for each $q > 2$, there exists $T = T_q$ such that $\|P_T\|_{2 \to q} \leq 1$.

Proof. If (P_t) is hypercontractive and $\|f\|_2 \leq 1$ then the semigroup property of (P_t) gives

$$\begin{aligned}
\|P_t f\|_q^q = \mathbb{E}\left(|P_T P_{t-T} f|^{q/4}\right)^4 \leq \mathbb{E}|P_T |P_{t-T} f|^{q/4}|^4 = \|P_T |P_{t-T} f|^{q/4}\|_4^4 &\leq \||P_{t-T} f|^{q/4}\|_2^4 \\
&= \left(\mathbb{E}|P_{t-T} f|^{q/2}\right)^2 \\
&= \|P_{t-T} f\|_{q/2}^q \ .
\end{aligned}$$

By iteration, for $k \geq 1$,

$$\|P_t f\|_q \leq \|P_{t-kT} f\|_{q/2^k} \ .$$

So given $q > 2$, choose $k = \lfloor \log_2 q \rfloor$, so that $q/2^k \leq 2$. If $t \geq kT$, one has by the contractive property of P_{t-kT},

$$\|P_t f\|_q \leq \|P_{t-kT}\|_{q/2^k} \leq \|P_{t-KT} f\|_2 \leq \|f\|_2 \leq 1 \ .$$

$\square$

Now we can prove the main result. It shows that LSI implies hypercontractivity. In fact, Gross proves the converse, establishing that LSI and hypercontractivity are equivalent. See, for example, Theorem 4.1 in Guionnet-Zegarlinski.

Theorem 3.19 (Gross)**.** *Suppose* (P_t) *satisfies the LSI*

$$Ent\ f^2 \leq c\,\mathcal{E}(f)\ .$$

Then, given $p > 1$ *and setting* $p(t) = 1 + (p-1)e^{4t/c}$,

$$\|P_t f\|_{p(t)} \leq \|f\|_p\ .$$

Remark. *In Theorem 4.1 of Guionnet-Zegarlinski, it is shown that if for all* $q > 2$, *one has* $\|P_t\|_{2\to q} \leq 1$ *for* $q = 1 + e^{4t/c}$ *then the LSI holds for constant* c.

Proof. The idea is to obtain a differential inequality by taking the derivative in t of $\|P_t f\|_{p(t)}$. Let $f \geq 0$ be with smooth with compact support (so that we may exchange derivatives and integrals, for instance – this is always possible in the case of the noise semigroup due to finiteness of the space), $p(t) = 1 + (p-1)e^{4t/c}$, and compute

$$\frac{\mathrm{d}}{\mathrm{d}p}\|f\|_p = \frac{\mathrm{d}}{\mathrm{d}p}\exp\left(\frac{\log \mathbb{E}f^p}{p}\right) = \|f\|_p\left[\frac{p\frac{\mathbb{E}f^p\log f}{\mathbb{E}f^p} - \log\mathbb{E}f^p}{p^2}\right] = \frac{\|f\|_p^{1-p}}{p^2}Ent\ f^p\ .$$

Furthermore, for fixed $p \geq 1$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|P_t f\|_p = \frac{\mathrm{d}}{\mathrm{d}t}\exp\left(\frac{1}{p}\log\mathbb{E}(P_t f)^p\right) = \frac{\|P_t f\|_p}{p}\cdot\frac{\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(P_t f)^p}{\mathbb{E}(P_t f)^p}\ .$$

Just as when we computed $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(P_t f)^2 = 2\mathbb{E}P_t f\mathcal{L}P_t f$ (in Proposition 3.11), we can find

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(P_t f)^p = p\mathbb{E}(P_t f)^{p-1}\mathcal{L}P_t f = -p\mathcal{E}(P_t f, (P_t f)^{p-1})\ .$$

Combining this with the above and using the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|P_t f\|_{p(t)} = p'(t)\frac{\|P_t f\|_{p(t)}^{1-p(t)}}{p(t)^2}Ent\ (P_t f)^{p(t)} - \|P_t f\|_{p(t)}^{1-p(t)}\mathcal{E}(P_t f, (P_t f)^{p(t)-1})\ ,$$

or

$$\frac{\mathrm{d}}{\mathrm{d}t}\log\|P_t f\|_{p(t)} = p'(t)\frac{Ent\ (P_t f)^{p(t)}}{p(t)^2\|P_t f\|_{p(t)}^{p(t)}} - \frac{1}{\|P_t f\|_{p(t)}^{p(t)}}\mathcal{E}(P_t f, (P_t f)^{p(t)-1})\ . \tag{14}$$

We would like to use the LSI, but the term on the right is not of the form $\mathcal{E}(g,g)$. So to remedy this, we inspect it further: if g is a nonnegative function with X_0 distributed as μ,

$$\begin{aligned}\mathcal{E}(g, g^{p(t)-1}) = -\mathbb{E}g^{p(t)-1}\mathcal{L}g &= -\lim_{h\downarrow 0}\frac{1}{h}\mathbb{E}g^{p(t)-1}(P_h g - g)\\ &= -\lim_{h\downarrow 0}\frac{1}{h}\mathbb{E}g^{p(t)-1}(X_0)(g(X_h) - g(X_0))\ .\end{aligned}$$

On the other hand, by stationarity,

$$\begin{aligned}\mathcal{E}(g,g^{p(t)-1}) = -\mathbb{E}g\mathcal{L}g^{p(t)-1} &= -\lim_{h\downarrow 0}\frac{1}{h}\mathbb{E}g(X_0)(g^{p(t)-1}(X_h)-g^{p(t)-1}(X_0)) \\ &= -\lim_{h\downarrow 0}\frac{1}{h}\left[\mathbb{E}g(X_0)g^{p(t)-1}(X_h)-\mathbb{E}g^{p(t)}(X_h)\right] \\ &= -\lim_{h\downarrow 0}\frac{1}{h}\mathbb{E}g^{p(t)-1}(X_h)(g(X_0)-g(X_h))\ .\end{aligned}$$

Combining these two expressions,

$$\mathcal{E}(g,g^{p(t)-1}) = \frac{1}{2}\lim_{h\downarrow 0}\frac{1}{h}\mathbb{E}\left[(g^{p(t)-1}(X_0)-g^{p(t)-1}(X_h))(g(X_0)-g(X_h))\right]\ .$$

Using the inequality (which we will justify later)

$$(x^{q-1}-y^{q-1})(x-y) \geq \frac{4(q-1)}{q^2}\left(x^{q/2}-y^{q/2}\right)^2\ , \tag{15}$$

which holds for $x,y\geq 0$ and $q\geq 0$, we obtain

$$\begin{aligned}\mathcal{E}(g,g^{p(t)-1}) &\geq \frac{2(p(t)-1)}{p(t)^2}\lim_{h\downarrow 0}\frac{1}{h}\mathbb{E}\left(g^{p(t)/2}(X_0)-g^{p(t)/2}(X_h)\right)^2 \\ &= \frac{2(p(t)-1)}{p(t)^2}\lim_{h\downarrow 0}\frac{1}{h}\mathbb{E}\left[g^{p(t)}(X_0)+g^{p(t)}(X_h)-2g^{p(t)/2}(X_0)g^{p(t)/2}(X_h)\right] \\ &= \frac{4(p(t)-1)}{p(t)^2}\lim_{h\downarrow 0}\frac{1}{h}\mathbb{E}\left[g^{p(t)/2}(X_0)(g^{p(t)/2}(X_0)-g^{p(t)/2}(X_h))\right] \\ &= -\frac{4(p(t)-1)}{p(t)^2}\langle\mathcal{L}g^{p(t)/2},g^{p(t)/2}\rangle \\ &= \frac{4(p(t)-1)}{p(t)^2}\mathcal{E}(g^{p(t)/2})\ .\end{aligned}$$

Returning to (14), one obtains

$$\frac{\mathrm{d}}{\mathrm{d}t}\log\|P_tf\|_{p(t)} \leq p'(t)\frac{Ent\ (P_tf)^{p(t)}}{p(t)^2\|P_tf\|_{p(t)}^{p(t)}} - \frac{4(p(t)-1)}{p(t)^2\|P_tf\|_{p(t)}^{p(t)}}\mathcal{E}((P_tf)^{p(t)/2})$$

Since $p'(t)=\frac{4}{c}(p(t)-1)$, this is

$$\frac{\mathrm{d}}{\mathrm{d}t}\log\|P_tf\|_{p(t)} \leq \frac{p'(t)}{p(t)^2}\left[\frac{Ent\ (P_tf)^{p(t)}}{\|P_tf\|_{p(t)}^{p(t)}} - c\ \mathcal{E}((P_tf)^{p(t)/2})\right]$$

Bound this using the LSI as

$$\frac{\mathrm{d}}{\mathrm{d}t}\log\|P_tf\|_{p(t)} \leq \frac{p'(t)}{p(t)^2\|P_tf\|_{p(t)}^{p(t)}}\left[Ent\ (P_tf)^{p(t)} - c\mathcal{E}((P_tf)^{p(t)/2})\right] \leq 0\ .$$

Therefore for $t_1 \le t_2$,

$$\|P_{t_2} f\|_{p(t_2)} \le \|P_{t_1} f\|_{p(t_1)} \ .$$

Taking $t_1 = 0$ gives the statement.

Now to justify (15), set $r = q/2$ and $z = x/y$ with $x \ge y$ for

$$\frac{(z^{2r-1}-1)(z-1)}{(z^r-1)^2} \ge \frac{2r-1}{r^2} \text{ for } r \ge 0, \ z \ge 1 \ .$$

Note that

$$\lim_{z \downarrow 1} \frac{(z^{2r-1}-1)(z-1)}{(z^r-1)^2} = \frac{2r-1}{r^2} \ ,$$

so we must show that the left side is non-increasing in z. To do this, make the substitution $z = e^x$ for $x \ge 0$ so the left side becomes

$$\frac{\sinh((r-\frac{1}{2})x)\sinh(\frac{x}{2})}{\sinh^2(\frac{r}{2}x)} \ .$$

We must show this is non-increasing in x. Taking log and writing $g(x) = \log \sinh x$, we must show that

$$\frac{g((r-\frac{1}{2})x) + g(\frac{x}{2})}{2} - g\left(\frac{r}{2}x\right)$$

is non-increasing. Taking the derivative, we want to show that

$$\frac{(r-1/2)g'((r-1/2)x) + (1/2)g'(x/2)}{2} \le (r/2)g'((r/2)x)$$

$$\left(1 - \frac{1}{2r}\right) g'((r-1/2)x) + \frac{1}{2r} g'(x/2) \le g'((r/2)x) \ .$$

Because g' is concave, the left side is bounded above by

$$g'\left(\left(1-\frac{1}{2r}\right)((r-1/2)x) + \frac{1}{2r}(x/2)\right) = g'\left(r - 1 + \frac{1}{2r}\right) \ .$$

However, as $r - 1 + \frac{1}{2r} \ge \frac{r}{2}$, concavity of g itself gives the upper bound $g'((r/2)x)$ and we have shown (15).

□

Corollary 3.20. *The noise semigroup and the OU semigroup are hypercontractive.*

Proof. We established the LSI for both of these semigroups, so the result follows from Gross.

□

The case of the noise semigroup is often stated as the Bonami-Beckner inequality.

Theorem 3.21 (Bonami-Beckner). *For $f : \{-1,+1\}^n \to \mathbb{R}$,*

$$\|P_t f\|_2 \le \|f\|_{1+e^{-2t}} \ .$$

Proof. Set $p = 1 + e^{-2t}$ in Gross's theorem to obtain

$$\|P_t f\|_{1+(1+e^{-2t}-1)e^{4t/c}} \le \|f\|_{1+e^{-2t}} \ .$$

As $c = 2$ in the Bernoulli LSI, this reduces to the Bonami-Beckner inequality. □

3.3 Applications of hypercontractivity

Rademacher sums. Let $\epsilon_1, \dots, \epsilon_n$ be i.i.d. Rademacher variables (in other words, $(\epsilon_1, \dots, \epsilon_n)$ is uniformly distributed on $\{-1, +1\}^n$). A homogeneous Rademacher chaos of order k is a sum of the form

$$Z = \sum_{S:\#S=k} a_S \left[\prod_{i \in S} \epsilon_i \right] .$$

In other words, an order 1 homogeneous chaos is a sum of the form $a_1\epsilon_1 + \cdots + a_n\epsilon_n$. Using hypercontractivity, we can get simple bounds for moments of such sums (and of course, these can be turned into concentration inequalities).

Hypercontractivity of the noise semigroup gives for $p > 1$

$$\|P_t Z\|_{p(t)} \le \|Z\|_p ,$$

where $p(t) = 1 + (p-1)e^{2t}$. However by homogeneity,

$$\|P_t Z\|_{p(t)} = \left\| \sum_{S:\#S=k} e^{-t\#S} a_S \chi_S \right\|_{p(t)} = e^{-tk} \|Z\|_{p(t)} .$$

Therefore

$$\|Z\|_{p(t)} \le e^{kt} \|Z\|_p .$$

Given $q > p > 1$ we can solve for t:

$$\begin{aligned} q = p(t) \Leftrightarrow e^{2t} &= \frac{q-1}{p-1} \\ \Leftrightarrow e^{kt} &= \left(\frac{q-1}{p-1} \right)^{k/2} . \end{aligned}$$

Therefore

$$\|Z\|_q \le \left(\frac{q-1}{p-1} \right)^{k/2} \|Z\|_p . \tag{16}$$

The case $k = 1$ gives with $q > 2$

$$\|a_1\epsilon_1 + \cdots + a_n\epsilon_n\|_q \le (q-1)^{1/2} \|Z\|_2 = \sqrt{(q-1)(a_1^2 + \cdots + a_n^2)} .$$

If $a_i = a$ for all i, we obtain

$$\|\epsilon_1 + \cdots + \epsilon_n\|_q \le \sqrt{(q-1)n} .$$

(This result is easy to derive anyhow, but is something we already know and love.)

In (16), we cannot take $p = 1$, but a trick will help. Taking $p = 3/2$ and $q = 2$ gives

$$\|Z\|_2 \le 2^{k/2} \|Z\|_{3/2} .$$

Next apply Cauchy-Schwarz for

$$\|Z\|_{3/2} = \left(\mathbb{E}Z\sqrt{Z}\right)^{2/3} \leq \left(\|\sqrt{Z}\|_2\|Z\|_2\right)^{2/3} = \|Z\|_1^{1/3}\|Z\|_2^{2/3} ,$$

which implies

$$\|Z\|_2^{1/3} \leq 2^{k/2}\|Z\|_1^{1/3} ,$$

or

$$\|Z\|_2 \leq 2^{3k/2}\|Z\|_1 .$$

Talagrand's inequality. Here we will prove an inequality due to Talagrand. It is a variance bound that incorporates influences in a similar way to the FS inequality at the beginning of this section. However, it predated the FS inequality by some time. The proof we will give is due to Benjamini-Kalai-Schramm (in their paper *First-passage percolation has sub-linear distance variance.)* They use this inequality to prove the sub-linear bound in FPP that we gave in Section 1.1 (and also on $\mathbb{Z}^d$). Indeed, you can check that on the d-dimensional torus in that example and f equal to the passage time T_n,

$$\frac{\|D_if\|_2}{\|D_if\|_1} \geq C\sqrt{\mathbb{P}(e_i \in Geo_n)} = C'/n^{d/2} ,$$

where D_i is the discrete derivative relative to the edge weight t_{e_i}, and $e_1, \ldots$ is an enumeration of the edges of the torus. Plugging this into Talagrand's inequality and using $\sum_i \|D_if\|_2 \leq Cn$ (which is the Efron-Stein upper bound on the variance of the passage time) gives the bound $\mathrm{Var}\ f \leq Cn/\log n$.

In applications in which $\|D_if\|_2$ is generally much larger than $\|D_if\|_1$, the following theorem gives an logarithmic improvement to Efron-Stein. This will happen when D_if has large probability to be very small (say 0) and small probability to be not-small. For example, in FPP, when edges have small probability to be in a geodesic, D_if is 0 with high probability and order 1 (bounded above by one edge weight) with low probability. In this situation, variables have small influence, and this is what gives us a better upper bound.

Theorem 3.22 (Talagrand)**.** *There exists $C > 0$ such that for all $f : \{-1, +1\}^n \to \mathbb{R}$,*

$$\mathrm{Var}\ f \leq C \sum_{i=1}^n \frac{\|D_if\|_2^2}{1 + \log \frac{\|D_if\|_2}{\|D_if\|_1}} .$$

Here $\{-1, +1\}^n$ is endowed with the uniform distribution $\mathbb{P}$.

Proof. For ease of notation, let $\rho = e^{-t}$, so that the semigroup P_t becomes

$$T_\rho f = \sum_S \rho^{\#S}\hat{f}(S)\chi_S \text{ for } \rho \in [0, 1]$$

(note here as t moves from 0 to ∞, the variable ρ goes from 1 to 0) and Bonami-Beckner becomes

$$\|T_\rho f\|_2 \leq \|f\|_{1+\rho^2} .$$

We begin with the observation

$$\int_0^1 \|T_\rho(D_i f)\|_2^2 \,\mathrm{d}\rho = \sum_{S:i\in S}\int_0^1 \rho^{2\#S}\hat{f}(S)^2 \,\mathrm{d}t = \sum_{S:i\in S}\frac{1}{2\#S+1}\hat{f}(S)^2$$
$$\geq \frac{1}{3}\sum_{S:i\in S}\frac{1}{\#S}\hat{f}(S)^2 \ .$$

This implies

$$\sum_{i=1}^n\int_0^1 \|T_\rho(D_i f)\|_2^2 \,\mathrm{d}\rho \geq \frac{1}{3}\sum_{i=1}^n\sum_S \mathbf{1}_{\{i\in S\}}\frac{1}{\#S}\hat{f}(S)^2 = \frac{1}{3}\mathrm{Var}\ f \ .$$

So by the Bonami-Beckner inequality,

$$\begin{aligned}\mathrm{Var}\ f \leq 3\sum_{i=1}^n\int_0^1 \|T_\rho(D_i f)\|_2^2 \,\mathrm{d}\rho &\leq 3\sum_{i=1}^n\int_0^1 \|D_i f\|_{1+\rho^2} \,\mathrm{d}\rho \\ &\leq 6\sum_{i=1}^n\int_{\frac{1}{2}}^1 \|D_i f\|_{1+\rho^2} \,\mathrm{d}\rho \ . \end{aligned} \tag{17}$$

Setting $\rho = e^{-t}$, use Hölder's inequality when $\|D_i f\|_2 > 0$.

$$\begin{aligned}\mathbb{E}|D_i f|^{1+\rho^2} = \mathbb{E}|D_i f|^{2\rho^2}|D_i f|^{1-\rho^2} &\leq \||D_i f|^{2\rho^2}\|_{\rho^{-2}}\||D_i f|^{1-\rho^2}\|_{(1-\rho^2)^{-1}} \\ &= \left(\mathbb{E}(D_i f)^2\right)^{\rho^2}\left(\mathbb{E}|D_i f|\right)^{1-\rho^2} \\ &= \|D_i f\|_2^{1+\rho^2}\left(\frac{\|D_i f\|_1}{\|D_i f\|_2}\right)^{1-\rho^2} \ .\end{aligned}$$

We obtain in the case $\|D_i f\|_2 > 0$

$$\begin{aligned}\int_{\frac{1}{2}}^1 \|D_i f\|_{1+\rho^2} \,\mathrm{d}\rho &\leq \|D_i f\|_2^2\int_{\frac{1}{2}}^1\left(\frac{\|D_i f\|_1}{\|D_i f\|_2}\right)^{\frac{2(1-\rho^2)}{1+\rho^2}} \,\mathrm{d}\rho \\ &= \|D_i f\|_2^2\int_0^{\frac{6}{5}}\left(\frac{\|D_i f\|_1}{\|D_i f\|_2}\right)^h \,\frac{\mathrm{d}h}{-h'(\rho)} \ ,\end{aligned}$$

where $h(\rho) = \frac{2(1-\rho^2)}{1+\rho^2}$. Since

$$h'(\rho) = 2\frac{-2\rho(1+\rho^2) - 2\rho(1-\rho^2)}{(1+\rho^2)^2} = -8\frac{\rho}{(1+\rho^2)^2} \ ,$$

one has $-h(\rho) \geq 1/2$ for $\rho \in [1/2, 1]$. Therefore

$$\int_{\frac{1}{2}}^1 \|D_i f\|_{1+\rho^2} \,\mathrm{d}\rho \leq 2\|D_i f\|_2^2\int_0^{\frac{6}{5}}\left(\frac{\|D_i f\|_1}{\|D_i f\|_2}\right)^h \,\mathrm{d}h \ .$$

The integral is of the type $\int_0^{6/5} a^h \, \mathrm{d}h$ for $a \in (0,1]$. This is evaluated as $(\log a)^{-1}(a^{6/5}-1)$ if $a<1$ and $6/5$ if $a=1$, so one obtains the upper bound for $a \in (0,1)$:

$$\int_0^1 \|T_\rho(D_i f)\|_2^2 \, \mathrm{d}\rho \leq 2\|D_i f\|_2^2 \frac{1-\left(\frac{\|D_i f\|_1}{\|D_i f\|_2}\right)^{6/5}}{\log \frac{\|D_i f\|_2}{\|D_i f\|_1}} \leq 2\frac{\|D_i f\|_2^2}{\log \frac{\|D_i f\|_2}{\|D_i f\|_1}}$$

and $(12/5)\|D_i f\|_2^2$ if $a=1$. If $\log(1/a) \geq 1$ then we can bound this above by $4\frac{\|D_i f\|_2^2}{1+\log \frac{\|D_i f\|_2}{\|D_i f\|_1}}$. If $\log(1/a) < 1$ then we can get the bound

$$\int_0^1 \|T_\rho(D_i f)\|_2^2 \, \mathrm{d}\rho \leq \int_0^1 \|D_i f\|_2^2 \, \mathrm{d}\rho \leq C\frac{\|D_i f\|_2^2}{1+\log \frac{\|D_i f\|_2}{\|D_i f\|_1}} \, .$$

In total, for $\|D_i f\|_2 > 0$, one has the bound

$$C\frac{\|D_i f\|_2^2}{1+\log \frac{\|D_i f\|_2}{\|D_i f\|_1}} \, .$$

In the case that $\|D_i f\|_2 = 0$, one has $D_i f = 0$ almost surely and so $\int_{\frac{1}{2}}^1 \|D_i f\|_{1+\rho^2} \, \mathrm{d}\rho = 0$. Placing both bounds into (17) gives the result. □

1 Isoperimetric inequalities

An isoperimetric inequality, as we stated in the section on entropy, gives an upper bound on the volume of a set by a function of its surface area:

$$\text{Vol } A \leq \phi\left(\text{Area } \partial A\right) \ . \tag{1}$$

In general metric spaces, it can be more convenient to work with an integrated version of such an inequality:

$$\mu(A_t) \geq \psi(\mu(A), t) \ ,$$

where $A_t = \{x : d(x,y) < t \text{ for some } y \in A\}$ and ψ is some function. In other words, we bound below the t-blowup of A by a function of the measure of A. Taking complements,

$$\mu(A_t^c) \leq 1 - \psi(\mu(A), t) \ .$$

We have seen something like this before: remember the concentration function

$$\alpha_\mu(t) = \sup\{\mu(A_t^c) : \mu(A) \geq 1/2\} \ .$$

In this way, isoperimetric inequalities are related to concentration. We will go through some of Ch. 7 in BLM to understand this connection in more detail. The simplest connection is due to Lévy. Let Mf be a median of a function f.

Theorem 1.1 (Lévy)**.** *Let μ be a Borel probability measure on a metric space (X, d). If $f : X \to \mathbb{R}$ is L-Lipschitz then for all $t > 0$,*

$$\mu(|f - Mf| \geq t) \leq 2\alpha_\mu(t/L) \ .$$

Conversely, if for some $\alpha : \mathbb{R}_+ \to \mathbb{R}$ and all 1-Lipschitz $f : X \to \mathbb{R}$, one has

$$\mu(f - Mf \geq t) \leq \alpha(t) \textit{ for all } t > 0 \ , \tag{2}$$

then $\alpha_\mu(t) \leq \alpha(t)$.

Proof. Supposing that f is Lipschitz, then take $A = \{x : f(x) \leq Mf\}$. Note that $\mu(A) \geq 1/2$ and that if $f(x) \geq Mf + t$ then $d(x, A) \geq t/L$. Therefore $\{x : f(x) \geq Mf + t\} \subset A_{t/L}^c$. Therefore

$$\mu(\{x : f(x) \geq Mf + t\}) \leq \mu(A_{t/L}^c) \leq \alpha(t/L) \ .$$

A similar argument holds for $\mu(\{x : f \leq Mf - t\})$ and the set $A = \{x : f(x) \geq Mf\}$. Combining the two bounds gives the first part of the theorem.

Suppose now that α satisfies (2). Then if $A \subset X$ satisfies $\mu(A) \geq 1/2$, define $f : X \to \mathbb{R}$ by $f(x) = d(x, A)$. Since f is $1-$Lipschitz, one has

$$\mu(\{x : d(x, A) \geq Mf + t\}) \leq \alpha(t) \ .$$

As $\mu(A) \geq 1/2$ and $x \in A$ implies $f(x) = 0$, the number 0 is a median of f. Therefore

$$\mu(A_t^c) = \mu(\{x : d(x, A) \geq t\}) \leq \alpha(t) \ .$$

Take infimum over A to get $\alpha_\mu(t) \leq \alpha(t)$. $\square$

1.1 Isoperimetric inequality in $\mathbb{R}^n$

The standard isoperimetric inequality in $\mathbb{R}^n$ is the fact that among regions with a fixed volume, Euclidean balls minimize surface area. The proof of this fact in BLM uses the Brunn-Minkowski inequality. Here we use the Minkowski sum notation

$$A + B = \{a + b : a \in A, b \in B\} .$$

Theorem 1.2 (Brunn-Minkowski). *Let A, B be nonempty bounded measurable sets in $\mathbb{R}^n$ such that $A + B$ is also measurable. Then*

$$\mathrm{Vol}(A + B)^{1/n} \geq \mathrm{Vol}(A)^{1/n} + \mathrm{Vol}(B)^{1/n} .$$

Before the proof, let's see why it implies the isoperimetric inequality. To state it, let us define the surface area of a set A by

$$\mathrm{Vol}(\partial A) = \lim_{t \downarrow 0} \frac{\mathrm{Vol}(A_t) - \mathrm{Vol}(A)}{t} \text{ if it exists,}$$

where $A_t = A + tB_1(0)$, and $B_1(0) = B_1 = \{x \in \mathbb{R}^n : \|x\|_2 < 1\}$.

Theorem 1.3 (Isoperimetric theorem). *Let $A \subset \mathbb{R}^n$ be such that $\mathrm{Vol}(A) = \mathrm{Vol}(B_1)$. Then for any $t > 0$, $\mathrm{Vol}(A_t) \geq \mathrm{Vol}((B_1)_t)$. Moreover, if $\mathrm{Vol}(\partial A)$ exists, then $\mathrm{Vol}(\partial A) \geq \mathrm{Vol}(\partial B_1)$.*

Proof.

$$\begin{aligned}
\mathrm{Vol}(A_t) = \mathrm{Vol}(A + tB_1) &\geq \left(\mathrm{Vol}(A)^{1/n} + \mathrm{Vol}(tB_1)^{1/n}\right)^n \\
&= \left(\mathrm{Vol}(B_1)^{1/n} + \mathrm{Vol}(tB_1)^{1/n}\right)^n \\
&= \left((t+1)\mathrm{Vol}(B_1)^{1/n}\right)^n \\
&= \mathrm{Vol}((t+1)B_1) \\
&= \mathrm{Vol}((B_1)_t) .
\end{aligned}$$

For the second statement,

$$\frac{\mathrm{Vol}(A_t) - \mathrm{Vol}(A)}{t} \geq \frac{(t+1)^n - 1}{t} \mathrm{Vol}(B_1) \to n\mathrm{Vol}(B_1) .$$

However this is equal to $\mathrm{Vol}(\partial B_1)$:

$$\mathrm{Vol}(\partial B_1) = \lim_{t \downarrow 0} \frac{\mathrm{Vol}((B_1)_t) - \mathrm{Vol}(B_1)}{t} = \mathrm{Vol}(B_1) \lim_{t \downarrow 0} \frac{(1+t)^n - 1}{t} = n\mathrm{Vol}(B_1) .$$

□

Before moving to the proof of Brunn-Minkowski, let us see why the formulation of the isoperimetric inequality can be put into the form (1). Letting A be a set of positive volume, set $c = \left(\frac{\mathrm{Vol}(B_1)}{\mathrm{Vol}(A)}\right)^{1/n} > 0$. Then

$$\mathrm{Vol}(cA) = c^n \mathrm{Vol}(A) = \mathrm{Vol}(B_1) .$$

Therefore by the isoperimetric inequality,

$$\mathrm{Vol}(\partial(cA)) \geq \mathrm{Vol}(\partial B_1) = n\mathrm{Vol}(B_1) = nc^n \mathrm{Vol}(A) \ .$$

However the left side is evaluated as

$$\begin{aligned}\lim_{t\downarrow 0} \frac{\mathrm{Vol}((cA)_t) - \mathrm{Vol}(cA)}{t} &= \lim_{t\downarrow 0} \frac{\mathrm{Vol}(cA_{t/c}) - \mathrm{Vol}(A)}{t} \\ &= c^{n-1} \lim_{t\downarrow 0} \frac{\mathrm{Vol}(A_{t/c}) - \mathrm{Vol}(A)}{t/c} \\ &= c^{n-1}\mathrm{Vol}(\partial A) \ .\end{aligned}$$

Therefore

$$\mathrm{Vol}(\partial A) \geq nc\mathrm{Vol}(A) = n(\mathrm{Vol}(B_1))^{1/n} \left[\mathrm{Vol}(A)\right]^{1-1/n} \ .$$

This is the more familiar form of the isoperimetric inequality: a lower bound of the form $C(\mathrm{Vol}(A))^{1-1/n}$.

Proof of Brunn-Minkowski. This proof is from the survey paper of R. J. Gardner (and originally due to Hadwiger-Ohmann). We will prove that the theorem holds if A and B are finite unions of disjoint cubes. By the π-λ theorem, this suffices. In this setting, we may assume that all three volumes in the statement are positive; otherwise, one of the sets must be empty and the theorem is trivial.

So let us begin with A a cube with side-lengths $x_1, \ldots, x_n$ and B a cube with side-lengths $y_1, \ldots, y_n$. Then $A+B$ is a cube with side-lengths $x_1+y_1, \ldots, x_n+y_n$ and so the inequality reads

$$\left(\prod_{i=1}^n (x_i+y_i)\right)^{1/n} \geq \left(\prod_{i=1}^n x_i\right)^{1/n} + \left(\prod_{i=1}^n y_i\right)^{1/n} ,$$

or

$$\left(\prod_{i=1}^n \frac{x_i}{x_i+y_i}\right)^{1/n} + \left(\prod_{i=1}^n \frac{y_i}{x_i+y_i}\right)^{1/n} \leq 1 \ .$$

However by concavity of log, if $a_1, \ldots, a_n$ are positive,

$$\left(\prod_{i=1}^n a_i\right)^{1/n} = \exp\left(\frac{1}{n}\sum_{i=1}^n \log a_i\right) \leq \exp\left(\log \frac{1}{n}\sum_{i=1}^n a_i\right) = \frac{1}{n}\sum_{i=1}^n a_i \ ,$$

which is the arithmetic-geometric inequality. Applying this above, we get the upper bound

$$\frac{1}{n}\sum_{i=1}^n \frac{x_i}{x_i+y_i} + \frac{1}{n}\sum_{i=1}^n \frac{y_i}{x_i+y_i} = 1 \ ,$$

proving this case.

For $n > 1$ we use induction. We may assume that the number of boxes in X is strictly bigger than 1. Note first that the inequality is invariant under translating either X or Y, or both, so we may translate X so that a coordinate hyperplane, say $\{x_n = 0\}$, separates two of the boxes of X. Let $X_\pm$ (respectively $Y_\pm$) denote the union of the boxes of X (respectively Y) formed by intersecting X with $\{\pm x_n \geq 0\}$. Next, translate Y so that

$$\frac{\mathrm{Vol}(X_\pm)}{\mathrm{Vol}(X)} = \frac{\mathrm{Vol}(Y_\pm)}{\mathrm{Vol}(Y)} \ . \tag{3}$$

Note that $X_+ + Y_+ \subset \{x_n \geq 0\}$ and $X_- + Y_- \subset \{x_n \leq 0\}$. Furthermore, the number of boxes in each of $X_+ \cup Y_+$ and $X_- \cup Y_-$ is strictly less than that in $X \cup Y$. Therefore if we use induction on the number of such boxes,

$$\begin{aligned}
\mathrm{Vol}(X+Y) &\geq \mathrm{Vol}(X_+ + Y_+) + \mathrm{Vol}(X_- + Y_-) \\
&\geq \left(\mathrm{Vol}(X_+)^{1/n} + \mathrm{Vol}(Y_+)^{1/n}\right)^n + \left(\mathrm{Vol}(X_-)^{1/n} + \mathrm{Vol}(Y_-)^{1/n}\right)^n \\
&= \mathrm{Vol}(X_+)\left(1 + \frac{\mathrm{Vol}(Y_+)^{1/n}}{\mathrm{Vol}(X_+)^{1/n}}\right)^n + \mathrm{Vol}(X_-)\left(1 + \frac{\mathrm{Vol}(Y_-)^{1/n}}{\mathrm{Vol}(X_-)^{1/n}}\right)^n \ .
\end{aligned}$$

By (3), this equals

$$\begin{aligned}
&\mathrm{Vol}(X_+)\left(1 + \frac{\mathrm{Vol}(Y)^{1/n}}{\mathrm{Vol}(X)^{1/n}}\right)^n + \mathrm{Vol}(X_-)\left(1 + \frac{\mathrm{Vol}(Y)^{1/n}}{\mathrm{Vol}(X)^{1/n}}\right)^n \\
&= \mathrm{Vol}(X)\left(1 + \frac{\mathrm{Vol}(Y)^{1/n}}{\mathrm{Vol}(X)^{1/n}}\right)^n \\
&= \left(\mathrm{Vol}(X)^{1/n} + \mathrm{Vol}(Y)^{1/n}\right)^n \ ,
\end{aligned}$$

completing the proof. □

1.2 Gaussian isoperimetric inequality

The Gaussian isoperimetric inequality deals with the standard Gaussian measure $\mathbb{P}$ on $\mathbb{R}^n$. It states that among all sets with a given measure, the half-space minimizes surface area. To state this analytically, given $A \subset \mathbb{R}^n$, a half-plane with the same measure as that of A is

$$\{(x_1, \ldots, x_n) : x_n \leq \Phi^{-1}(\mathbb{P}(A))\} \ .$$

The t-blowup of this set is

$$\{(x_1, \ldots, x_n) : x_n \leq \Phi^{-1}(\mathbb{P}(A)) + t\} \ .$$

The Gaussian measure of this set is $\Phi\left(\Phi^{-1}(\mathbb{P}(A)) + t\right)$. Therefore the Gaussian isoperimetric inequality states

$$\mathbb{P}(A_t) \geq \Phi\left(\Phi^{-1}(\mathbb{P}(A)) + t\right) \tag{4}$$

with equality if and only if A is a half-space.

An equivalent statement is that the concentration function satisfies

$$\alpha(t) = 1 - \Phi(t) \text{ for } t > 0 \ .$$

That is, for any A with $\mathbb{P}(A) \geq 1/2$,

$$\mathbb{P}(A_t^c) \leq \frac{1}{\sqrt{2\pi}} \int_t^\infty e^{-x^2/2} \, \mathrm{d}x \; .$$

For large t, this behaves like

$$\mathbb{P}(A_t^c) \lesssim \frac{1}{\sqrt{2\pi} t} e^{-t^2/2} \; .$$

We will prove (4) in the section on Bobkov's inequality, but for now, let us remark that the Gaussian concentration inequality we have already derived (using the LSI) implies an inequality that is close.

Proposition 1.4. *For any A with $\mathbb{P}(A) \geq 1/2$,*

$$\mathbb{P}(A_t^c) \leq e^{-(t-1)^2/2} \; .$$

Proof. Define $f(x) = d(x, A)$. Then f is 1-Lipschitz, so

$$\mathbb{P}(f(X) - \mathbb{E}f(X) \geq t) \leq e^{-t^2/2} \text{ for } t > 0 \; .$$

To estimate the expected value, use the Gaussian Poincaré inequality for

$$\text{Var } f(X) \leq 1, \text{ so } \mathbb{E}f(X) \leq 1 \; .$$

Therefore

$$\mathbb{P}(d(x, A) \geq t + 1) \leq \mathbb{P}(d(x, A) \geq t + \mathbb{E}f(X)) \leq e^{t^2/2} \; .$$

Because $\{x : d(x, A) \geq t\} = A_t$, this implies the proposition. $\square$

1.3 The hypercube

Using only the bounded differences inequality, we can derive a simple bound on the concentration function for the hypercube endowed with the Hamming metric. Let $\Omega = \{-1, +1\}^n$ and set $d_H(x, y) = \#\{i = 1, \ldots, n : x_i \neq y_i\}$. Then if $A \subset \Omega$ with $\mathbb{P}$ the uniform measure, the function $f : \Omega \to \mathbb{R}$ given by $f(x) = d_H(x, A)$ is 1-Lipschitz, so by the bounded differences Gaussian concentration inequality we have already derived,

$$\mathbb{P}(|d_H(x, A) - \mathbb{E}d_H(x, A)| \geq t) \leq e^{-2t^2/n} \; .$$

If we take $t = \mathbb{E}d_H(x, A)$ then we obtain

$$\mathbb{P}(A) \leq e^{-2(\mathbb{E}d_H(x,A))^2/n} \; ,$$

or

$$\mathbb{E}d_H(x, A) \leq \sqrt{\frac{n}{2} \log \frac{1}{\mathbb{P}(A)}} \; .$$

Therefore

$$\mathbb{P}\left(d_H(x, A) \geq \sqrt{\frac{n}{2} \log \frac{1}{\mathbb{P}(A)}} + t \right) \leq e^{-2t^2/n} \; .$$

Now let's try to list the results in the book on isoperimetry for the hypercube.

- Edge isoperimetry

  1. These results can be stated in terms of influences, since one may write

  $$I(A) = \sum_{i=1}^{n} \mathbb{P}(X_i \text{ is pivotal for } A) = 2\frac{\#\partial_E(A)}{2^n} \ ,$$

  where $\mathbb{P}$ is the uniform measure and

  $$\partial_E(A) = \{(x, x') : x \in A, x' \in A^c, d_H(x, x') = 1\}$$

  is the edge boundary of A. The first result is from Efron-Stein:

  $$\mathbb{P}(A)(1 - \mathbb{P}(A)) \leq \frac{I(A)}{4} \ ,$$

  or written in terms of the boundary,

  $$\frac{\#A}{2^n}\left(1 - \frac{\#A}{2^n}\right) \leq \frac{1}{2 \cdot 2^n}\#\partial_E(A) \ .$$

  Written more simply,

  $$\#A \leq \#\partial_E(A) \text{ if } \mathbb{P}(A) \leq 1/2 \ . \tag{5}$$

  2. Using entropy, one can obtain

  $$2\mathbb{P}(A)\log_2 \frac{1}{\mathbb{P}(A)} \leq I(A) \ ,$$

  or again in terms of the boundary,

  $$\#A \log_2 \frac{2^n}{\#A} \leq \#\partial_E(A) \ . \tag{6}$$

  Note that this inequality is better when $\#A$ is small, and approaches the bound $n\#A \leq \#\partial_E(A)$ as $\#A$ goes to 1.

  3. We have shown using the FS inequality that

  $$\max_{i=1,\ldots,n} I_i(A) \geq \frac{\log n}{n}\mathbb{P}(A)(1 - \mathbb{P}(A)) \ .$$

  Therefore if all influences are the same,

  $$I(A) \geq \log n \ \mathbb{P}(A)(1 - \mathbb{P}(A)) \ .$$

  In terms of isoperimetry,

  $$\frac{2}{2^n}\#\partial_E(A) \geq \log n \frac{\#A}{2^n}\left(1 - \frac{\#A}{2^n}\right) \ .$$

  So

  $$\frac{\log n}{4}\#A \leq \#\partial_E(A) \text{ if } \mathbb{P}(A) \leq 1/2 \text{ and } A \text{ is symmetric} \ . \tag{7}$$

- Vertex isoperimetry

1. The vertex boundary of a set $A \subset \{-1,+1\}^n$ is defined as

$$\partial A = \{x : d_H(x, A) = 1\} \ .$$

The simplicial order on $\{-1,+1\}^n$ is defined as $x < y$ if either $\|x\| < \|y\|$ (where $\|x\| = \#\{i : x_i = 1\}$) or $\|x\| = \|y\|$ and $x_i = 1$ and $y_i = -1$ for the first i such that $x_i \neq y_i$. Then Harper's isoperimetric inequality states

$$\#\partial A \geq \#\partial S_{\#A} \ , \tag{8}$$

where S_k is the set of the first k elements in the simplicial order. We will not prove this, but its proof is outlined in the exercises of Ch. 7.

2. Using Harper's inequality, one can show:

Proposition 1.5. *Let $A \subset \{-1,+1\}^n$ be such that $\#A \geq \sum_{i=0}^k \binom{n}{i}$. Then for any $t = 1, 2, \ldots, n-k+1$,*

$$\#A_t \geq \sum_{i=0}^{k+t-1} \binom{n}{i} \ .$$

Furthermore, if $\mathbb{P}(A) \geq 1/2$ then

$$\mathbb{P}(A_t) \geq 1 - \exp\left(-\frac{2t^2}{n}\right) \ ,$$

where $\mathbb{P}$ is the uniform measure.

Proof. Let $N = \sum_{i=0}^k \binom{n}{i}$. Then $S_N \subset A$ and therefore $(S_N)_t \subset A_t$. However $(S_N)_t = S_{N_t}$, where $N = \sum_{i=0}^{k+t-1} \binom{n}{i}$. Therefore $\#A_t \geq \#(S_N)_t = N_t$.
If X is a binomial random variable with parameters n and $1/2$ then

$$\mathbb{P}(A_t) = 2^{-n} \sum_{i=0}^{k+t-1} \binom{n}{i} = \mathbb{P}(X \leq k+t-1) \ ,$$

giving

$$\mathbb{P}(A_t^c) = \mathbb{P}(X > k+t-1) \ .$$

As $\mathbb{P}(A) \geq 1/2$ we must have $k \geq n/2$. Therefore

$$\mathbb{P}(X > k+t-1) \leq \mathbb{P}(X \geq \mathbb{E}X + t) \leq e^{-2t^2/n}$$

by Hoeffding. □

3. A version of Harper's inequality remains valid in the p-biased case, but for monotone sets. If other words, if $\mathbb{P}$ is i.i.d. with $\mathbb{P}(X_i = +1) = p$ then for each k, set $B(k) = \{x : d_H((-1,\ldots,-1), x) \leq k\}$. If A is monotone with $\mathbb{P}(A) \geq \mathbb{P}(B(k))$ then $\mathbb{P}(\partial A) \geq \mathbb{P}(\partial B(k))$. (This was proved by Bollobás and Leader.)

2 Talagrand's convex distance inequality

If $\alpha = (\alpha_1, \ldots, \alpha_n)$ is a vector of nonnegative numbers (written $\alpha \geq 0$), we can generalize the Hamming distance to

$$d_\alpha(x, y) = \sum_{i=1}^{n} \alpha_i \mathbf{1}_{\{x_i \neq y_i\}} .$$

Talagrand introduced the following *convex distance*: if $\mathcal{X}$ is a measurable set and $A \subset \mathcal{X}^n$ then

$$d_T(x, A) = \sup_{\alpha : \|\alpha\|_2 = 1} d_\alpha(x, A) ,$$

where $d_\alpha(x, A) = \min_{y \in A} d_\alpha(x, y)$. His famous convex distance inequality reads:

Theorem 2.1. *Let $X = (X_1, \ldots, X_n)$ be independent variables taking value in a measurable set $\mathcal{X}^n$. For any $t > 0$,*

$$\mathbb{P}(A)\mathbb{P}(d_T(X, A) \geq t) \leq e^{-t^2/4} .$$

Why is this inequality so useful? We will try to understand by looking at some examples. Then we will give the proof. First, if we do not use supremum in the definition of d_T, then concentration is not so hard to show.

Proposition 2.2. *For $t > 0$,*

$$\sup_{\alpha : \|\alpha\|_2 = 1} \Big[\mathbb{P}(A)\mathbb{P}(d_\alpha(X, A) \geq t) \Big] \leq e^{-t^2/2} .$$

Proof. The proof is similar to that which we derived for the Hamming distance. Use the bounded differences inequality with constants $\alpha_1, \ldots, \alpha_n$:

$$\mathbb{P}\left(|d_\alpha(X, A) - \mathbb{E}d_\alpha(X, A)| \geq t\right) \leq \exp\left(-\frac{2t^2}{\|\alpha\|_2^2}\right) = e^{-2t^2} .$$

Again taking $t = \mathbb{E}d_\alpha(X, A)$, one has

$$\mathbb{P}(A) \leq e^{-2(\mathbb{E}d_\alpha(X,A))^2} ,$$

so

$$\mathbb{E}d_\alpha(X, A) \leq \sqrt{\frac{1}{2}\log\frac{1}{\mathbb{P}(A)}} ,$$

and last

$$\mathbb{P}\left(d_\alpha(X, A) \geq t + \sqrt{\frac{1}{2}\log\frac{1}{\mathbb{P}(A)}}\right) \leq e^{-2t^2} .$$

Putting $u = \sqrt{\frac{1}{2}\log\frac{1}{\mathbb{P}(A)}}$, one then has for $t \geq 2u$

$$\mathbb{P}(d_\alpha(X, A) \geq t) \leq e^{-2(t-u)^2} \leq e^{-2(t/2)^2} = e^{-t^2/2} .$$

For $t \le 2u$, however, $t \le \sqrt{2 \log \frac{1}{\mathbb{P}(A)}}$, or $\mathbb{P}(A) \le e^{-t^2/2}$. Therefore

$$\min \{\mathbb{P}(A), \mathbb{P}(d_\alpha(X, A) \ge t)\} \le e^{-t^2/2} \ ,$$

and this implies the result. $\square$

2.1 Examples

To use the convex distance inequality, we must derive a saddle point representation for it.

Proposition 2.3. *For $A \subset \mathcal{X}^n$ measurable, let $\mathcal{M}(A)$ be the set of probability measures on A. Then for fixed $x \in \mathcal{X}$,*

$$\begin{aligned} d_T(x, A) &= \sup_{\alpha : \|\alpha\|_2 = 1} \inf_{\nu \in \mathcal{M}(A)} \sum_{j=1}^n \alpha_j \nu(Y_j \neq x_j) \\ &= \inf_{\nu \in \mathcal{M}(A)} \sup_{\alpha : \|\alpha\|_2 = 1} \sum_{j=1}^n \alpha_j \nu(Y_j \neq x_j) \ , \end{aligned}$$

where Y has distribution ν.

Before moving to the proof, let us discuss how this proposition relates our first definition of convex distance to another. By Cauchy-Schwarz, any vector $a \in \mathbb{R}^n$ satisfies $\|a\|_2 = \sup_{\alpha : \|\alpha\|_2 = 1} \langle a, \alpha \rangle$. Therefore we can write the second line above as

$$\inf_{\nu \in \mathcal{M}(A)} \left(\sum_{j=1}^n \left(\mathbb{E}_\nu \mathbf{1}_{\{Y_j \neq x_j\}} \right)^2 \right)^{1/2} = \inf_{\nu \in \mathcal{M}(A)} \left\| \mathbb{E}_\nu \left(\mathbf{1}_{\{Y_1 \neq x_1\}}, \ldots, \mathbf{1}_{\{Y_n \neq x_n\}} \right) \right\|_2 \ . \tag{9}$$

This quantity can be interpreted as follows. For $x \in \mathcal{X}$ fixed, we can measure the error in approximating x by a point $y \in A$ using the vector of "misses"

$$h(x, y) = (\mathbf{1}_{\{y_1 \neq x_1\}}, \ldots, \mathbf{1}_{\{y_n \neq x_n\}}) \ .$$

The number of 1's here is equal to the number of coordinates in which y fails to approximate x. The Hamming distance from x to A equals

$$d_H(x, A) = \min_{y \in A} \|h(x, y)\|_1 \ .$$

However the Hamming distance does not capture the fact that some points outside of A can be approximated by many different points in A, often in very different coordinates. If we would like to design a distance that includes this information (that is, it makes such points "closer" to A), one possibility is to average the vector $h(x, y)$ over $y \in A$. For example, if $x = (0, 0) \in \mathbb{R}^2$ and $A = \{(5, 0), (0, 5)\}$, then the Hamming distance from x to A is 1, whereas if we take an equal average of the vector $h(x, y)$ over $y \in A$, we obtain $(1/2, 1/2)$. In this case we would take Euclidean norm of the vector (since ℓ^1 norm will still give a distance of 1), so we obtain a "distance" from x to A of $\|(1/2,1/2)\|_2 = 1/\sqrt{2} < 1$.

In our averaging above, there is no good reason to average uniformly over A, so we define the distance as

$$\inf_{\nu \in \mathcal{M}(A)} \|\mathbb{E}_\nu h(x,Y)\|_2 \ .$$

This is equal to the right side of (9). In fact, we can interpret this quantity geometrically. The set

$$U_A(x) = \{h(x,y) : y \in A\} \subset \mathbb{R}^n$$

is our set of "error" vectors in approximating by A. For each $\nu \in \mathcal{M}(A)$, the vector $\mathbb{E}_\nu h(x,y)$ lies in the convex hull of $U_A(x)$. Conversely, each point in the convex hull can be represented in this form. Therefore if we set $V_A(x)$ to be the convex hull of $U_A(x)$, then

$$d_T(x) = \inf_{v \in V_A(x)} \|v\|_2$$

is the minimal distance from the origin in $\mathbb{R}^n$ to the convex hull of error vectors. This is actually the definition given in Talagrand's paper *A new look at independence.* It seems that one of the beauties of this definition is that we have transformed the problem of measuring distances on $\mathcal{X}^n$, which has no structure whatsoever, to a problem of measuring distances on $\mathbb{R}^n$, where we can take combinations, convex hulls, etc, only using the Hamming metric.

Proof of Proposition 2.3. For the first equality, write for $\|\alpha\|_2 = 1$ and $\nu \in \mathcal{M}(A)$

$$\begin{aligned}\sum_{j=1}^n \alpha_j \nu(Y_j \neq x_j) = \mathbb{E}_\nu \left(\sum_{j=1}^n \alpha_j \mathbf{1}_{\{Y_j \neq x_j\}} \right) = \mathbb{E}_\nu d_\alpha(x,Y) &\geq \min_{y \in A} d_\alpha(x,y) \\ &= d_\alpha(x,A) \ .\end{aligned}$$

This shows $d_T(x,A) \geq \sup_{\alpha : \|\alpha\|_2 = 1} \inf_{\nu \in \mathcal{M}(A)} \sum_{j=1}^n \alpha_j \nu(Y_j \neq x_j)$. On the other hand, if $\|\alpha\|_2 = 1$ then let $y \in A$ be such that $d_\alpha(x,A) = d_\alpha(x,y)$. Putting $\nu = \delta_y$, one has $\mathbb{E}_\nu d_\alpha(x,Y) = d_\alpha(x,A)$. Therefore

$$\inf_{\nu \in \mathcal{M}(A)} \sum_{j=1}^n \alpha_j \nu(Y_j \neq x_j) \leq d_\alpha(x,A) \ .$$

Now take supremum over α to get the other inequality.

For the second equality, it will be useful to rewrite the optimization problem. For our fixed $x \in \mathcal{X}^n$, let $\chi : A \to \{0,1\}^n$ be given by

$$\chi(y_1, \ldots, y_n) = (\mathbf{1}_{\{y_1 \neq x_1\}}, \ldots, \mathbf{1}_{\{y_n \neq x_n\}}) \ .$$

Then writing $\omega = (\omega_1, \ldots, \omega_n)$ for the image $\chi(y) = \chi(y_1, \ldots, y_n)$, one has

$$\nu(Y_j \neq x_j) = (\nu \circ \chi^{-1})(\omega_j = 1) \ .$$

So if we put $\mathcal{M}(A) \circ \chi^{-1}$ for the image measures under the map χ, we can rewrite

$$\sup_{\alpha:\|\alpha\|_2=1} \inf_{\nu \in \mathcal{M}(A)} \sum_{j=1}^n \alpha_j \nu(Y_j \neq x_j) = \sup_{\alpha:\|\alpha\|_2=1} \inf_{\mu \in \mathcal{M}(A)\circ\chi^{-1}} \sum_{j=1}^n \alpha_j \mu(\omega_j = 1) \ .$$

To interchange the sup and inf, use Sion's theorem:

Lemma 2.4. *Let $\mathcal{X}$ be a compact convex subset of a topological vector space and $\mathcal{Y}$ be a subset of a topological vector space. If $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ has the properties:*

- *$f(x, \cdot)$ is upper semi-continuous and concave on $\mathcal{Y}$ for all $x \in \mathcal{X}$ and*
- *$f(\cdot, y)$ is lower semi-continuous and convex on $\mathcal{X}$ for all $y \in \mathcal{Y}$,*

then

$$\inf_x \sup_y f(x,y) = \sup_y \inf_x f(x,y) \ .$$

Proof. See H. Komiya, 1988. □

To complete the proof we must verify the assumptions of the lemma. Take $\mathcal{X} = \mathcal{M}(A) \circ \chi^{-1}$, which we claim is compact and convex in the space of measures on $\{0,1\}^n$. To show this, let $\mu_1, \mu_2 \in \mathcal{M}(A) \circ \chi^{-1}$ and $\lambda \in [0,1]$, so that there are $\nu_1, \nu_2 \in \mathcal{M}(A)$ with $\mu_i = \nu_i \circ \chi^{-1}$. Then $(\lambda\nu_1 + (1-\lambda)\nu_2) \circ \chi^{-1} = \lambda\mu_1 + (1-\lambda)\mu_2$, so $\mathcal{M}(A) \circ \chi^{-1}$ is convex. For compactness, let (ν_n) be a sequence in $\mathcal{M}(A)$. Then by compactness of the space of measures on $\{0,1\}^n$, there is a subsequence $\nu_{n_k} \circ \chi^{-1}$ that converges to some μ. Enumerate the elements of $\{0,1\}^n$ as $\tau_1, \ldots, \tau_{2^n}$ and write $p_i = \mu(\{\tau_i\})$. We must write μ as the image of some $\nu \in \mathcal{M}(A)$ under χ, so for each i such that $p_i > 0$, define $A_i = \chi^{-1}(\{\tau_i\})$. Since A_i is nonempty (this follows from weak convergence to μ, so $\nu_{n_k}(A_i) = \nu_{n_k} \circ \chi^{-1}(\{\tau_i\}) > 0$ for large k), we may choose $y_i \in A_i$ and define $\nu = \sum_{i:p_i>0} p_i \delta_{y_i}$. This is a probability measure on A and completes the proof of compactness.

Next, take $\mathcal{Y} = \{\alpha \in \mathbb{R}^n : \|\alpha\|_2 = 1,\ \alpha_i \geq 0 \text{ for all } i\}$ as a subset of $\mathbb{R}^n$. The function f is then

$$f(\nu, \alpha) = \sum_{j=1}^n \alpha_j \mathbb{E}_\nu \mathbf{1}_{\{Y_j \neq x_j\}} \ .$$

For fixed $\nu \in \mathcal{M}(A)$, this is just a linear function of the α_j's, so it is upper semi-continuous and concave. For fixed $\alpha \in \mathcal{Y}$, we must show it is lower semi-continuous and convex in ν. It suffices to show this for the function $\nu \mapsto \mathbb{E}_\nu \mathbf{1}_{\{Y_j \neq x_j\}}$ for a fixed j. Since $\mathbf{1}_{\{Y_j \neq x_j\}}$ is a (bounded) continuous function on $\{-1,+1\}^n$, any sequence (ν_n) in $\mathcal{M}(A)$ converging weakly to ν must have $\mathbb{E}_{\nu_n}(Y_j \neq x_j) \to \mathbb{E}_\nu(Y_j \neq x_j)$. This shows continuity. Furthermore $\nu \mapsto \mathbb{E}_\nu \mathbf{1}_{\{Y_j \neq x_j\}}$ is simply linear in ν so this proves convexity.

By Sion's theorem, we obtain

$$\sup_{\alpha:\|\alpha\|_2=1} \inf_{\mu\in\mathcal{M}(A)\circ\chi^{-1}} \sum_{j=1}^n \alpha_j \mu(\omega_j = 1) = \inf_{\mu\in\mathcal{M}(A)\circ\chi^{-1}} \sup_{\alpha:\|\alpha\|_2=1} \sum_{j=1}^n \alpha_j \mu(\omega_j = 1)$$
$$= \inf_{\nu\in\mathcal{M}(A)} \sup_{\alpha:\|\alpha\|_2=1} \sum_{j=1}^n \alpha_j \mu(Y_j \neq x_j) \ .$$

□

Convex Lipschitz functions. Using Talagrand's inequality, we can strengthen the previous concentration inequality for convex lipschitz functions.

Theorem 2.5. *Let $X = (X_1, \ldots, X_n)$ be independent with values in $[0,1]$. Suppose $f : [0,1]^n \to \mathbb{R}$ is 1-Lipschitz and quasi-convex: the set $\{x : f(x) \leq s\}$ is convex for each $s \in \mathbb{R}$. Then $Z = f(X)$ satisfies*

$$\max\{\mathbb{P}(Z - MZ > t), \mathbb{P}(Z - MZ < -t)\} \leq 2e^{-t^2/4} \ ,$$

where MZ is a median of Z.

Proof. Let $s \in \mathbb{R}$ and define $A_s = \{x : f(x) \leq s\}$. Then A_s is convex. We first claim that for any convex $A \subset [0,1]^n$, one has

$$d(x, A) \leq d_T(x, A) \ ,$$

where $d(x, A) = \inf_{y\in A} \|x - y\|_2$. To see why, first note that

$$d(x, A) = \inf_{\nu\in\mathcal{M}(A)} \|x - \mathbb{E}_\nu Y\|_2 \ ,$$

where Y has law ν. The inequality $\leq$ follows from the fact that since A is convex and $Y \in A$ almost surely, also $\mathbb{E}_\nu Y \in A$. The other inequality follows from using measures $\nu = \delta_y$ for $y \in A$. We can rewrite the above as

$$\inf_{\nu\in\mathcal{M}(A)} \sqrt{\sum_{j=1}^n (\mathbb{E}_\nu(x_j - Y_j))^2} \leq \inf_{\nu\in\mathcal{M}(A)} \sqrt{\sum_{j=1}^n (\mathbb{E}_\nu \mathbf{1}_{\{x_j \neq Y_j\}})^2}$$

since x_j and Y_j are in $[0,1]$. For any vector $a = (a_1, \ldots, a_n)$, Cauchy-Schwarz says $\|a\|_2 = \sup_{\alpha:\|\alpha\|_2=1} \langle a, \alpha \rangle$, so the above is equals to

$$\inf_{\nu\in\mathcal{M}(A)} \sup_{\alpha:\|\alpha\|_2=1} \sum_{j=1}^n \alpha_j \mathbb{E}_\nu \mathbf{1}_{\{x_j \neq Y_j\}} = d_T(A) \ .$$

We now apply the convex distance inequality: for $t \geq 0$,

$$\mathbb{P}(d(X, A_s) \geq t) \leq \mathbb{P}(d_T(X, A_s) \geq t) \leq (\mathbb{P}(A_s))^{-1} e^{-t^2/4} \ .$$

By the Lipschitz condition, if $f(x) > s + t$ then $d(x, A_s) \geq t$. Therefore taking $s = Mf$, a median of f,

$$\mathbb{P}(f(X) > t + Mf) \leq 2e^{-t^2/4} \ .$$

For the other bound, we choose $s = Mf - t$ and note that $f(x) \geq Mf$ implies that $d(x, A_s) \geq t$, so

$$1/2 \leq \mathbb{P}(f(X) \geq Mf) \leq \mathbb{P}(d_T(X, A_s) \geq t) \leq (\mathbb{P}(A_s))^{-1} e^{-t^2/4} \ .$$

So

$$\mathbb{P}(f(X) < Mf - t) \leq \mathbb{P}(A_s) \leq 2e^{-t^2/4} \ .$$

□

Longest increasing subsequence. Let $X = (X_1, \ldots, X_n)$ be i.i.d. uniform $[0, 1]$ random variables and let $L_n = L_n(X)$ be the length of the longest increasing subsequence of X.

Theorem 2.6. *For all $t > 0$,*

$$\mathbb{P}(L_n \geq M + t) \leq 2 \exp\left(-\frac{t^2}{4(M+t)}\right)$$

and

$$\mathbb{P}(L_n \leq M - t) \leq 2 \exp\left(-\frac{t^2}{4M}\right) \ ,$$

where M is a median of L_n.

Proof. Define $A_s = \{y \in [0, 1]^n : L_n(y) \leq s\}$. We first relate the length of the longest increasing subsequence to the convex distance from A_s. We claim that

$$L_n(x) \geq s + u \Rightarrow d_T(x, A_s) \geq \frac{u}{\sqrt{s + u}} \ . \tag{10}$$

To show this, pick a set of indices I of cardinality $L_n \geq s + u$ which correspond to a longest increasing subsequence of x. Defining

$$\alpha_i = \begin{cases} 1 \text{ if } i \in I \\ 0 \text{ otherwise} \end{cases} \ ,$$

one has $\|\alpha\|_2 = \sqrt{L_n}$, and so

$$d_T(x, A_s) \geq L_n^{-1/2} \inf_{y \in A_s} \sum_{i=1}^{n} \alpha_i \mathbf{1}_{\{x_i \neq y_i\}} = L_n^{-1/2} \inf_{y \in A_s} \#\{i \in I : x_i \neq y_i\} \ .$$

This infimum is of a finite set, so it is attained:

$$d_T(x, A_s) \geq L_n^{-1/2} \#\{i \in I : x_i \neq y_i\} \text{ for some } y \in A_s \ .$$

Call J the set on the right, so that $(y_i : i \in I \setminus J)$ is an increasing subsequence. As $y \in A_s$, one has $\#I - \#J \leq s$. In other words

$$L_n \leq \#J + s \leq L_n^{1/2} d_T(x, A_s) + s \ ,$$

giving $d_T(x, A_s) \geq \frac{L_n - s}{\sqrt{L_n}} \geq \frac{u}{\sqrt{s+u}}$ (since $v \mapsto \frac{v-a}{\sqrt{v}}$ is monotone for $v > 0$ and $a \geq 0$). This shows (10).

By (10) with $s = M - t$ and $u = t$ and Talagrand's inequality,

$$1/2 \leq \mathbb{P}(L_n \geq M) \leq \mathbb{P}(d_T(X, A_s) \geq t/\sqrt{M}) \leq (\mathbb{P}(A_s))^{-1} e^{-t^2/(4M)} \ ,$$

or

$$\mathbb{P}(A_s) \leq 2e^{-t^2/(4M)} \ .$$

For the other inequality, let $s = M$ and use (10) again:

$$\mathbb{P}(L_n \geq M + t) \leq \mathbb{P}\left(d_T(X, A_s) \geq \frac{t}{\sqrt{M+t}}\right) \leq 2\exp\left(-\frac{t^2}{4(M+t)}\right) \ .$$

□

Bin Packing. If $X = (X_1, \ldots, X_n)$ are independent variables taking values in $[0, 1]$ then we define $N_n = N_n(X)$ as the minimal number of bins of size 1 needed to pack n objects whose sizes are $X_1, \ldots, X_n$. We will now prove a concentration inequality for N_n from Talagrand's inequality. Recall that if we apply the bounded differences inequality, we obtain

$$\mathbb{P}(|N_n - \mathbb{E}N_n| > t) \leq 2e^{-t^2/n} \ .$$

We, however, expect this to be suboptimal in certain cases. For example, assume that the x_i's are small, say x_i is uniformly distributed on $[0, 1/i]$. Then simply by packing objects consecutively in bins (deterministically), we can get a bound of $N_n \leq C \log n$. Here the concentration inequality above is not very good. Talagrand gives the following improvement.

Theorem 2.7. *Define $\sigma_n^2 = \sum_{i=1}^n \mathbb{E}X_i^2$. Then*

$$\mathbb{P}(|N_n - MN_n| \geq t + 1) \leq 8\exp\left(-\frac{t^2}{16(2\sigma_n^2 + t)}\right) \ \textit{for } t > 0 \ ,$$

where MN_n is a median of N_n.

Proof. We start with a simple bound:

$$N_n \leq 2\sum_{i=1}^n x_i + 1 \ .$$

To see this, start with each item in a separate box. Then iteratively take any two boxes than are no more than half-full and dump them into one box. When this procedure ends, we will have all boxes that are at least half-full, with the exception of possibly one. If we have M_n boxes at the end with boxes $1, \ldots, M_n - 1$ at least half full, then

$$M_n = 2 \sum_{i=1}^{M_n-1} \frac{1}{2} + 1 \leq 2 \sum_{i=1}^{M_n-1} \left[\sum_{j:x_j \text{ in box } i} x_j \right] + 1 \leq 2 \sum_{i=1}^{n} x_i + 1 \ .$$

As a consequence of this bound, we note that

$$N_n(x) \leq N_n(y) + 2 \sum_{i:x_i \neq y_i} x_i + 1 \ . \tag{11}$$

This is because to pack the objects $x_1, \ldots, x_n$, we can pack the objects $y_1, \ldots, y_n$ and then remove only the ones that are not equal to any x_i. Then pack the remaining x_i's.

To prove the concentration inequalities, we again estimate the distance from the set $A_s = \{x : N_n(x) \leq s\}$. Given x, define $\alpha(x) = x/\|x\|_2$, one has

$$d_T(x, A_s) \geq d_{\alpha(x)}(x, A_s) = \inf_{y \in A_s} \sum_{i=1}^{n} (\alpha(x))_i \mathbf{1}_{\{x_i \neq y_i\}} = \|x\|_2^{-1} \inf_{y \in A_s} \sum_{i:x_i \neq y_i} x_i \ .$$

As y ranges over A_s, the sum on the right takes only finitely many values. So choosing a minimizing $y \in A_s$ and using (11),

$$N_n(x) \leq N_n(y) + 2\|x\|_2 d_T(x, A_s) + 1 \leq s + 2\|x\|_2 d_T(x, A_s) + 1 \ . \tag{12}$$

Typically $\|x\|_2$ will not be much larger than order σ_n, so we split into two events: for any s,

$$\begin{aligned} \mathbb{P}(N_n(X) \geq s + t + 1) &\leq \mathbb{P}(N_n(X) \geq s + t + 1, \|X\|_2 \leq \sqrt{2\sigma_n^2 + t}) + \mathbb{P}(\|X\|_2 \geq \sqrt{2\sigma_n^2 + t}) \\ &\leq \mathbb{P}\left(N_n(X) \geq s + t \frac{2\|X\|_2}{2\sqrt{2\sigma_n^2 + t}} + 1 \right) + \mathbb{P}(\|X\|_2 \geq \sqrt{2\sigma_n^2 + t}) \ . \end{aligned}$$

(The precise order $\sqrt{2\sigma_n^2 + t}$ is chosen to optimize the following bounds.) The first term is bounded using (12) and Talagrand's inequality as

$$\begin{aligned} \mathbb{P}\left(N_n(X) \geq s + t \frac{2\|X\|_2}{2\sqrt{2\sigma_n^2 + t}} + 1 \right) &\leq \mathbb{P}\left(d_T(X, A_s) \geq \frac{t}{2\sqrt{2\sigma_n^2 + t}} \right) \\ &\leq (\mathbb{P}(A_s))^{-1} \exp\left(-\frac{t^2}{16(2\sigma_n^2 + t)} \right) \ . \end{aligned} \tag{13}$$

For the second term, we will show that

$$\mathbb{P}(\|X\|_2 \geq \sqrt{2\sigma_n^2 + t}) \leq \exp\left(-\frac{3}{8}(\sigma_n^2 + t) \right) \ . \tag{14}$$

To do this, we use Bernstein's inequality. Recall that it states that if $Y_1, \ldots, Y_n$ are independent with $Y_i \leq b$ and $\mathbb{E}Y_i^2 < \infty$ for all i then for $t \geq 0$,

$$\mathbb{P}(Y_1' + \cdots + Y_n' \geq t) \leq \exp\left(-\frac{t^2}{2(\nu + bt/3)}\right) ,$$

where $Y_i' = Y_i - \mathbb{E}Y_i$ and $\nu = \sum_{i=1}^n \mathbb{E}Y_i^2$. We apply this to the variables X_i^2 with $b = 1$ and $\nu = \sum_{i=1}^n \mathbb{E}X_i^4 \leq \sigma_n^2$ to obtain

$$\begin{aligned}\mathbb{P}(X_1^2 + \cdots + X_n^2 \geq 2\sigma_n^2 + t) &= \mathbb{P}((X_1^2)' + \cdots + (X_n^2)' \geq \sigma_n^2 + t) \\ &\leq \exp\left(-\frac{(\sigma_n^2 + t)^2}{2(\nu + (\sigma_n^2 + t)/3)}\right) \\ &\leq \exp\left(-(\sigma_n^2 + t) \cdot \frac{\sigma_n^2 + t}{2(\sigma_n^2 + (\sigma_n^2 + t)/3)}\right) .\end{aligned}$$

Because

$$\frac{\sigma_n^2 + t}{2(\sigma_n^2 + (\sigma_n^2 + t)/3)} = \frac{1}{2\left(\frac{\sigma_n^2}{\sigma_n^2 + t} + 1/3\right)} \geq 3/8 ,$$

this shows (14).

Putting together (13) and (14), we obtain

$$\mathbb{P}(N_n(X) \geq s + t + 1) \leq (\mathbb{P}(A_s))^{-1} \exp\left(-\frac{t^2}{16(2\sigma_n^2 + t)}\right) + \exp\left(-\frac{3}{8}(\sigma_n^2 + t)\right) . \tag{15}$$

Taking $s = MN_n$, one has $\mathbb{P}(A_s) \geq 1/2$ so one has

$$\begin{aligned}\mathbb{P}(N_n(X) - MN_n(X) \geq t + 1) &\leq 2\exp\left(-\frac{t^2}{16(2\sigma_n^2 + t)}\right) + \exp\left(-\frac{3}{8}(\sigma_n^2 + t)\right) \\ &\leq 3\exp\left(-\frac{t^2}{16(2\sigma_n^2 + t)}\right) .\end{aligned}$$

□

2.2 Proof of the convex distance inequality

Talagrand's initial proof was by induction on n. We will give the proof of BLM (2003). It is by the entropy method but gives an upper bound of $e^{-t^2/10}$ instead of $e^{-t^2/4}$.

The first step is to show the "(4, 0)-weakly self-bounding property" of $f(x) = d_T^2(x, A)$:

$$\sum_{i=1}^n \left(f(x) - f_i(x^{(i)})\right)^2 \leq 4f(x) \text{ with } 0 \leq f(x) - f(x^{(i)}) \leq 1 \tag{16}$$

where

$$f_i(x^{(i)}) = \inf_{x_i'} f(x_1, \ldots, x_{i-1}, x_i', x_{i+1}, \ldots, x_n) .$$

(For simplicity, BLM assume that this infimum is achieved. Otherwise one can do a limiting argument.) It is clear that $f(x) - f(x^{(i)}) \geq 0$. To show it is bounded by 1, first recall that we can write

$$d_T(x, A) = \inf_{\nu \in \mathcal{M}(A)} \|v_\nu(x)\|_2 \ ,$$

where $v_\nu(x) = (\mathbb{E}_\nu \mathbf{1}_{\{Y_1 \neq x_1\}}, \ldots, \mathbb{E}_\nu \mathbf{1}_{\{Y_n \neq x_n\}})$. Furthermore by compactness of the image space $\mathcal{M}(A) \circ \chi^{-1}$ (see the application of Sion's theorem in the saddle point representation in Section 2.1), we can find $\nu \in \mathcal{M}(A)$ such that

$$f(x) = d_T^2(x, A) = \|v_\nu(x)\|_2^2 \ .$$

Write $\nu^{(i)}$ for the element of $\mathcal{M}(A)$ that in this way corresponds to $f_i(x^{(i)})$ (and ν is the one corresponding to $f(x)$). Then

$$\begin{aligned}
f(x) - f(x^{(i)}) &\leq \left[\sum_{j \neq i} \left(\mathbb{E}_{\nu^{(i)}} \mathbf{1}_{\{Y_j \neq x_j\}}\right)^2 + \left(\mathbb{E}_{\nu^{(i)}} \mathbf{1}_{\{Y_i \neq x_i\}}\right)^2\right] \\
&\quad - \left[\sum_{j \neq i} \left(\mathbb{E}_{\nu^{(i)}} \mathbf{1}_{\{Y_j \neq x_j\}}\right)^2 + \left(\mathbb{E}_{\nu^{(i)}} \mathbf{1}_{\{Y_i \neq x_i^{(i)}\}}\right)^2\right] \\
&= \left(\mathbb{E}_{\nu^{(i)}} \mathbf{1}_{\{Y_i \neq x_i\}}\right)^2 - \left(\mathbb{E}_{\nu^{(i)}} \mathbf{1}_{\{Y_i \neq x_i^{(i)}\}}\right)^2 \\
&\leq 1 \ .
\end{aligned}$$

Last we focus on the inequality $\sum_{i=1}^n (f(x) - f(x^{(i)}))^2 \leq 4f(x)$. First using $f(x^{(i)}) \leq f(x)$, we first bound

$$\begin{aligned}
\sum_{i=1}^n (f(x) - f(x^{(i)}))^2 &= \sum_{i=1}^n \left[(\sqrt{f(x)} - \sqrt{f(x^{(i)})})^2 (\sqrt{f(x)} - \sqrt{f(x^{(i)})})^2\right] \\
&\leq 4 \sum_{i=1}^n f(x) \left(\sqrt{f(x)} - \sqrt{f(x^{(i)})}\right)^2 \ .
\end{aligned}$$

So once we show

$$\sqrt{f(x)} - \sqrt{f(x^{(i)})} \leq \alpha_i \ , \tag{17}$$

where

$$\alpha_i = \frac{\mathbb{E}_\nu \mathbf{1}_{\{Y_i \neq x_i\}}}{\|v_\nu(x)\|_2}$$

is the i-th coordinate of the minimizing vector α in the definition of the convex distance, we will be done with the first step. (ν is still the minimizing measure from above for $f(x)$, so that here $d_T(x, A) = \langle \alpha, v_\nu(x) \rangle$.) Let $\nu^{(i)}$ and $\alpha^{(i)}$ be the corresponding measure and vector for the quantity $f(x^{(i)})$. Then

$$\sqrt{f(x^{(i)})} = \langle \alpha^{(i)}, v_{\nu^{(i)}}(x^{(i)}) \rangle = \inf_{\hat{\nu}} \sup_{\hat{\alpha}: \|\hat{\alpha}\|_2 = 1} \langle \hat{\alpha}, v_{\hat{\nu}}(x^{(i)}) \rangle \geq \inf_{\hat{\nu}} \langle \alpha, v_{\hat{\nu}}(x^{(i)}) \rangle \ .$$

Picking ν' that minimizes this expression (again this uses compactness), we also have

$$\sqrt{f(x)} = \inf_{\hat{\nu}} \langle \alpha, v_{\hat{\nu}}(x) \rangle \le \langle \alpha, v_{\nu'}(x) \rangle \ .$$

Since $v_{\nu'}(x)$ and $v_{\nu'}(x^{(i)})$ are equal except perhaps in the i-th coordinate,

$$\sqrt{f(x)} - \sqrt{f(x^{(i)})} \le \alpha_i \mathbb{E}_{\nu'}(\mathbf{1}_{\{Y_i \neq x_i\}} - \mathbf{1}_{\{Y_i \neq x_i^{(i)}\}}) \le \alpha_i \ .$$

This proves (17).

We now must prove a concentration inequality for $(4,0)$-weakly self-bounding functions. The result is:

Lemma 2.8. *Suppose that f is $(4,0)$-weakly self-bounding. Then*

$$\mathbb{P}(f - \mathbb{E}f > t) \le \exp\left(-\frac{t^2}{8\mathbb{E}f + 4t}\right) \text{ for } t > 0 \ .$$

Furthermore,

$$\mathbb{P}(f - \mathbb{E}f < -t) \le \exp\left(-\frac{t^2}{8\mathbb{E}f}\right) \ .$$

Proof. We will use the modified LSI that we derived in the section on the entropy method: setting $\psi(x) = e^x - x - 1$, then for all $t \in \mathbb{R}$,

$$Ent \ e^{tf(x)} \le \sum_{i=1}^n \mathbb{E}\left[e^{tf(x)} \psi(-t(f(x) - f_i(x^{(i)})))\right] \ .$$

For $u \le 0$, one has $\psi(u) \le u^2/2$, so we obtain the upper bound

$$\frac{t^2}{2}\mathbb{E}\left[e^{tf(x)} \sum_{i=1}^n (f(x) - f_i(x^{(i)}))^2\right] \le 2t^2 \mathbb{E}\left[f(x) e^{tf(x)}\right] \ .$$

Setting $G(t) = \log \mathbb{E} e^{t(f - \mathbb{E}f)}$, this inequality is rewritten as

$$\left(\frac{1}{t} - 2\right) G'(t) - \frac{G(t)}{t^2} \le 2\mathbb{E}f \text{ for } t \in [0, 1/2) \ .$$

The left side is the derivative of $(1/t - 2)G(t)$, so if we integrate this, we obtain

$$G(t) \le \frac{2t^2}{1 - 2t} \text{ for } t \in [0, 1/2) \ . \tag{18}$$

Applying the Chernoff bound to this results in the upper-tail inequality.

For the lower-tail inequality, we skip the details. You can find them on p. 192-194 of the text. The main idea is to obtain a differential inequality, but this time it is

$$(t - 4\psi(-t))G'(t) - G(t) \le 4\psi(-t)\mathbb{E}f \ .$$

This can be integrated after some work and a technical lemma for

$$G(t) \leq \frac{t^2}{2(1-\frac{11}{6}t)} \cdot 4\mathbb{E}f \text{ for } t \in (-1/4, 0) \ .$$

One then argues that the restriction on t can be removed and applies Chernoff for the bound in the lemma. $\square$

Last we use the concentration bounds to complete the proof of the convex distance inequality. It is now similar to the bound we gave on d_α for fixed α. First note that $X \in A$ if and only if $d_T(X, A) = 0$. So use the lower tail bound from the previous lemma with $t = \mathbb{E}d_T^2(X, A)$ for

$$\mathbb{P}(X \in A) = \mathbb{P}(d_T^2(X, A) - \mathbb{E}d_T^2(X, A) \leq -t) \leq \exp\left(-\frac{\mathbb{E}d_T^2(X, A)}{8}\right) \ ,$$

which is the same as

$$\mathbb{P}(X \in A) \exp\left(\frac{\mathbb{E}d_T^2(X, A)}{8}\right) \leq 1 \ .$$

Now use the upper tail bound (specifically the bound (18)) for

$$\log \mathbb{E}e^{t(d_T^2(X,A) - \mathbb{E}d_T^2(X,A))} \leq \frac{2t^2 \mathbb{E}d_T^2(X, A)}{1-2t} \text{ for } t \in [0, 1/2) \ .$$

Setting $t = 1/10$,

$$\mathbb{E}e^{d_T^2(X,A)/10} \leq e^{\mathbb{E}d_T^2(X,A)/10} e^{\mathbb{E}d_T^2(X,A)/50} \leq e^{\mathbb{E}d_T^2(X,A)/8} \ .$$

Therefore

$$\mathbb{P}(X \in A) \mathbb{E}e^{d_T^2(X,A)/10} \leq 1 \ .$$

By Markov,

$$\mathbb{P}(d_T(X, A) \geq t) \leq e^{-t^2/10} \mathbb{E}e^{d_T^2(X,A)/10} \leq (\mathbb{P}(X \in A))^{-1} e^{-t^2/10} \ .$$